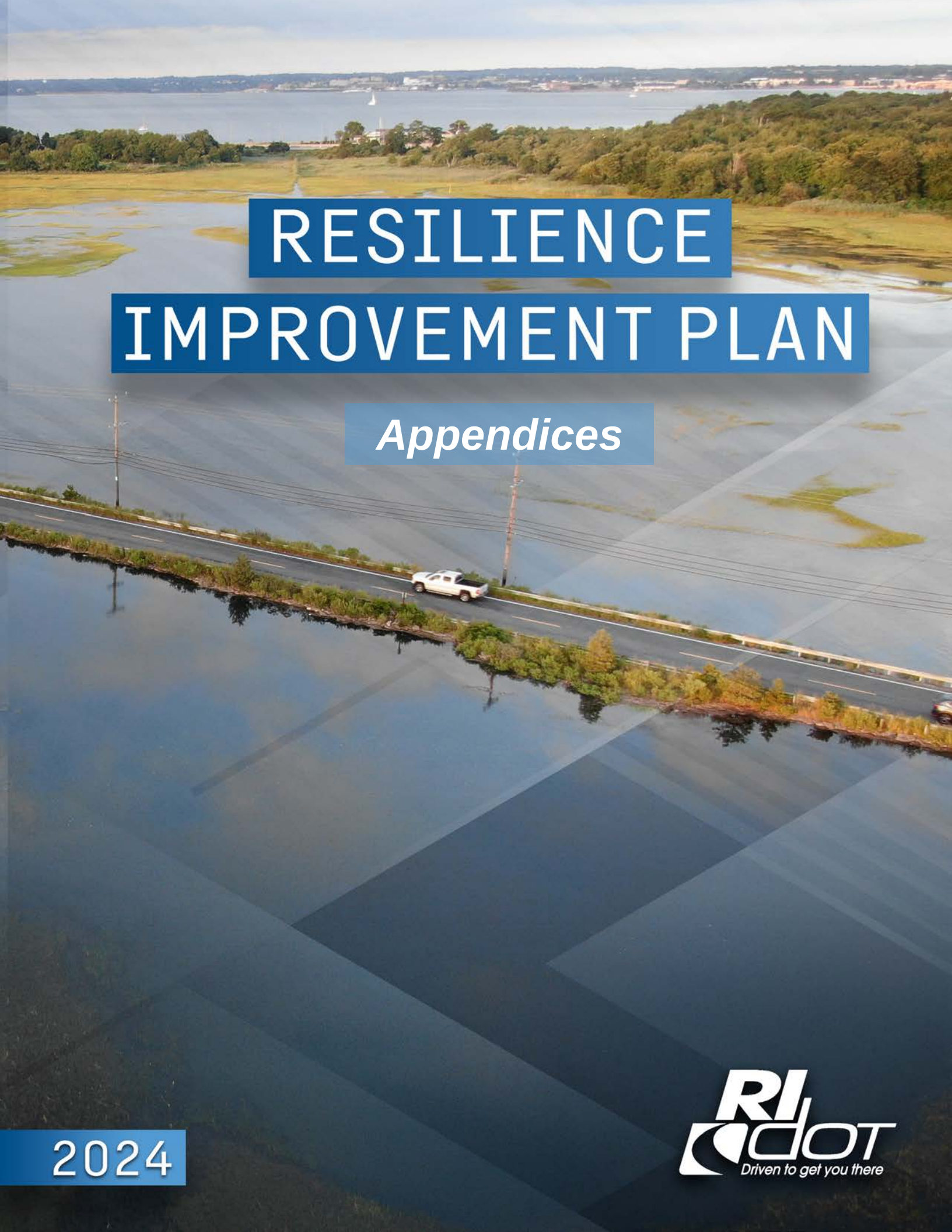


An aerial photograph of a coastal area. In the foreground, a road runs diagonally across the frame, with a white pickup truck driving on it. The road is flanked by flooded fields and marshes. In the background, there is a large body of water, likely a bay or harbor, with a distant shoreline visible under a cloudy sky.

RESILIENCE IMPROVEMENT PLAN

Appendices

2024

Rhode Island Department of Transportation Resilience Improvement Plan

Appendices

Rhode Island Department of Transportation

date

June 2024

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APPENDIX A REVIEW OF RESILIENCE BEST PRACTICES AND DEVELOPMENT OF RESILIENCE IMPROVEMENT PLAN FRAMEWORK



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A.1 Introduction

To support Rhode Island Department of Transportation (RIDOT) in developing a Resilience Improvement Plan (RIP), Cambridge Systematics (CS) undertook a review of best practices in improving resilience in state departments of transportation (DOT) and conducted interviews with eight peer agencies. This tech memo summarizes this review and subsequently provides a framework to guide development of the RIP.

A.1.1 Background

Transportation assets in Rhode Island are facing many current and future climate stressors such as sea level rise, storm surge, flooding, coastal erosion, and other extreme weather events. According to NOAA National Centers for Environmental Information, Rhode Island is projected to experience continued increases in frequency and intensity of extreme precipitation events, sea level rise, and coastal flooding, which could damage infrastructure, increase repair and maintenance costs, and disrupt normal operations of transportation systems.

RIDOT's Long Range Transportation Plan (LRTP), Moving Rhode Island 2040, identifies coastal resiliency as a concern for the state's coastal counties, and highlights the need to prepare for more frequent flooding and thawing cycles. Improving asset resilience to climate change, storm surge, and sea level rise is also a key component of asset management; a priority focus of the State Transportation Improvement Program (STIP). In addition, RIDOT has conducted several studies and assessments to understand transportation assets' vulnerability to sea level rise and storm surge, and to develop strategies to incorporate resilience into project planning.

To further its effort to improve resilience of transportation assets and processes within the agency, RIDOT is developing a RIP that will evaluate transportation infrastructure vulnerabilities and risks to natural and climate stressors, establish resilience strategies, and guide the implementation of resilient projects.

The recent passing of the Infrastructure Investment and Jobs Act (IIJA) transportation funding bill in 2021 provides RIDOT an opportunity to leverage additional funding when integrating resilience. As part of the IIJA, by developing RIPs, DOTs and metropolitan planning organizations (MPOs) can reduce local funding match requirements by up to 10 percent for Promoting Resilient Operations for Transformative, Efficient, and Cost-Saving Transportation (PROTECT) Formula and Discretionary Program applications.¹

At a minimum, the RIPs must accomplish the following objectives:

Exhibit 1-1 Resilience Improvement Plan Components

PROTECT - Resilience Improvement Plan

¹ Applicants are typically required to account for a 20 percent non-federal match for capital projects. Through the development of a RIP however, for state DOTs and MPOs this non-federal match can drop to 13 percent. Furthermore, if the RIP is integrated into the agency's long-range transportation plan (LRTP), the non-federal match drops to 10 percent.



- Define the objectives and scope of the RIP by taking a long-term planning and a system wide approach to achieving system resilience.
- Include a risk-based assessment of vulnerabilities of transportation assets and systems to current and future weather events and natural disasters, such as severe storms, flooding, drought, levee and dam failure, wildfire, rockslides, mudslides, sea-level rise, extreme weather, extreme temperatures, and earthquakes.
- Develop strategies that include both immediate and long-range planning activities and resilience investments. These strategies could include the benefits of natural infrastructure.
- Ensure that the RIP is ready for integration and implementation if it's consistent with and complements state and local hazard mitigation plans and incorporates codes, standards, and regulatory framework to ensure improvements; and

Tasked with maintaining the transportation infrastructure network of coastal and dense state, the development of a RIP will allow RIDOT to best position itself to address resilience needs.

The Plan Shall..	
1	Encompass immediate and long-range planning activities and resilience investments
2	Demonstrate a system-wide approach to transportation system resilience
3	Consistent with and complement State and local hazard mitigation plans
4	Include a risk-based assessment of vulnerability to current and future weather events and natural disasters
Shall, as appropriate..	
5	Describe ways to improve response to impacts and changes
6	Describe the codes, standards, and regulatory framework to ensure improvements
7	Consider benefit of natural Infrastructure
8	Assess community infrastructure resilience
9	Use a long-term planning period
May also..	
10	Designate evacuation routes and strategies
11	Plan for response to anticipated emergencies
12	Describe the resilience improvement policies
13	Include investment plan & priority projects
14	Use science and data

A.1.2 Purposes

Despite the well-documented and growing impacts of climate change on transportation infrastructure, guidance to develop a transportation resilience improvement plan and comprehensively integrate related decision making into DOT processes have been limited. On the other hand, in recent years, a number of state DOTs have established sustainability and resilience departments or initiatives aimed at better protecting transportation assets from climate change and extreme weather impacts, as well as reducing overall carbon footprints. Many State DOTs are also in the process of developing their RIPs.



Therefore, as an initial step in developing the RIDOT RIP, the project team determined that it would be beneficial to conduct interviews with peer agencies facing similar extreme weather and climate challenges as RIDOT to better understand the best practices in improving transportation resilience and summarize lessons learned from peer agencies' experience. Findings from the review and interviews will help develop a RIP framework to guide RIDOT RIP development.

This tech memo is aimed to:

- Assess the current state of practice in relation to resilience and the development of RIPs by state DOTs.
- Identify implications and takeaways for RIDOT's RIP development process.
- Develop a framework that outlines key steps to guide RIDOT RIP development.

A.1.3 Research & Interview Process

To accomplish the objectives above, the project team identified eight peer DOT agencies for the review of relevant resilience materials and for interview outreach, including Colorado DOT (CDOT), Delaware DOT (DelDOT), Florida DOT (FDOT), Maryland DOT (MDOT), Minnesota DOT (MnDOT), New Hampshire DOT (NH DOT), North Carolina DOT (NCDOT), and Virginia DOT (VDOT). These peer agencies were chosen based on a combination of physical and operational similarities in resilience needs to RIDOT and/or familiarity of resiliency specialization staff or initiatives at these agencies. These states also face similar climate and extreme weather challenges as Rhode Island. The selected peer agencies are shown in Exhibit 2.1.

The best practice review and interviews center around assessing the following topics:

- What is the current state of practice in relation to resilience and the development of RIPs by state DOTs?
- What are the main goals of state DOTs for developing RIPs and how do they envision RIPs to support their existing practices?
- What are best practices in measuring transportation vulnerability, risk, and resilience?
- What are the resilience strategies identified in best practice programs and initiatives?
- How do peer agencies integrate resilience considerations and adaptation strategies into project development process in these agencies?
- What is the best way to engage stakeholders and the public for resilience implementation?

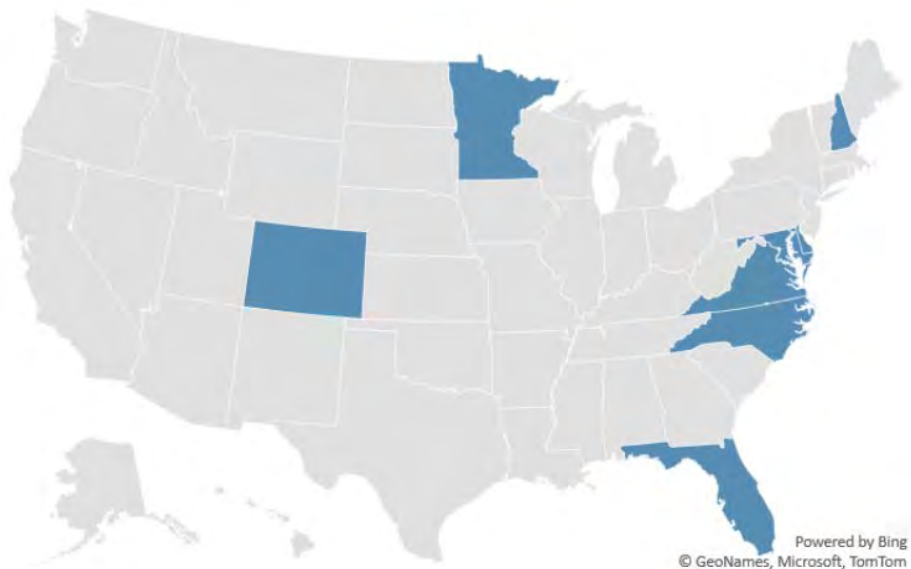
These topic areas were selected by considering the key components of plans and project development process in state DOTs and analyzing the requirements for RIP in the PROTECT funding implementation



guidance, which are related to establishing RIP objectives and study scope, assessing vulnerability and risks, developing strategies, integration, implementation, and coordination.

Findings and key takeaways from the interviews are summarized in Chapter 2.0.

Exhibit 1-2 Peer State DOT Agencies Engaged with for Best Practice Interviews



Following the peer agency interviews, the project team proceeded with the development of the RIP framework. The development of RIP framework is based on synthesizing best available frameworks for improvement resilience in transportation and other industries, lessons learn from peer agencies, and understanding RIDOT's current practices and previous resilience effort in Rhode Island, including data, methodology, tools, and initiatives. The RIP framework is aimed to establish an initial architecture for the RIP document and is described in Chapter 3.0.

A.2 Best Practice Findings & Key Takeaways

A.3 Resilience Progress and RIP Status

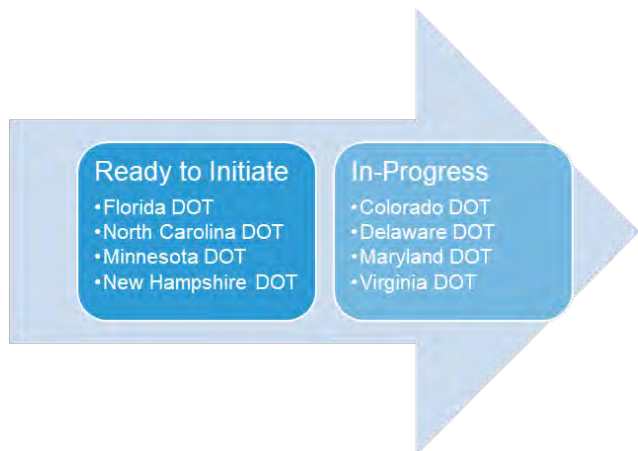
Similar to RIDOT, peer agencies are in the early phases of developing RIPs. Of eight agencies interviewed, half of them is in the process of developing their RIPs, and the other half of them are getting ready to initiate their RIP development.



Of all the peer agencies that were engaged with, Virginia DOT (VDOT) appears to have one of the more established RIP frameworks. Published in November 2022, VDOT's Resilience Plan meets approximately 80 percent of FHWA requirements. Although the document is a broader directional framework rather than a direct resilience plan, the document does set the stage for future resilience and sustainability efforts. The plan is broken out into the following objectives:

- Promote Data Driven Decisions
- Engage & Partner with Stakeholders
- Identify At-Risk Infrastructure & Prioritize Needs
- Utilize Feasibility & Cost Effectiveness Analysis
- Incorporate Resilience into Current Funding Policies

Exhibit 2-3 Peer Agency RIP Status



The document additionally includes appendices detailing current VDOT resilience efforts and an inventory of existing datasets. On the other hand, VDOT also indicated that the formal RIP will not include a prioritized project list, or evaluation of community infrastructure vulnerability.

CDOT is in the early stages of developing their RIP and shared a portion of the preliminary outline of the document. The document includes in the following components, expanding upon the Resilience Appendix of the Colorado Statewide Transportation Plan (STP):

- First section defining resiliency and highlighting the importance of CDOT's Resilience Program. CDOT plans to expand this section to include sections on responses to extreme weather (including snowplow routes and detour routes), recovery (information on the Colorado Recovery Office), and a broad introduction to the tradeoffs associated with risk management.
- CDOT plans to add an entirely new section which will cover resiliency in the transportation planning process. This will include information on CDOT's Risk Assessment Program, Risk Register, detour identification, resiliency mapping, and project scoring.
- The plan will also include recommendations, an assessment of resiliency in transportation planning at CDOT, and prioritization.

MDOT, which is also in the early stages of the development, indicated that the RIP will build upon multiple previous studies undertaken by the agency, including MDOT's coastal vulnerability assessments.



Getting ready to start the RIP development, NCDOT indicated that it will leverage the agency's resilience strategy adopted in 2021. The policy includes the following components:

- Define the core components of a policy framework.
- Assess the vulnerability and risk of key infrastructure in North Carolina's multimodal transportation network.
- Identify and pilot risk profiles to guide resilience efforts across the spectrum of NCDOT activities.
- Identify current and future opportunities to coordinate with federal and state agencies and local planning partners.

The remaining peer agencies indicated they have only very recently begun to commence development of their RIPs or have yet to formally begin. Nevertheless, every peer agency identified the planned inclusion of a risk assessment component, either directly as a future dedicated RIP chapter, or as the primary factor influencing overall RIP decision making. Best practices in relation to risk management are discussed in further detail in section 2.3.

A.4 Establish Objectives and Scope

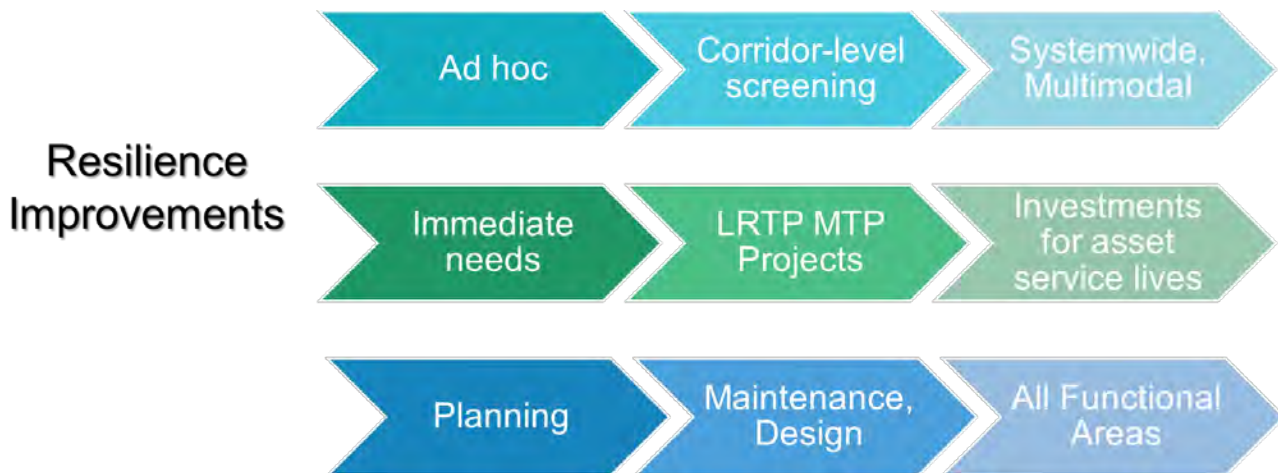
The ability to leverage PROTECT funding is a common motivation for peer agencies to develop RIPs. In addition, many of them also want to use RIP as an opportunity to improve resilience practice within the agencies, such as getting a better understanding of the vulnerability and risk to their transportation systems, enhancing planning and decision making for resilience projects to address not only immediate issues but also long-term resilience needs, and integrating and mainstreaming resilience across functional areas/departments within the agencies. These objectives are inline with the PROTECT funding requirements for a RIP that it should “*support efforts to identify vulnerabilities, develop proposed resilience solutions, and schedule and prioritize resilience improvements*”.

Resilience Progress & RIP Status Key Takeaways:

- Largely early stages of RIP development.
- Use of previous studies, initiatives, and frameworks as guidance can set a very strong foundation and starting point for RIP development.
- Risk assessment is a key component to be included in RIPs and added to agencies' current resilience efforts.



Exhibit 2-4 RIP Objectives from Best Practices



A.4.1 Expand Scope of Vulnerability Assessments

All peer agencies interviewed have conducted some levels of vulnerability assessments to their assets. However, many of the assessments are only performed at key corridors or for major projects as they come up and are mostly focused on assets for automobile travel (roads and bridges). To meet the PROTECT funding requirements, a RIP should “*demonstrate a system-wide approach*” and consider multimodal facilities. As such, many peer agencies would like to use RIPs to expand the scope of vulnerability assessments from corridor-level to system-wide, to consider additional multimodal facilities, and to add evaluation of risk by taking into account the likelihood and consequence of extreme weather and climate hazards. This will help the agencies develop a more comprehensive understanding of the vulnerability and risk facing their systems, assets, and operations.

In addition to incorporating vulnerability and risk in their plans, many peer agencies discussed the topic of criticality. Criticality is the degree to which a given asset is important in fulfilling the mission and goals of an agency. VDOT considers criticality and accessibility to physically vulnerable communities. NHDOT identified its most critical transportation infrastructure in terms of their importance and combined the asset criticality with asset risk tolerance to come with a prioritized list of assets. CDOT developed a criticality framework that considers factors related to socioeconomic, usage and operation, and health and safety.

A.4.2 Improve Resilience Planning and Decision Making

The PROTECT funding requirements also state that RIPs should “*use a long-term planning period*” and provide additional incentives (three percent further reduction of non-federal cost share) for RIPs to be incorporated into LRTP. Therefore, several peer agencies state in the interviews that they would like to enhance resilience practices from currently focusing on improvements for immediate needs (such as maintenance activities) to integrating resilience projects with long-term planning, and ultimately evaluating the benefit and cost of resilience



investments and developing strategies based on assets' full service lives (or longer). For example, CDOT currently assigns funding on a 'first come, first served' basis, assuming that the benefit-cost ratio of the proposed project is above 1.0. Moving forward, CDOT envisions further expanding its resilience funding strategy to include three separate funding sources for short-term and small-scale projects, projects on highly critical corridors, and longer-term proactive projects to address high vulnerability needs.

A.4.3 Better Integration of Resilience

Integrating resilience considerations into all DOT functional areas is another RIP objective discussed during the interviews. Similar to RIDOT, many peer agencies' resilience effort are led by one or two departments, usually by planning or asset management staff. Some agencies have adjusted maintenance schedules or design standards in response to extreme weather or climate challenges, but resilience has not been fully mainstreamed. Reviewing the agencies' missions and visions is helpful in establishing resilience that are closely related to activities in all DOT functional areas, including planning, environmental review, design, construction, operation, and maintenance. This is particularly important for identify system-wide resilience strategies and obtain staff buy-ins for implementation.

Objectives and Scopes Key Takeaways:

- Have a better understanding of the vulnerability and risk.
- Enhance planning and decision making for resilience projects to address immediate and long-term resilience needs.
- Integrate resilience considerations into all DOT functional areas

A.5 Assess Vulnerability and Risk

A.5.1 Vulnerability

FHWA defines **vulnerability** as the extent to which a transportation asset or system is susceptible to sustaining damage from hazards during extreme events. Vulnerability is a function of the extent to which an asset or system is exposed to damaging forces; its sensitivity to those forces; and its adaptive capacity².

- **Exposure** – The degree to which a transportation asset (road, bridge, etc.) experiences a hazard.
- **Sensitivity** – The degree to which an asset is damaged or service is interrupted by a climatic hazard, usually measures how the asset or system fares when exposed to coastal flooding.

² <https://highways.dot.gov/research/infrastructure/resilient-pavements/definitions>



- **Adaptive capacity** – The degree to which the system containing the asset (road, bridge, etc.) can adjust or mitigate the potential for damage or service interruption by the hazards, usually reflected by system redundancy or the availability of alternative options.

In the early stages of the RIP development process, peer agencies have varying degrees of experience in previous planning efforts to measure vulnerability. Most agencies have measured vulnerability using only exposure analysis and have not considered adaptive capacity and sensitivity of the transportation assets to the climate stressors in their vulnerability calculation.

There are three common approaches that can be used to assess vulnerability: stakeholder input approach, indicator – based approach, and engineering-informed approach. The first two are primarily used for system level analysis whereas the engineering informed approach focuses on specific transportation assets.

NCDOT and NHDOT used FHWA's Vulnerability Assessment tool (VAST), an indicator-based approach to measure vulnerability of transportation assets to climate stressors. Their vulnerability calculation included the three components of exposure, sensitivity, and adaptive capacity. USDOT developed VAST as an indicator-based, Microsoft-excel desktop application that enables users to produce a set of vulnerability scores for multiple types of transportation assets (such as airports, bridges, and roadways) and climate stressors (such as temperature, inland flooding, and sea-level rise) that can be compared across two scenarios. The project team will use it to assign vulnerability scores to transportation assets for various sea level rise scenarios. This tool, whose process is outlined in Exhibit 2-4, begins by selecting climate stressors and entering data for different assets in the study area. Then, different indicators are used to represent the exposure, sensitivity, and adaptive capacity of those assets, including assigning weights between the three components of vulnerability and within the indicators chosen for each component. Using these inputs, the tool calculates a vulnerability score ranging from 0 (not vulnerable at all) to 4 (highest possible vulnerability). Scores that range from 0-2 are considered low vulnerability, from 2-3 are medium vulnerability, and from 3-4 are high vulnerability.

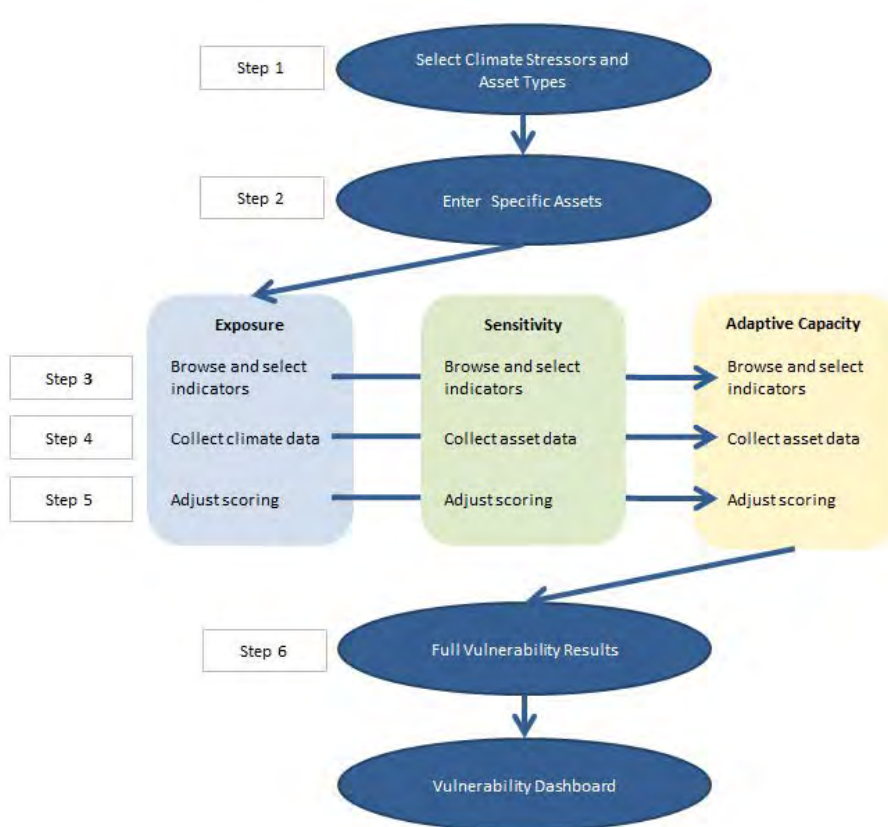
Exhibit 2-5 Components of Vulnerability



Source: FHWA Vulnerability Assessment Tool



Exhibit 2-6 FHWA's Vulnerability Assessment Tool



Source: FHWA Vulnerability Assessment Tool

A.5.2 Risk

As indicated in the previous section, risk is expected to be a major component of the RIP development process, especially based on the availability of detailed and accurate data. Although efforts to measure risk are still in the early stages, it is the intention of these peer agencies to formalize each of these best practices as key components of their RIPs.

CDOT has undertaken multiple steps to incorporate risk and resilience across its operations. Utilizing the Risk Analysis and Management for Critical Asset Protection (RAMCAP)³ approach, CDOT formalized its *Risk and Resilience Analysis Procedure*. CDOT uses the following factors to measure asset risk:

- Vulnerability, which measures an asset's susceptibility to damage from a natural hazard.

³ <https://files.asme.org/asmeiti/ramcap/12604.pdf>



- Threat likelihood, which is derived from historical records, such as government records, agency maintenance records, news articles, etc.
- Consequence – which includes both owner consequence and user consequence. Owner consequence is the direct consequences inflicted upon an asset due to a realized threat, and is calculated based on asset repairs and replacement, as well as cleanup costs when applicable. User consequence refers to the indirect costs imposed upon the public due to interruptions of the transportation network and is quantified by the additional cost of operating vehicles and value of time lost to the additional travel time and/or distance necessitated by detours and delays.

During the interview process, CDOT further elaborated on its approach to measuring risk. CDOT adheres to a scalable Risk Management Plan (RMP) which includes a Risk Workbook. To complete an RMP, the project manager for the proposed project and project team need to conduct risk identification, analysis, and planning using the following framework:

- Determine risks that could affect the project. This is known as Risk Identification.
- Perform a qualitative assessment of the probability and impact associated with a risk. This is known as the Qualitative Risk Assessment.
- Develop risk response strategies and assigning owners to monitor risks. This is known as Risk Response Planning.
- Track risk status, as well as implementation of response strategies, and identification of new risks. This is known as Risk Monitoring & Control.

Exhibit 2-7 CDOT's Risk and Resilience Analysis Procedure Guidebook

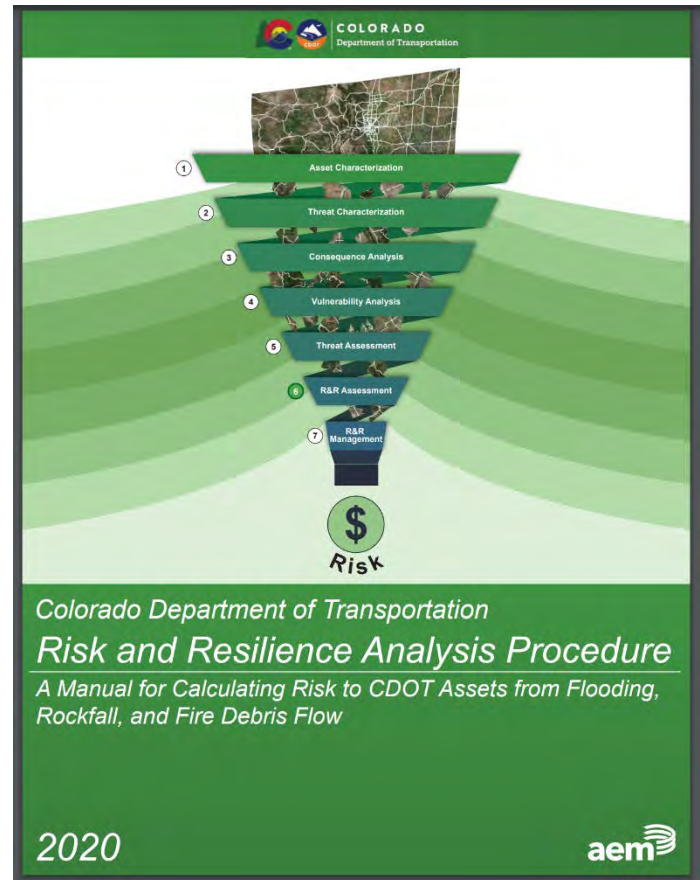




Exhibit 2-8 CDOT Risk Management Plan Process



Source: CDOT

The Qualitative Risk Assessment (Step 2) includes what is known as the Risk Register. The Risk Register (a worksheet in the Risk Workbook) uses the likelihood and impact to rank each risk by calculating a heat map. Red, yellow, and green indicate priority, with red indicating risks that warrant the most attention. The Risk Register is a key part of CDOT's risk and resilience strategy which directly considers insight from multiple agency departments, as well as quantitative and qualitative components. It was noted that DeIDOT is currently using this process, adopted from CDOT, to guide strategy development. Although completed prior to the formalization of the agency's risk & resiliency methodology, CDOT used a similar procedure to identify risks and resilience needs along the Interstate 70 corridor within Colorado. This included the qualitative process of criticality scoring by Interstate segment, and a vulnerability assessment based on quantified risks.

NHDOT elaborated on their experiences in identifying and translating risk to be applied across all agency departments. This is especially relevant for NHDOT's engineers who require a more refined format of data rather than predicted flood probabilities. NHDOT accomplished this process by keeping its engineering divisions looped into the resiliency planning process, while providing opportunities to provide feedback.

Nearby, MDOT uses National Oceanic and Atmospheric Administration (NOAA) Atlas 14 Precipitation Frequency Data to estimate frequency, intensity, and flood event duration, as part of a larger risk management procedure that considers flooding, as well as excessive heat and wind.

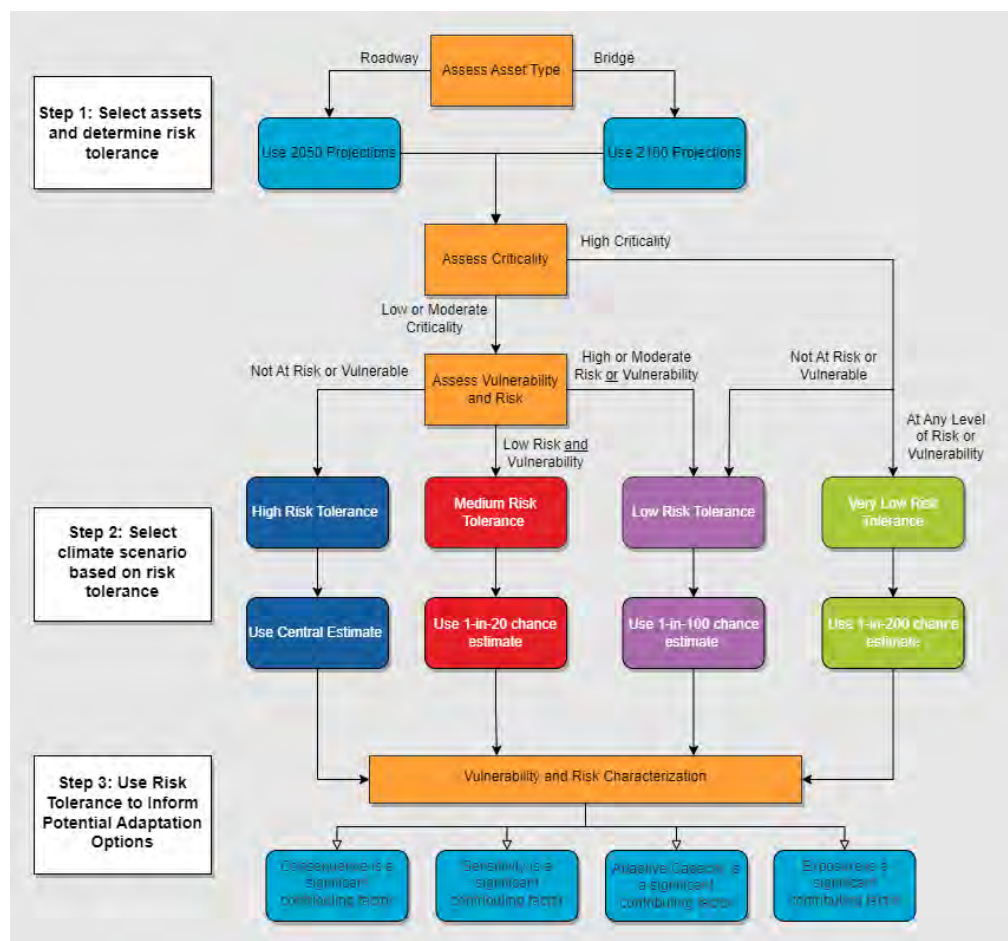


A.5.3 Risk Tolerance

Risk tolerance is an important topic discussed during the interview. With the potential impact from hazards identified agencies usually need to consider at what point do the key assets or functional areas become threatened and require actions, which will then determine the needs for resilience improvement. The level of acceptable change, or risk thresholds, are often different for different assets.

NHDOT developed a coastal risk tolerance framework that considers asset types, assets' criticality, and the vulnerability and risk that assets are facing to determine risk tolerance, which then can be used to inform the climate stressors or scenarios that the assets should be protected from, and the potential adaptation options.

Exhibit 2-9 NHDOT Coastal Risk Tolerance Framework



Source: NHDOT

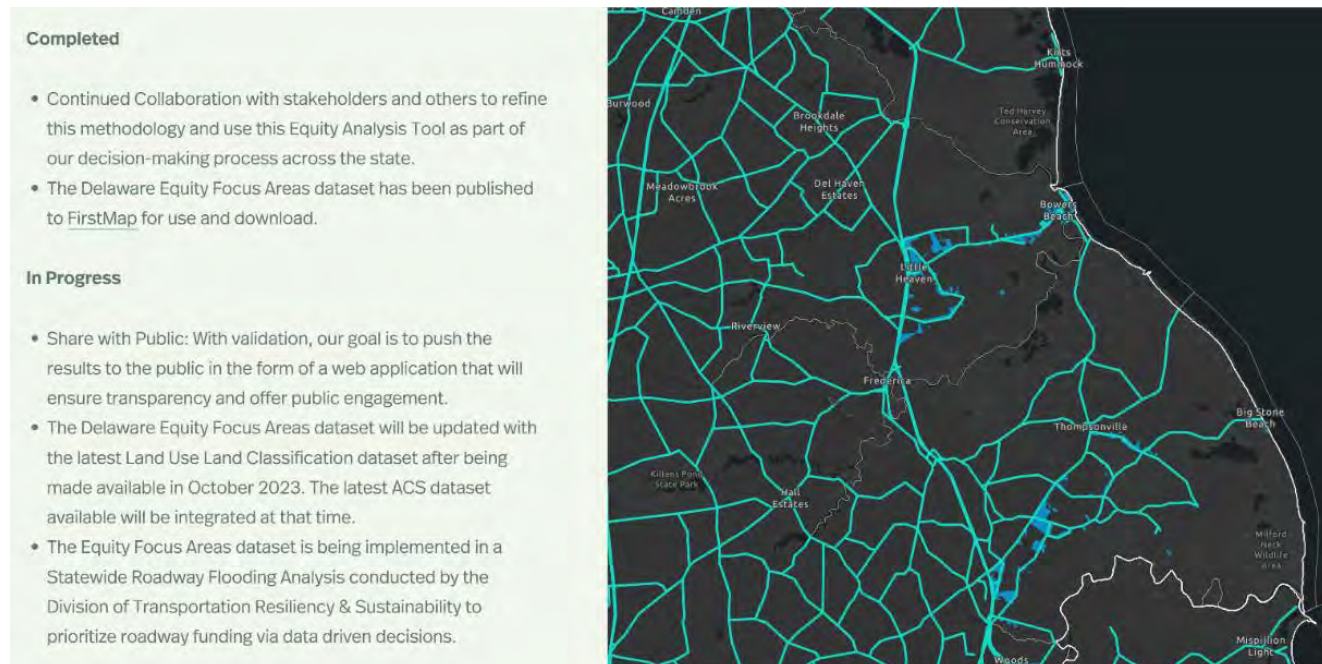
Equity is another factor that can help determine risk tolerance. DeIDOT provided insight on their 'Equity Analysis Tool' which aims to integrate equity into resilience planning. The mapping tool considers traffic, infrastructure



criticality, transit availability, evacuation routes, and physical accessibility (given that some coastal Delaware communities are only accessible via a single route) overlaid to a number of environmental justice (EJ) indicators.

Determining that USDOT-designated EJ indicators aggregated to the census tract level are too geographically broad to be useful for resilience planning, and not necessarily representative of underserved communities in Delaware, DelDOT used a combination of residential land use cover data and block group census data to identify equity focus areas. The tool has been rolled out, with DelDOT actively working on dissemination to the general public, updating the map to include the most recent land use data, and further integrating roadway flooding analyses features. Overall, DelDOT plans to use this tool to target potential locations for PROTECT project funding. The use of the tool will continue to provide DelDOT with a more accurate and targeted approach to targeted resilience investments, especially for vulnerable communities.

Exhibit 2-10 Delaware Equity Analysis Tool Screenshot



Source: DelDOT



Assess Vulnerability and Risk Key Takeaways:

- Risk-based vulnerability is a required component of RIPs based on PROTECT funding guidance.
- Many agencies have conducted vulnerability assessment that measures the extent to which an asset or system is exposed to hazards/threads, some have also included sensitivity and adaptive capacity in their vulnerability assessments.
- A comprehensive assessment of risk should consider the likelihood and consequence of hazards/threads.
- Other factors such as criticality and equity can be used to determine risk tolerance which can help inform adaptation strategies and priorities.

A.6 Developing Effective Resilience Strategies

A.6.1 Resilience Strategies

Peer agencies offered multiple ideas for developing resilience strategies, covering actions that could be taken in planning, environmental review, design, construction, operation, maintenance, and asset management.

VDOT is conducting a literature review to assess best practices in relation to strategy development, which will include best strategies in coordination with local agencies to develop adaptive design, physical enhancement, nature-based, operational, and administrative/policy-based strategies.

During the peer agency interviews, further guidance on strategy development was provided. VDOT for example, has expanded findings from the literature into a strategy toolbox which includes the themes highlighted above. VDOT further indicated that strategy development should use a data-driven approach that considers benefits and costs analysis (BCA), lifecycle analysis, and feasibility analysis, and should ultimately focus on cost effective options.

To develop strategies, MDOT is planning to leverage its State Hazard Mitigation Plan and Transportation Asset Management Plan (TAMP) which acknowledge resilience and form a strong foundation by identifying critical asset classes.

NCDOT referenced the National Cooperative Highway Research Program (NCHRP) *Project 25-64: Considering Greenhouse Gas Emissions and Climate Change in Environmental Reviews: Resources for State DOTs* as part of its resilience strategy.

FDOT indicated the use of multiple sources to help guide strategy development. This includes incorporating guidance from FHWA's Drainage Design Manual and NCHRP Project 15-61: *Applying Climate Change Information to Hydrologic and Hydraulic Design of Transportation Infrastructure*. FDOT also discussed the idea



of managed retreat, a strategy which has already been applied in one community in the state, as well as broader strategies in relation to backflow prevention, green space, freeboard requirements, shoulder and base protection, routine maintenance and inspection, pavement enhancement, and planning for road alternatives during emergencies. FDOT's formal strategy development process will focus on actionable strategies with a comprehensive context and appropriate projection of future conditions.

A.6.2 Evaluating Benefits & Costs

Peer agencies identified multiple methods for evaluating benefits & costs, both through the review of published literature, and through interviews. Overall, the evaluation of benefits & costs for agencies follows a typically standard process. However, given that the formalization of resilience as an agency initiative is a relatively new strategy, peer agencies cited challenges and future needs related to integration into the cost-benefit analysis (CBA) process. DeIDOT provided valuable insight regarding the evaluation of resilience benefits against larger, more prominent capital projects. As the smallest state in the U.S., Delaware is somewhat unique in that it also carries one of the most critical national highway links (Interstate 95) within the center of the Northeast Corridor. As a result, using a traditional CBA alone would result in the vast majority of funding going to projects which support the major corridor, as opposed to resilience upgrades which could improve access to vulnerable communities. Similarly, MDOT iterated the need for a dynamic CBA system that better highlights and promotes resilience.

VDOT discussed the idea of project economic competitiveness of those projects which have a benefit – cost (BC) ratio of less than 1.0. For these projects, total costs are expected to be larger than anticipated benefits, based on the agency's scoring system. In some of these instances, and given limited funding availability, operational and maintenance measures can be implemented as a shorter-term substitute. As an example, VDOT cited using permeable pavement treatments to strengthen infrastructure that frequently floods where it wouldn't be feasible to elevate the system.

Develop Resilience Strategies Key Takeaways:

- Strategy development can leverage TAMP, Hazard Mitigation Plans, as well as industry research and guidelines.
- RIPs should consider the benefit of combining physical infrastructure improvement with natural and nature-based solutions.
- Evaluation of benefits & costs should be a dynamic process that accurately reflects resilience component benefits.
- There is always a need to ensure that larger capital projects don't overshadow smaller capital projects which may address resilience needs.



A.7 Implementation and Integration

A.7.1 Integrating Resilience into Agency Project Development

Integration of resilience into the project development phase of DOT operations represents an important step of the RIP development process. As with all planning documents, strategies, and initiatives, there is always a risk that a plan will simply be ‘shelved’. This occurs when a plan, regardless of ambitiousness, fails to meaningfully impact agency strategy, direction, and operations. In the case of the RIP development process, successful implementation will consist of a unified resiliency mindset across the entire agency, including in the development of a Long-Range Transportation Plan and a fiscally constrained STIP project list.

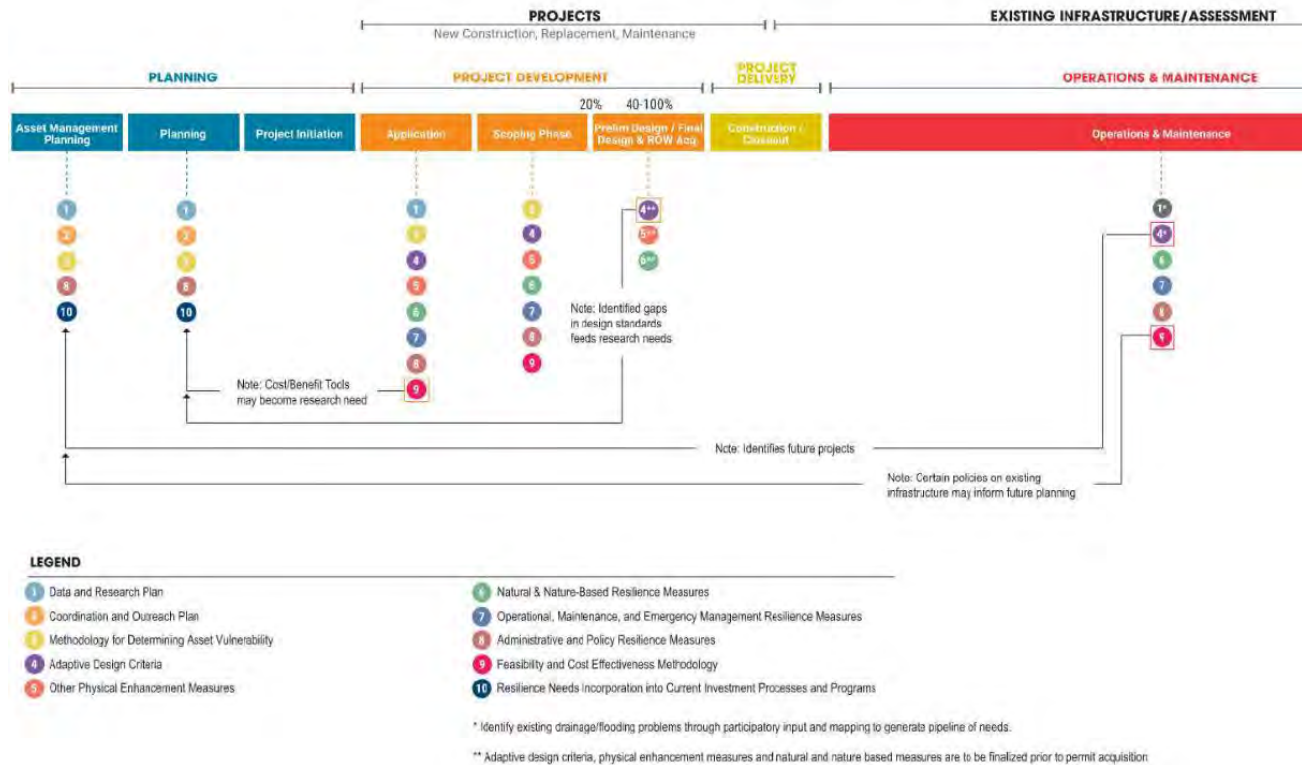
Given that most peer agencies are still in the early stages of RIP development, best practices regarding implementation are still somewhat limited. Nevertheless, peer agencies still offered insight on how to successfully integrate resilience into agency-wide decision making.

VDOT has developed a process to integrate resilience into its current business practices by looking at decision points where resilience can be considered, and identified potential actions that could improve resilience practices in each phase of project planning, development and management.



Exhibit 2-11 Integrate Resilience into VDOT Existing Business Practices

Figure 3. Integration of Resilience Plan Strategies into VDOT's Existing Business Practices



Source: VDOT.

When asked about any challenges so far in promoting and integrating resiliency, NHDOT staff indicated that they were pleasantly surprised that many at the agency were supportive of the ongoing resiliency efforts. When pressed a further on the reasons for this success, NHDOT offered the following thoughts:

- NHDOT has a strong understanding of what resiliency data is available and how it can be used. This includes NHDOT engineers who help monitor and manage infrastructure integrity. As an example, sea level rise data is often provided in the form of probabilities, given the uncertainties associated with the magnitude of climate change impacts. NHDOT undertook a process to help translate this data into a format much more compatible and useful to engineers who work best with defined values and quantities, rather than probabilities.
- In general, NHDOT has a strong understanding of where and when to make assumptions, and where to compromise when promoting resiliency decision making across the agency.
- In relation to the development of the RIP, NHDOT has undertaken the mindset of developing a reasonably strong and implementable final product, rather than an overly ambitious and drawn-out strategy that would be difficult to implement.



Other peer agencies added further thought to the discussion of integration and implementation. FDOT indicated that a number of agency-wide practices already support and promote resiliency. In this case, it's just a matter of formalizing these strategies in a unified and singular format. On the other hand, FDOT also indicated that full buy-in to an official agency-wide and overarching resilience strategy is an ongoing challenge.

MDOT indicated that the agency already vet city- and county-submitted projects to the consolidated transportation program (CTP), for resilience and carbon reduction components. According to MDOT, this process meets RIP and coastal resilience plan (CRP) requirements. As a result, MDOT aims to develop a data-driven process for its RIP to help guide and meet implementation requirements.

On the other hand, peer agencies expressed potential concern about implementation. VDOT indicated some unclarity regarding FHWA's specific requirements implementing the RIP through the lens of project alignment. NCDOT expressed some degree of concern in that resilience can increase the costs of capital projects, potentially making these projects less competitive in a fiscally constrained capital program. To develop their implementation strategy, NCDOT has a goal of better integrating resilience into its cost-benefit analysis process. DelDOT highlighted similar concerns and envisions developing project lists exclusively related to resilience for a separate funding pool.

A.7.2 Alignment with LRTP and STIP

Without clear guidance from FHWA, many peer agencies have expressed concern about how to effectively align RIP with LRTP and STIP, particularly on what potential alignment approaches will meet the PROTEC funding requirements for RIP. Including resilience in LRTP goals and objectives is usually the first step to align RIP with LRTP, many peer agencies have already done that. However, peer agencies indicated that it could be challenging to align resilience projects within LRTP projects and the fiscally constrained STIP project lists. This is especially the case when the agencies want to avoid excessive LRTP iterations and updates as RIP projects progress and are altered. Although VDOT is relatively far along in the development of their RIP, their document does not yet include alignment with the LRTP. NCDOT also cited that although the agency is actively working to incorporate risk and resilience into all day-to-day organizational activities, integrating resilience features into capital projects is challenging due to the additional costs. Outside of these challenges, MDOT indicated that incorporation of resilience projects into the LRTP should include short-term (5 to 10 year), and long-term (10+ years) time periods.

One potential strategy for aligning the RIP with the fiscally constrained STIP was identified by CDOT. In the development of their most recent LRTP (2045 Statewide Transportation Plan), CDOT included an appendix devoted entirely to resilience. The appendix includes an introduction to CDOT's resilience program and methodology (explained in Section 3.2) used to assess resiliency. Moving forward, integration of the RIP into the LRTP document as an appendix could be an effective method of alignment. This appendix would take the form of an expanded summary of necessary RIP components, and a corresponding project list.



Resilience Integration & Implementation Key Takeaways:

- Having a strong understanding of data sources and capabilities is key to successful implementation.
- Agency-wide acceptance is importance for successful implementation of resilience strategies. This can be achieved through effective engagement of staff from all functional areas or departments across the entire RIP development process.
- Concerns exist about successfully aligning the RIP with the LRTP planning process. Integration of the RIP into the LRTP document as an appendix could be an effective method.

A.8 Stakeholder Engagement

Multiple strategies were discussed in relation to stakeholder engagement. In the early stages of the RIP development process, stakeholder engagement has primarily applied to inter-agency collaboration and outreach to MPOs and FHWA. In terms of inter-agency stakeholder engagement, MNDOT has an existing resilience adaptation team that meets monthly to discuss challenges in addressing resilience vulnerabilities. NCDOT similarly has a climate advisory council which meets monthly. Meanwhile, as part of the discussion on stakeholder engagement, MDOT and FDOT highlighted the importance of creating a definition of resilience that considers functionality in the transportation planning process, in order to get support and buy-in across all agencies and stakeholder groups.

Since 2015, has held CDOT holds a monthly Resiliency Working Group to help further advance resiliency efforts. This working group includes members of CDOT, the Colorado Resiliency Office, FHWA/FTA decision makers, MPO staff, transit agency officials, local agency staff, federal land management agencies, FEMA, emergency responders, stormwater management agencies, fire prevention districts, state and university climatologists, community members, and transportation specialists.

Stakeholder Engagement Key Takeaways:

- Agency-wide buy-in is important for successful RIP implementation.
- Engaging stakeholders during the process of RIP development is critical, which includes coordination with FHWA Division Office, state or local resilience committees/working groups, and representatives from MPOs, counties, and municipalities.



A.9 Performance Monitoring

Like many other plans, developing resilience related performance measures and targets is an important step to measure performance and track progress on achieving the goals and objectives in RIPs. Although there is still no standard guidance or process for creating resilience specific performance measures for state DOTs, many peer agencies have started to incorporate risk and resilience into their performance measures, however, most of them are not directly related to impact from natural or climate hazards, instead are linked to safety and security or system preservation. For example, MDOT has “percent of compliance on erosion and sediment control ratings” as a measure for its goal “institutionalize the consideration of future sea levels and storm conditions in prioritizing infrastructure investments in coastal areas”. MnDOT has “annual percentage of routine culvert inspections completed on time” as a measure for its goal “increase the resilience of the transportation system and adapt to changing needs”.

Performance Monitoring Key Takeaways:

- Most peer agencies are still in early stage of developing resilience related performance measures.
- Some agencies have identified resilience measures that are linked to their safety or system preservation goals.



A.10 Framework Development

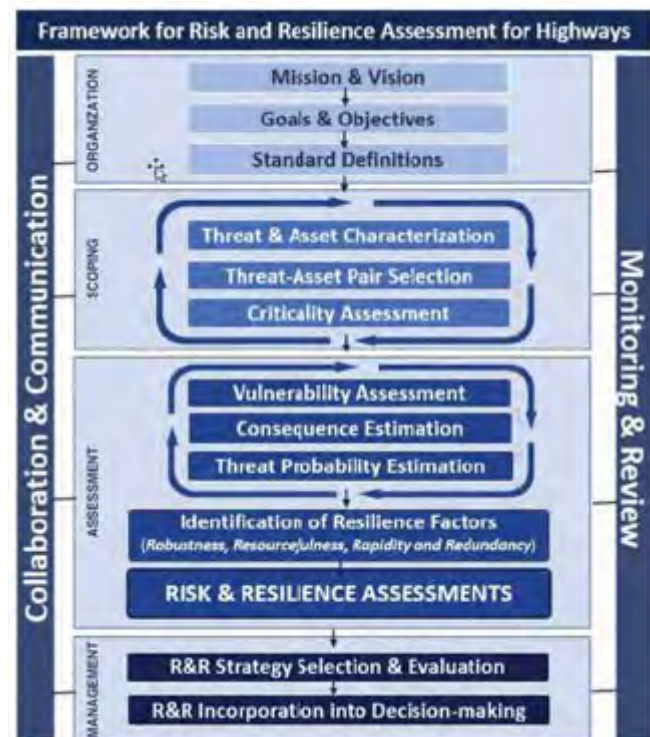
A.10.1 Development Approach

Following the review of internal and peer agency documentation and general approaches, the research team proceeded with the development of the RIP framework. The purpose of this process is to provide an initial architecture for the RIP document, based on best available practices, including from national sources. Framework development was guided based on insight from the following sources:

- **NCHRP Project 23-09 – Developing a Highway Framework to Conduct an All-Hazards Risk and Resilience Analysis (2023):** This project provides a research roadmap to develop a comprehensive manual, tools, training, and implementation guidelines for quantitative risk and resilience assessment that satisfies new federal requirements related to risk and resilience. The document recommends a framework consisting of the following components:

- *Organizational Components:* Agency-wide statements which include defined mission and goal statements, goals and objectives, and standardized definitions.
- *Scoping:* Including threat and asset characterization, threat-asset pair selection, and criticality assessment.
- *Assessment:* The risk assessment portion of the framework should include vulnerability assessment, consequence estimation, threat probability estimation, identification of resilience factors, and general risk and resilience assessments.
- *Management:* Includes the design and selection of mitigation and adaptation strategies with sections for risk and resilience strategy selection and evaluation, incorporation into decision-making.
- *Collaboration and Communication:* Provision of a stakeholder engagement strategy.

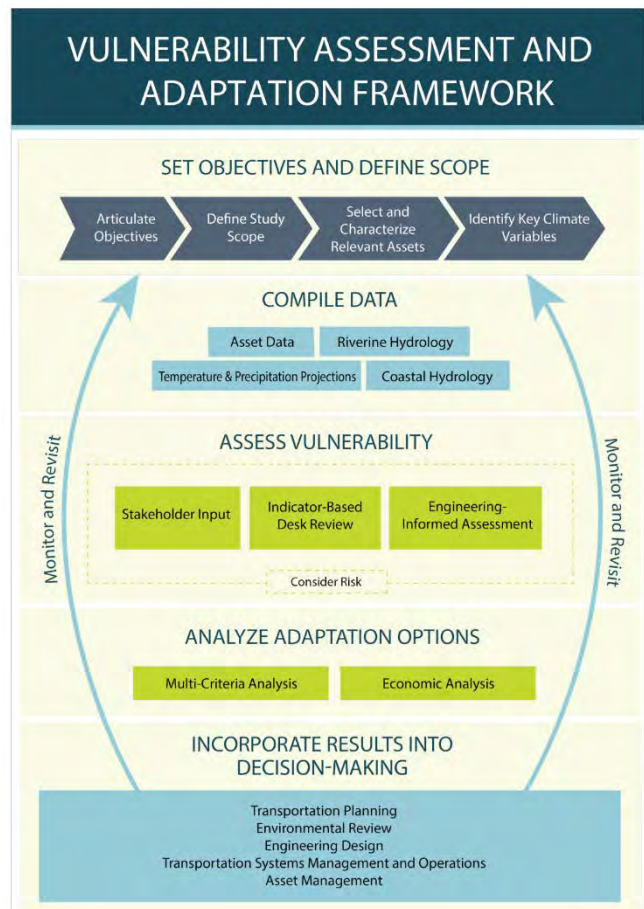
Exhibit 3-13 NCHRP 23-09 Recommended Risk and Resilience Framework





- *Monitoring and Review*: Method for ensuring continued compliance and adherence to framework strategies.
- **FHWA Vulnerability Assessment and Adaptation Framework**: Developed by FHWA, the purpose of this reference guide is to assess the vulnerability of transportation infrastructure and systems to extreme weather and climate effects. This includes the following framework for conducting vulnerability assessments:
 - *Set Objectives and Define Scope*: Establishing a clear and defined objective will help minimize extraneous data collection and analysis activities, while increasing the likelihood of successful buy-in and implementation, taking into consideration resource constraints.
 - *Compile Data*: Identification of asset data such as infrastructure systems, and climate data such as changes in temperature, precipitation, hydrology, floodplains, sea level, and storm surge.
 - *Assess Vulnerability*: Including considerations for vulnerability, exposure, and risk. This framework includes insight on stakeholder, indicator-based, and engineering-informed assessments for assessing asset vulnerability.
 - *Analyze Adaptation Options*: Including natural, structural, and policy-based mitigation strategies. This framework includes insight on multi-criteria and economic analysis strategies for analyzing adaptation options.
 - *Incorporation of Results into Decision-Making*: Includes strategies to effectively incorporate results into transportation planning, project development and environmental review, project level design and engineering, transportation systems management, operations, and emergency management, and asset management.
 - *Monitoring*: Including establishing monitoring and evaluation processes to assess the success of adaptation strategies and other initiatives that were established based on assessment findings. The

Exhibit 3-14 FHWA Vulnerability Assessment Framework





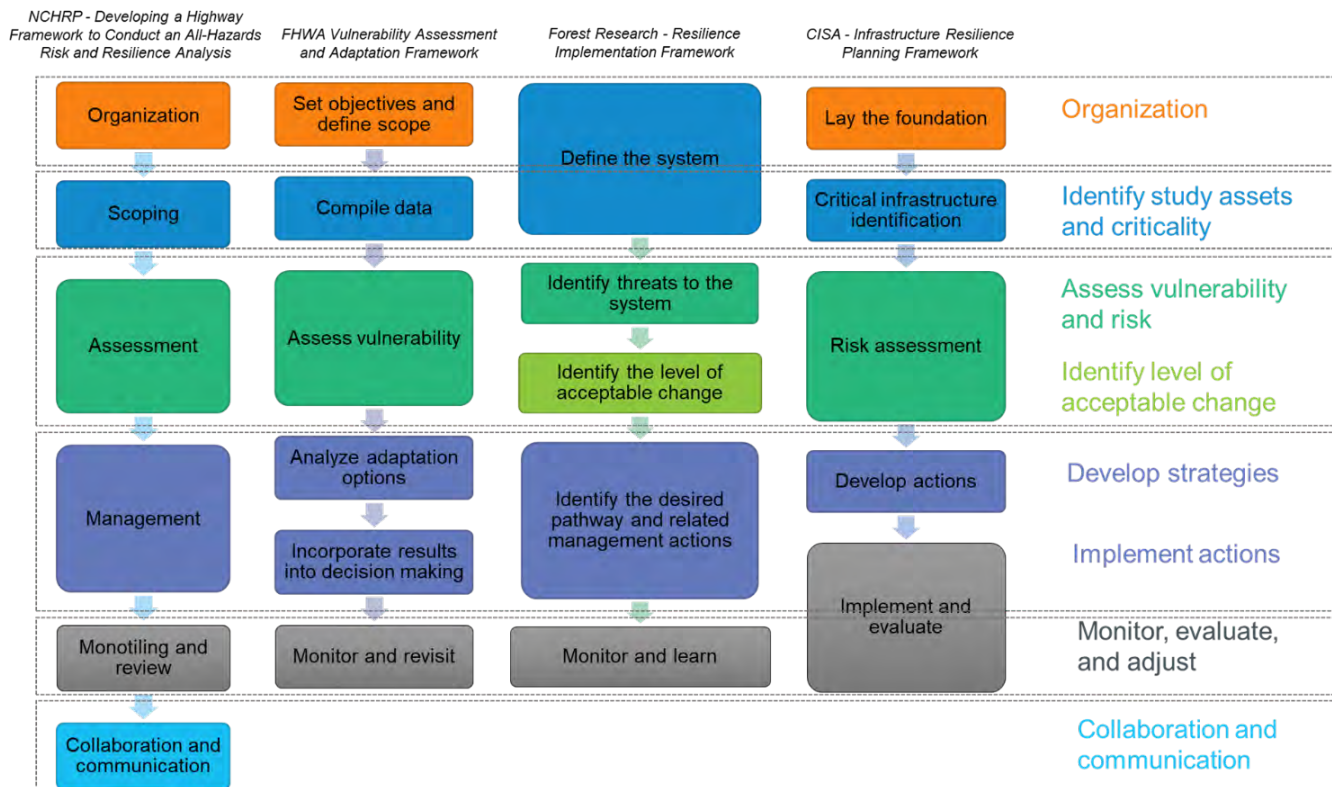
monitoring and evaluation process may identify the need to revisit the assumptions, underlying data, or approaches used in the original vulnerability assessment.

- **Forest Research Resilience Implementation Framework:** Developed by the United Kingdom's forestry research organization, this framework includes a methodology for strategic, tactical or operational resilience management. The Forest Research resilience framework includes the following components:
 - *Define the System:* This includes considerations for system type, functions, spatial scale, and timeframe.
 - *Identify System Threats:* This includes considerations for biological, physical, chemical, and social/economic/political threats.
 - *Identify Level of Acceptable Change:* Guiding themes include level of acceptable change (such as defined habitat loss) as well as boundaries, thresholds, and trade-offs.
 - *Identify the Desired Pathway and Related Management Actions:* Guiding themes include potential resistance actions, recovery actions, adaptation actions, and transformation actions.
 - *Monitor and Learn:* Including considerations for agencies to adapt to the sequence of events that occur following strategy implementation.
- **Cybersecurity & Infrastructure Security Agency Infrastructure Resilience Planning Framework:** Developed by the Cybersecurity & Infrastructure Security Agency (CISA), the purpose of this reference guide is to provide an approach for localities, regions, and the private sector to work together to plan for the security and resilience of critical infrastructure services. The Framework includes the following components:
 - *Lay the Foundation:* Including steps for a project leader, defining a scope, collecting and reviewing existing information, collaboration, and setting goals and objectives.
 - *Critical Infrastructure Identification:* Including steps for identifying and prioritizing infrastructure and identifying dependencies.
 - *Risk Assessment:* Outlines steps for identifying threats and hazards, assessing vulnerability, assessing potential consequences, and delineating infrastructure system risks.
 - *Develop Actions:* Including steps for refining goals and objectives, identifying risk solutions, assessing existing resources and capabilities, selecting resilience solutions, and developing implementation strategies.
 - *Implementation & Evaluation:* Including steps for implementation through existing planning measures, monitoring and evaluating action effectiveness.



The figure below shows the four frameworks reviewed and highlights their common components, which are the foundation of the development of RIDOT RIP Framework.

Exhibit 3-15 Resilience Frameworks' Common Components



The project team also identified and reviewed existing Rhode Island Department of Transportation (RIDOT) resilience guidance and planning efforts. The purpose of this review was to establish a baseline for existing data, methodology, tools, and initiatives which could eventually be leveraged to formally develop the agency's RIP. Additionally, this review aims to set a foundation for aligning and incorporating resilience strategies and improvement into existing plans and processes, especially for the Rhode Island's stormwater and congestion management planning processes. The review of internal literature covered the following documents:

- MLD 601 - RIDOT Final Memo (2022):** Developed as a research paper by the Harvard Operations Management Team, and focusing primarily on bridge infrastructure, this memo aims to assist RIDOT in developing a framework to better account for resiliency in capital planning, given that the current approach doesn't sufficiently account for general effects from climate change. The memo includes a review of potential incorporation strategies, such as repairs, managed retreat, alternative routings, and nature-based solutions. It also includes an implementation plan for RIDOT which includes extensive stakeholder engagement.



- **RIDOT State Transportation Improvement Plan, 2022 – 2031 (2021, Revisions through 2023):** The STIP is a list of transportation projects the State of Rhode Island intends to implement using United States DOT funds. The STIP includes projects for the upcoming ten years and is reauthorized every four years. Overall, improving asset resiliency to climate change, storm surge, and sea level rise is a key component of asset management, a priority focus of the STIP.
- **Moving Rhode Island 2040 (2020):** As required by the Federal Highway Administration (FHWA), RIDOT's Long Range Transportation Plan (LRTP) provides a framework for statewide transportation goals and objectives and identifies strategies to help Rhode Island facilitate the efficient movement of people and goods and promote economic development through transportation investments. RIDOT's LRTP identifies coastal resiliency as a concern for the state's coastal counties and highlights the need to prepare for more frequent flooding and thawing cycles. The plan includes a vision for enhance transportation network resiliency in part by encouraging state agencies and municipalities to work together within the Municipal Resilience Program at the RI Infrastructure Bank.
- **Rhode Island Congestion Management Process (2020):** As required by the FHWA for metropolitan areas with a population greater than 200,000, the Rhode Island Congestion Management Process (CMP) aims to identify traffic congestion and its causes, develop monitoring processes to measure transportation system performance and reliability, and develop congestion management strategies while also moving them into the funding and implementation stages. Given that Department of Administration's Division of Statewide Planning serves as staff to the State Planning Council, which serves as Rhode Island's metropolitan planning organization (MPO), the state agency is the developer of this plan.
- **RIDOT Stormwater Control Plan Approach (2019):** This document summarizes the approach and procedures used during the development and implementation of Stormwater Control Plans. This includes information on infrastructure site descriptions, the process to develop stormwater reduction targets, existing stormwater controls, potential enhanced non-structural and structural stormwater controls, and implementation mechanisms.
- **State of Rhode Island Hazard Mitigation Plan Update (2019):** Published in 2019, this plan update provides a broad assessment of all hazards that pose a potentially significant risk to Rhode Island. Not limited to transportation and climate resilience, the plan covers biological incidents, chemical incidents, civil disturbances, cybersecurity, dam failure, drought, earthquake, extreme cold, extreme heat, fire, flood, high winds, infectious diseases, infrastructure failure, radiological incidents, sea level rise, severe winter weather, terrorism, thunderstorms, tornadoes, and tropical and extratropical storms. The plan includes an assessment of key geographic, infrastructure, and socioeconomic characteristics of the state, as well as a methodology for identifying hazards and risk, occurrence history, relevant stakeholders, and broad mitigation strategies.
- **STORMTOOLS Website (2019):** Developed by in collaboration between the Rhode Island Coastal Resources Management Council and the University of Rhode Island, STORMTOOLS maps storm

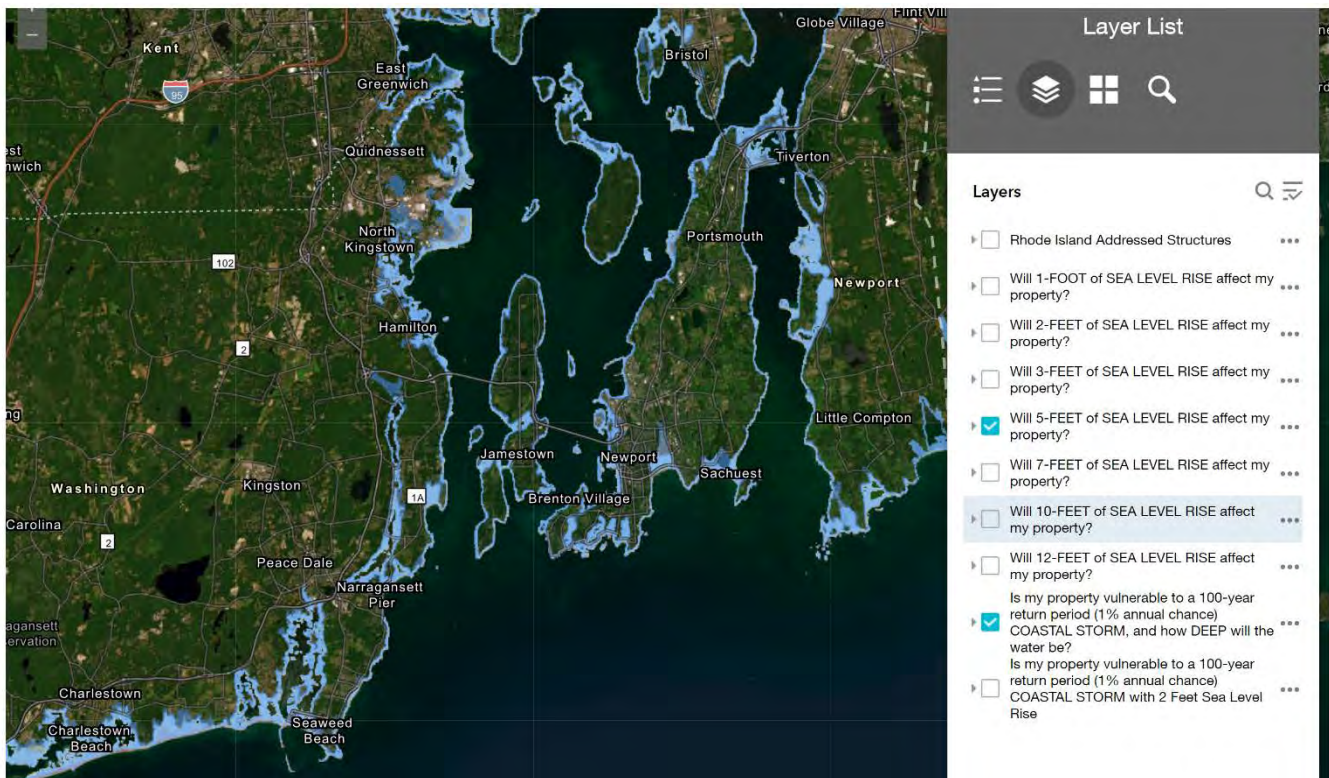


inundation, with and without sea level rise, for varying return period storms that covers all of Rhode Island's coastal waters. Predictions are provided which show water extent and depth at any given point for nuisance floods (1-, 3-, 5-, and 10-year recurrence intervals) and 25-, 50-, 100-, and 500-year storm scenarios at a 95% confidence interval. Sea level rise of 1, 2, 3, 5, and 7 feet on their own as well as combined with each storm scenario are also modeled.

- **Resilient Rhode Impact Report (2018):** Published as a three-year update on statewide climate resilience goals, the document highlights coordination efforts and tools, as well as current and recently completed projects and initiatives. Key thematic areas include critical infrastructure and utilities, natural systems, emergency preparedness, community health & resilience, and financing.
- **Technical Paper 167 – Vulnerability of Municipal Transportation Assets to Sea Level Rise and Storm Surge (2016):** Developed by the RIDOT Division of Planning in 2016, this document provides a comprehensive assessment of statewide asset vulnerability to sea level rise scenarios. The assessment uses STORMTOOLS data, developed in collaboration between the Rhode Island Coastal Resources Management Council and the University of Rhode Island.

Exhibit 3-16 STORMTOOLS Mapping Feature

Exhibit 3-17 STORMTOOLS Mapping Feature



Source: Rhode Island Coastal Resources Management Council



A.10.2 Resilience Improvement Plan Framework

In consideration of the requirements for RIP in the PROTECT Funding Implementation Guidance, existing RIDOT documentation and based on peer agency and national best practices in RIP development, resilience management, and framework design, the project team developed a framework for RIDOT's RIP as shown in Exhibit 4.4. The following section discusses its key components.

The RIP Framework is designed to enable RIDOT to address the RIP requirements to be eligible for in the PROTECT Formula and Discretionary Funds. The orange circles on the framework are the connections of each step of the framework with the required and optional elements of a RIP, which is detailed in Exhibit 4.5.



Exhibit 3-18 Resilience Improvement Plan Framework

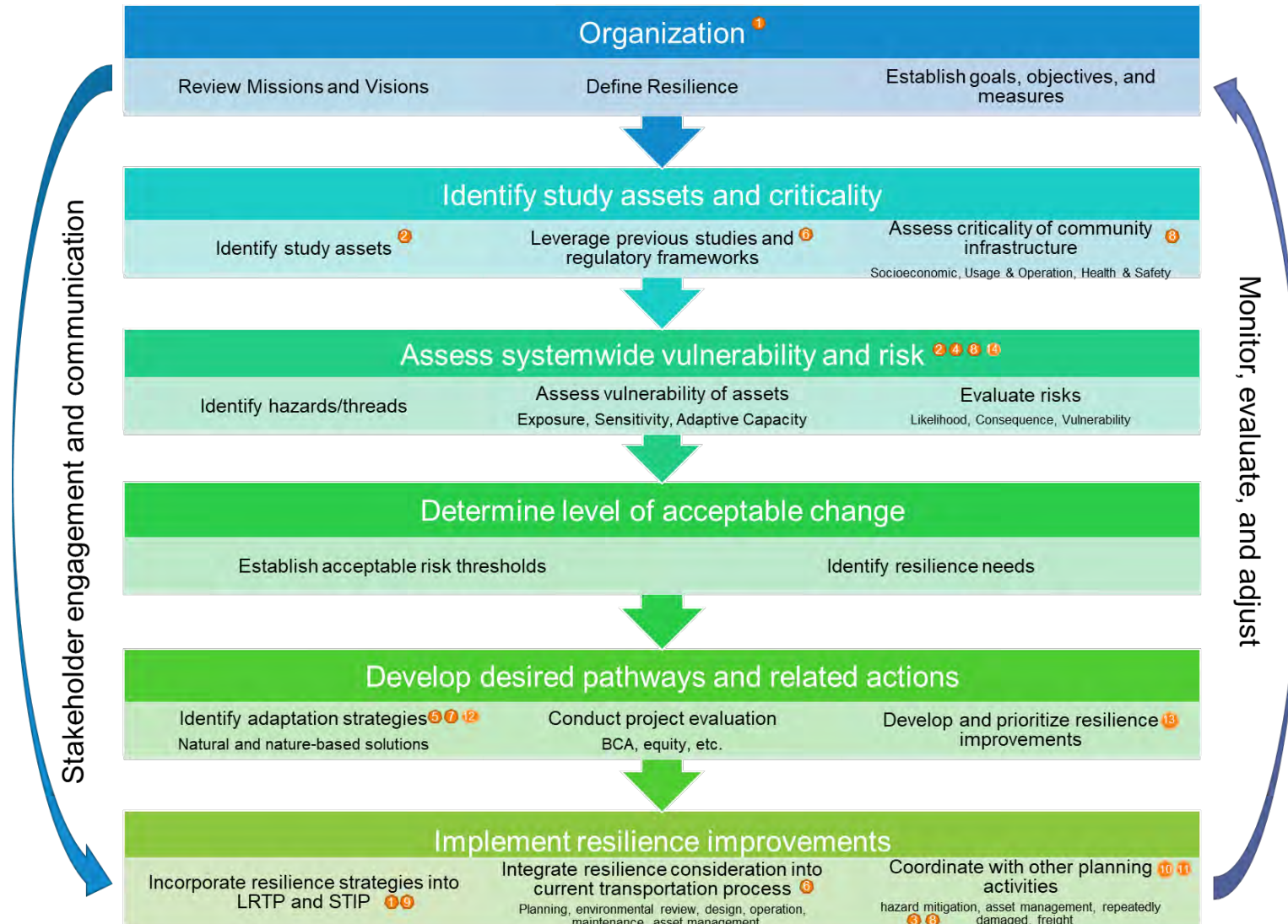




Exhibit 3-19 Required and Optional Elements of RIP

PROTECT Formular Program Implementation Guide - Resilience Improvement Plan Content		
The Plan Shall..		
1	Encompass immediate and long-range planning activities and resilience investments	(A) shall be for the immediate and long-range planning activities and investments of the State or metropolitan planning organization with respect to resilience of the surface transportation system within the boundaries of the State or metropolitan planning organization, as applicable
2	Demonstrate a system-wide approach to transportation system resilience	(B) shall demonstrate a systemic approach to surface transportation system resilience, and
3	Consistent with and complement State and local hazard mitigation plans	(B) ...be consistent with and complementary of the State and local mitigation plans required under section 322 of the Robert T. Stafford Disaster Relief and Emergency Assistance Act (42 U.S.C. 5165)
4	Include a risk-based assessment of vulnerability to current and future weather events and natural disasters	(C) shall include a risk-based assessment of vulnerabilities of transportation assets and systems to current and future weather events and natural disasters, such as severe storms, flooding, drought, levee and dam failures, wildfire, rockslides, mudslides, sea level rise, extreme weather, including extreme temperatures, and earthquakes (23 U.S.C.176(e)(2)(A-C)).
Shall, as appropriate..		
5	Describe ways to improve response to impacts and changes	...Include a description of how the agency is prepared to respond to the impacts of weather events, natural disasters and is prepared for changing conditions;
6	Describe the codes, standards, and regulatory framework to ensure improvements	...Describe the codes, standards, and regulatory framework, adopted and enforced by the agencies, to ensure that resilience improvements within the impacted area of proposed projects that are included in the plan;
7	Consider benefit of natural Infrastructure	...Consider the benefits of combining hard surface transportation assets, and natural infrastructure, through coordinated efforts by the Federal Government and the States;
8	Assess community infrastructure resilience	...Assess the resilience of other community assets, including buildings and housing, emergency management assets, and energy, water, and communication infrastructure;
9	Use a long-term planning period	...use a long-term planning period...A Resilience Improvement Plan should cover a period at least as long as the relevant SLRTP, MTP, or asset management plan. A longer period that considers the service lives of relevant assets is recommended.
	Other	...Include such other information as the State or metropolitan planning organization considers appropriate
May also..		
10	Designate evacuation routes and strategies	...Designate evacuation routes and strategies, including multimodal facilities, designated with consideration for individuals without access to personal vehicles;
11	Plan for response to anticipated emergencies	...Plan for response to anticipated emergencies, including plans for the mobility of emergency response personnel and equipment and access to emergency services, including for vulnerable or disadvantaged populations;
12	Describe the resilience improvement policies	...Describe the resilience improvement policies, including strategies, land-use and zoning changes, investments in natural infrastructure, or performance measures that will inform the transportation investment decisions of the State or metropolitan planning organization with the goal of increasing resilience;
13	Include investment plan & priority projects	...Include an investment plan that includes a list of priority projects and describes how funds apportioned to the State under section 104(b)(8), or provided by a grant under the PROTECT program would be invested and matched, which shall not be subject to fiscal constraint requirements;
14	Use science and data	... Use science and data and indicate the source of data and methodologies

Sources: PROTECT Formula Funding Implementation Guidance, AASHTO Informal Worksheet with RIP Checklist



Organization

RIDOT RIP Framework starts with a review of the organization's mission and vision. This will help develop a resilience definition that supports RIDOT overall missions, aligns with how RIDOT envision their system and service will be, and relates to staffs and activities across all functional areas. This step will also establish the goals, objectives, and performance measures for RIDOT's resilience improvement plan, which should encompass immediate and long-rang planning activities and investments with respect to the resilience of RIDOT's transportation system (*PROTECT Formula Funding Implementation Guidance*).

Identify study assets and criticality

Guided by the resilience goals and objectives established in the previous step, the second step of the RIP framework will identify transportation assets to be analyzed in the RIP and evaluate their criticality for system resilience. The study assets should be identified with a systemic approach and include multi-modal transportation infrastructure. The assessment of transportation assets' criticality should leverage any previous studies and regulatory frameworks and consider their importance to the community with respect to socioeconomic, usage and operation, and health and safety.

Assess systemwide vulnerability and risk

This step will conduct a risk-based assessment of vulnerabilities of transportation assets and systems to current and future weather events and natural disasters (*PROTECT Formula Funding Implementation Guidance*). This include identifying hazards to be analyzed in the RIP, which could be all or high priority current and future extreme weather events, natural disasters, and changing conditions that are projected to impact RIDOT's transportation system and service. The vulnerability assessment should evaluate transportation assets' exposure, sensitivity, and adaptive capabilities to each hazard. The risk assessment should consider the vulnerability of different assets to the impacts of a hazard, the likelihood of that hazard event occurring, and the consequences that may result from that hazard should it occur. The assessment can leverage state and local hazard mitigation plans to obtain information about the likelihood and consequence of hazards impacting the study area, and their impacts to transportation system and other community assets, including buildings and housing, emergency management assets, and energy, water, and communication infrastructure.

Determine level of acceptable change

With the potential impact from hazards identified previously, this step will consider at what point do the key assets or functional areas of RIDOT become threatened and require actions, which will then determine the needs for resilience improvement. The level of acceptable change, or risk thresholds, are often different for different assets. RIDOT should consider the criticality of each of the study assets for supporting its missions and for achieving its



visions when determining their risk thresholds. Equity is another important factor to consider identifying community resilience needs. The level of acceptable change, or risk thresholds, can inform the pathways to improve assets and system resilience, including the selection of adaptation strategies and future climate scenarios and timelines.

Develop desired pathways and related actions

Based on the resilience needs identified, this step describe how RIDOT *is prepared to respond to the impacts of weather events, natural disasters and is prepared for changing conditions*. The adaptation strategies should consider *the benefits of combining hard surface transportation assets, and natural infrastructure, through coordinated efforts by the Federal Government and the States*. Optionally, the step can describe *the resilience improvement policies, including strategies, land-use and zoning changes, investments in natural infrastructure, or performance measures that will inform the transportation investment decisions of the State or metropolitan planning organization with the goal of increasing resilience (PROTECT Formula Funding Implementation Guidance)*.

Benefit and cost analysis can be conducted to evaluate the benefit of resilience improvements and the cost of no-action. RIDOT should work with its stakeholders to develop criteria to prioritize resilience improvements, taking into account both the quantitative results from the risk-based vulnerability assessment, and other measures that reflects the agencies priorities, such as equity, safety, and mobility. Optionally, an investment plan can be developed to *describes how funds apportioned to the State under section 104(b)(8) or provided by a grant under the PROTECT program would be invested and matched, which shall not be subject to fiscal constraint requirements (PROTECT Formula Funding Implementation Guidance)*.

Implement resilience improvements:

To implement the RIP, RIDOT can incorporate the plan into its Long-Range Transportation Plan and State Transportation Improvement Program, which can be done through the alignment of goals and objectives, projects, priorities, and/or performance measures. Resilience strategies should also be integrated into the existing practices and processes across all functional areas in RIDOT, including Planning, environmental review, design, operation, maintenance, and asset management. *This step should also describe any codes, standards, and regulatory framework adopted and enforced by the agencies to ensure implementation of resilience improvements (PROTECT Formula Funding Implementation Guidance)*. Coordination with other planning activities is important for implementing resilience improvements, including the state and local hazard mitigation plans, asset management plans, any plans for repeatedly damaged areas, and freight plans. This framework will deploy decision-support approaches that are both needs-based (e.g., critical and vulnerable facilities) and opportunistic - resilience investments may line-up with maintenance schedules, RIDOT's TIP, or other existing work programs.



Stakeholder engagement and communication

Stakeholder engagement and communication will be conducted in all steps of the framework. As many peer agencies indicated, agency-wide buy-in is important for successful implementation of resilience strategies. This starts with engaging staff internally to establish a resilience definition that collectively reflect each RIDOT department's mission and vision, and develop RIP goals, objectives, and measures are relatable to all functional areas in RIDOT. Proactive communication is also critical during the process of identifying critical assets, assessing vulnerability and risk, and determining the level of acceptable change. Building a clear understanding of the underlying data and methodology among staff will not only help obtain the best available data for the analysis, but also support improvement of future data collection and management effort, especially for collaboration with asset management and emergency operation. This understanding will also help inspire and encourage staff from different departments to come up with resilience strategies building upon their existing practices, or consider how resilience strategies that are identified by others can be integrated to enhance their day-to-day work. Such active engagement will make it easier for resilience improvements to be incorporated into LRTP and STIP along with other RIDOT priorities. Coordination is also important to incorporate equity to evaluate risk tolerance and determine community resilience needs. Lastly, the RIP should be coordinated with other planning activities, including the State Hazard Mitigation Plan, Transportation Asset Management Plan, Freight and Goods Movement Plan, and any other planning effort for areas that were repeatedly damaged. Engaging external stakeholders during the process of RIP development is also critical, which includes coordination with FHWA Division Office, state or local resilience committees/working groups, and representatives from MPOs, counties, and municipalities.

Monitor, evaluate, and adjust the plan

This step will develop performance metrics to monitor and evaluate the progress of the RIP and adjust and update the plan periodically. The RIP is only the first step in making RIDOT's asset and system more resilient. It will be a living document with metrics identified by RIDOT to monitor the progress of achieving its goals. The RIP can be updated periodically to assess the impact of additional hazards (including natural, technological, and human-caused), evaluate additional assets, or to redo the risk-based vulnerability analysis when conditions have changed, and the previous analysis results no longer apply.



A.11 Summary

This tech memo has summarized results of the peer agency interview and identified best practice of improving resilience and developing RIPs by state DOTs. The key findings and takeaways from this process was synthesized and used to develop the RIDOT RIP framework. Both the design of the best practice interviews and the development of the framework are closely guided by the RIP requirements described in the PROTECT funding guidance. Therefore, the RIP framework includes detail steps to guide RIDOT RIP development and provides linkages to the RIP requirements with approaches to address the required and optional elements of the plan. The findings and examples from the best practice interview are also organized by key RIP elements and will serve as insightful references for RIDOT RIP development.

A.12 Presentation to American Association of State Highway and Transportation Officials (AASHTO) Committee - Risk and Resilience Working Group Meeting



CAMBRIDGE
SYSTEMATICS

Think  Forward

Resilience Improvement Plan Review of Best Practices

Rhode Island DOT - Peer Agency Interview Summary

presented to

*AASHTO Committee - Risk and Resilience
Working Group Meeting*

presented by

*Cambridge Systematics, Inc.
Yingfei Huang, Suseel Indrakanti*

5/10/2023

Introduction

Christos Xenophontos

Assistant Director

Rhode Island Department of Transportation (RIDOT)

Suseel Indrakanti

Principal

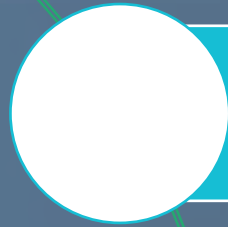
Cambridge Systematics

Yingfei Huang

Senior Associate

Cambridge Systematics

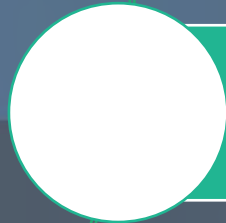
Agenda



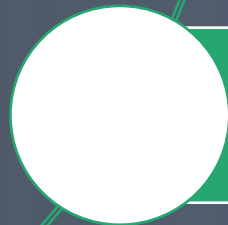
Motivation



Requirements



Peer Agency Interviews

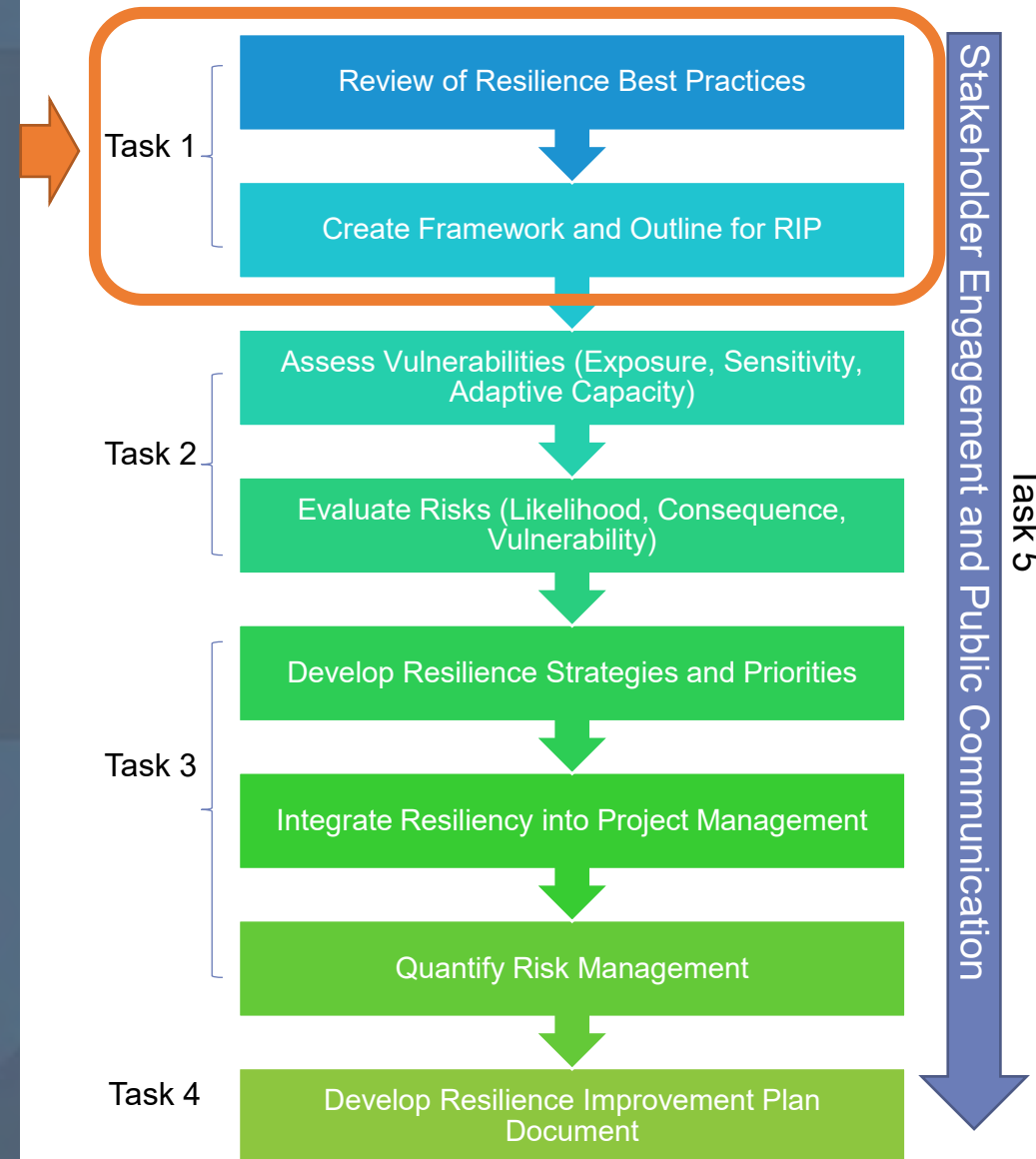


Findings & Best Practices

Motivation



Rhode Island DOT Resilience Improvement Plan



Promoting Resilient Operations for Transformative, Efficient, and Cost-Saving Transportation (PROTECT) Program

*“Support efforts to
identify vulnerabilities,
develop proposed resilience solutions,
and schedule and prioritize resilience improvements”*

PROTECT

- Released July 29th, 2022, as part of PROTECT Funding Implementation Guidance
- Resilience Improvement Plan (RIP) can reduce the non-Federal cost share for a project by up to 10 percent.
 - 7% reduced if develop a RIP and prioritize projects on RIP;
 - 3% further reduction if plan provisions incorporated into LRTP.

Sources:

PROTECT Formula Funding Implementation Guidance
AASHTO Informal Worksheet with RIP Checklist

PROTECT - Resilience Improvement Plan

The Plan Shall..

- 1 Encompass **immediate and long-range planning activities** and **resilience investments**
- 2 Demonstrate a **system-wide** approach to transportation **system resilience**
- 3 Consistent with and complement **State and local hazard mitigation plans**
- 4 Include a **risk-based assessment of vulnerability** to current and future weather events and natural disasters

Shall, as appropriate..

- 5 Describe ways to improve **response to impacts** and changes
- 6 Describe the codes, standards, and regulatory framework to **ensure improvements**
- 7 Consider benefit of **natural Infrastructure**
- 8 Assess **community infrastructure resilience**
- 9 Use a **long-term** planning period

May also..

- 10 Designate evacuation routes and strategies
- 11 Plan for response to anticipated emergencies
- 12 Describe the resilience improvement policies
- 13 Include investment plan & priority projects
- 14 Use science and data

Objectives and Scope

PROTECT - Resilience Improvement Plan	
<i>The Plan Shall..</i>	
1	Encompass immediate and long-range planning activities and resilience investments
2	Demonstrate a system-wide approach to transportation system resilience
3	Consistent with and complement State and local hazard mitigation plans
4	Include a risk-based assessment of vulnerability to current and future weather events and natural disasters
<i>Shall, as appropriate..</i>	
5	Describe ways to improve response to impacts and changes
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Sources:
 PROTECT Formula Funding Implementation Guidance
 AASHTO Informal Worksheet with RIP Checklist

Assess Vulnerability and Risk

PROTECT - Resilience Improvement Plan	
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Sources:
 PROTECT Formula Funding Implementation Guidance
 AASHTO Informal Worksheet with RIP Checklist

Develop Strategies

PROTECT - Resilience Improvement Plan	
<i>The Plan Shall..</i>	
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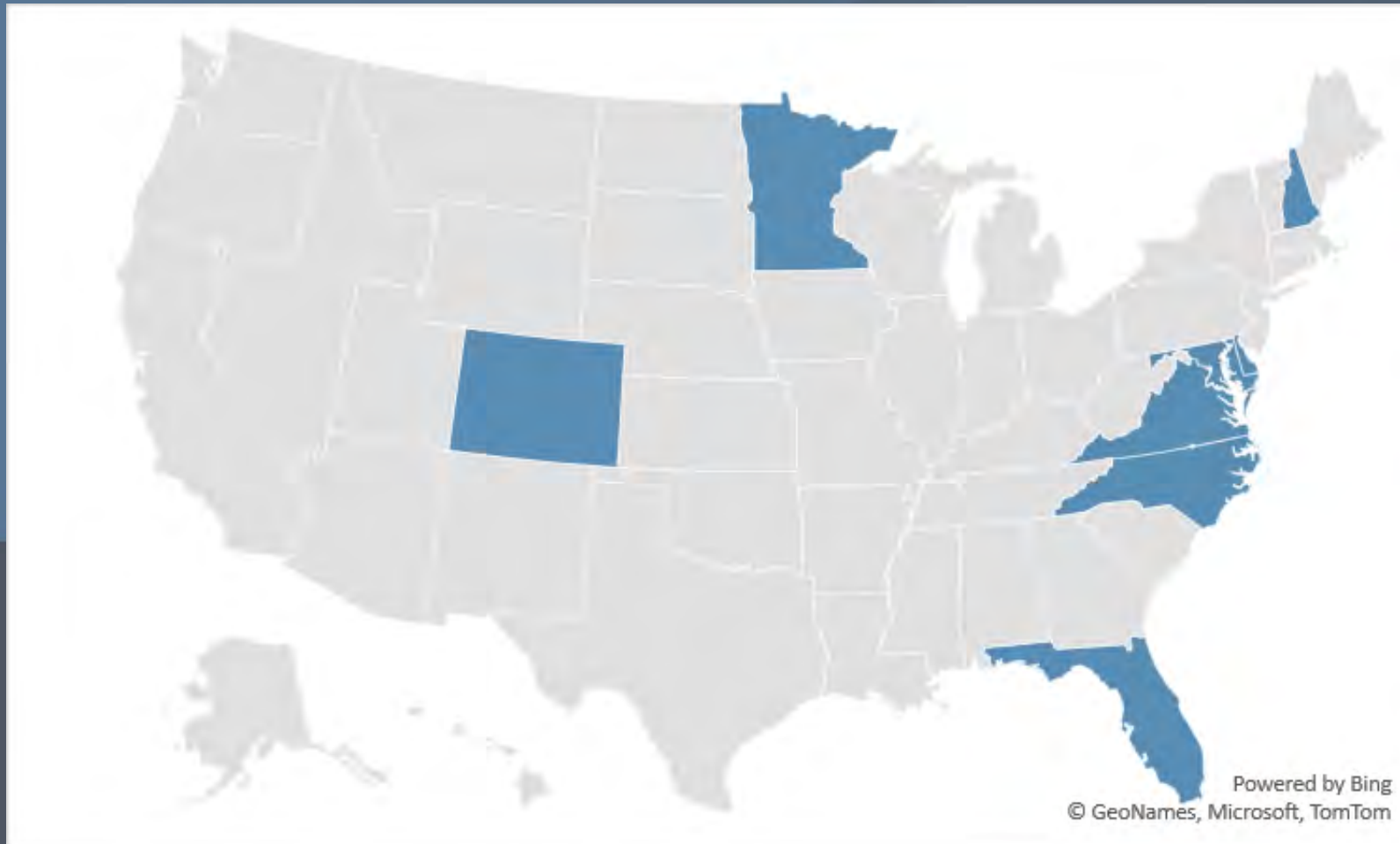
Sources:
 PROTECT Formula Funding Implementation Guidance
 AASHTO Informal Worksheet with RIP Checklist

Integration and Implementation

PROTECT - Resilience Improvement Plan	
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1	Encompass immediate and long-range planning activities and resilience investments
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12	Describe the resilience improvement policies
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Sources:
 PROTECT Formula Funding Implementation Guidance
 AASHTO Informal Worksheet with RIP Checklist

Peer Agency Interviews



Ready to Initiate

- Florida DOT
- North Carolina DOT
- Minnesota DOT
- New Hampshire DOT

In-Progress

- Colorado DOT
- Delaware DOT
- Maryland DOT
- Virginia DOT

Peer Agency Interviews

Establish
Objectives and
Scope

Assess
Vulnerability and
Risk

Develop
Strategies

Integration and
Implementation

Objectives and Scope - Requirements

2

Demonstrate a **system-wide** approach to transportation **system resilience**

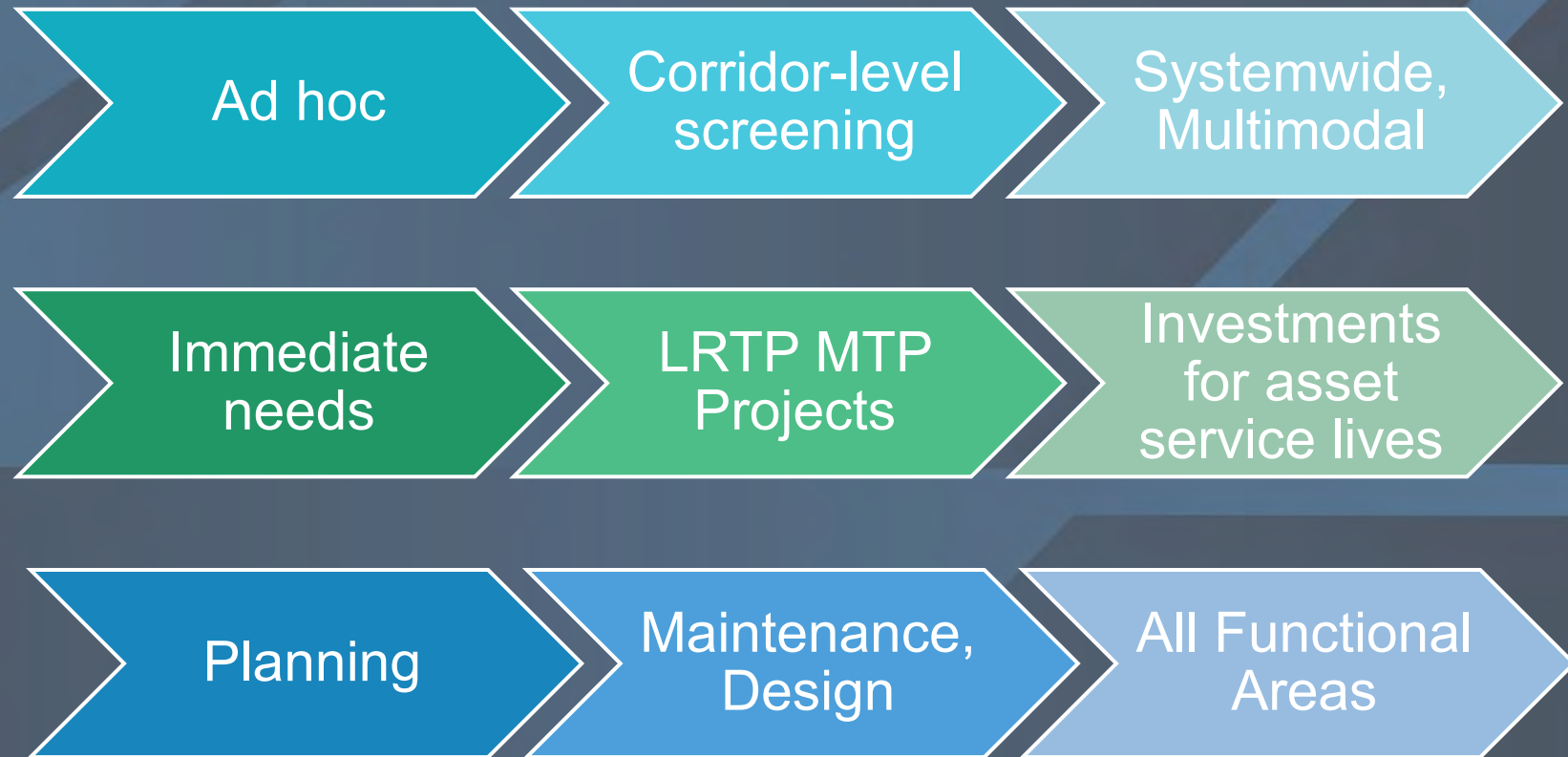
9

Use a **long-term** planning period

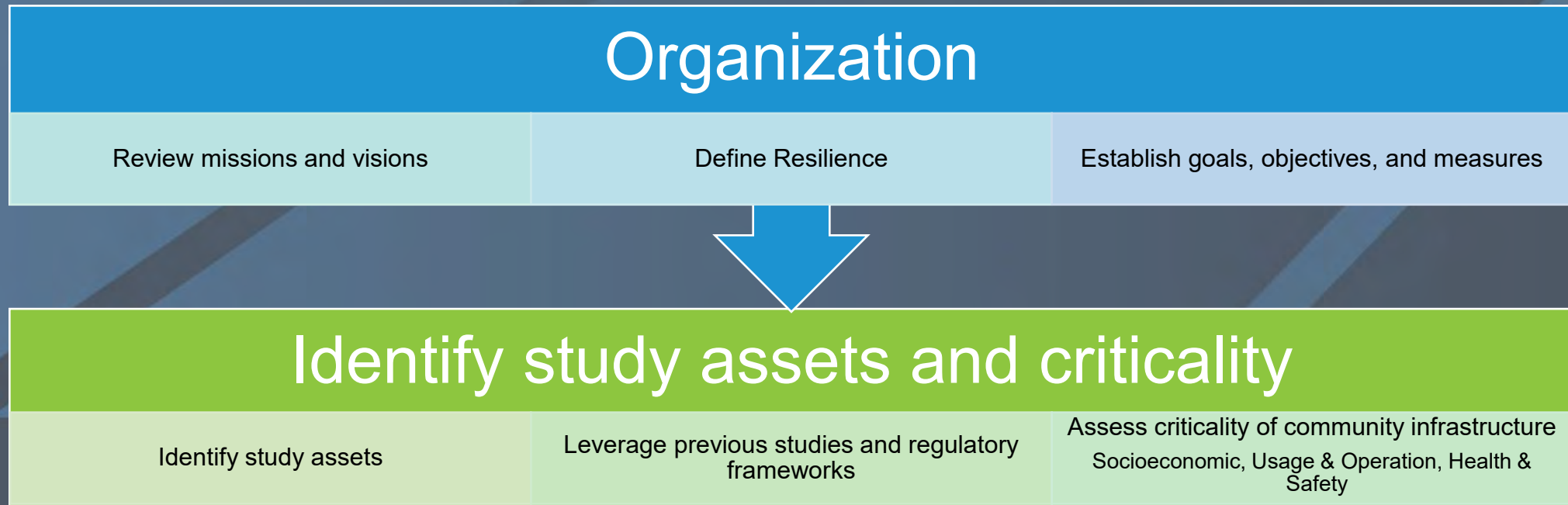


Objectives and Scope – Interview Findings

Resilience Improvements



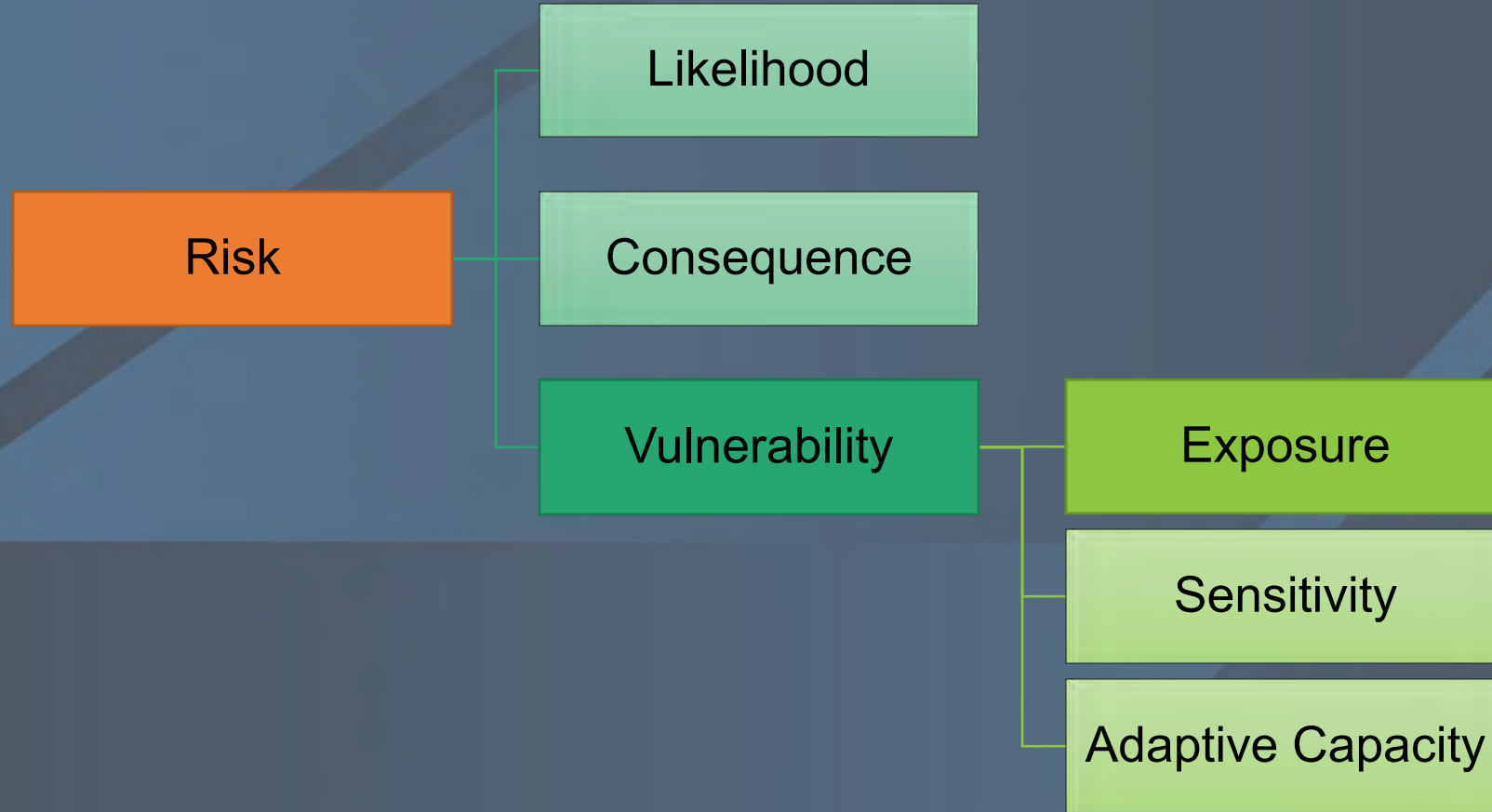
Objectives and Scope – Takeaways



Assess Vulnerability and Risk - Requirements

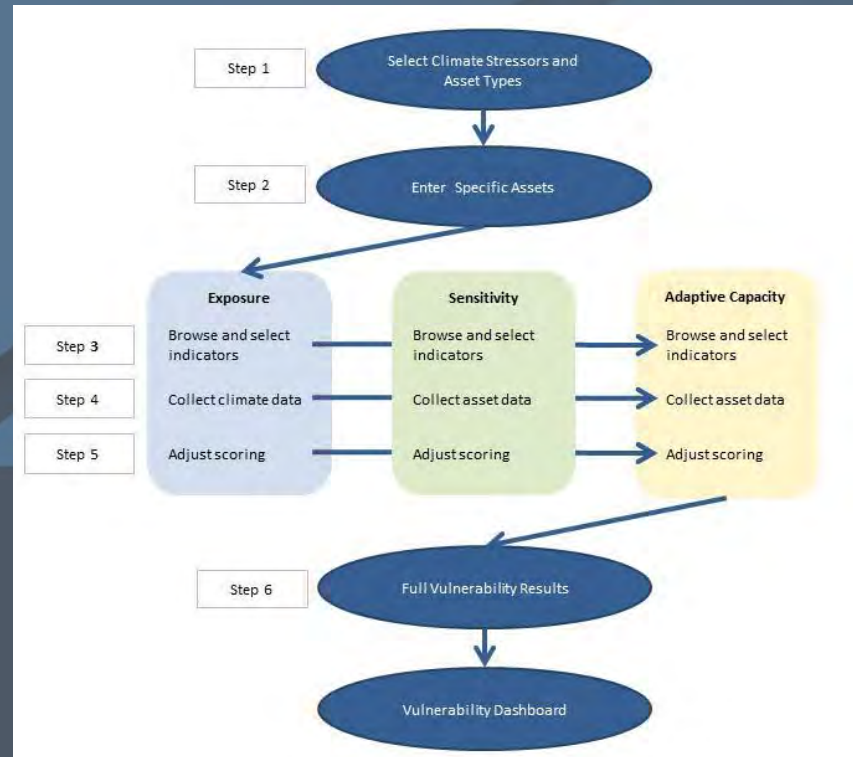
4	Include a risk-based assessment of vulnerability to current and future weather events and natural disasters
8	Assess community infrastructure resilience
14	Use science and data

Assess Vulnerability and Risk – Interview Findings



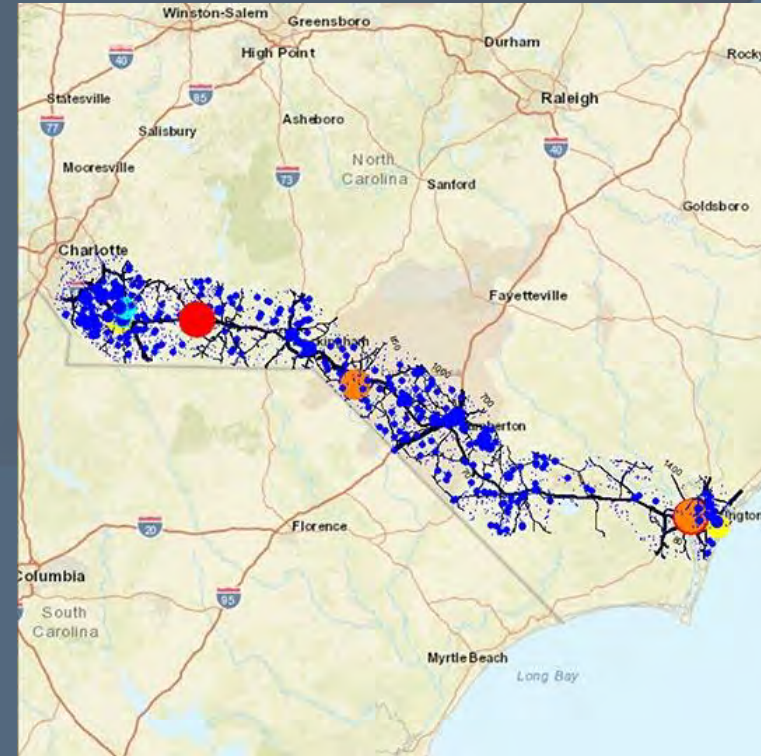
NCDOT Vulnerability Assessments for Strategic Corridors

➤ US 70 Vulnerability and Risk Assessment using VAST



FHWA Vulnerability Assessment Scoring Tool (VAST)

➤ US 74 Resilience Study using City Simulator model



<https://www.snclavalin.com/en/projects/ncdot-us74-corridor-resilience-study>

CDOT Risk and Resilience Analysis Procedure

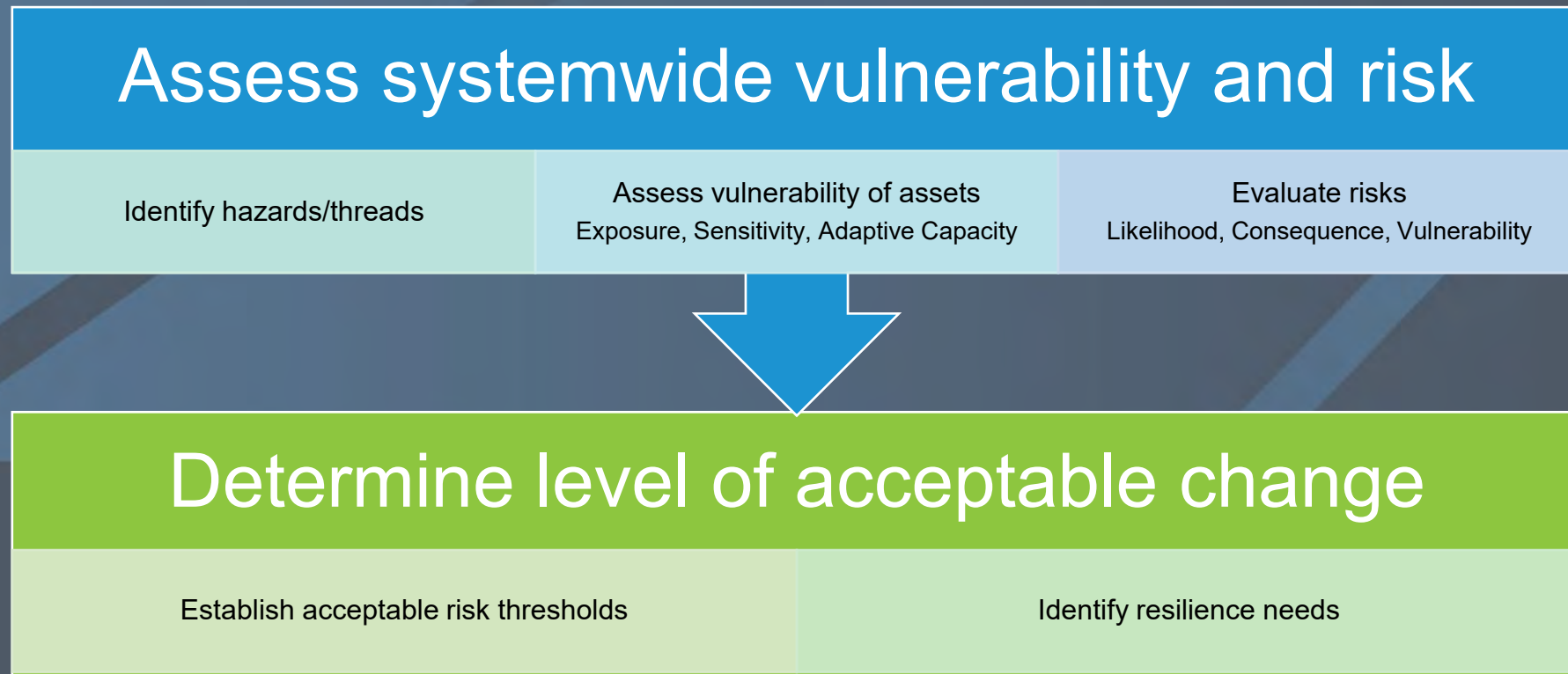
- Consequence Calculation:
 - » Owner consequences
 - unit costs for each asset type
 - % of replacement cost + clean up
 - » User consequences
 - additional vehicle operation cost
 - value of time lost to additional travel time and distance by detours and delays



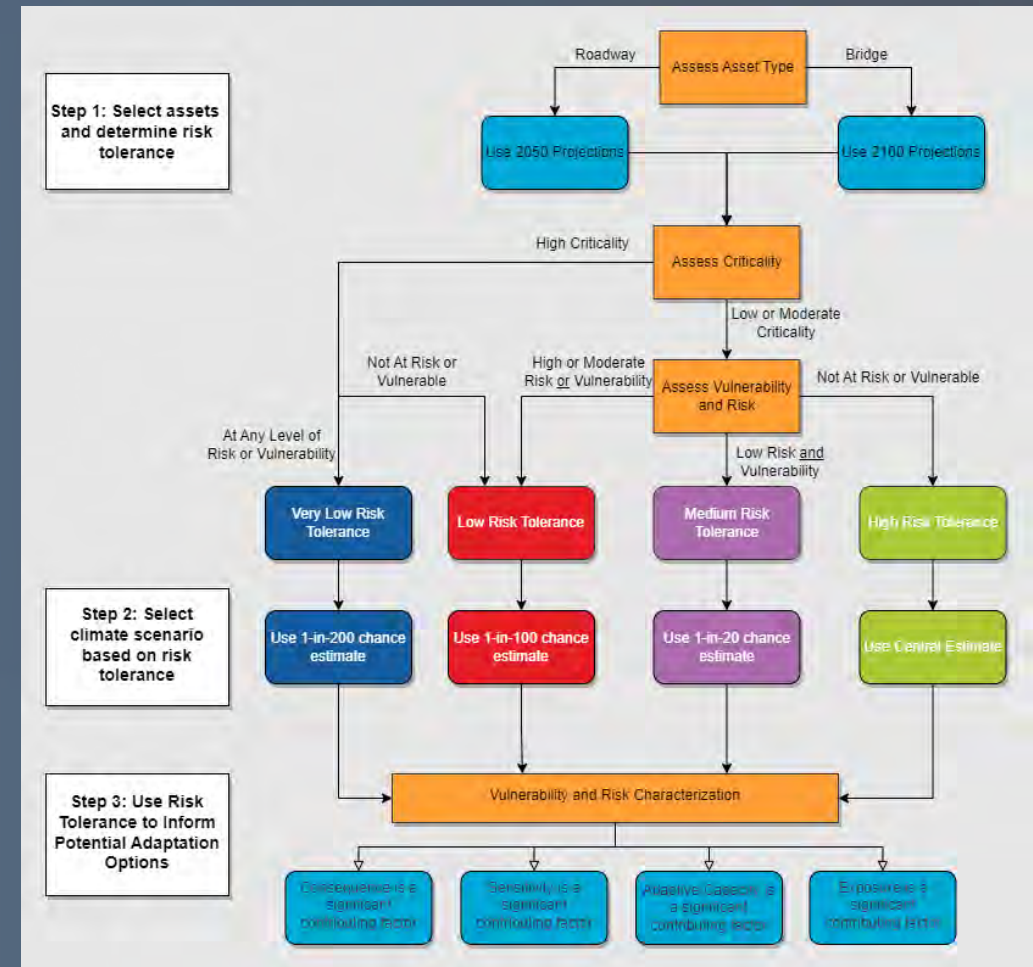
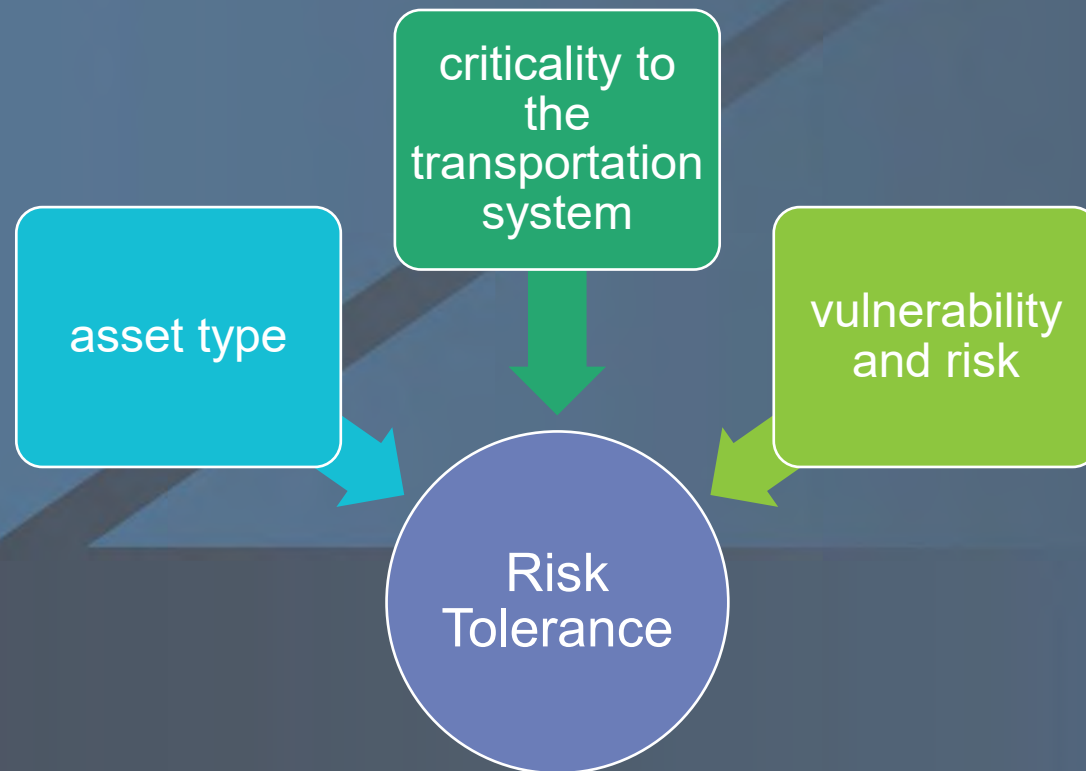
$$\text{Risk} = \text{Consequence} \times \text{Vulnerability} \times \text{Likelihood}$$



Assess Vulnerability and Risk – Takeaways



NHDOT Coastal Risk Tolerance Framework



NHDOT Coastal Risk Tolerance Framework Process

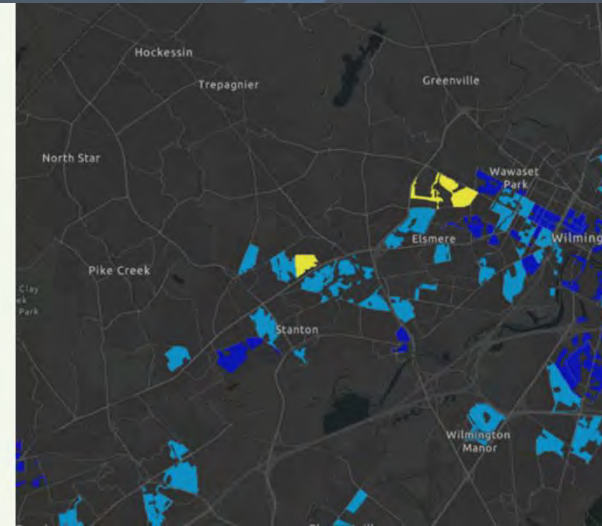
Delaware Equity Analysis Tool

- Integrate equity into resilience planning
- Identify Equity Focus Areas to:
 - » inform investment decisions
 - » support public outreach and engagement strategies
 - » identify disadvantaged communities

Delaware Equity Analysis Tool

Mapping Equity Focus Areas in the First State

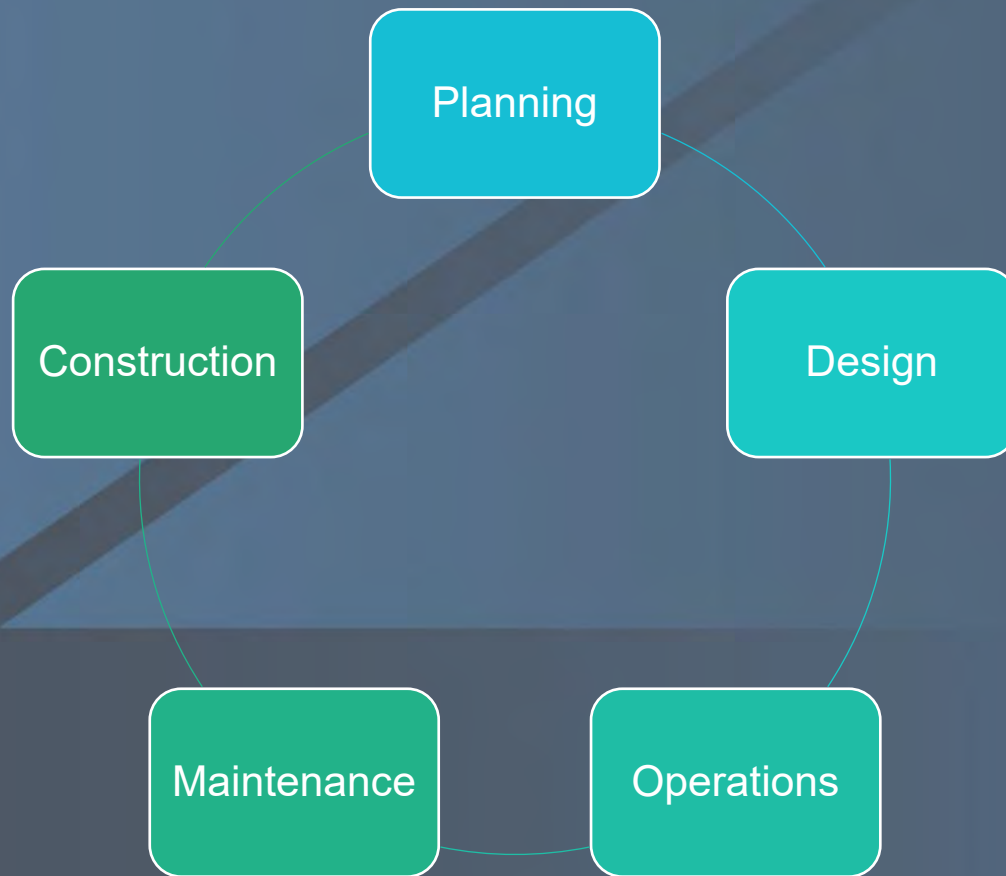
Delaware Department Of Transportation



Develop Strategies - Requirements

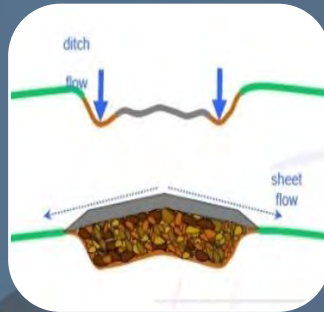
1	Encompass immediate and long-range planning activities and resilience investments
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7	Consider benefit of natural Infrastructure
10	Designate evacuation routes and strategies
11	Plan for response to anticipated emergencies
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Develop Strategies – Interview Findings

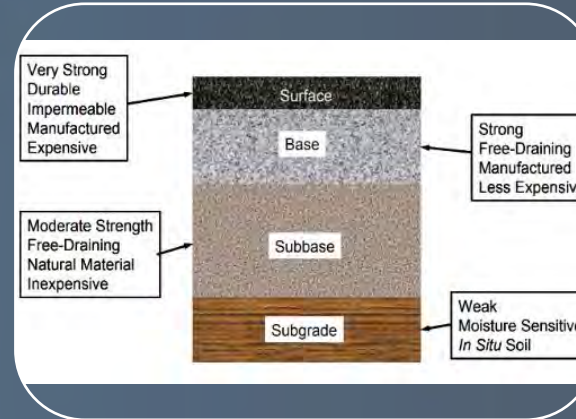


- *FDOT Capability Maturity Framework*
- *NCDOT Resilience Strategy Report*
- *DelDOT Incorporate Resilience into Design Manual*
- *VDOT Strategy Toolbox*

Develop Strategies – Interview Findings



Raise Road Profile



Enhance Road Surface or Subbase



Enhance Drainage



Protect Medians Shoulders

Develop Strategies – Interview Findings



Hardened
Shoulders



Seawalls
Revetments



Wave
Attenuation

Living
Shorelines



Sources:

- <https://hiveminer.com/User/Aaron%20Volkering>
- https://www.dirtandgravel.psu.edu/sites/default/files/General%20Resources/Technical%20Bulletins/TB_Raising_Road_Profile.pdf
- "Design Considerations for Embankment Protection During Road Overtopping Events," Marr et al. University of Minnesota, MN/RC 2017-21, 2017.
- Geotechnical Aspects of Pavements, Publication No. FHWA-NHI-10-009, Federal Highway Administration
- <http://www.tbo.com/news/business/pyramid-key-to-saving-egmont-key-20140526/>
- <http://www.connectionnewspapers.com/news/2012/may/23/street-runs-through-it/>
- Virginia DCR Stormwater Design Specification No 7 <http://vwrrc.vt.edu/swc/NonPBMPSpecsMarch11/VASWMBMPSpec7PERMEABLEPAVEMENT.html>
- Maccaferri Gabion Mat brochure
- Restore America's Estuaries, Living Shorelines – From Barriers to Opportunities,

Develop Strategies – Takeaways

Develop desired pathways and related actions

Identify adaptation strategies
Natural and nature-based solutions

Conduct project evaluation
BCA, equity, etc.

Develop and prioritize resilience
improvements

Integration and Implementation - Requirements

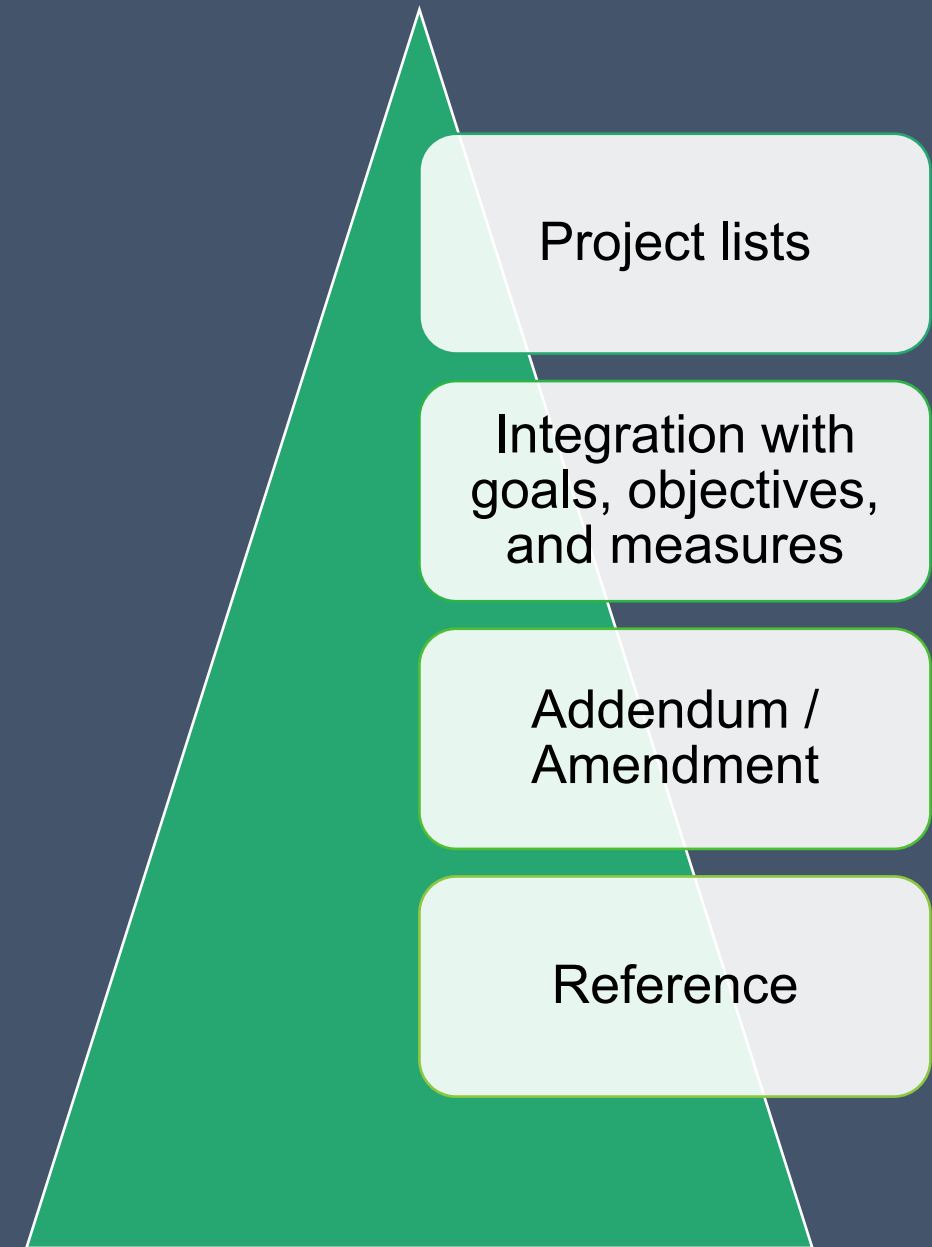
3	Consistent with and complement State and local hazard mitigation plans
6	Describe the codes, standards, and regulatory framework to ensure improvements

Integration and Implementation – Interview Findings

- Leverage existing practices
 - » Environmental review, design manual, TAMP, maintenance, operations
- Overlay Resilience Needs with STIP
 - » Consider criticality, equity, BCA, funding match availability
- Develop framework to incorporate resilience into all aspects of the transportation /DOT functions

Alignment with LRTP

- Aligning RIP with LRTP is a challenge
 - » Unclear about FHWA's specific requirements
 - » Prefer not to do a project-base alignment to avoid frequent update
 - » Consider adding RIP as an addendum to LRTP instead

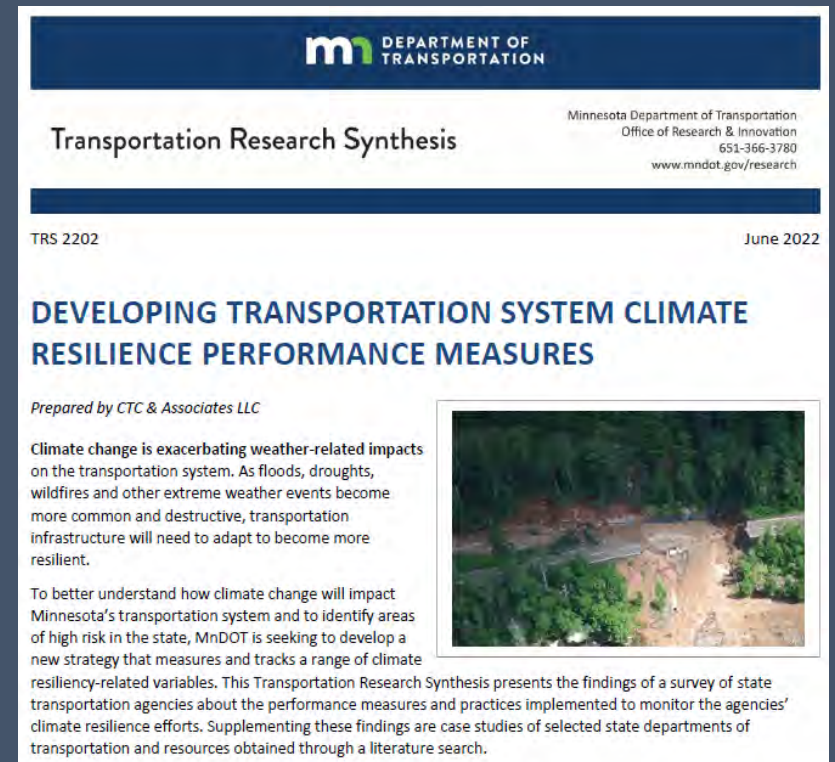


Performance Monitoring

- Develop resilience performance measures and link them to agency goals and objectives
 - » Such as safety, mobility, and level of service
 - emergency/incident responsiveness, road closures clearance time, etc.

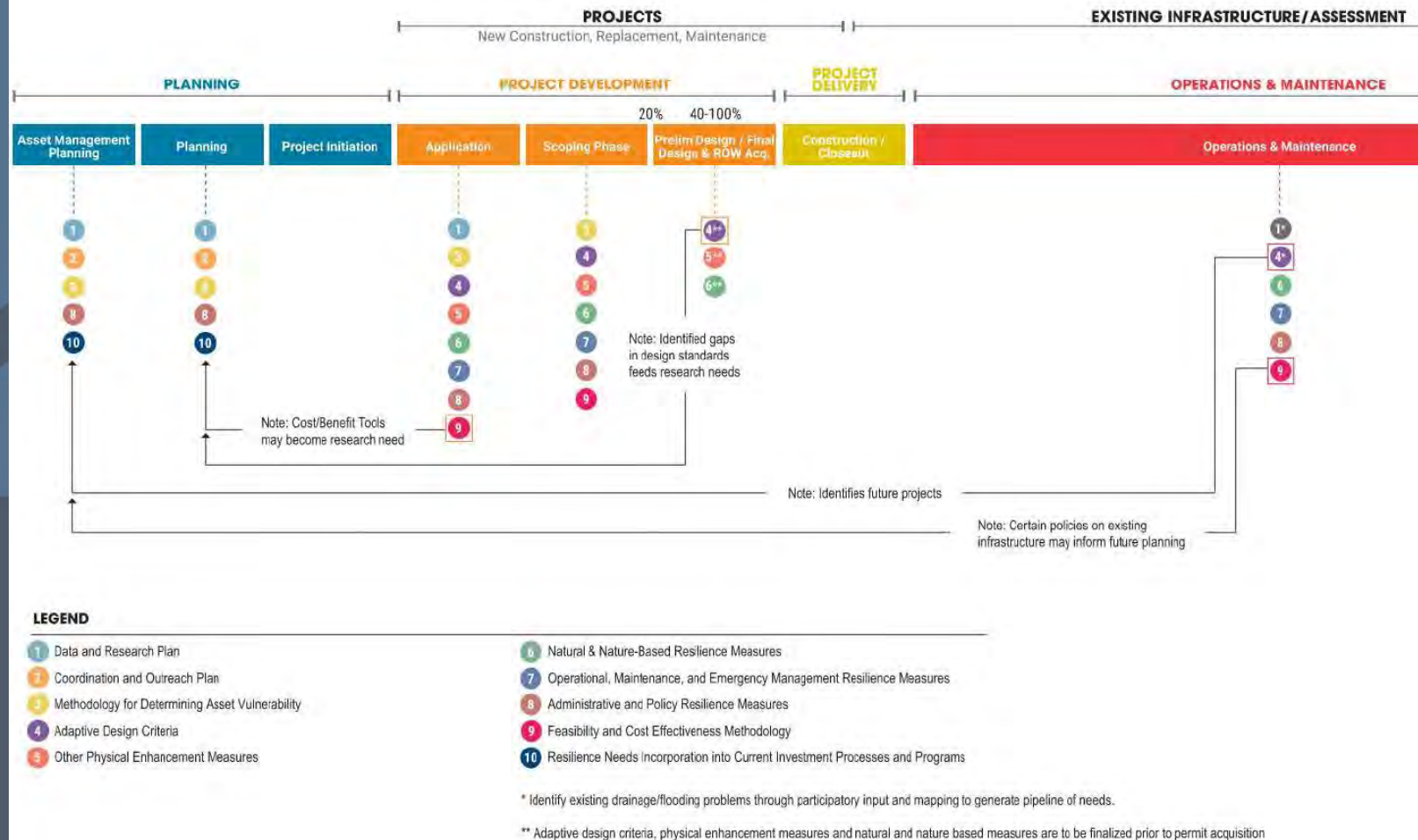
MnDOT Transportation Research Synthesis –

Developing Transportation System Climate Resilience Performance Measures



VDOT Resilience Plan

Figure 3. Integration of Resilience Plan Strategies into VDOT's Existing Business Practices



Integration and Implementation - Takeaways

Implement resilience improvements

Incorporate resilience strategies
into LRTP and STIP

Integrate resilience consideration
into current transportation process

Planning, environmental review, design,
operation, maintenance

Coordinate with other planning
activities

hazard mitigation, asset management,
repeatedly damaged, freight

Stakeholders Engagement and Communication

- It is important to create a definition of resilience and make it relatable to each functional areas, to get buy-in and engage staff
- Coordinate regularly
 - » FHWA Division Office (RIP development and approval)
 - » Resilience Steering committee
 - » Resilience Working Group
 - » MPO, Counties, Cities

Other Lessons Learned

- Workforce development is important
 - » education, outreach, and trainings on resilience
- Having good data is critical

Framework





APPENDIX B REVIEW OF PREVIOUS RESILIENCE-THEMED LITERATURE IN RHODE ISLAND

The project team's review of existing resilience guidance and planning efforts included the following publications:

- **MLD 601 - RIDOT Final Memo (2022):** Developed as a research paper by the Harvard Operations Management Team, and focusing primarily on bridge infrastructure, this memo aims to assist RIDOT in developing a framework to better account for resiliency in capital planning, given that the original approach doesn't sufficiently account for general effects from climate change. The memo includes a review of potential incorporation strategies, such as repairs, managed retreat, alternative routings, and nature-based solutions. It also includes an implementation plan for RIDOT which was developed through extensive stakeholder engagement.
- **RIDOT State Transportation Improvement Plan, 2022 – 2031 (2021, Revisions through 2023)⁴:** The STIP is a list of transportation projects the State of Rhode Island intends to implement using United States DOT funds. The STIP includes projects for the upcoming ten years and is reauthorized every four years. Overall, improving asset resiliency to climate change, storm surge, and sea level rise is a key component of asset management, a priority focus of the STIP.
- **Moving Rhode Island 2040 (2020)⁵:** As required by the Federal Highway Administration (FHWA), Rhode Island State LRTP provides a framework for statewide transportation goals and objectives, and identifies strategies to help Rhode Island facilitate the efficient movement of people and goods. Rhode Island State's LRTP identifies coastal resiliency as a concern for the state's coastal counties and highlights the need to prepare for more frequent flooding and thawing cycles. The plan includes a vision for enhancing transportation network resiliency in part by encouraging state agencies and municipalities to work together within the Municipal Resilience Program at the RI Infrastructure Bank.

⁴ <https://planning.ri.gov/stip>

⁵ <https://planning.ri.gov/sites/g/files/xkqbur826/files/documents/trans/2020/Final-LRTP-December-2020.pdf>



- **Rhode Island Congestion Management Process (2020)⁶:** As required by the FHWA for metropolitan areas with a population greater than 200,000, the Rhode Island Congestion Management Process (CMP) aims to identify traffic congestion and its causes, develop monitoring processes to measure transportation system performance and reliability, and develop congestion management strategies while also moving them into the funding and implementation stages. Given that the Department of Administration's Division of Statewide Planning serves as staff to the State Planning Council, which serves as Rhode Island's metropolitan planning organization (MPO), the state agency is the developer of this plan.
- **RIDOT Stormwater Control Plan Approach (2019)⁷:** This document summarizes the approach and procedures used during the development and implementation of Stormwater Control Plans. This includes information on infrastructure site descriptions, the process to develop stormwater reduction targets, existing stormwater controls, potential enhanced non-structural and structural stormwater controls, and implementation mechanisms.
- **State of Rhode Island Hazard Mitigation Plan Update (2024)⁸:** Updated in 2024, this plan (referred to as the SHMP) provides a broad assessment of all hazards that pose a potentially significant risk to Rhode Island. Not limited to transportation and climate resilience, the plan covers biological incidents, hazardous materials incidents, civil disturbances, cybersecurity, transmissible disease, dam failure, drought, earthquake, extreme temperatures, wildfire, flood, , infrastructure failure, radiological incidents, sea level rise, severe winter weather, terrorism, severe thunderstorms, tornadoes, and tropical and extratropical storms. The plan includes an assessment of key geographic, infrastructure, and socioeconomic characteristics of the state, as well as a methodology for identifying hazards and risk, occurrence history, relevant stakeholders, and broad mitigation strategies.
- **STORMTOOLS⁹ :** Developed by in collaboration between the Rhode Island Coastal Resources Management Council and the University of Rhode Island, STORMTOOLS maps storm inundation, with and without sea level rise, for varying return period storms that covers all of Rhode Island's coastal waters. Predictions are provided which show water extent and depth at any given point for nuisance floods (1-, 3-, 5-, and 10-year recurrence intervals) and 25-, 50-, 100-, and 500-year storm scenarios at a 95% confidence interval. Sea level rise of 1, 2, 3, 5, and 7 feet on their own as well as combined with each storm scenario are also modeled.

⁶ https://planning.ri.gov/sites/g/files/xkgbur826/files/documents/trans/2020/CMP_Full_Report_Jun20.pdf

⁷ https://www.dot.ri.gov/documents/about/protecting/stormwater/SCP/SCP_Approach_Document.pdf

⁸ https://riema.ecms.ri.gov/sites/g/files/xkgbur671/files/2024-02/2024%20RI%20Hazard%20Mitigation%20Plan%20FINAL%20_Reduced%20size.pdf

⁹ <https://stormtools-mainpage-crc-uri.hub.arcgis.com/>



- **Resilient Rhody Impact Report (2018)¹⁰**: Published as a three-year update on statewide climate resilience goals, the document highlights coordination efforts and tools, as well as current and recently completed projects and initiatives. Key thematic areas include critical infrastructure and utilities, natural systems, emergency preparedness, community health & resilience, and financing.
- **Technical Paper 167 – Vulnerability of Municipal Transportation Assets to Sea Level Rise and Storm Surge (2016)¹¹**: Developed by the Department of Administration Division of Planning in 2016, this document provides a comprehensive assessment of statewide asset vulnerability to sea level rise scenarios. The assessment uses STORMTOOLS data, developed in collaboration between the Rhode Island Coastal Resources Management Council and the University of Rhode Island.

¹⁰ <https://climatechange.ri.gov/resilient-rhody>

¹¹ https://planning.ri.gov/sites/g/files/xkqbur826/files/documents/sea_level/2016/TP167.pdf



APPENDIX C REVIEW OF NATIONAL RESILIENCE DEFINITIONS AND EXISTING RIDOT LITERATURE WHERE RESILIENCE IS MENTIONED

C.1 USDOT Climate Action Plan for Resilience (2021)¹²

The USDOT Climate Action Plan for Resilience was developed in 2021 for the purposes of bolstering adaptation and increasing resilience for the nation's transportation system. It builds on previous USDOT Climate Action Plans developed in 2012 and 2014. It includes five objectives:

- Incorporate Resilience into DOT Grant and Loan Programs;
- Enhance Resilience Throughout the Project Planning and Development Process;
- Ensure Resiliency of DOT Facilities and Operational Assets;
- Ensure Climate-ready Services and Supplies; and
- Improve Climate Education and Research on Resilience

As per document footnote, USDOT's definition of resilience is taken from the U.S. Global Change Research Program's Fourth National Climate Assessment and CEQ Instructions for Preparing Draft Climate Action Plans under Executive Order 14008.

Resilience: *The ability to anticipate, prepare for, and adapt to changing conditions and withstand, respond to, and recover rapidly from disruptions.*

FHWA Order 5520¹³

Established in 2014, the purpose of this order is to establish Federal Highway Administration (FHWA) policy on preparedness and resilience to climate change and extreme weather events. This FHWA Order uses the same definition for resiliency.

¹² https://www.transportation.gov/sites/dot.gov/files/2022-04/Climate_Action_Plan.pdf

¹³ <https://www.fhwa.dot.gov/legregs/directives/orders/5520.cfm>



AASHTO Resiliency Peer Exchange on Extreme Weather and Climate Impacts¹⁴

The American Association of State Highway and Transportation Officials (AASHTO) convened the Resiliency Peer Exchange on Extreme Weather and Climate Impacts in 2017. The goal of the exchange was to bring together transportation professionals from across the United States to discuss how resiliency is being addressed in state Departments of Transportation (DOTs). In the context of this peer exchange, and as the organization's formal definition, AASHTO defines resilience as follows:

Resilience: *The ability to prepare and plan for, absorb, recover from, or more successfully adapt to adverse events.*

National Academies of Science (TRB, NCHRP)

In addition, the National Academies of Science defined resilience as:

Resilience: *The ability to prepare and plan for, absorb, recover from, or more successfully adapt to actual or potential adverse events.*

C.2 Rhode Island Efforts to Define Resilience

At the state level, and recognizing the risks associated with climate change on Rhode Island, the state has highlighted resilience across multiple documents and planning efforts. These efforts include the following:

Resilient Rhody (2018)¹⁵:

Resilient Rhody was developed in 2018, following the signing of an Executive Order to spearhead climate resilience efforts across the state. In addition to developing the Resilient Rhody climate action plan, the Executive Order also appoints a Chief Resilience Officer. Resilient Rhody defines climate resilience as follows:

Climate Resilience: *The capacity of individuals, institutions, businesses, and natural systems within Rhode Island to survive, adapt, and grow no matter what chronic stresses and weather events they experience.*

Resilience Rhody also indicates:

While the effects of climate change are felt across the state, these impacts are not equally distributed. Effective climate resilience requires a focus on environmental justice and equity to support local leadership for sustained interaction between community, business, and government.

¹⁴ https://environment.transportation.org/wp-content/uploads/2021/05/summary_report_aashto_resiliency_peer_exchange.pdf

¹⁵ <https://climatechange.ri.gov/sites/g/files/xkgbur481/files/documents/resilientrhody18.pdf>



State of Rhode Island State Hazard Mitigation Plan Update (2024)¹⁶

The purpose of this document is to provide comprehensive guidance for hazard mitigation statewide. This includes guidance for making homes, businesses, and communities more resilient to the impacts of climate change, hurricanes, floods, tornadoes, earthquakes, winter storms, wildfires, and other natural hazards. The document expands to include additional hazards as well.

Resilience is mentioned throughout the plan as its primary purpose is to provide the impetus for making homes, businesses, and communities more resilient to the impacts of hurricanes, floods, tornadoes, earthquakes, winter storms, wildfires, and other natural hazards as well as cybersecurity and civil disturbance incidents, dam and infrastructure failures, terrorism, and other human- and technologically-caused hazards. The document however does not formally resilience. Additionally, resilience is not just limited to physical infrastructure and ecosystems, but is also expanded to include technology, cybersecurity, and terrorism challenges.

Additional efforts and responses to climate change and resilience needs include the following the following:

- **2021 Act on Climate:** Set mandatory, enforceable climate emissions reduction goals culminating in net-zero emissions by 2050.
- **2017 Executive Order 17-06:** Reaffirmed Rhode Island's commitment to the principles of the Paris Climate Agreement.
- **2015 Executive Order 15-17:** Required that, by 2025, 25% of new light-duty state fleet purchases and leases must be zero-emissions vehicles.
- **2014 Resilient Rhode Island Act:** Created the Rhode Island Executive Climate Change Coordinating Council (EC4).

¹⁶ https://riema.ecms.ri.gov/sites/g/files/xkgbur671/files/2024-02/2024%20RI%20Hazard%20Mitigation%20Plan%20FINAL%20_Reduced%20size.pdf



C.3 A.2.2 Review of Existing RIDOT Literature Where Resilience is Mentioned

Although RIDOT does not have a formally established definition of resilience, the term and concept has already been integrated into multiple planning efforts. This includes:

Rhode Island Long Range Transportation Plan (2040)¹⁷:

Resilience is mentioned throughout the document, but not officially defined. Mentions include:

- *Improving the resiliency and reliability of the transportation system and reducing or mitigating stormwater impacts of surface transportation; identified as a national goal for long range transportation planning.*
- *Vision for the plan includes a safe and resilient transportation system, including by means of providing affordable, effective, and supportive transportation choices.*
- *Climate Impacts & Resiliency is listed as a Challenge/Opportunity. 'Rising sea levels, extreme flooding, and hotter temperatures will stress transportation infrastructure in the future differently than today. These environmental effects require changes in design specifications for system resiliency. Long range planning will be required to adapt to more frequent freeze thaw cycles and precipitation events.'*
- *'Enhancing Transportation Network Resilience' is an objective listed for the goal of 'Maintaining Transportation Infrastructure'.*
- *Coastal resiliency is listed as a concern for Rhode Island's coastal communities.*

Rhode Island Transportation Asset Management Plan (2022)¹⁸:

Resilience is mentioned throughout the plan. In addition, environmental resiliency is listed as a principal asset risk. Listed under this section, RIDOT has a plan to make the state more resilient. As per the plan, this includes:

Research: *RIDOT has partnered with the Harvard Kennedy School (HKS) to develop a matrix of environmental resiliency treatment options for vulnerable assets. Future SPR projects will advance this matrix to create a set of planning tools for the state and local communities to leverage in transportation planning.*

Planning: *Beginning in 2023, RIDOT will advance a Carbon Reduction Plan (CRP) to expand resiliency efforts and comply with IIJA requirements. RIDOT and the Division of Statewide Planning (RIDSP) are currently collaborating to expand the scope of services supporting the Statewide Travel Demand Model to conduct mesoscale analyses documenting the impact of RIDOT projects on greenhouse gas emissions. RIDOT is also considering adopting a Resilience Improvement Plan (RIP) to support USDOT's objectives under the PROTECT Program.*

Mapping: *RIDOT is integrating GIS data on stormwater with its larger GIS platform to incorporate drainage, water quality, and stormwater treatment elements into capital projects. When fully implemented, this will reduce the risk*

¹⁷ <https://planning.ri.gov/sites/g/files/xkgbur826/files/documents/trans/2020/Final-LRTP-December-2020.pdf>

¹⁸ http://www.dot.ri.gov/documents/RhodeWorks/RIDOT_TAMP_2022.pdf



of budget and schedule overruns while reducing pollution from runoff and constructing a resilient transportation system.

Technology: Since 2019, RIDOT and RIDSP have collaborated with Esri to develop the Statewide Intake Framework for Transportation (SWIFT), a digital portal to manage transportation project proposals. SWIFT contains a “Project Readiness” module which leverages geospatial data to flag items of complexity for applicants, including sea-level rise. Expected to be fully deployed by 2024, SWIFT will completely digitize the project intake workflow and inform more reliable schedule estimates informed by asset management data.

Investment: RIDOT has expanded its investments in congestion mitigation and greenhouse gas emissions reduction and will continue to do so. In 2022, RIDOT deployed IJA funds to create new programs integrating advanced technologies into capital projects and support modeling initiatives at the project, local, and state levels. Projects have already begun to reduce congestion at 44 locations around the state and resurface miles of shared-use path across the state, supporting greener forms of transportation. And, of course, RIDOT is committing more than \$100 million over a 10-year period to ensure compliance with the Clean Water Act and a number of remedial measures under a consent decree with the Environmental Protection Agency (EPA).

Rhode Island Freight & Goods Movement Plan (2022)¹⁹

Resilience is mentioned throughout the plan. Similarly, resiliency is not formally defined. Mentions of resiliency include:

- ‘Ensure a Resilient Post-Disaster Freight Network in Rhode Island’ is listed as a goal of the plan. This includes the following strategies:
- Strengthen Storm Resilience and PostStorm Recovery at the Ports Through Strategic Partnerships and Planning
- Collaborate at All Levels of Government for Planning Efforts to Ensure Efficient and Coordinated Response to Emergency and Disaster Events
- Establish New Collaborative Partnerships Between the State and Port Community to Improve Storm Resilience Planning
- Utilize the Transportation Asset Management Plan (TAMP) to Make Data Driven Decisions

‘Climate Change & Port Resiliency’ is discussed for ‘Maritime Port Conditions’

Resilience is discussed as a key freight policy recommendation. The following points are also included:

- Identify critical components of the RI freight network and freight facilities that are located within the 1% annual chance flood zone (100-year flood) and determine measures to protect or mitigate from flooding and storm surges.
- For critical supplies and commodities such as fuel and food, assess the resilience of these supply chains to ensure these commodities can continue to be distributed to RI residents after catastrophic and infrastructure damaging events.

¹⁹ https://planning.ri.gov/sites/g/files/xkqbur826/files/2022-06/2022_RI_Interim_Freight_Plan_Update_6_1_22.pdf



- *Consider undertaking scenario planning with appropriate public agencies and private companies to plan contingencies (e.g., park fuel trucks away from flood prone areas). Identify key freight equipment and assets that can assist with contingency and response operations.*
- *Improve redundancy of freight transportation network.*
- *Use STORMTOOLS and other existing data to identify specific freight assets vulnerable to the impacts of flooding, storm surges, and sea level rise.*
- *Work with RIEMA to develop specific resiliency actions plans for the most important/vulnerable state assets.*

In addition to these factors, resilience is indicated as part of the RIDOT framework for assessing project management risk. Resiliency is additionally mentioned throughout Rhode Island's STIP²⁰.

The definition of resilience for this RIP, is as follows:

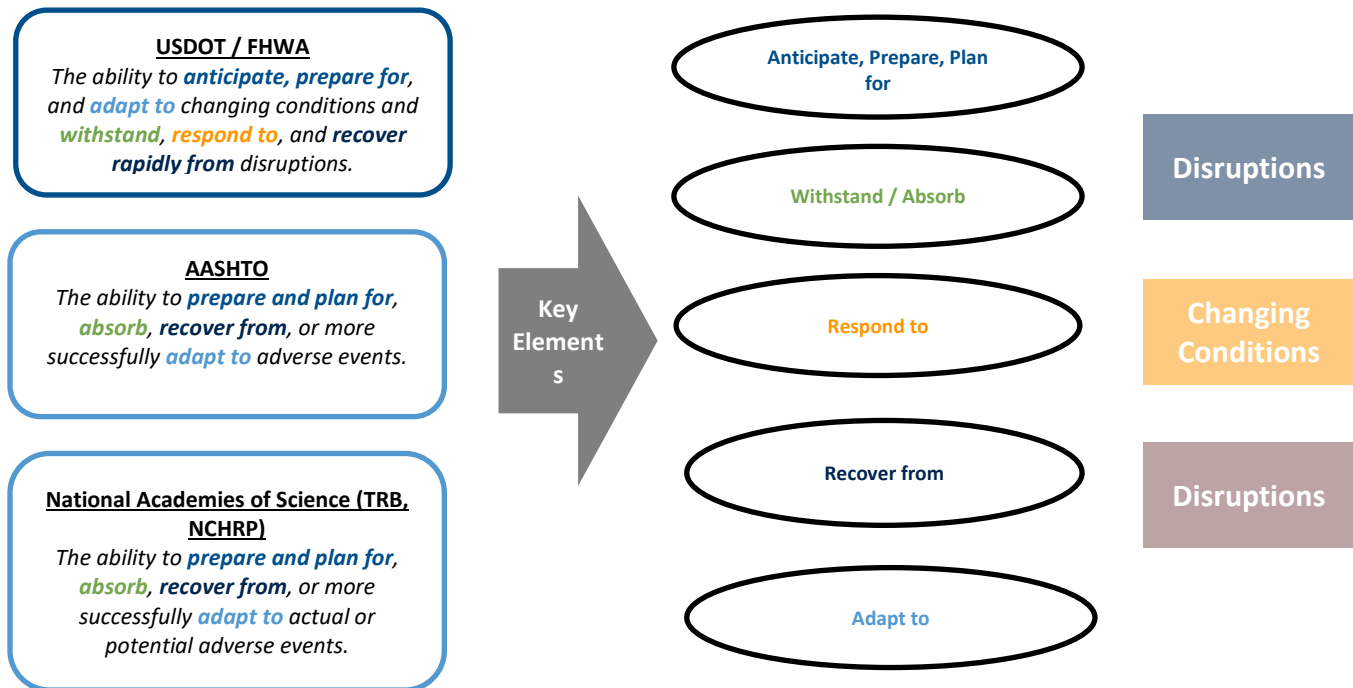
Resilience is defined by RIDOT as the ability of the Rhode Island transportation system to anticipate, prepare for, and adapt to changing conditions; and withstand, respond to, and recover rapidly from any disruptions.

This definition was developed synthesizing the comment elements in the resilience definitions described by USDOT/FHWA, AASHTO, the National Academies of Science/TRB, as illustrated in the graphic below, and considering RIDOT and its agency partners' continuous effort to promote resilience in the transportation system.

²⁰ <https://planning.ri.gov/stip>



Figure B.1 Synthesizing Key Elements of Resilience Definitions



Connecting a Formalized Definition of Resilience Back to Rhode Island State's LRTP

In aligning with Rhode Island's LRTP, the RIP, and this definition of resilience will support RIDOT's goal to maintain its transportation infrastructure by assessing the vulnerability of RIDOT's multimodal transportation system and developing adaptation and mitigation strategies to protecting asset integrity and minimize damage due to extreme weather or climate events. These proactive measures will enable more accurate projection of maintenance needs and reduce repair and maintenance costs, as well as reducing network disruptions or closures, and enhancing emergency management. The multimodal analysis conducted in the RIP will enhance RIDOT's ability to provide safe, reliable, and efficient travel options to all residents in the state across all modes, which are critical to supporting its goals of Connect People and Places and Support Economic Growth. Following the PROTECT guidelines and the Framework developed in Chapter 1, the RIP identifies adaptation and mitigation strategies with emphasis on natural or nature-based solutions, coordinating with RIDOT's Carbon reduction plans to Promote Environmental Sustainability. In addition, the RIP develops resilience strategies responding to needs of communities, especially those that are socially vulnerable, to Strengthen the Communities by enhancing their travel, place, and quality of life in dealing with the impact from extreme weather and climate change.



APPENDIX D CRITICALITY COMPONENTS

This Appendix describes the individual components of these three categories, including individual maps for each criticality factor. As identified in Section 3.3, criticality for each asset is comprised of the three categories:

- **Usage & Operational Importance:** A measure of how important the asset is to the overall function of the transportation system.
- **Socioeconomic Importance:** A measure of the extent to which an asset serves people and businesses in Rhode Island, including underserved or disadvantaged communities.
- **Health & Safety Importance:** A measure of the access provided by an asset to essential locations and services related to health and safety, including emergency response and healthcare facilities.

The individual factors comprising usage & operational importance criticality are visualized geographically across the Rhode Island road network in the following figures. Evacuation and lifeline routes are shown in Figure C.1. and are located in and around the southern, eastern, and coastal portions of the state. This includes thoroughfares into and out of Rhode Island's coastal islands. Road functional class is shown in Figure C.2, which assigns the greatest weight to interstates and expressways (such as I-95, I-295, US-1, RI-138, and RI-146) and arterials (such as US-6 and RI-114). Average annual daily traffic (AADT) is shown Figure C.3 with the greatest weighting assigned to the portion of the statewide road network carrying most traffic. In Rhode Island, this is primarily comprised of the I-95 corridor between Warwick, Providence, and Pawtucket, which carries at least 84,000 vehicles per day on average. The state's freight network (Figure C.4) and broadband network (Figure C.6) are comprised of a large proportion of the state's expressways and arterials, as well as a sizable portion of its collectors. Proximity to the Northeast Corridor (Figure C.5) is limited to those roads within half a mile of the rail line, which stretches between Westerly, South Kingstown, Providence, and points northeast.

The summation of usage & operational importance criticality is shown in Figure C.7. Usage & operational criticality is highest amongst many of Rhode Island's major thoroughfares, including in and around the state's major population centers. This includes nearly every portion of I-95, I-195, and RI-4, which each generated a usage & operational criticality score of at least (9). In addition to these major thoroughfares, the major crossings spanning Rhode Island's coastal communities generated high scores, including RI-24, RI-114, and RI-138. These results are expected given the relatively high weight assigned to highway characteristics, combined with designation of coastal evacuation routes.



Figure C.2 Road Functional Classification

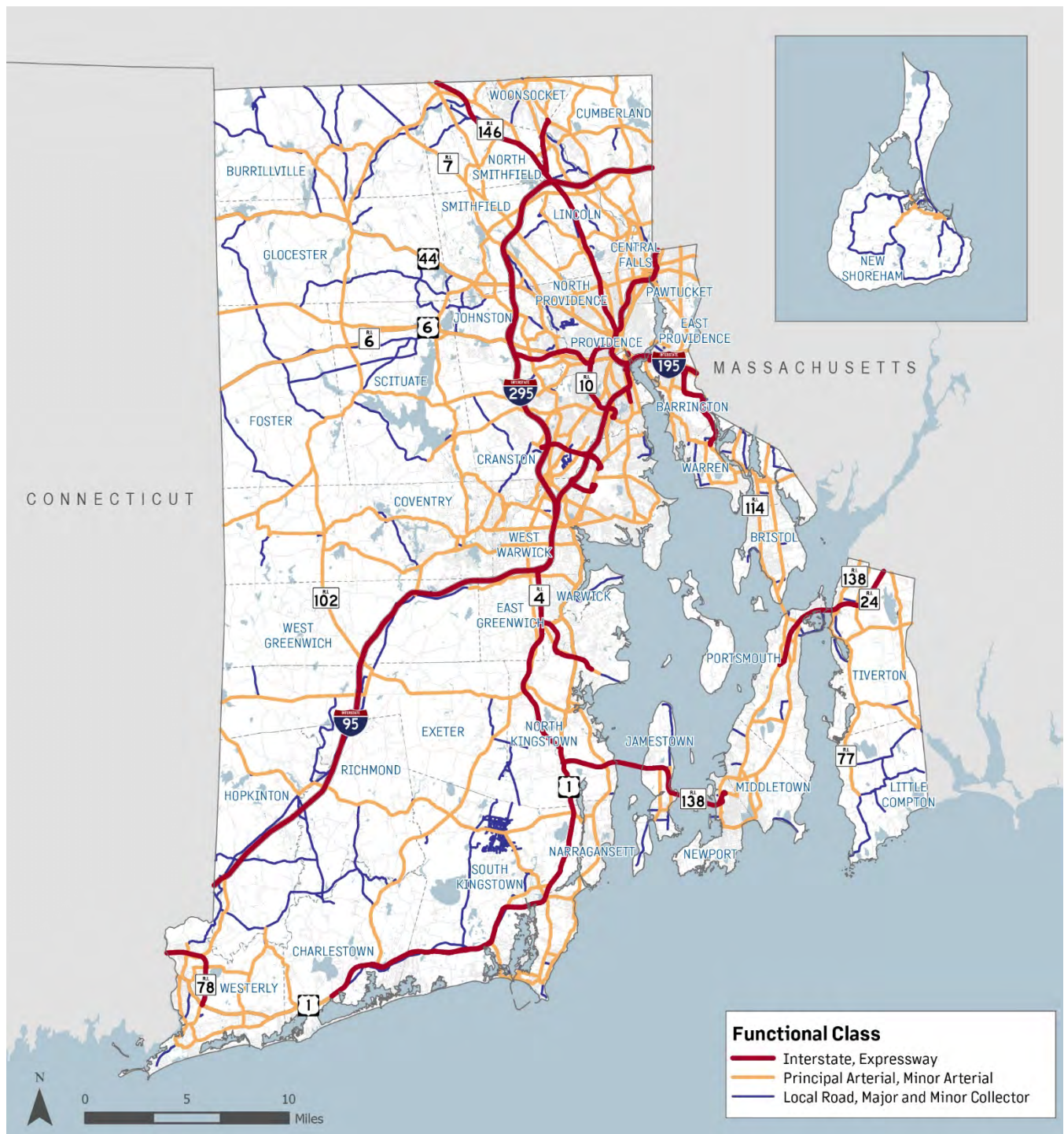
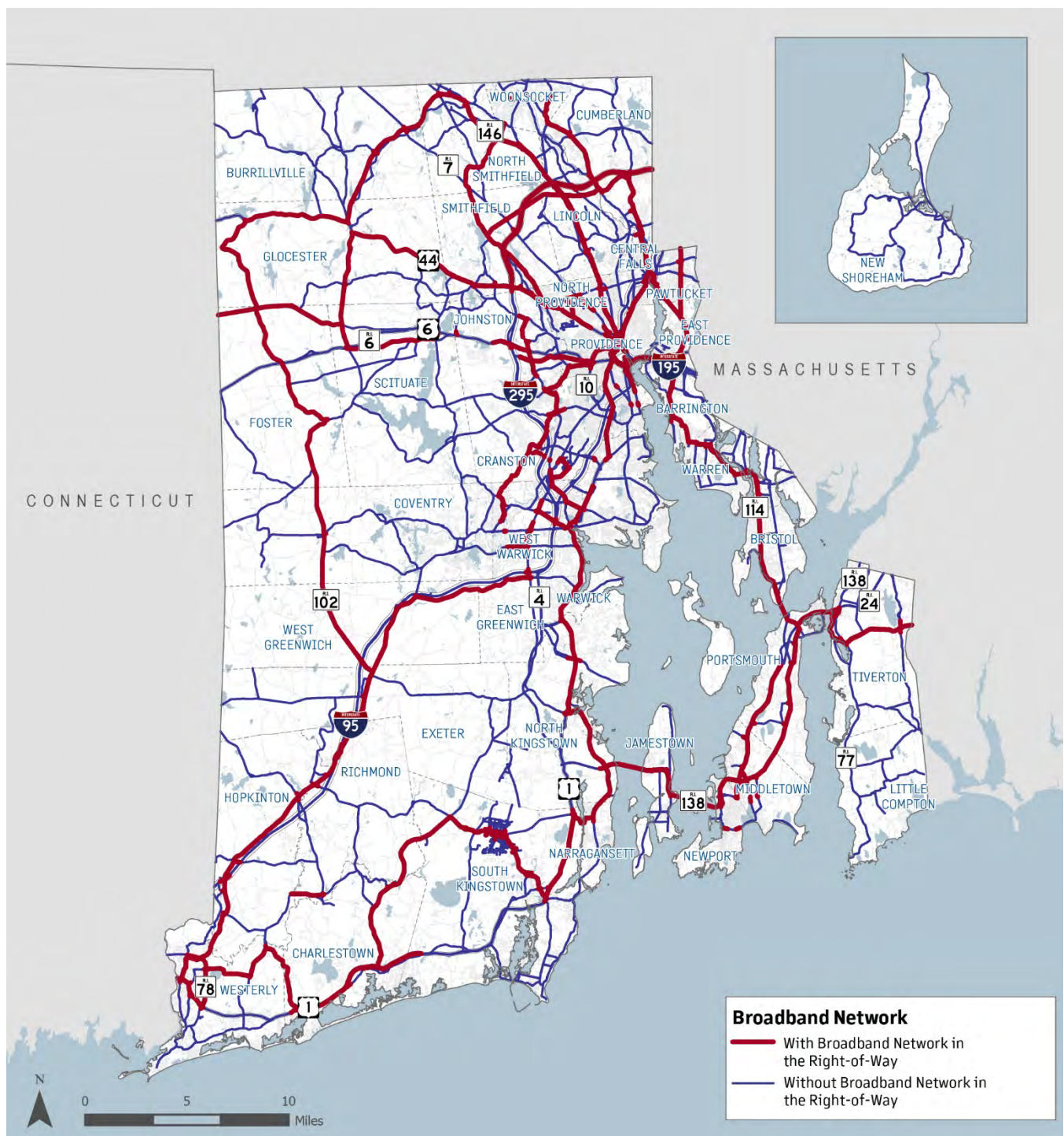




Figure C.6 Broadband Network





The individual factors comprising socioeconomic importance are visualized geographically across the Rhode Island road network in the following figures. As shown in Figure C.8, road criticality based on presence in equity areas is locally concentrated around population centers such as Providence, Woonsocket, Cranston, and a few other communities. Equity areas were defined by using a combination of three datasets, the Disadvantaged/Justice 40 communities identified by USDOTs transportation disadvantage tool, Climate and Economic Justice Screening tool and RIDOTs Environmental Justice Areas.

Population Density, measured by population per square mile (Figure C.9), the most critical roads are located in the northeast quadrant of Rhode Island, especially along the Warwick – Cranston – Providence – Woonsocket corridor. Employment density (Figure C.10), measured by jobs per square mile, somewhat mirrors population density, with additional areas of concentration located to the south and west of this corridor.

The summation of socioeconomic importance criticality is shown in Figure C.11. Socioeconomic criticality is highest amongst arterials and highways immediately in and around Providence, followed by Pawtucket and Woonsocket. Additional arterials and highways with criticality scores of at least (6) can be found to the south and west of Providence, as well as in Westerly, South Kingstown, Middletown, and Bristol.



Figure C.8 Equity Areas

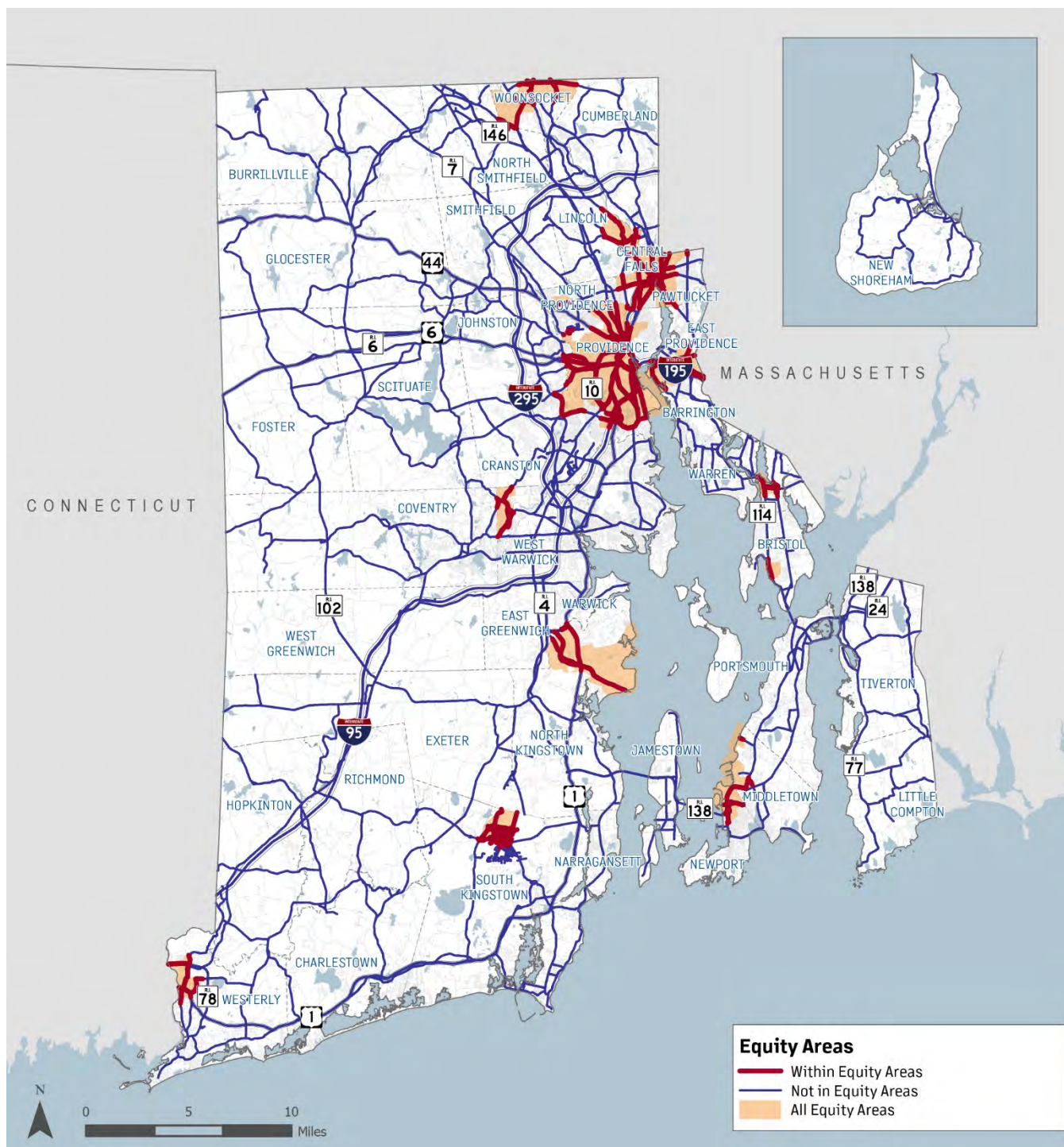




Figure C.9 Population Density

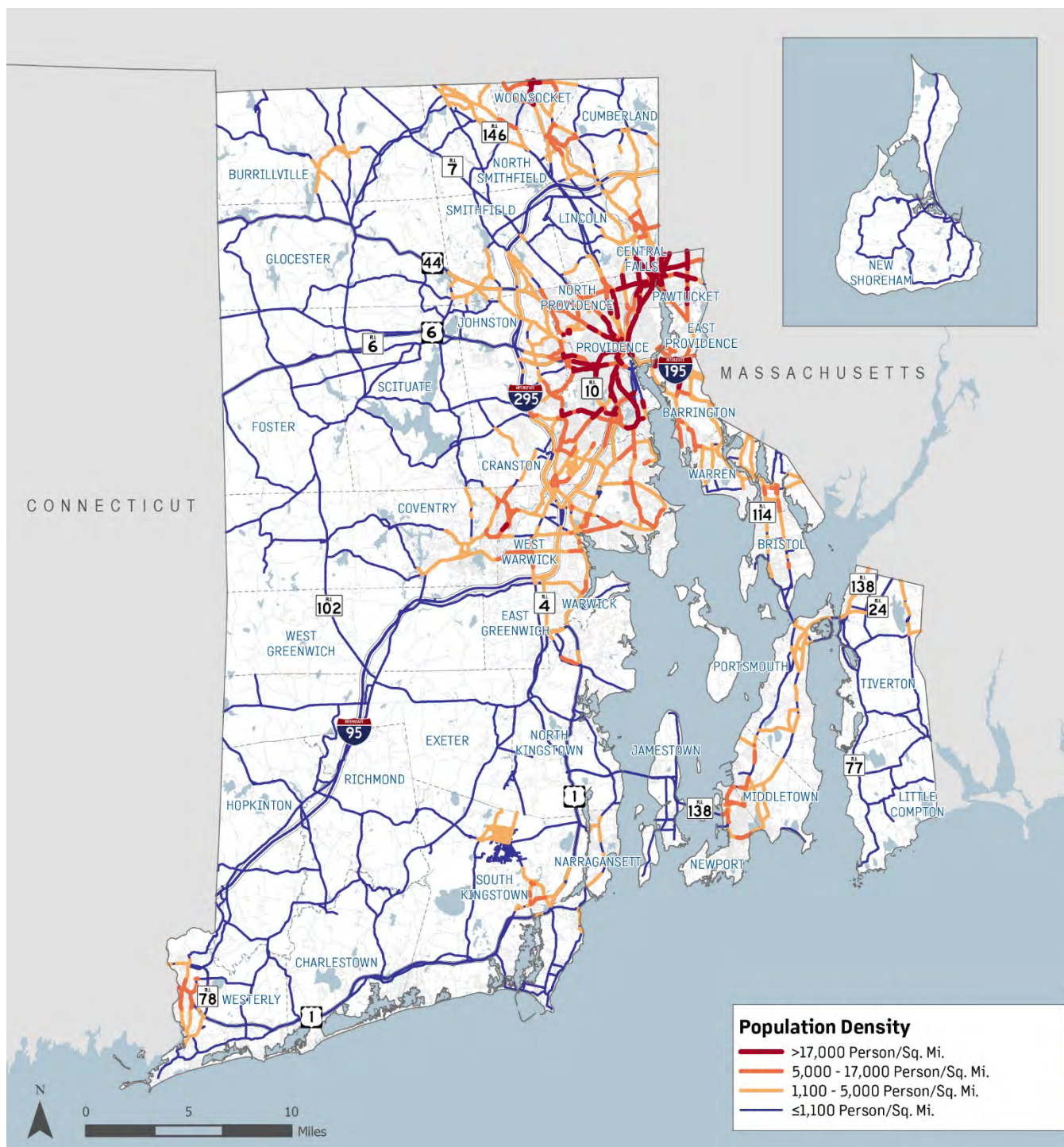




Figure C.10 Employment Density Criticality

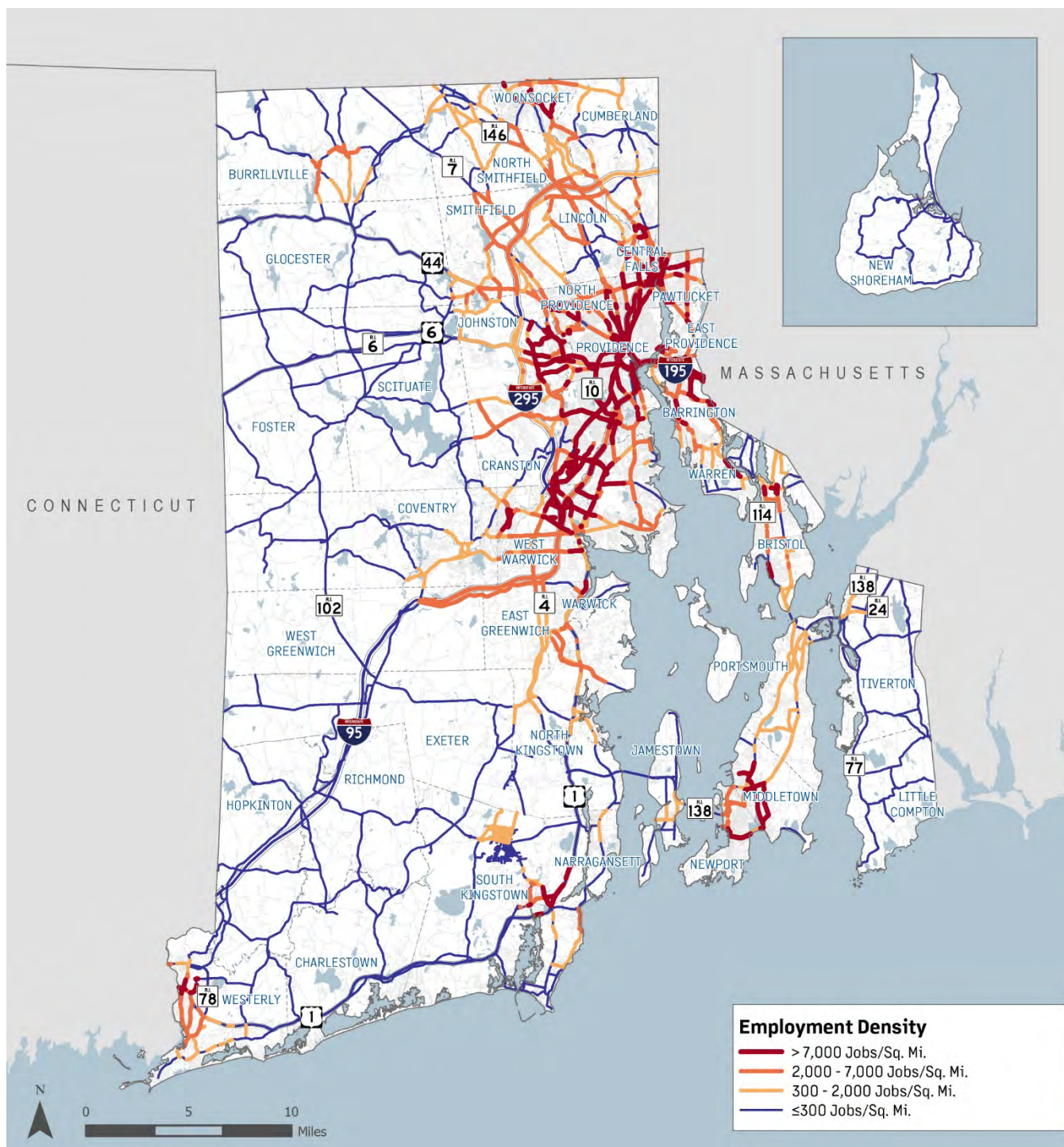
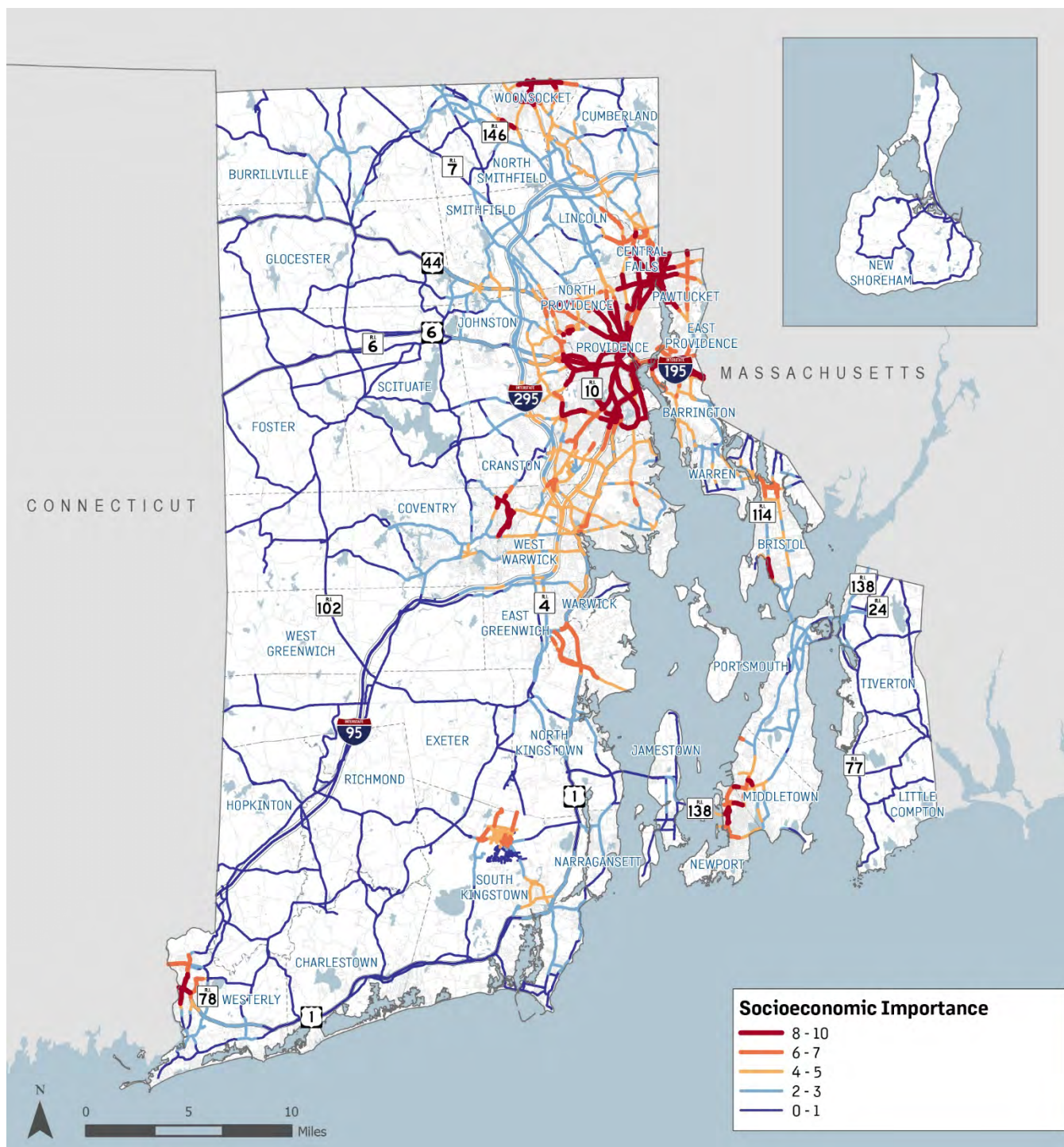




Figure C.11 Socioeconomic Importance Scores





The individual factors comprising health & safety importance are visualized geographically across the Rhode Island road network in the following figures. Accessibility to transit centers, airports, and seaports is all shown in Figure C.12. These types of facilities can be found across Rhode Island. Key locations where one or more of these types of facilities are found include Westerly, Point Judith / Wakefield, New Shoreham (Block Island), Newport, Cranston, and Providence. Accessibility to dams, powerplants, and maintenance facilities (Figure C.13) follows a similar pattern of geographic distribution. The same can be said of fire or police stations, hospitals, schools, and emergency shelters (Figure C.14) which also includes additional critical roads in the suburbs of Providence. Lastly, accessibility to military installations is shown in Figure C.15. This includes portions of the road network in Point Judith, Newport, East Greenwich, Cranston, and Providence.

The summation of health & safety criticality is shown in Figure C.16. Those portions of the road network with criticality scores of at least (8) can be found in and around Providence, as well as in Newport / Middletown, Wakefield, and along New Shoreham. Additional portions of the road network with large concentrations of criticality scores of at least (6) can also be found in and around Westerly, Warwick, Cranston, and Woonsocket.



Figure C.12 Access to Transit Centers, Airports, and Seaports

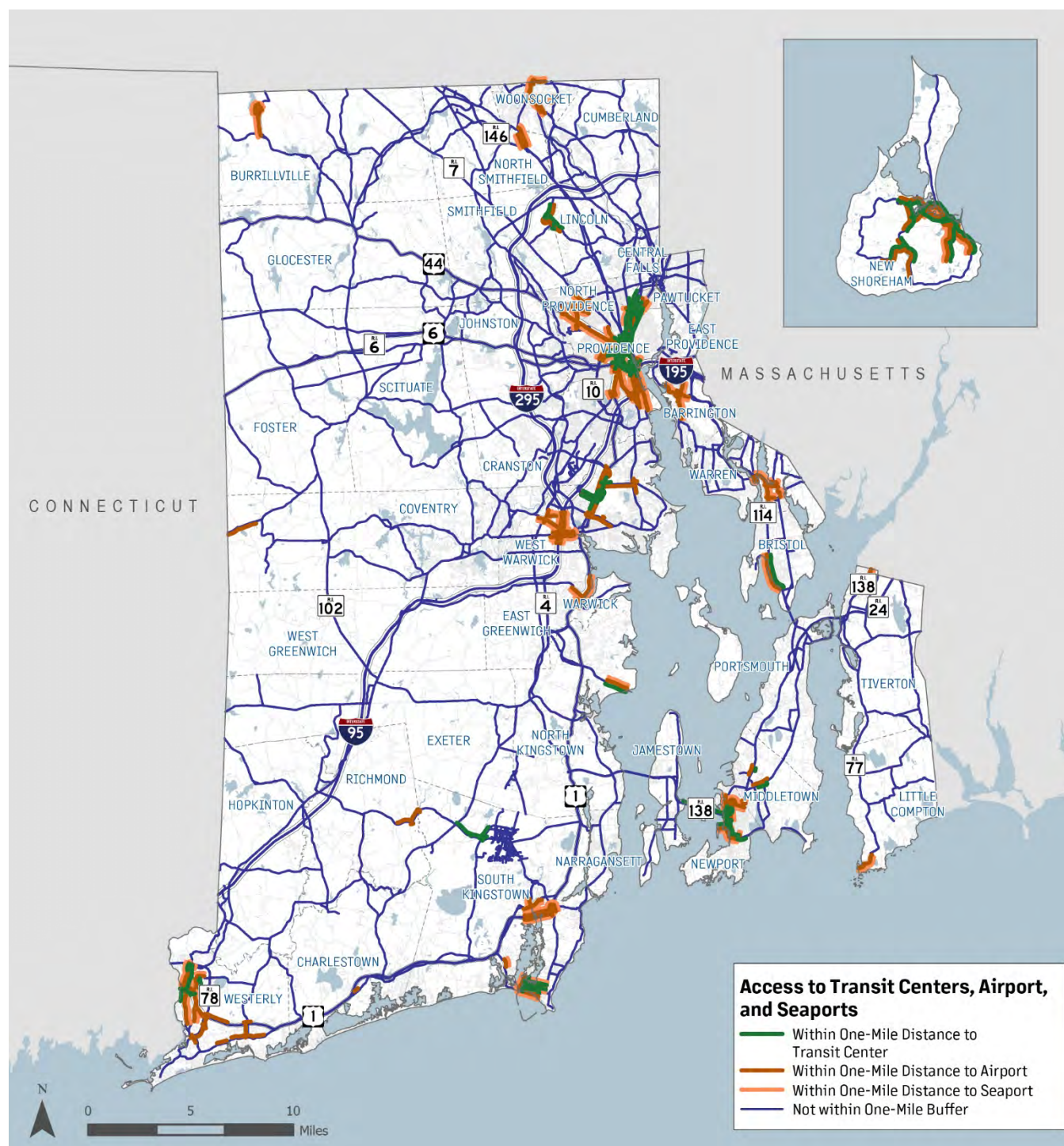




Figure C.13 Access to Dams, Powerplants, and Maintenance Facilities

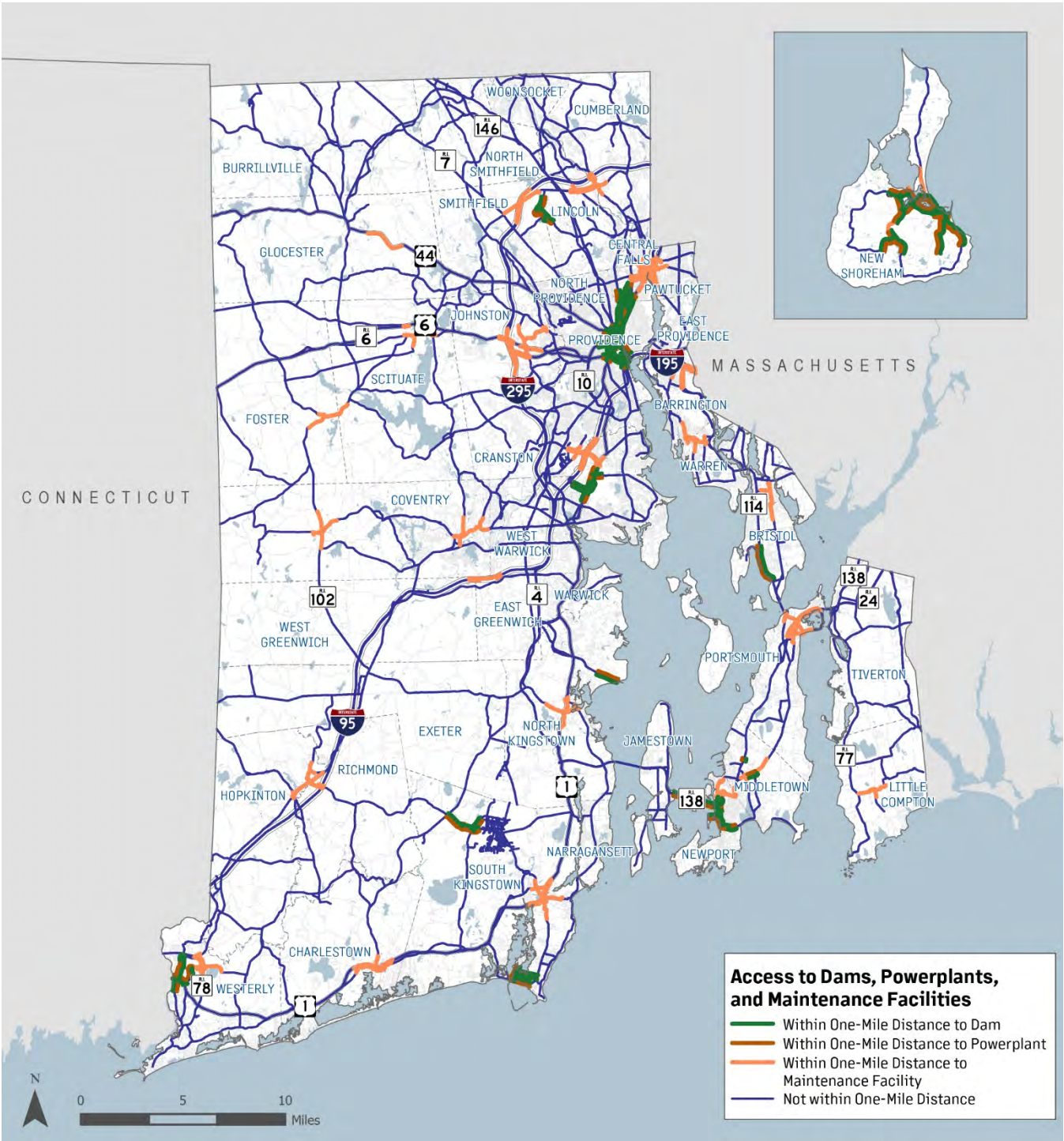




Figure C.14 Access to Fire or Police Stations, Schools, and Emergency Shelters

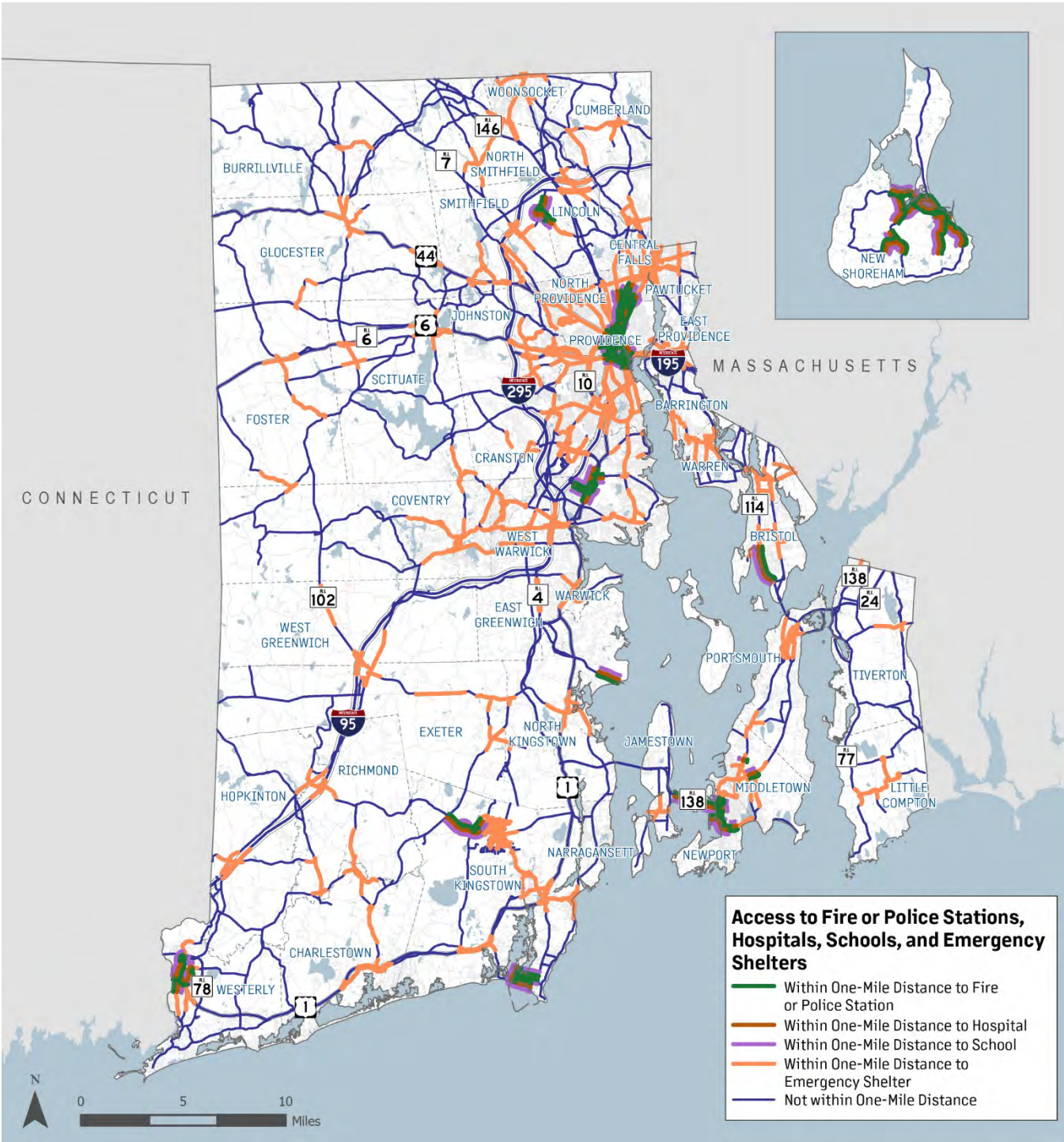




Figure C.15 Access to Military Installations

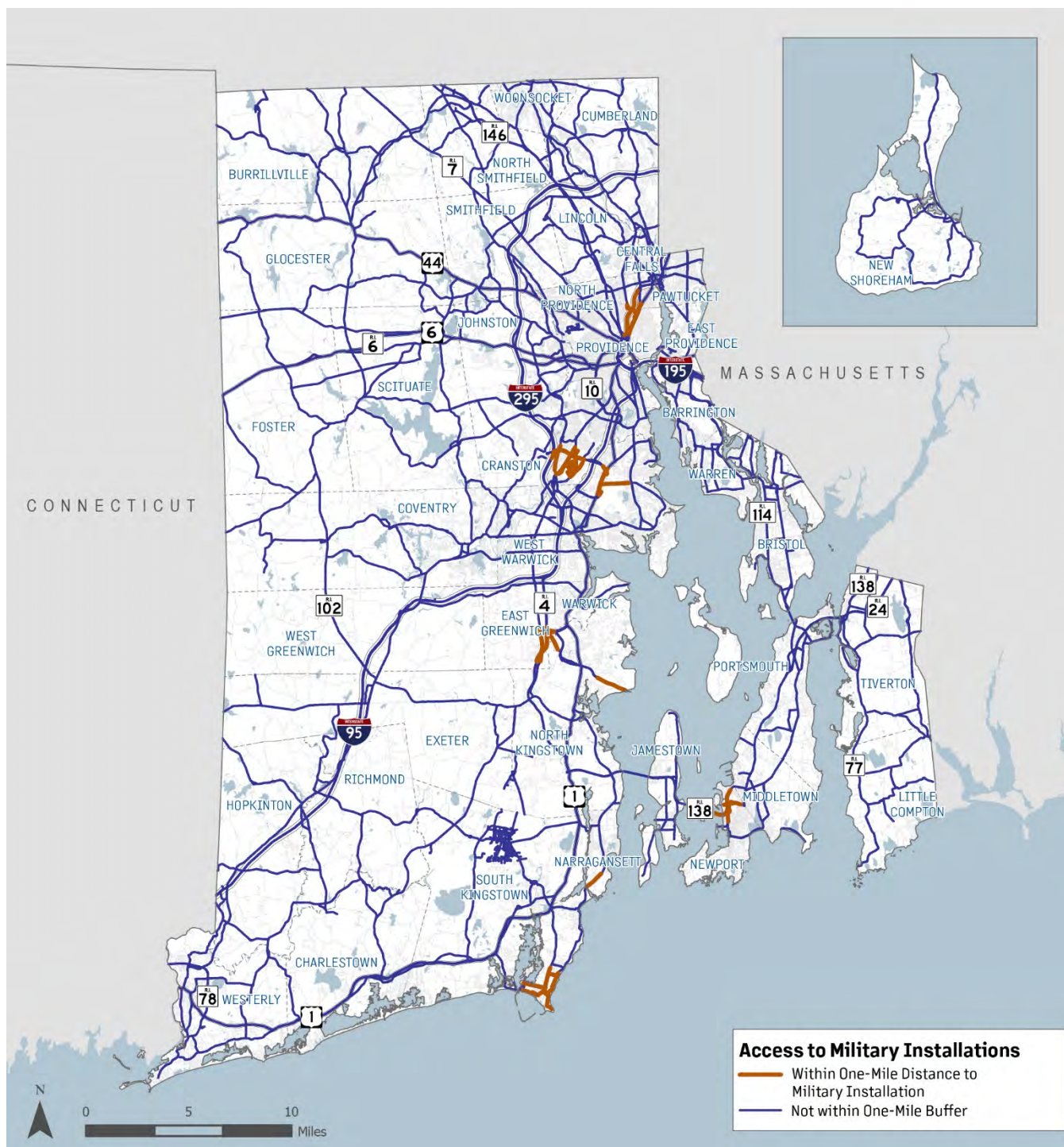
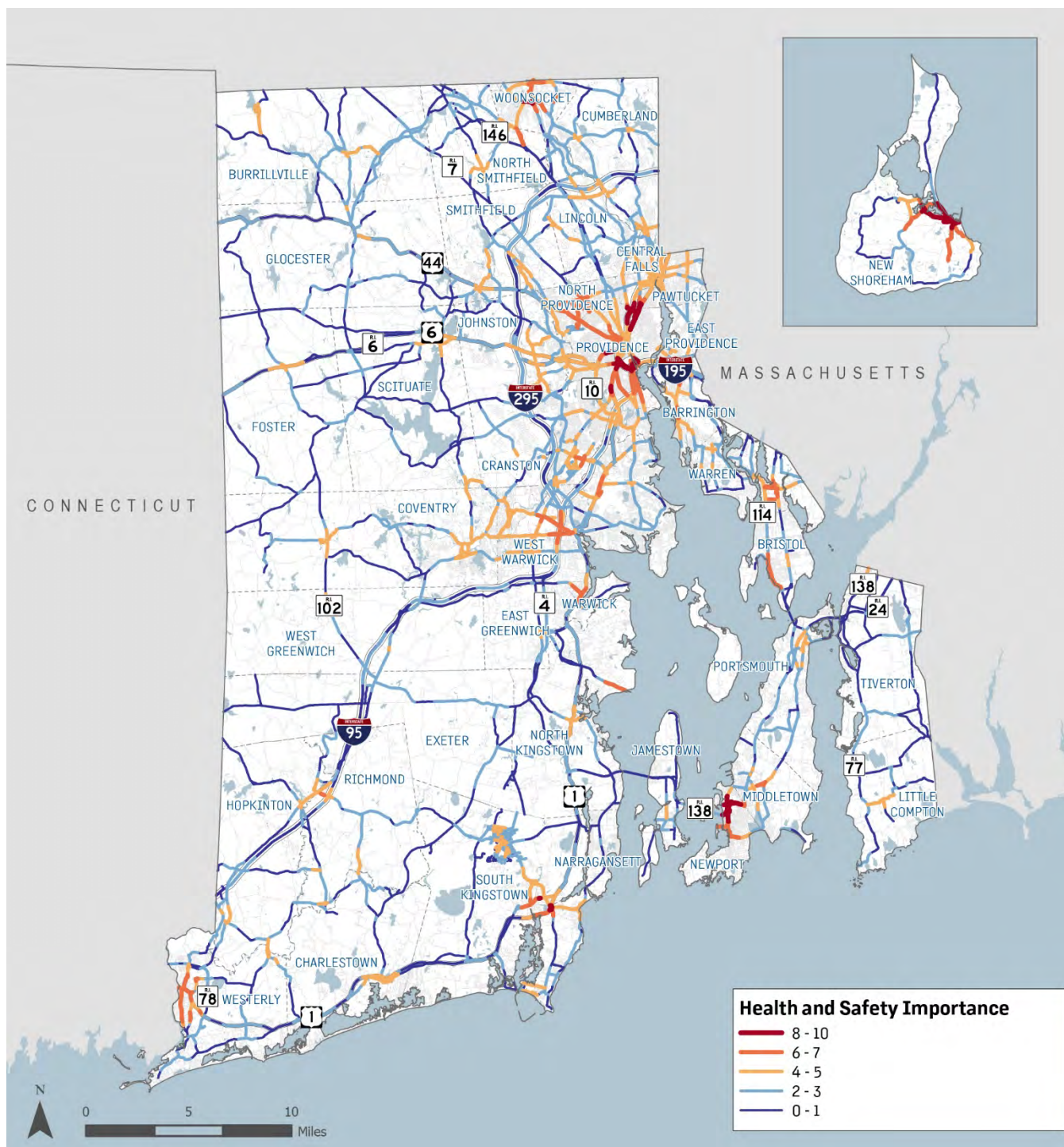




Figure C.16 Health & Safety Importance Scores





APPENDIX E GEOGRAPHIC VISUALIZATION OF EXPOSURE

Total degree of exposure, including exposure scores, is visualized across the Rhode Island's transportation assets in the following figures, primarily comprised of the road network. Figure E.11, Figure E.12 and Figure E.13 model total exposure under scenarios of 1 foot, 2 foot and 7 foot sea level rise. The impacts of 1 foot of sea level rise produce a local effect of up to 2 feet of inundation (exposure score of 3) at the northern end of Bristol. As shown in Figure E.12, a 2 foot sea level rise scenario produces localized inundation impacts of at least 2 feet or more (including an exposure score of 4) in one location each in Newport and on Block Island. At this stage, RI-114 between Warren and Barrington becomes inundated up to 2 feet (exposure score of 3). At 7 feet of sea level rise (Figure E.13), additional inundation of up to 2 feet is expected in very localized portions of Matunuck, Point Judith, Newport, Jamestown, Middletown, Tiverton, Bristol, Warren, and Barrington, as well as in a slightly expanded portion of New Shoreham on Block Island. This would yield an exposure score of (3) for these expanded portions of the Rhode Island transportation network.

The next figures examine the impacts of sea level rise with the added impact of storm surge. Figure E.14 shows exposure impacts across from 50-year storm surge overlaid to 1 foot of sea level rise. Although 1 foot of sea level rise alone does not produce any widespread impacts to the transportation network, the addition of 50-year storm surge is enough to produce impacts across a number of areas of coastal Rhode Island. Storm surge impacts of up to 10 feet (exposure score of 3) are expected in portions of Westerly, Block Island, Point Judith, Jamestown, Newport, Middletown, Portsmouth, Tiverton, Bristol, Barrington, and Providence. The greatest concentration of exposed assets can be found along the Bristol – Warren – Barrington corridor of RI-114, where scattered portions of the corridor's arterials are expected to see storm surge of up to, and greater than 10 feet, under this initial scenario. As shown in Figure E.15, a scenario of 100-year storm surge overlaid to 2 feet of sea level rise leads to a slightly increased number of exposed assets, in comparison to a scenario of 50-year storm surge over 1 foot of sea level rise. However, a greater portion of these impacted assets would be expected to see storm surge impacts of up to 10 feet (exposure score of 3 or higher). A scenario of 100-year storm surge overlaid to 7 feet of sea level rise (Figure E.16) leads to the majority of impacted assets, found across most of coastal Rhode Island, inundated by over 10 feet of water, with an exposure score of (4).

Lastly, Figure E.17 shows the impacts from floodplain exposure. A relatively high proportion of the Rhode Island road network is located within 100-year or current base floodplain. This includes assets scattered around Rhode Island's most populated corridors, including coastal portions of the state and within the Providence area. The current floodplain also includes portions of a number of Rhode Island's major thoroughfares including I-95, I-195, RI-114, RI-138, and RI-146, in addition to arterials located throughout the state.



Figure E.2 Exposure Scores of Assets: 2-Foot Sea Level Rise Scenario

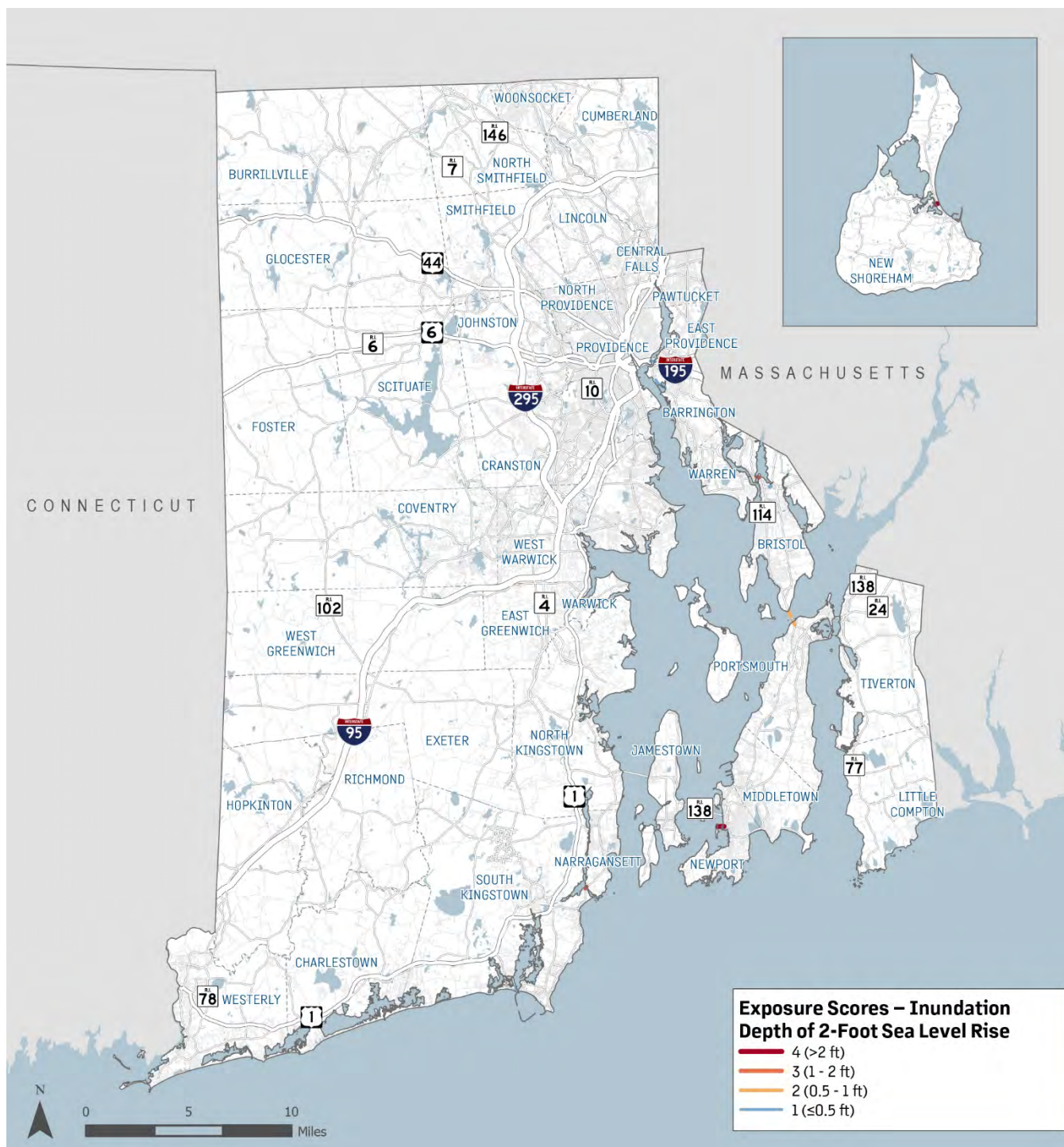




Figure E.3 Exposure Scores of Assets: 7-Foot Sea Level Rise Scenario

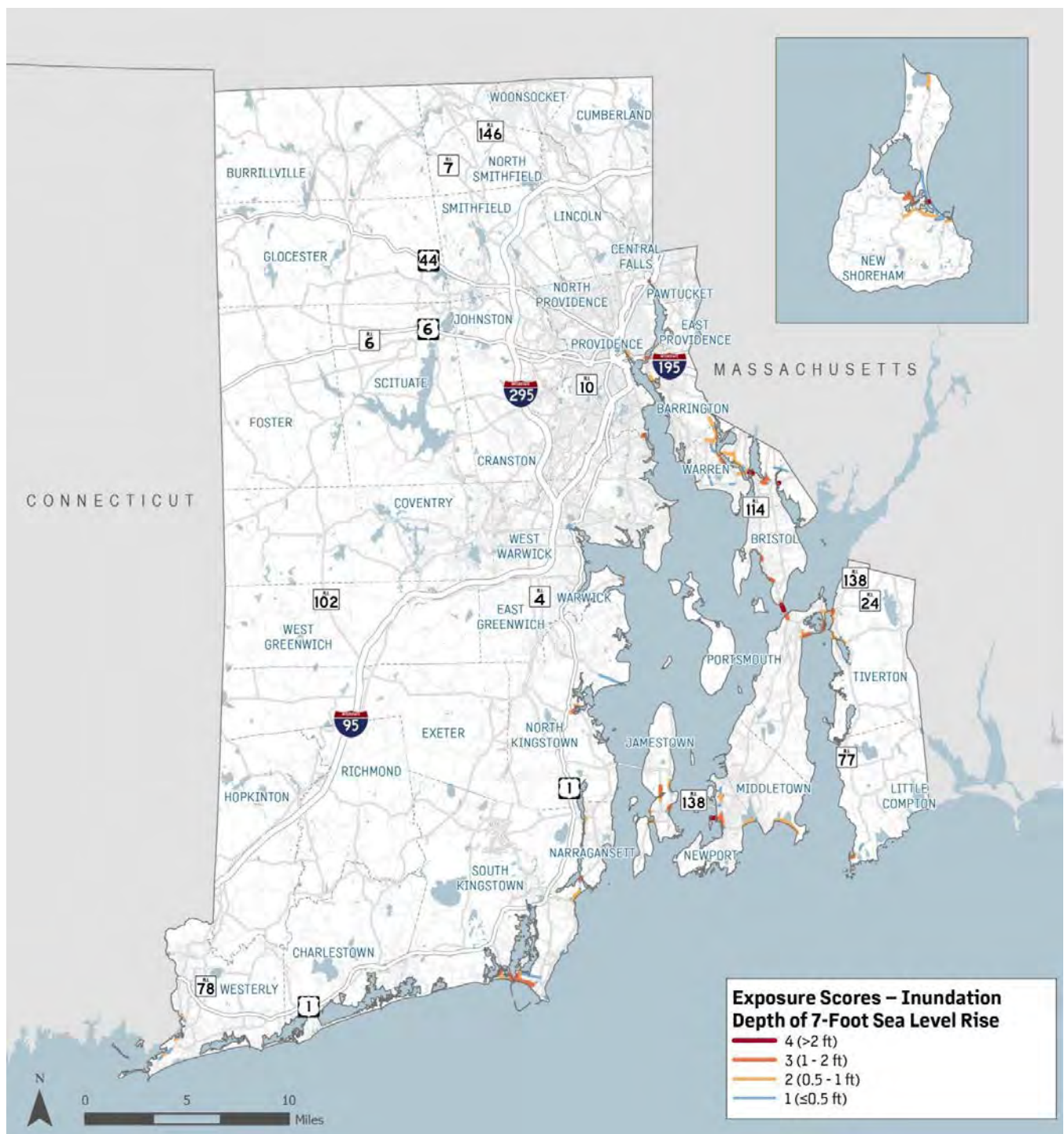




Figure E.4 Exposure Scores of Assets: Inundation of 50-Year Storm + 1 Foot Sea Level Rise Scenario





Figure E.5 Exposure Scores of Assets: Inundation of 100-Year Storm + 2 Foot Sea Level Rise Scenario

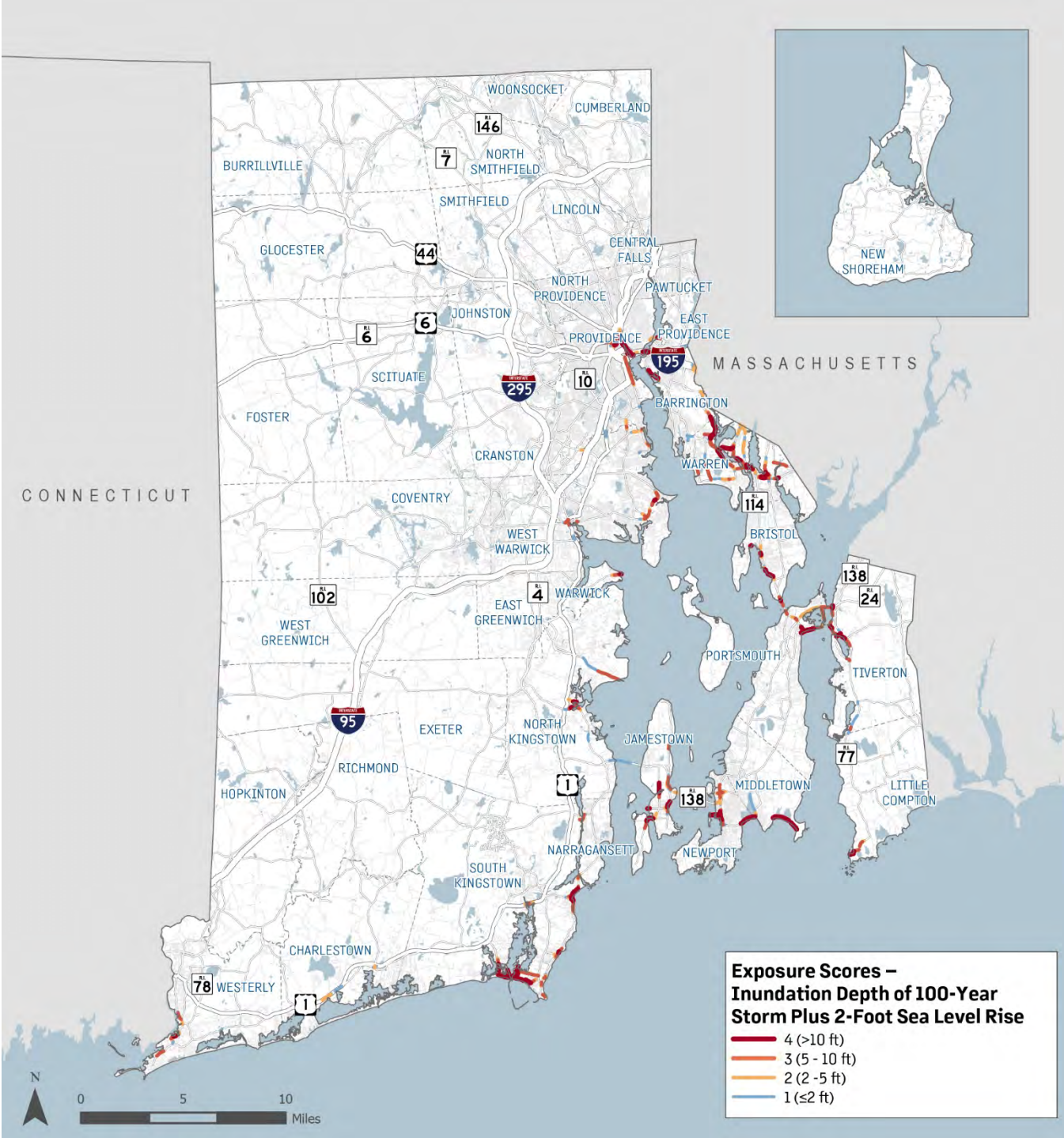




Figure E.6 Exposure Scores of Assets: Inundation of 100-Year Storm + 7 Foot Sea Level Rise Scenario

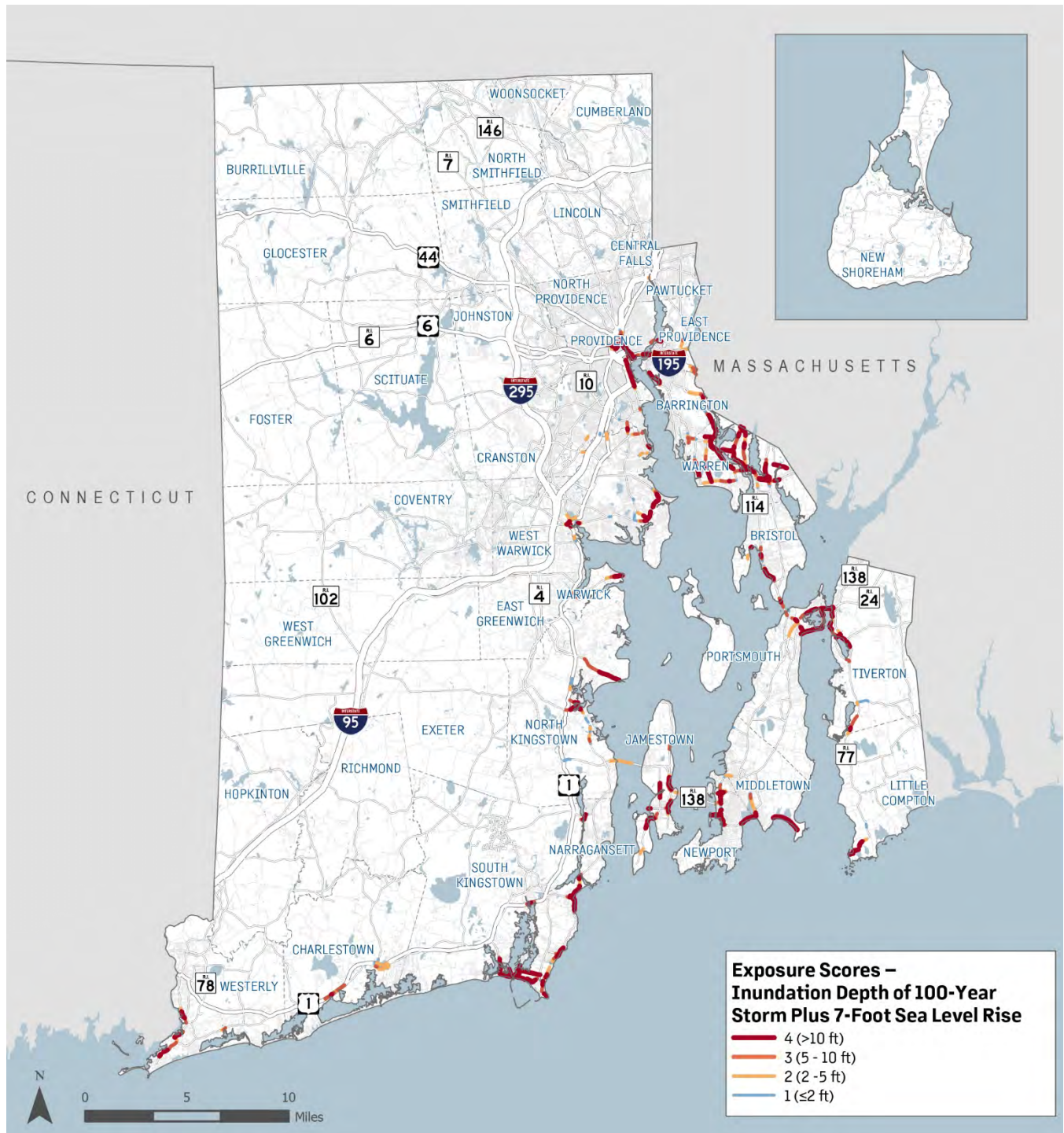
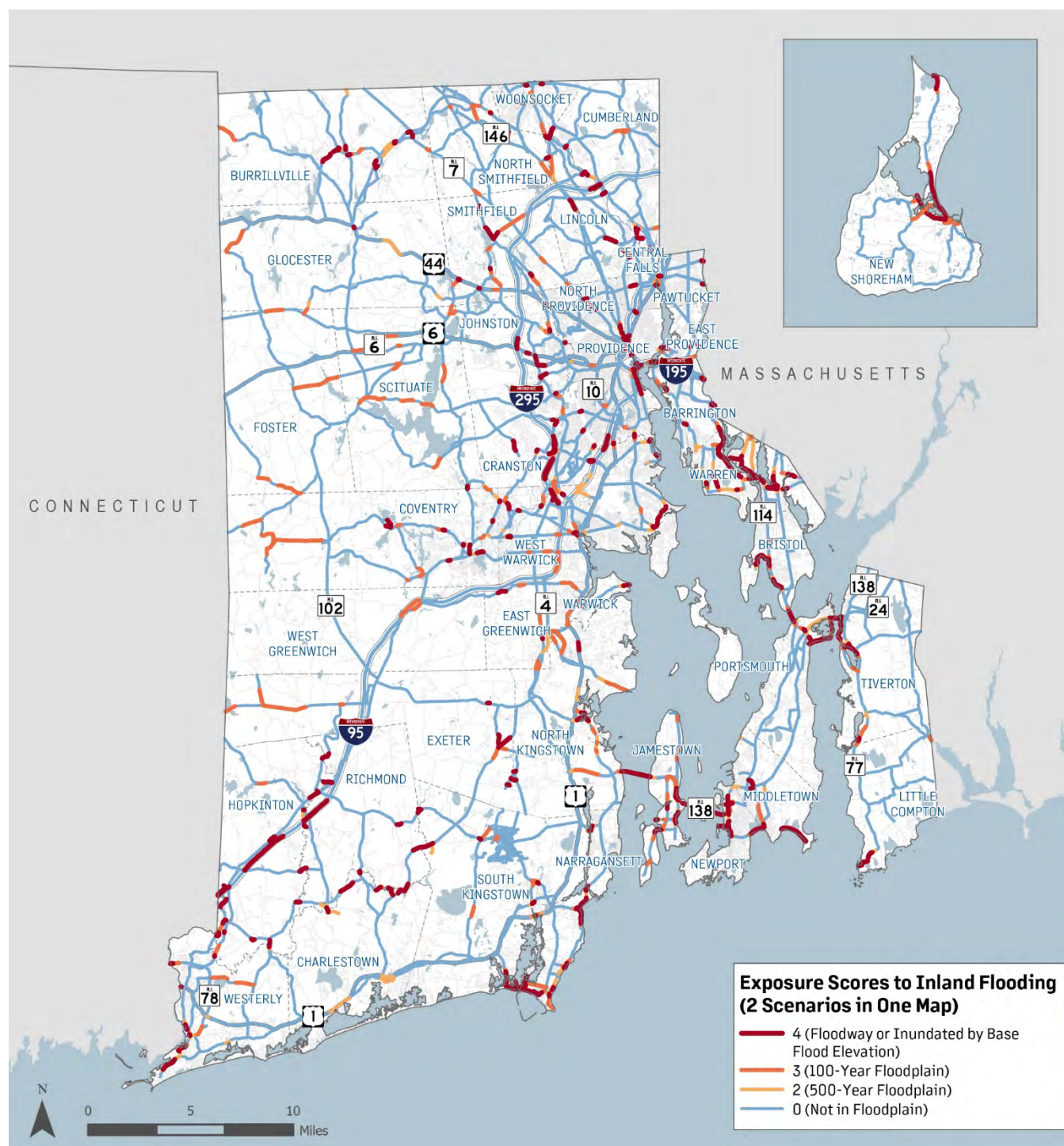




Figure E.7 Exposure Scores of Assets: Location within Severity of Floodplain





APPENDIX F VULNERABILITY OUTPUTS

The following section displays additional vulnerability maps for the 1-foot and 2-foot sea level rise, and 1-foot sea level rise, and 100-year storm plus 2-foot sea level rise in addition to maps shown in Chapter 4.2.4.



Figure D.2 Vulnerability of Roads & Bridges to 2-Foot Sea Level Rise Scenario

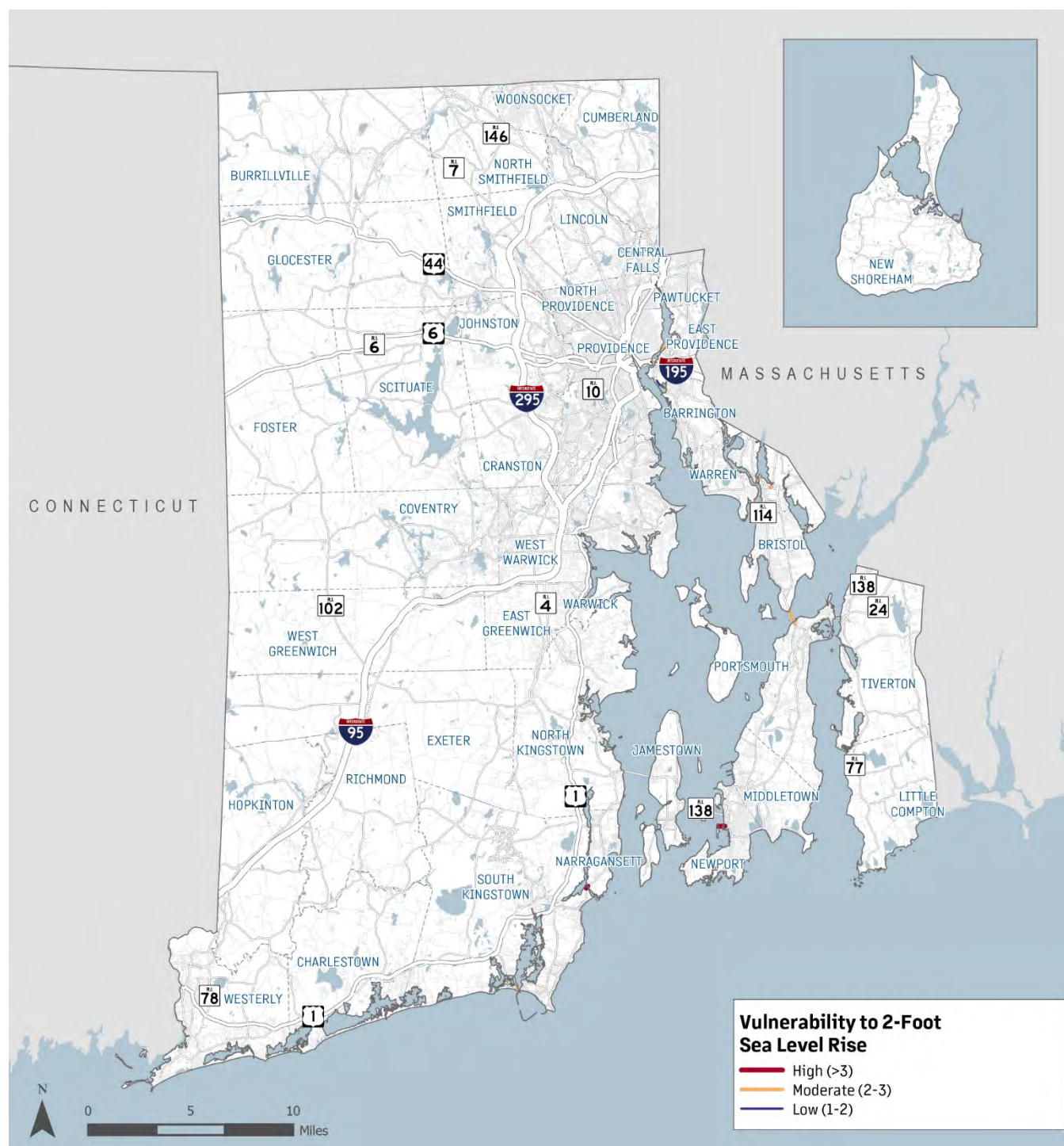
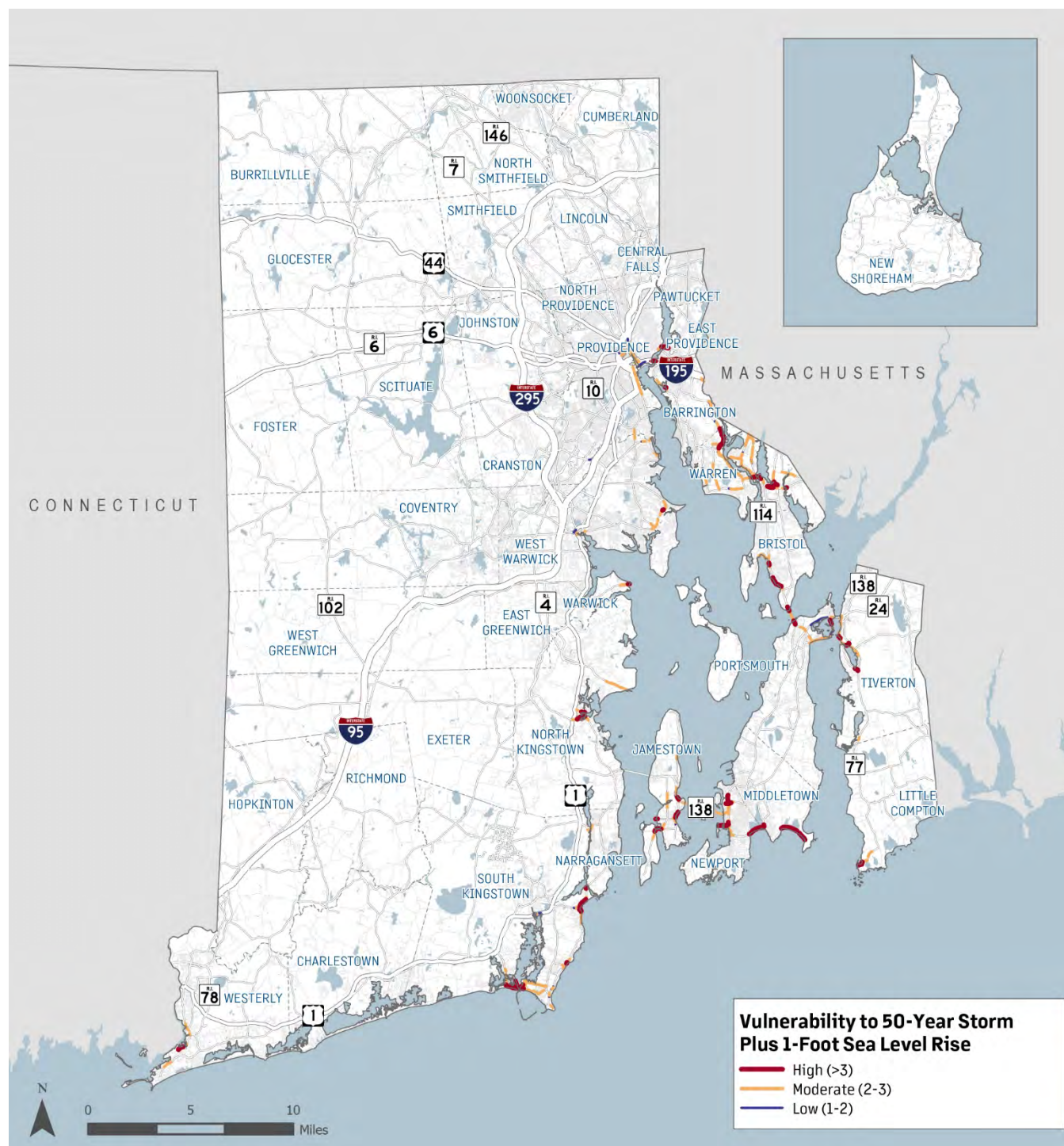




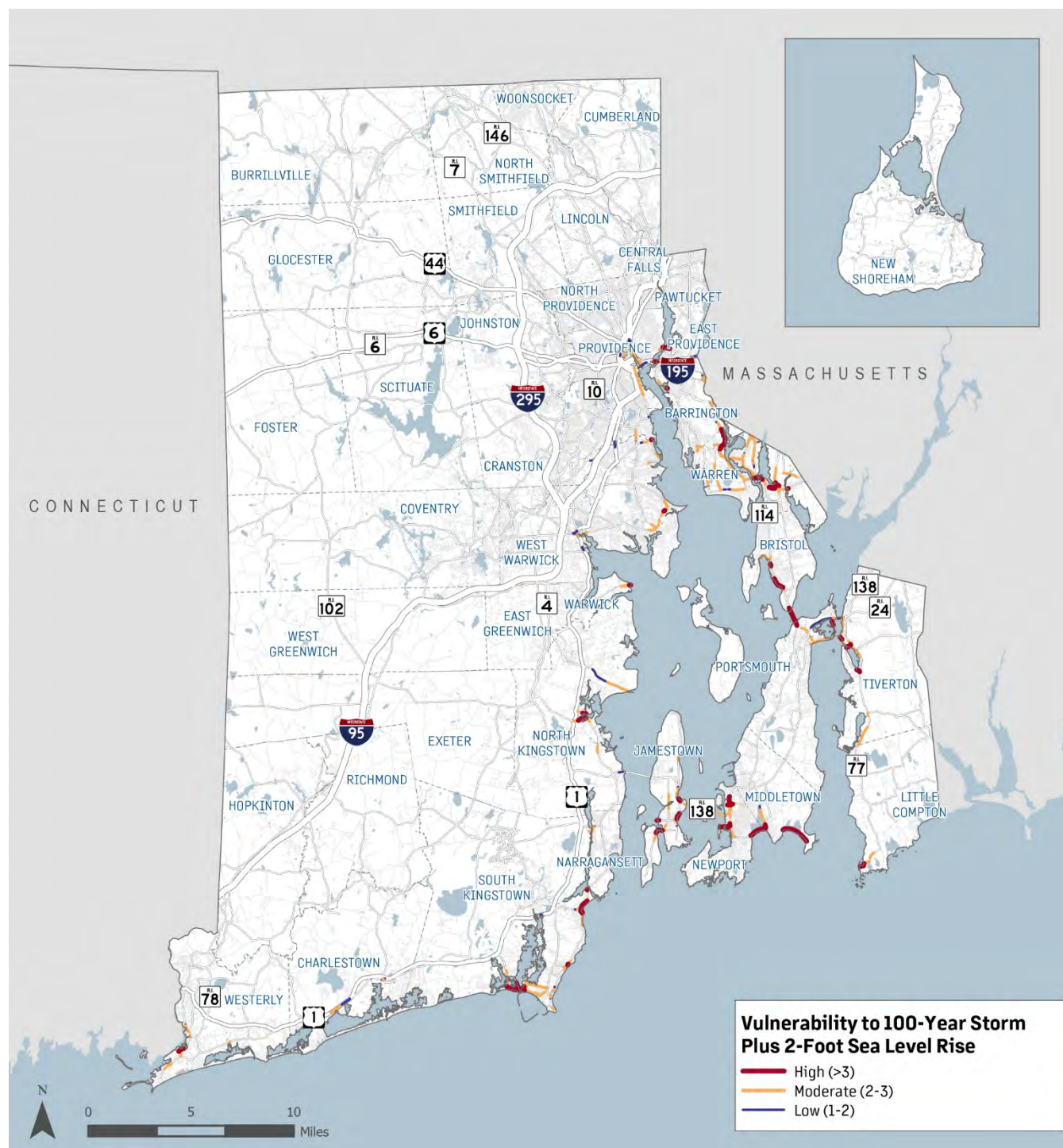
Figure D.3 Vulnerability of Roads & Bridges: Inundation of 50-Year Storm + 1 Foot Sea Level Rise Scenario



Note: Storm Surge data is not available for New Shoreham.



Figure D.4 Vulnerability of Roads & Bridges: Inundation of 100-Year Storm + 2 Foot Sea Level Rise Scenario



Note: Storm Surge data is not available for New Shoreham



Figure D.5 Vulnerability of Sidewalks to 1-Foot Sea Level Rise Scenario

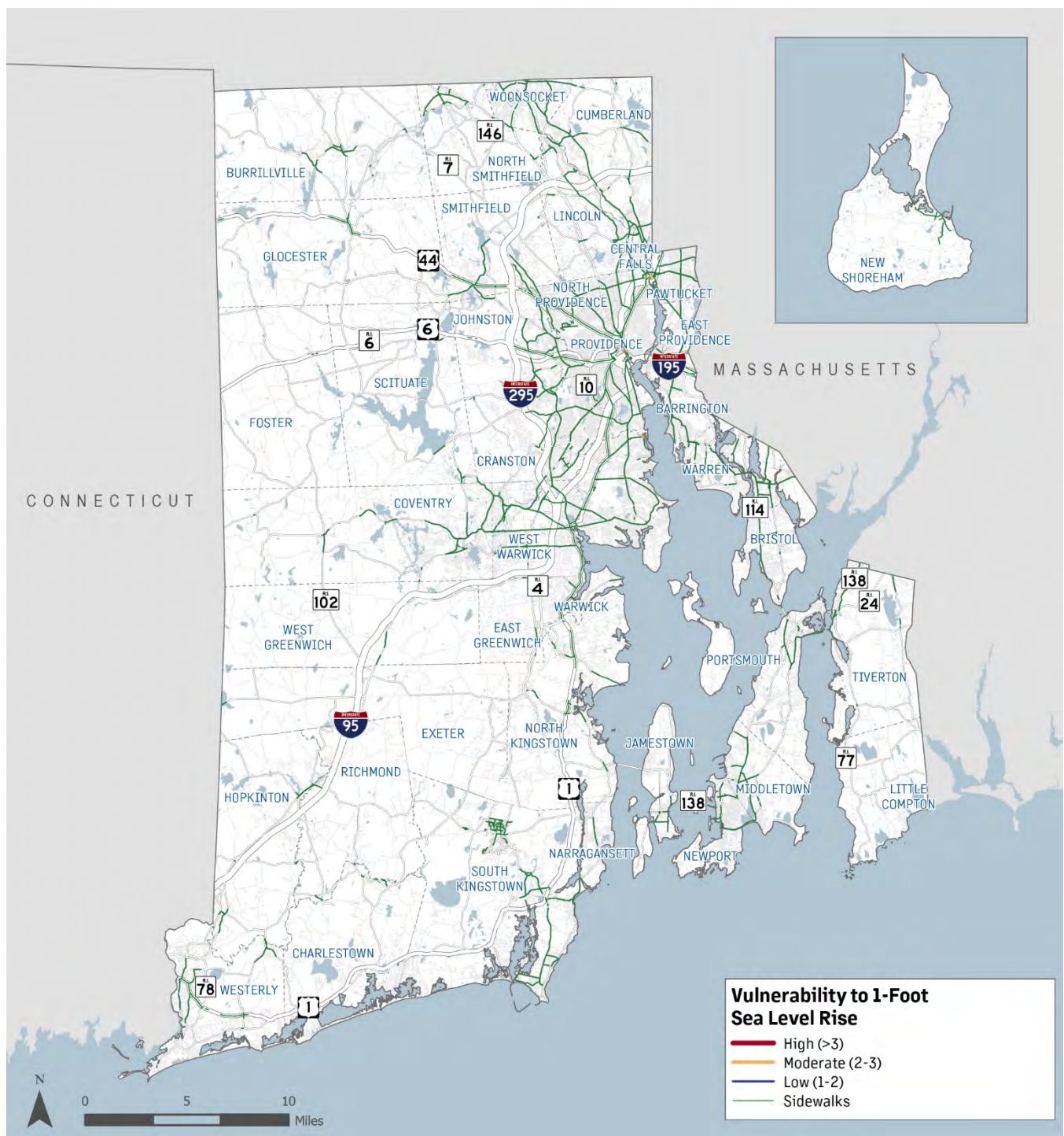




Figure D.6 Vulnerability of Sidewalks to 2-Foot Sea Level Rise Scenario

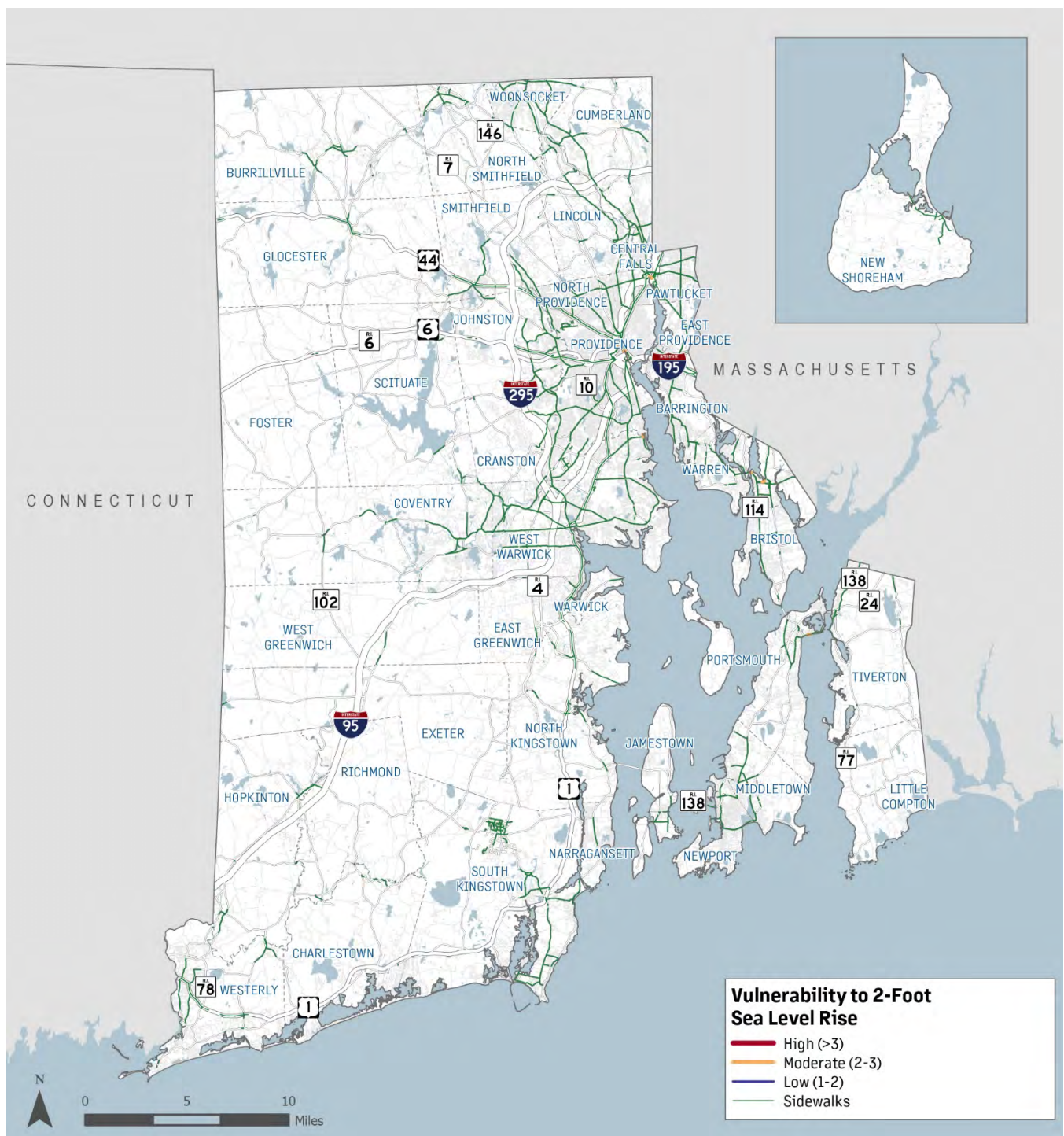




Figure D.7 Vulnerability of Sidewalks: Inundation of 50-Year Storm + 1 Foot Sea Level Rise Scenario

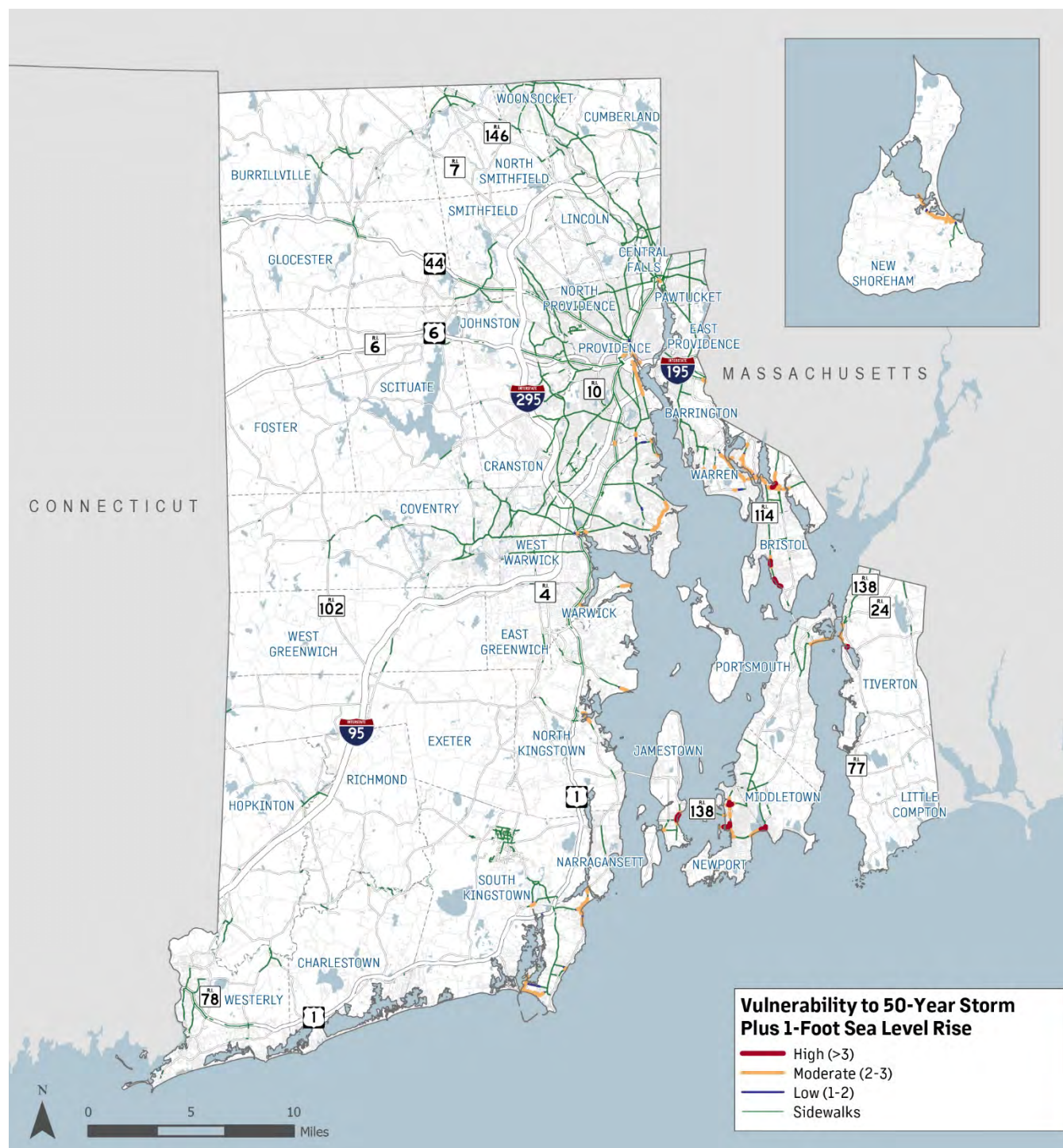
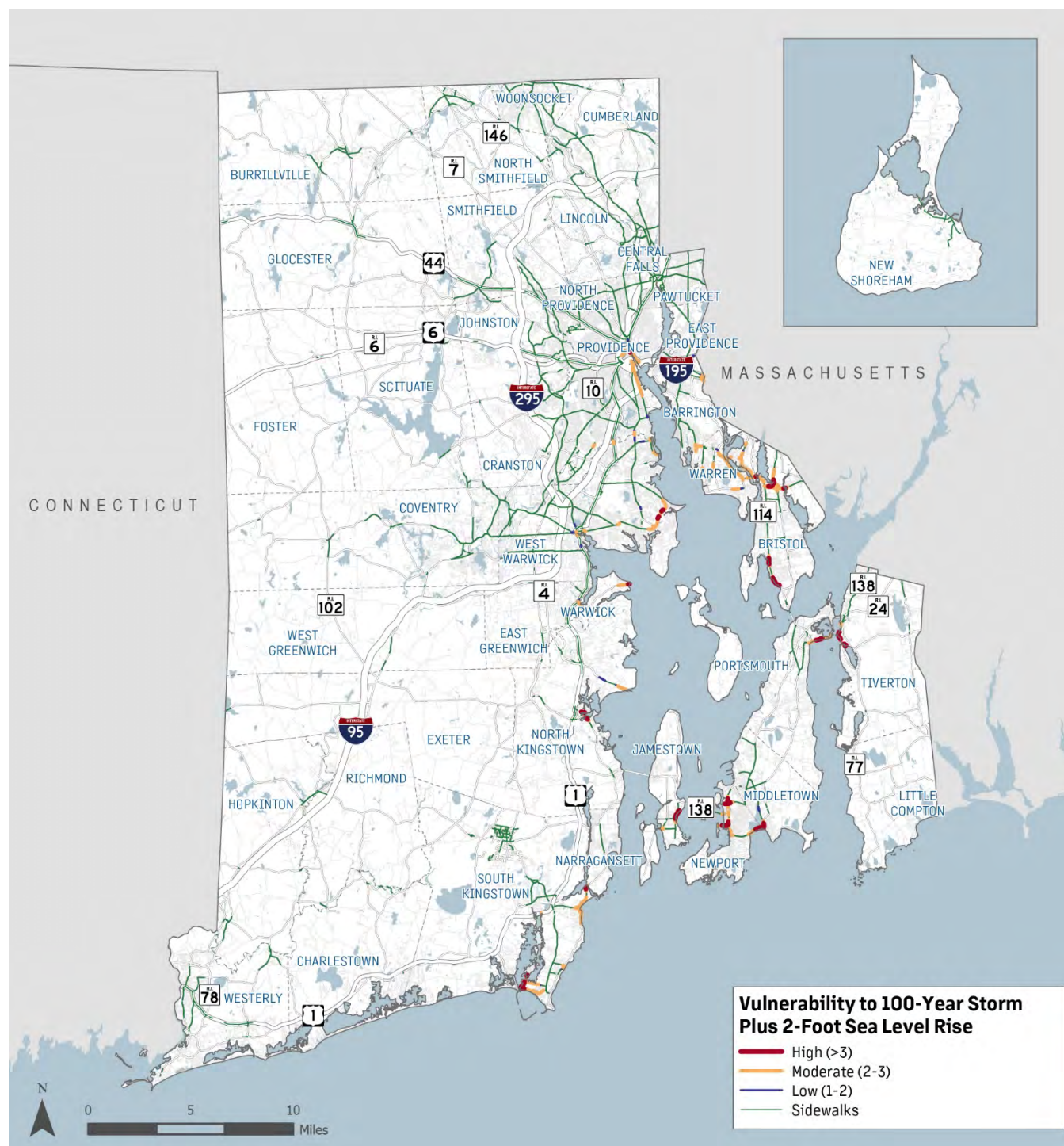




Figure D.8 Vulnerability of Sidewalks: Inundation of 100-Year Storm + 2 Foot Sea Level Rise Scenario



Note: Storm Surge data is not available for New Shoreham



Figure D.9 Vulnerability of Shared-Use Paths to 1-Foot Sea Level Rise Scenario

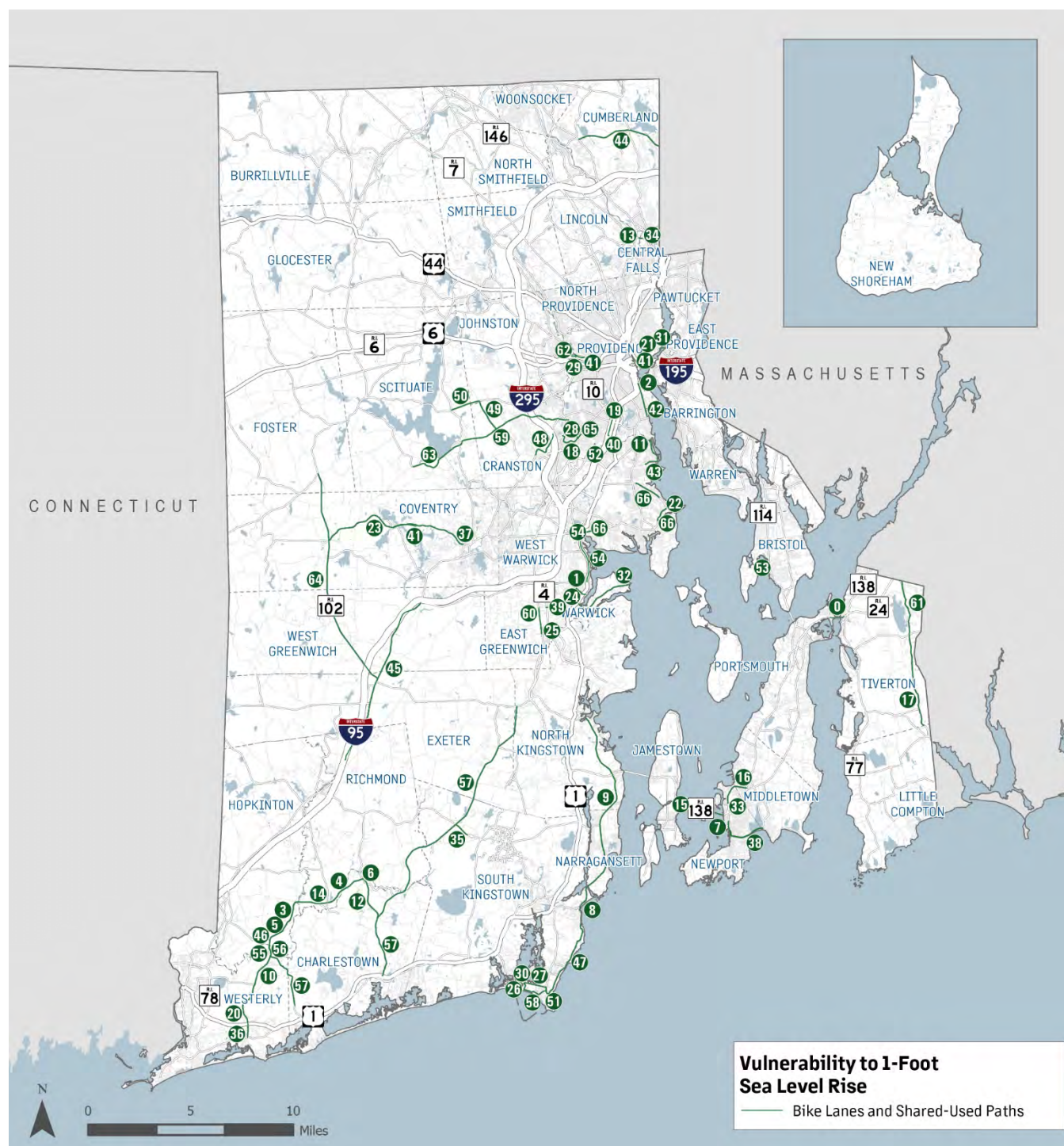




Figure D.10 Vulnerability of Shared-Use Paths to 2-Foot Sea Level Rise Scenario

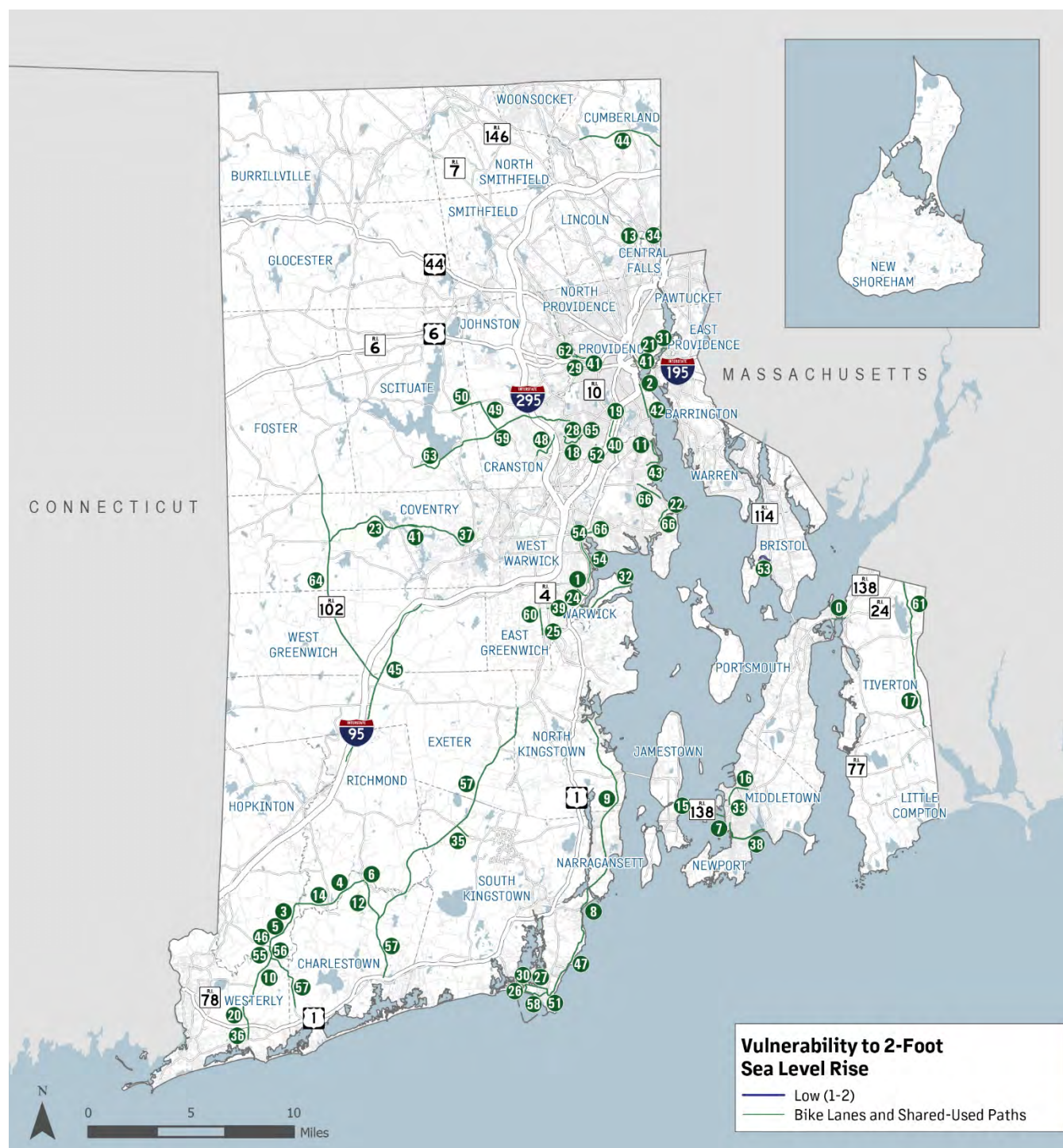




Figure D.11 Vulnerability of Shared-Use Paths: Inundation of 50-Year Storm + 1 Foot Sea Level Rise Scenario

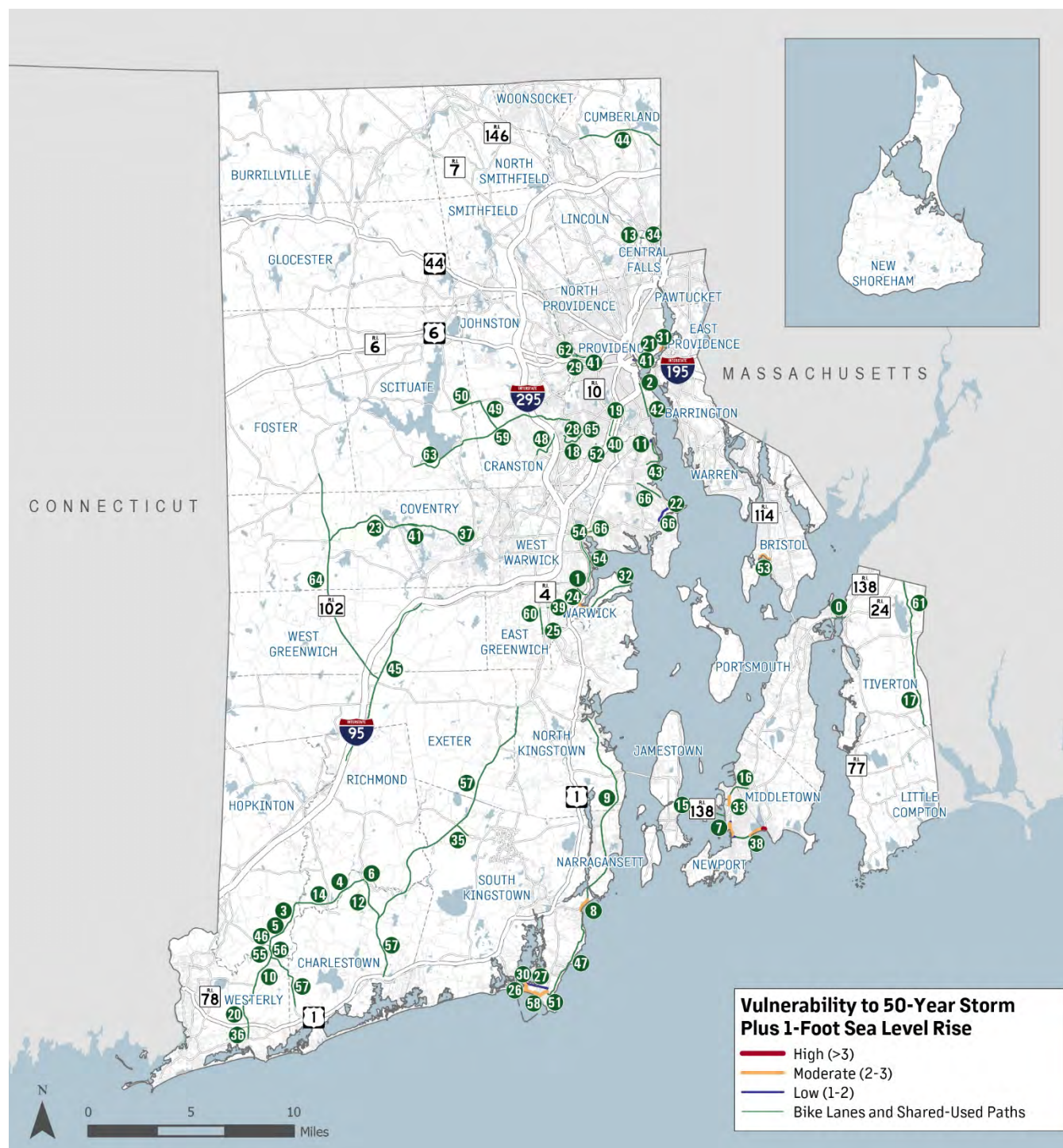




Figure D.12 Vulnerability of Shared-Use Paths: Inundation of 100-Year Storm + 2 Foot Sea Level Rise Scenario

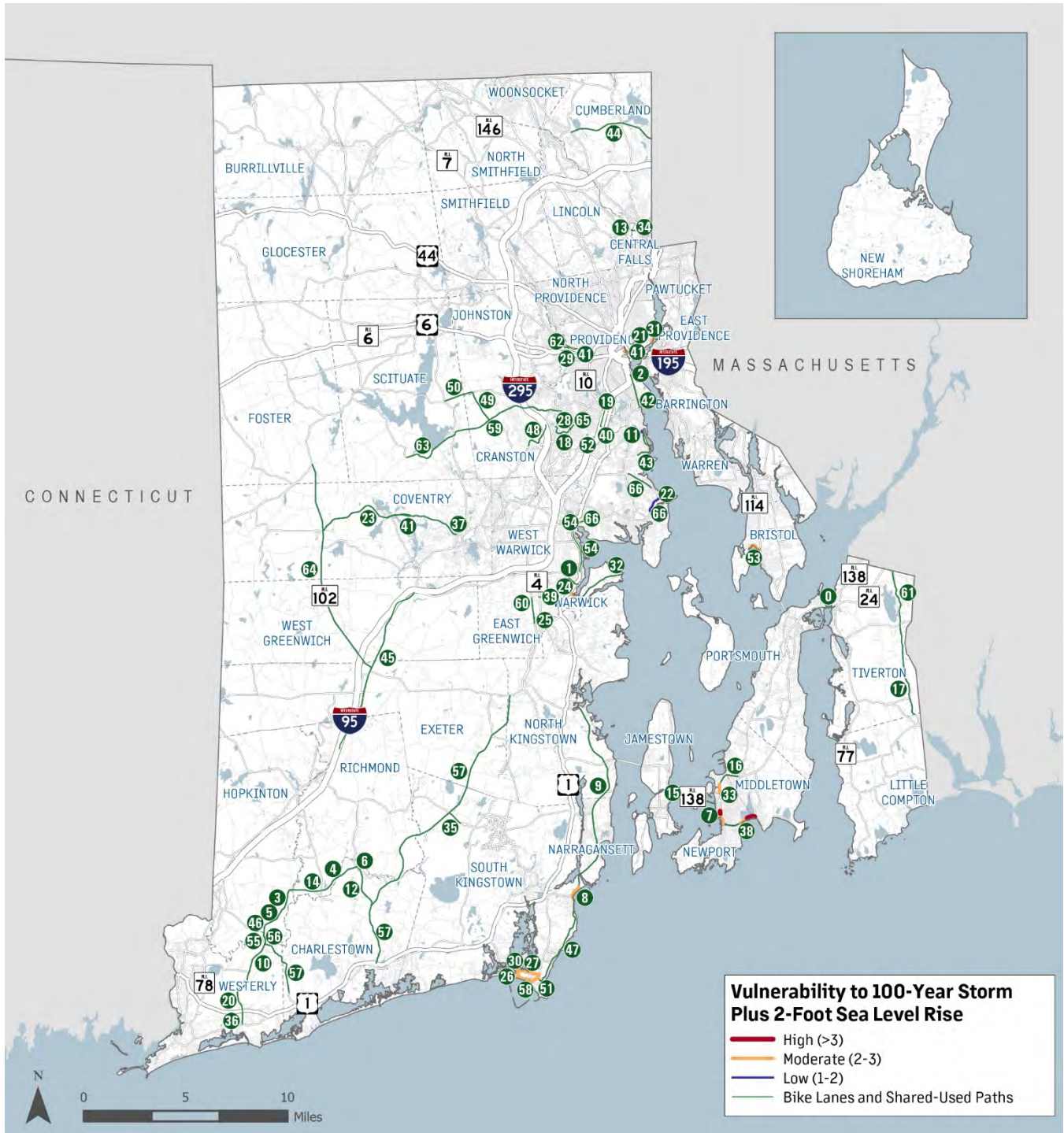




Figure D.13 Vulnerability of Stormwater Pipes to 1-Foot Sea Level Rise Scenario





Figure D.14 Vulnerability of Stormwater Pipes to 2-Foot Sea Level Rise Scenario





Figure D.16 Vulnerability of Stormwater Pipes: Inundation of 100-Year Storm + 2 Foot Sea Level Rise Scenario

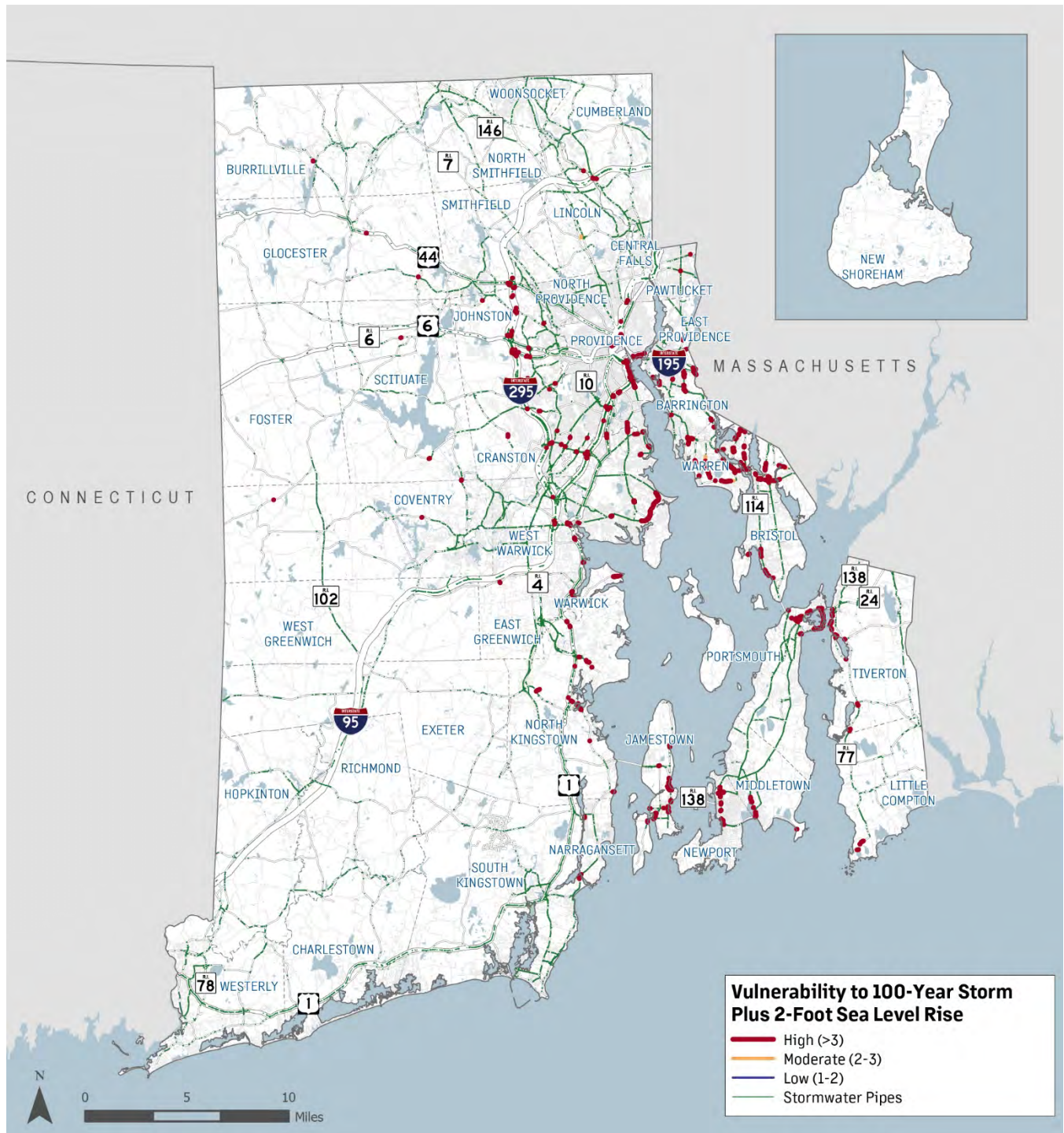




Figure D.17 Vulnerability of Stormwater Treatment Units to 1-Foot Sea Level Rise Scenario

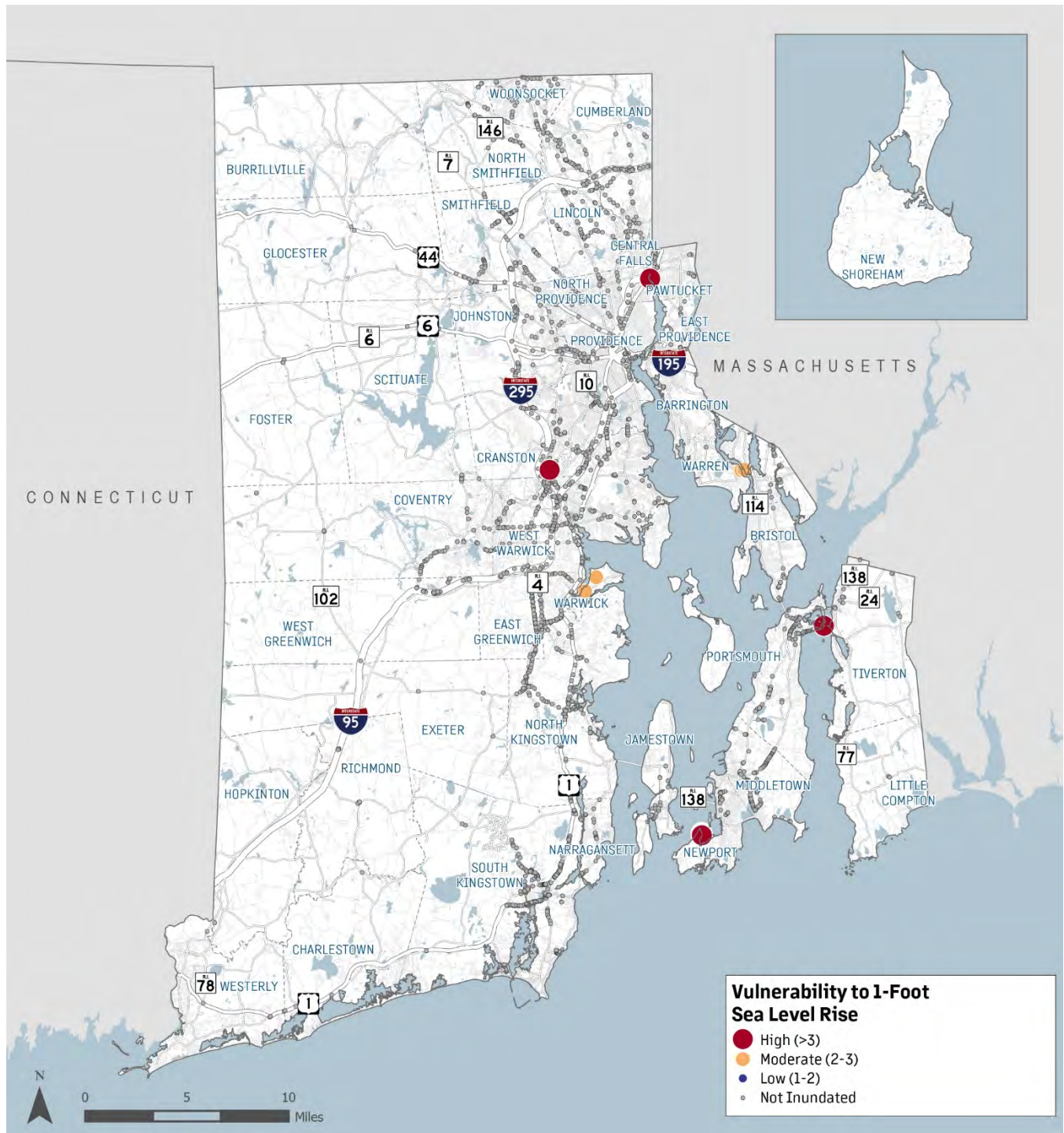




Figure D.18 Vulnerability of Stormwater Treatment Units to 2-Foot Sea Level Rise Scenario

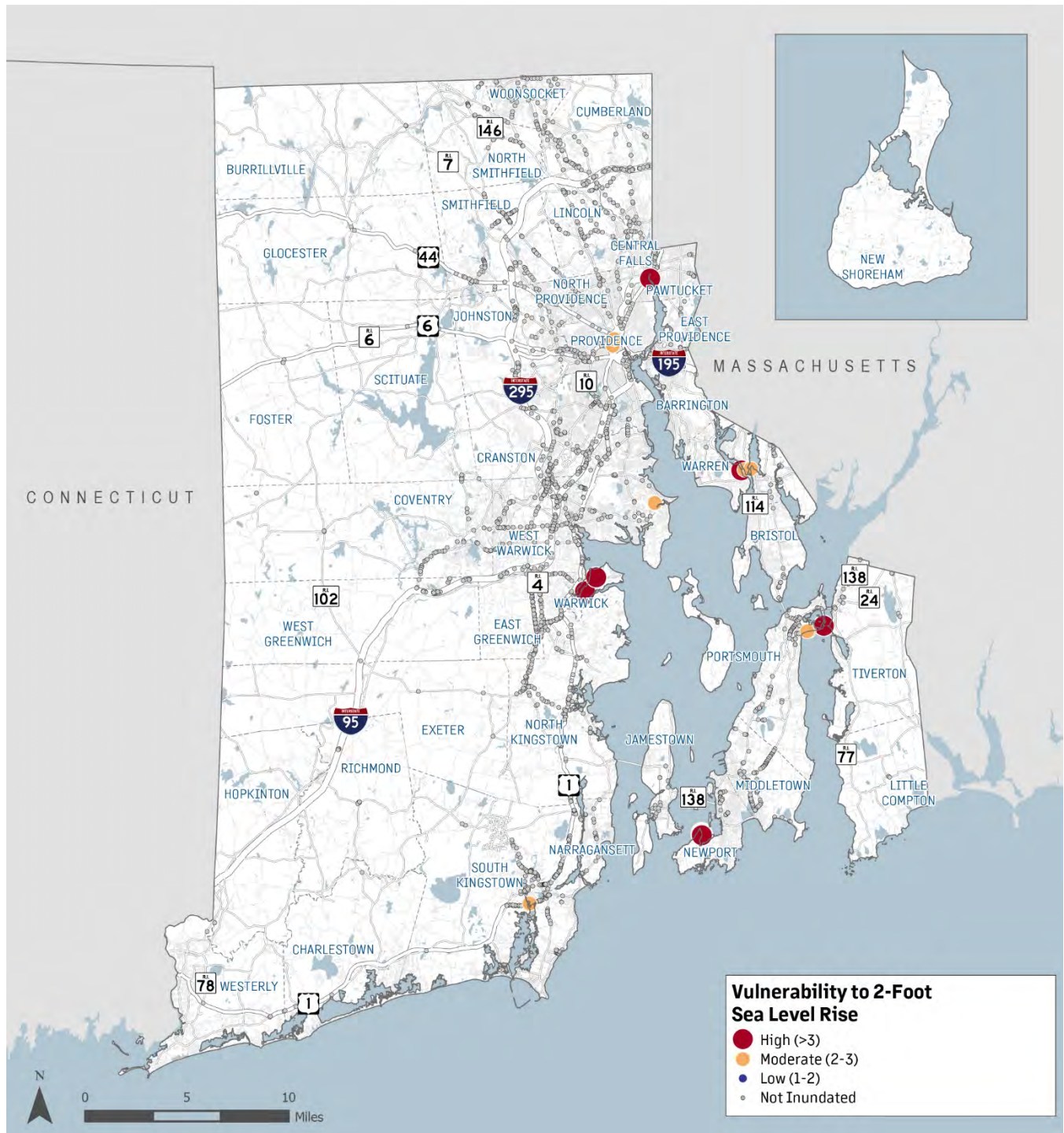




Figure D.19 Vulnerability of Stormwater Treatment Units: Inundation of 50-Year Storm + 1 Foot Sea Level Rise Scenario

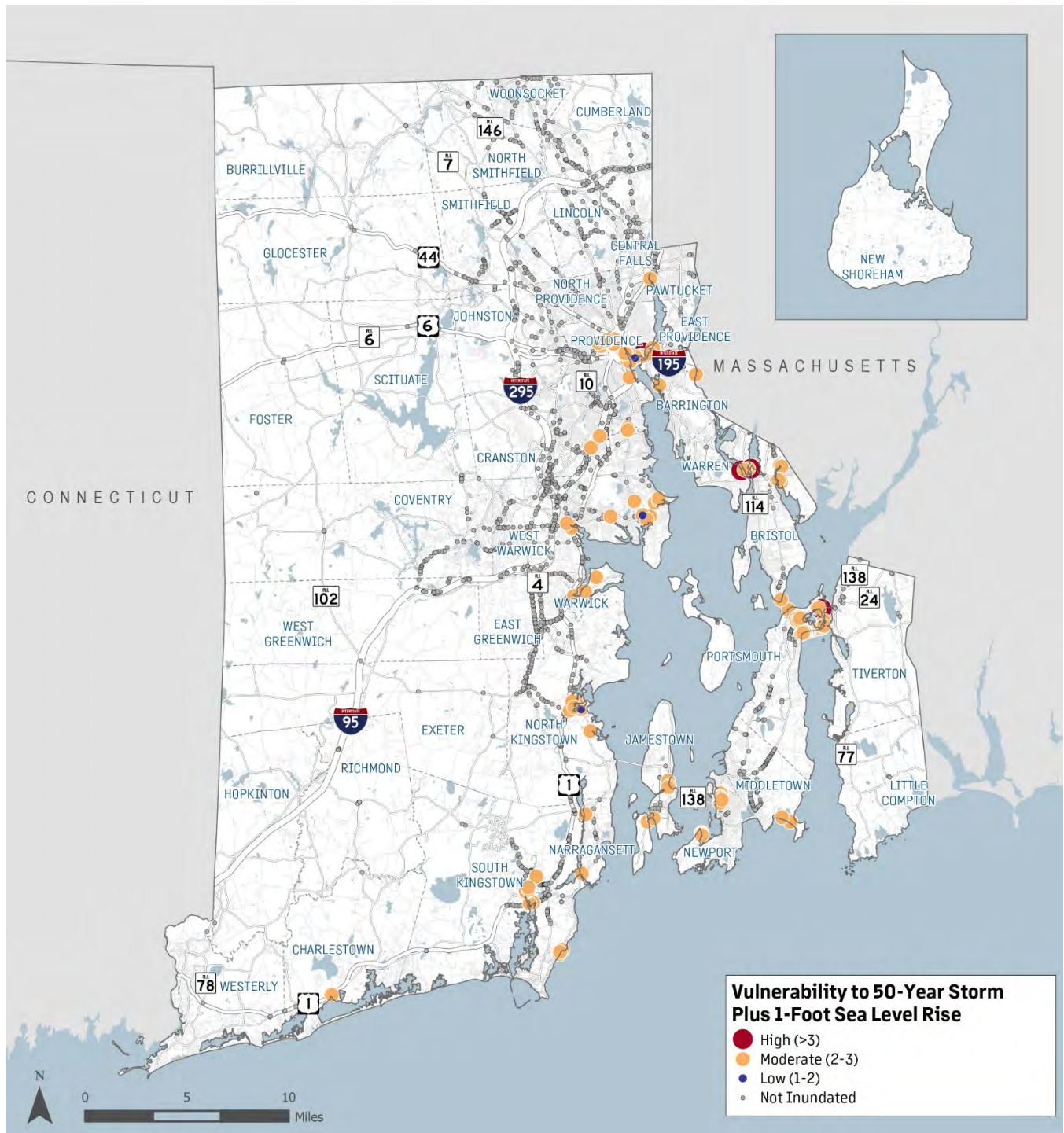
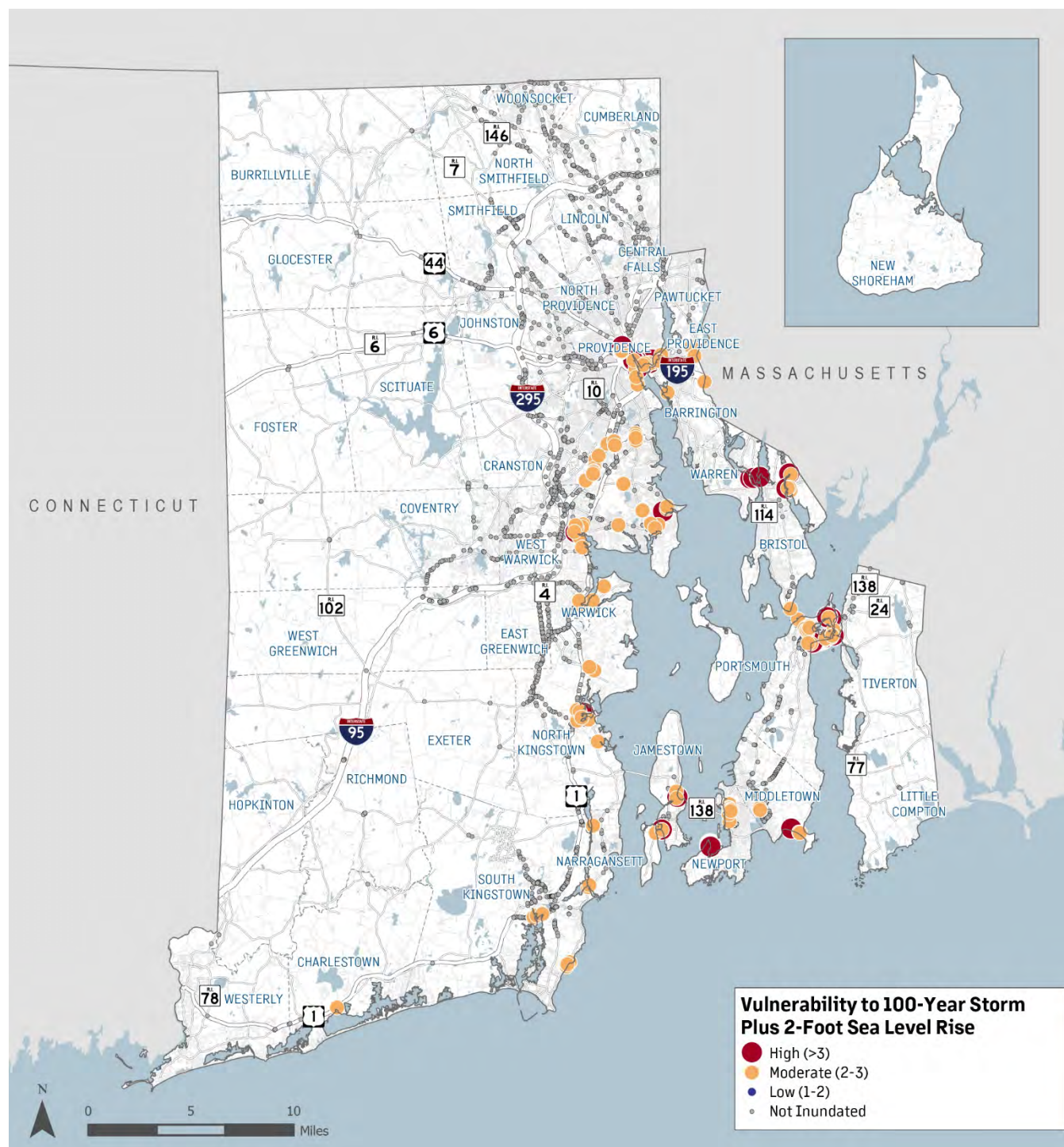




Figure D.20 Vulnerability of Stormwater Treatment Units: Inundation of 100-Year Storm + 2 Foot Sea Level Rise Scenario





APPENDIX G GEOGRAPHIC VISUALIZATION OF CONSEQUENCE

Total consequence costs, comprised of owner and user costs are visualized across Rhode Island's transportation assets in the following figures, comprised of the road network. Figure E.18, Figure E.19, and Figure E.20 model total consequence under scenarios of 1 foot, 2 foot and 7 foot sea level rise. Under scenarios of 1 foot and 2 foot sea level rise, consequence is limited to localized road segments in Newport, Portsmouth, and East Providence. On the other hand, consequence is expected to surpass \$3 million on these road segments following a hazard. As shown in Figure E.23, a scenario of 7 foot sea level rise includes additional locations of consequence of up to \$3 million sparsely scattered throughout coastal Rhode Island from Point Judith and Block Island north through Barrington and East Providence.

The next figures examine the consequences of sea level rise with the added impact of storm surge. Text on storm surge maps.

Lastly, Figure E.24 visualizes consequence as a result of flood. Consequence from flooding is primarily localized along a portion of I-95, as well as I-295, and portions of the RI-114 corridor. Consequence from flooding is also evident in Point Judith and Newport. For most of these locations, consequence ranges from \$1 million to \$5 million.



Figure E.1 Total Consequence in 1 Foot Sea Level Rise Scenario

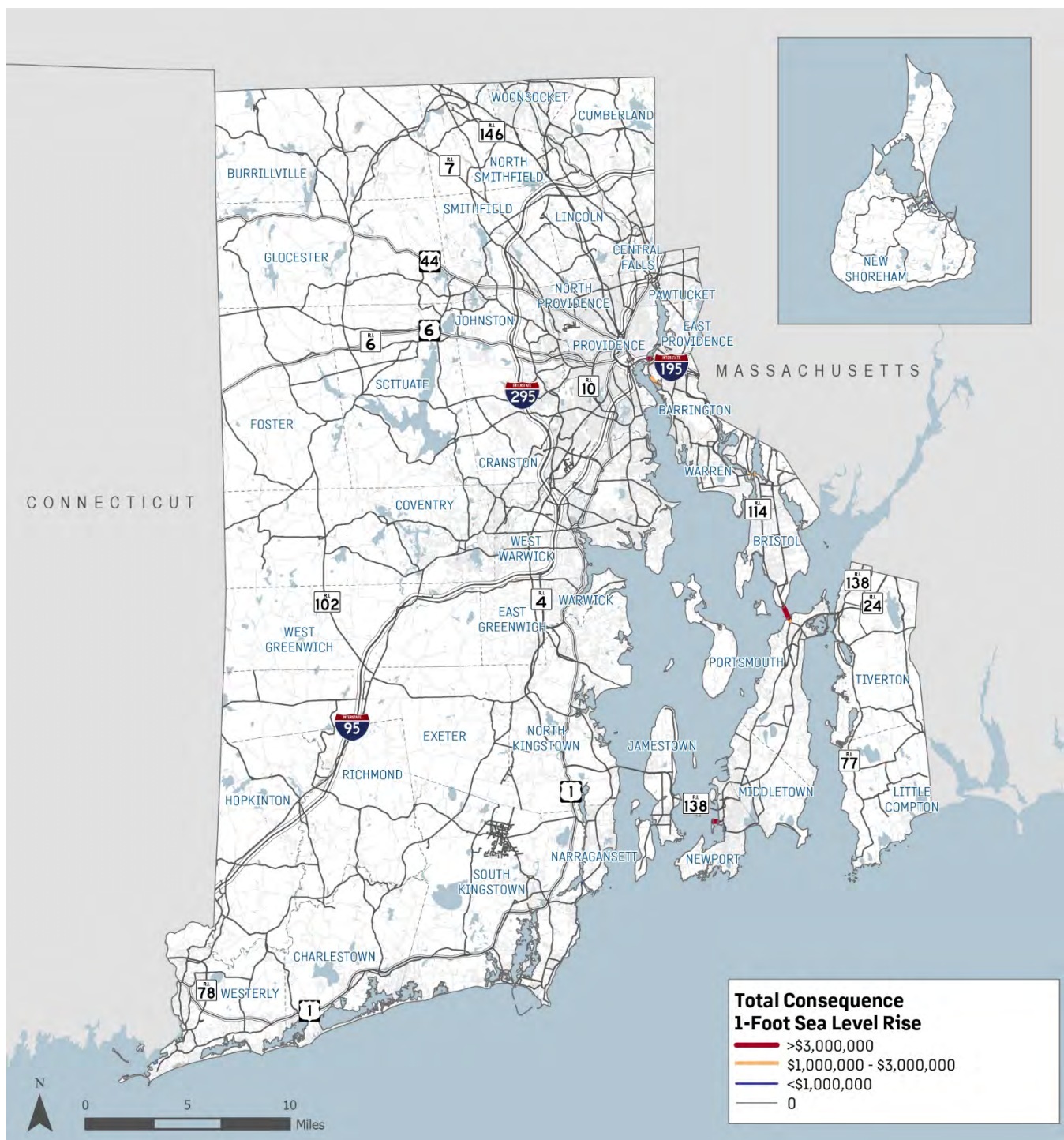




Figure E.2 Total Consequence in 2 Foot Sea Level Rise Scenario

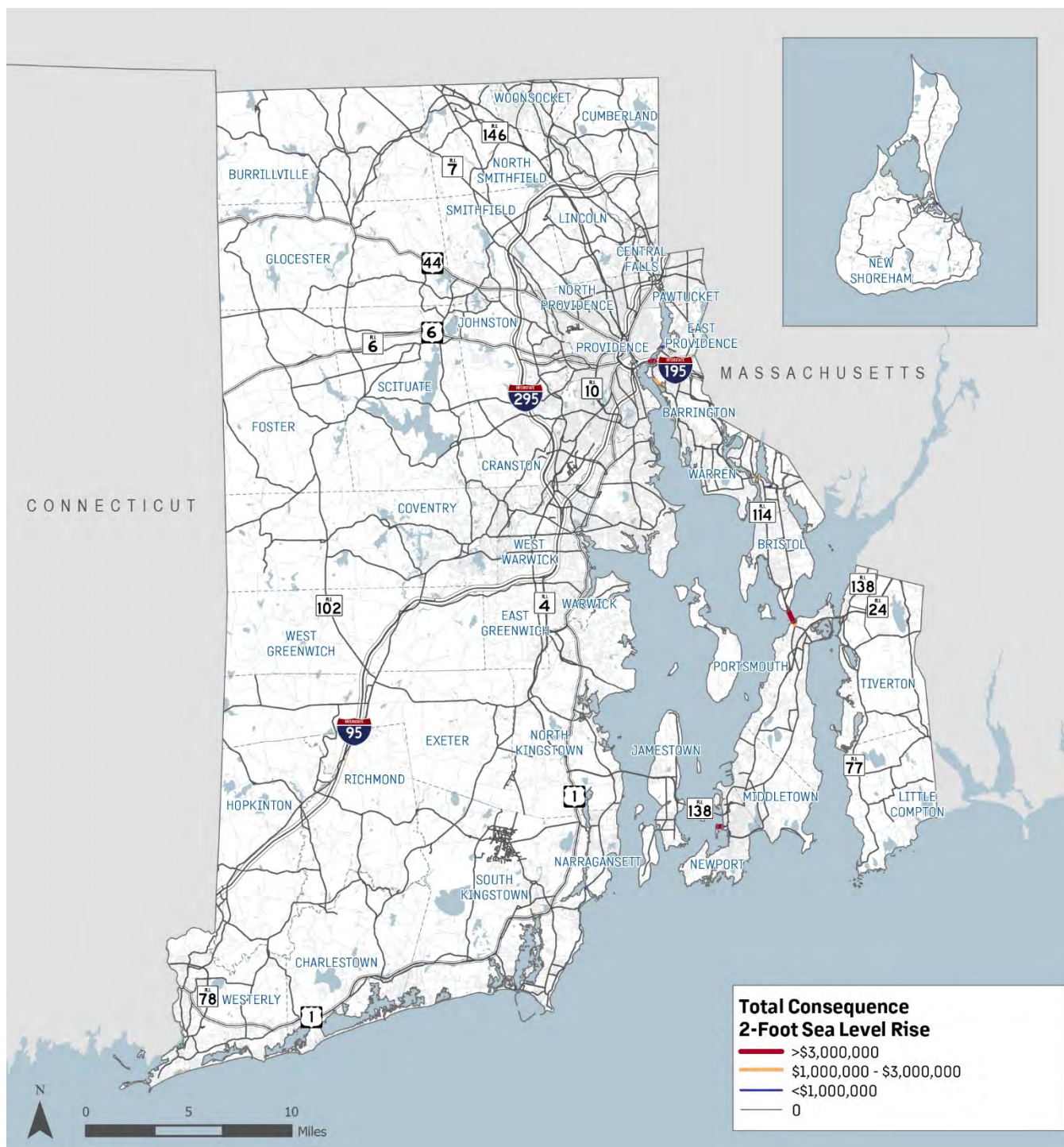




Figure E.3 Total Consequence in 7 Foot Sea Level Rise Scenario

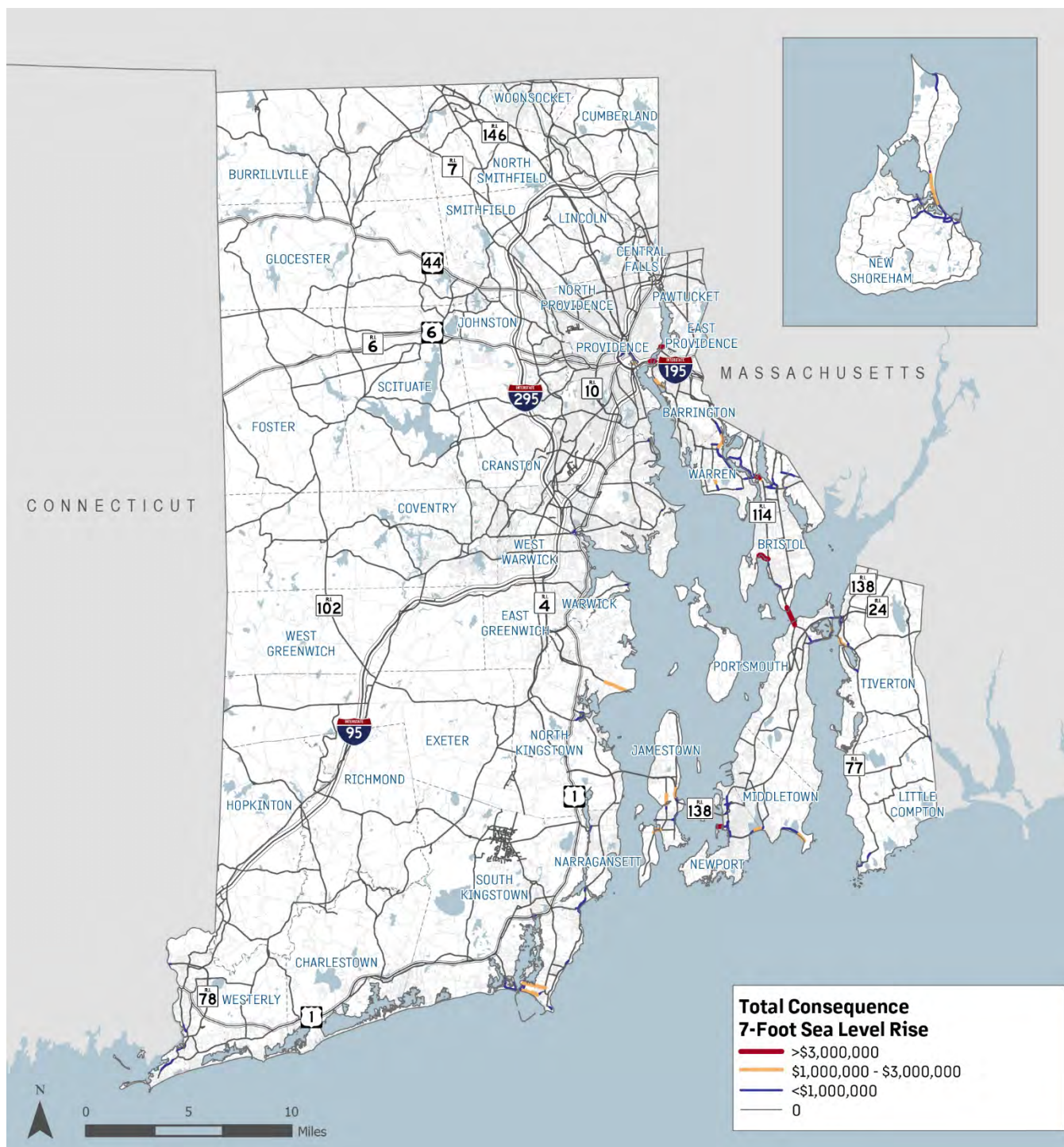




Figure E.4 Total Consequence in 1 Foot Storm Surge Scenario

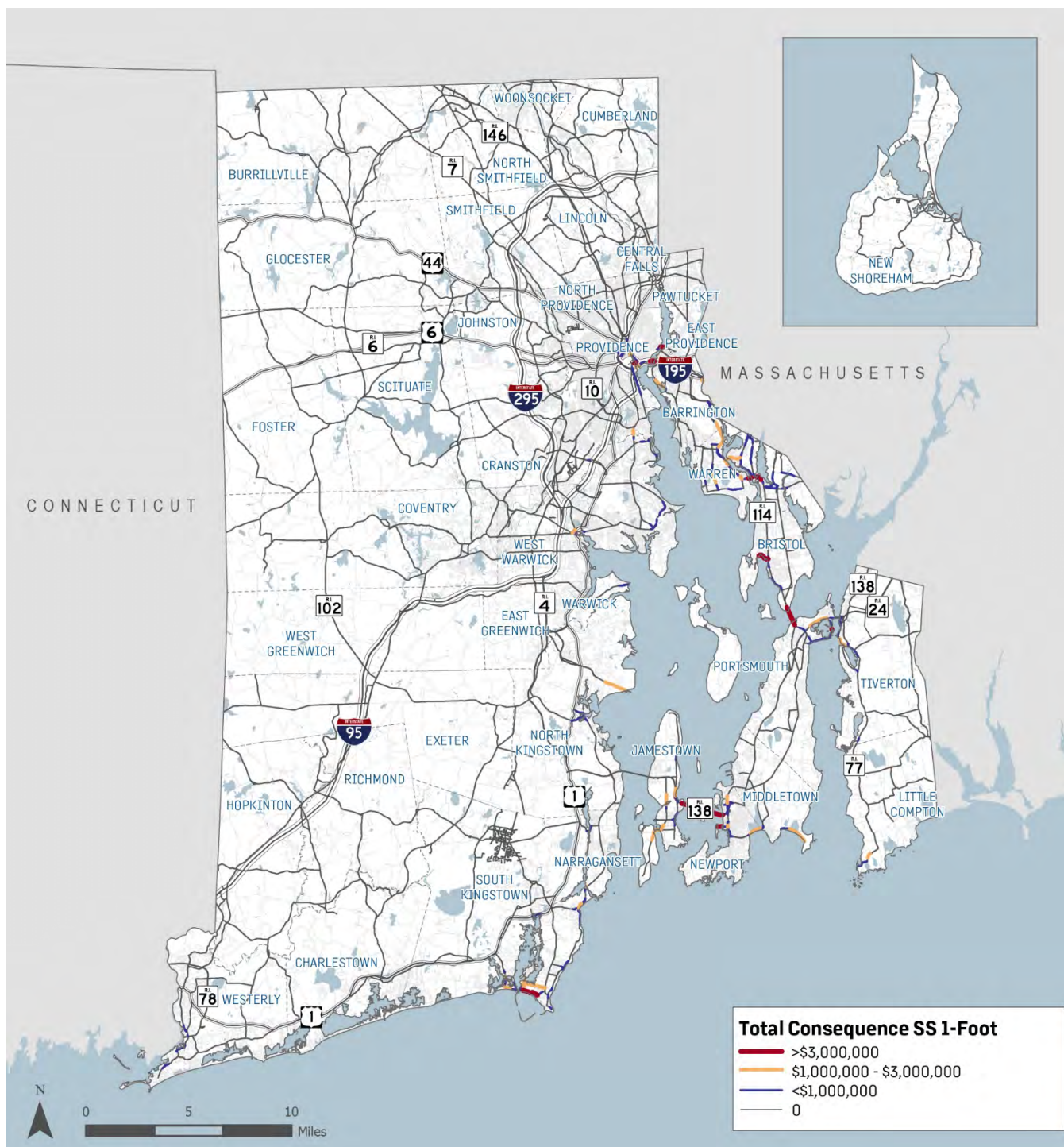




Figure E.6 Total Consequence in 7 Foot Storm Surge Scenario





Figure E.7 Total Consequence in a Flood





APPENDIX H RISK OUTPUTS

The following figures display individualized and composite risk across the Rhode Island road and bridge network.



Figure E.1 Risk of Sea Level Rise in 2035

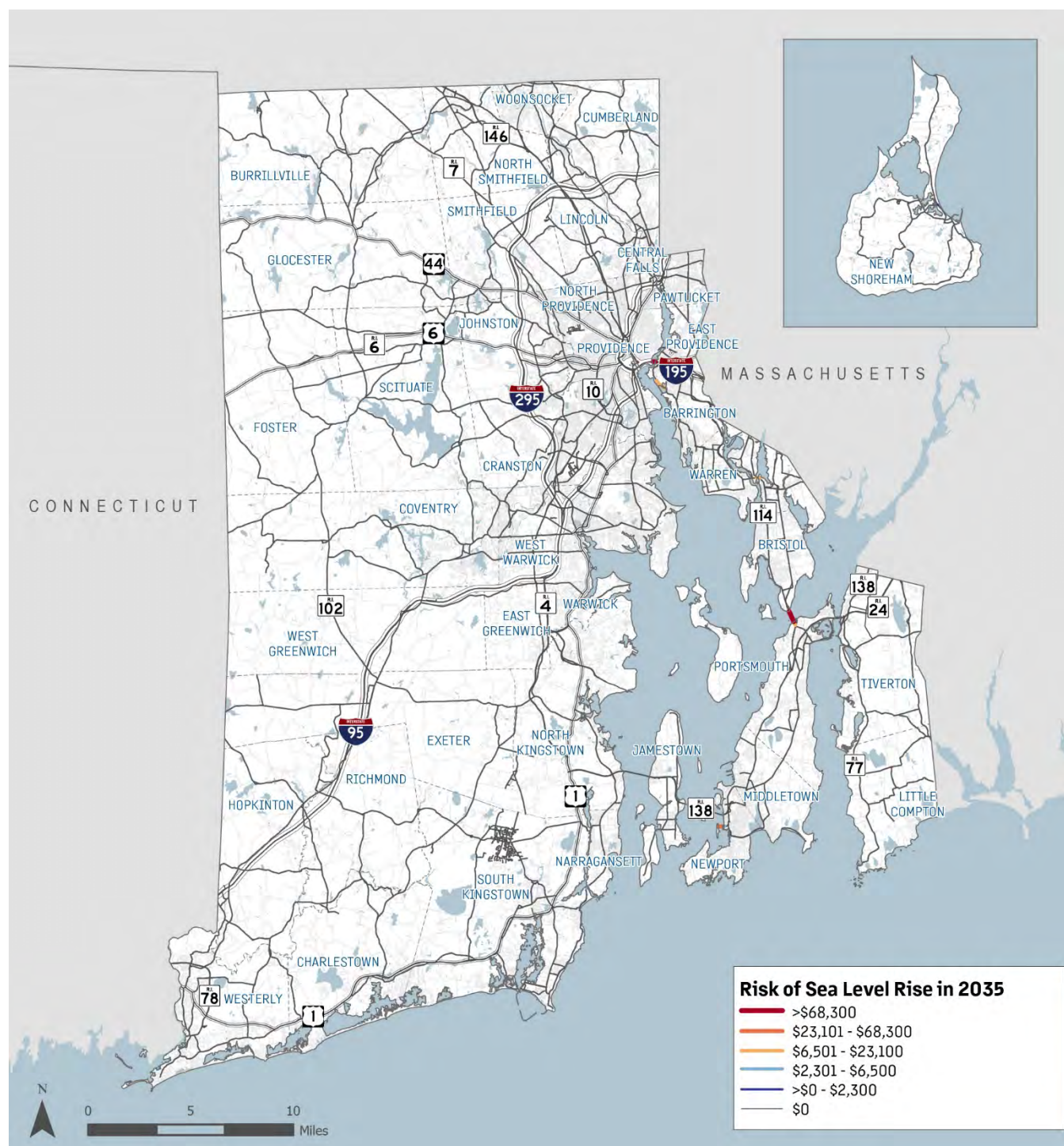




Figure E.2 Risk of Sea Level Rise in 2050

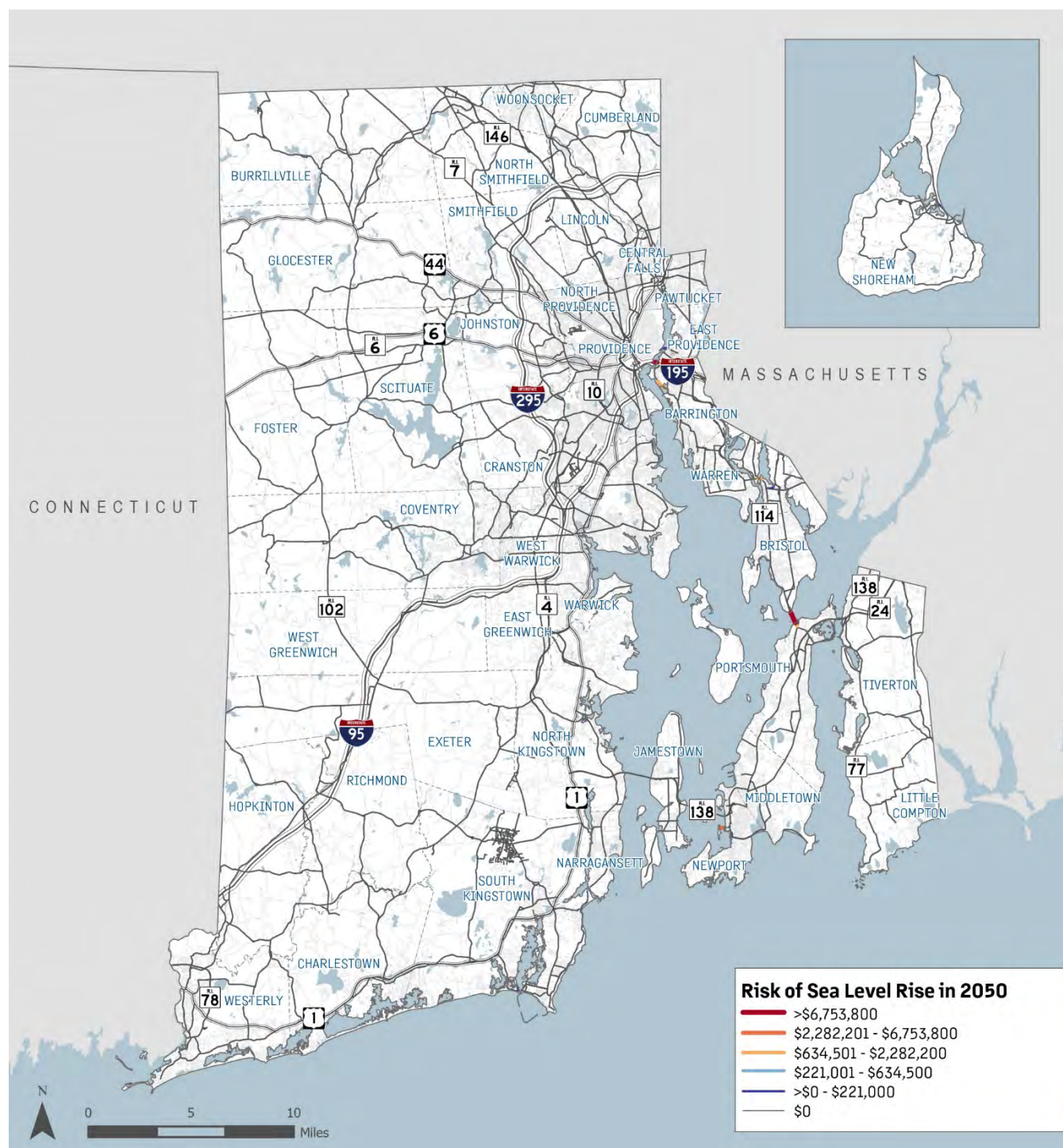




Figure E.3 Risk of Sea Level Rise in 2100

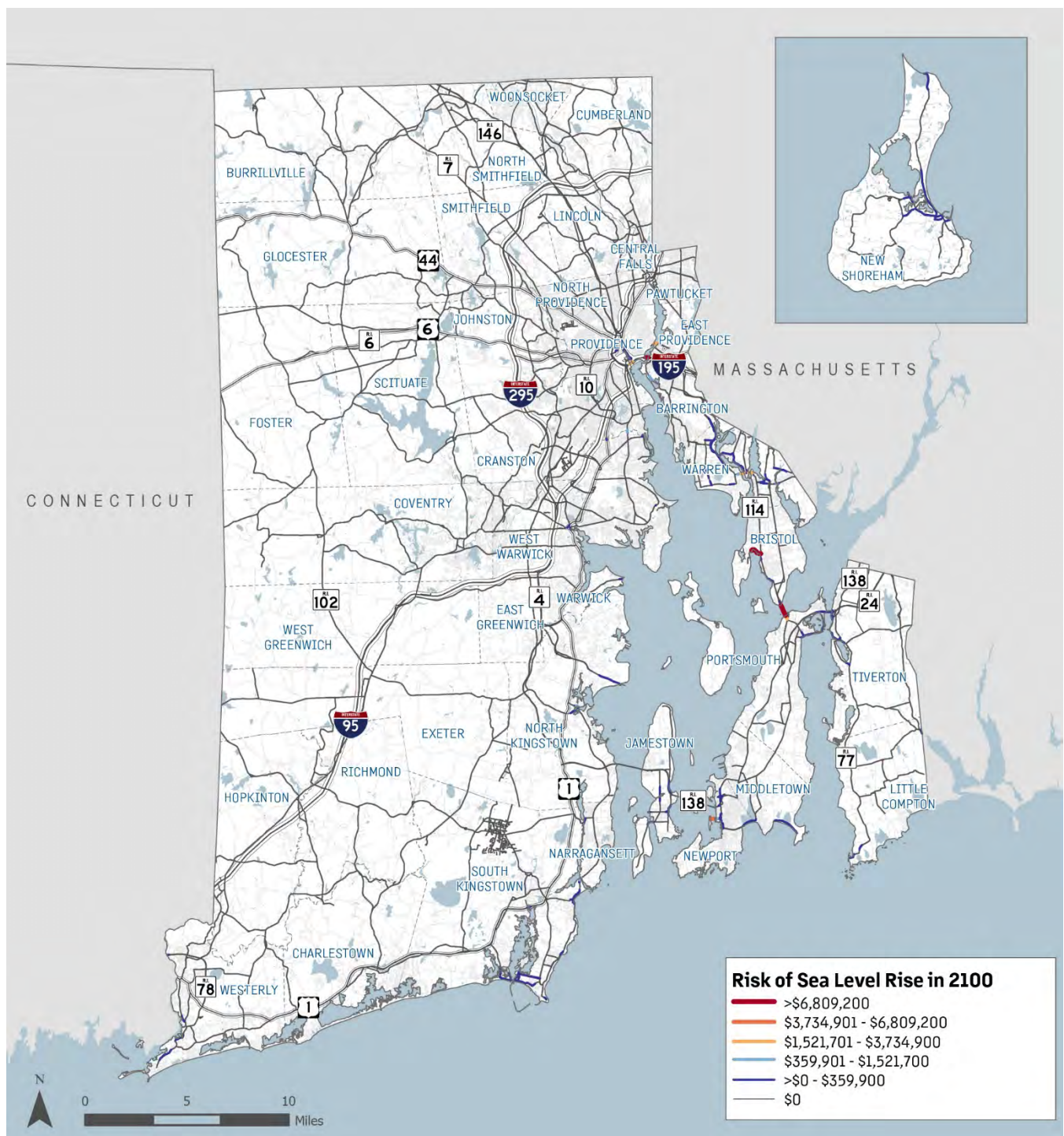




Figure E.4 Risk of Storm Surge + Sea Level Rise in 2035

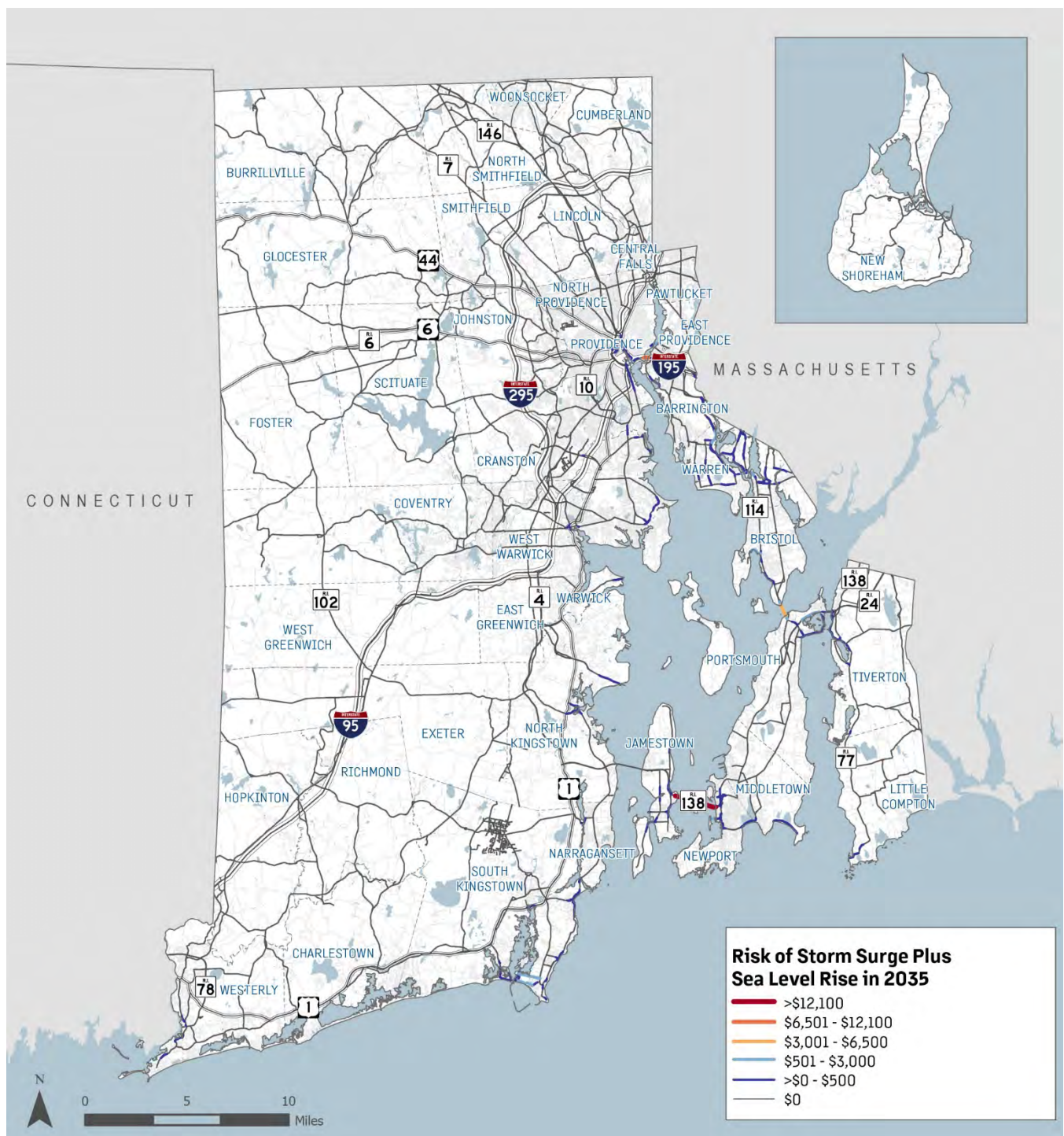




Figure E.5 Risk of Storm Surge + Sea Level Rise in 2050

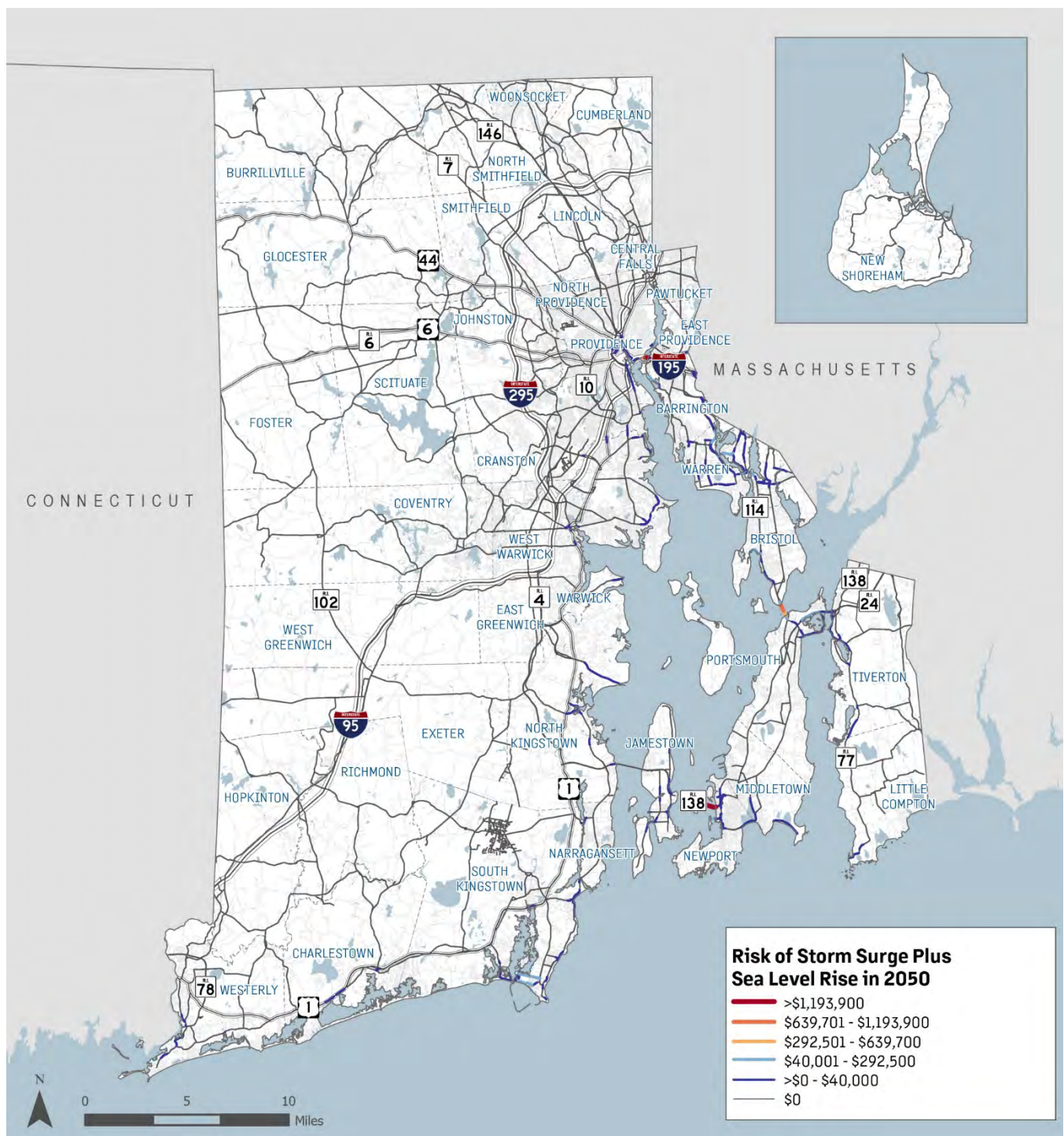




Figure E.6 Risk of Storm Surge + Sea Level Rise in 2100

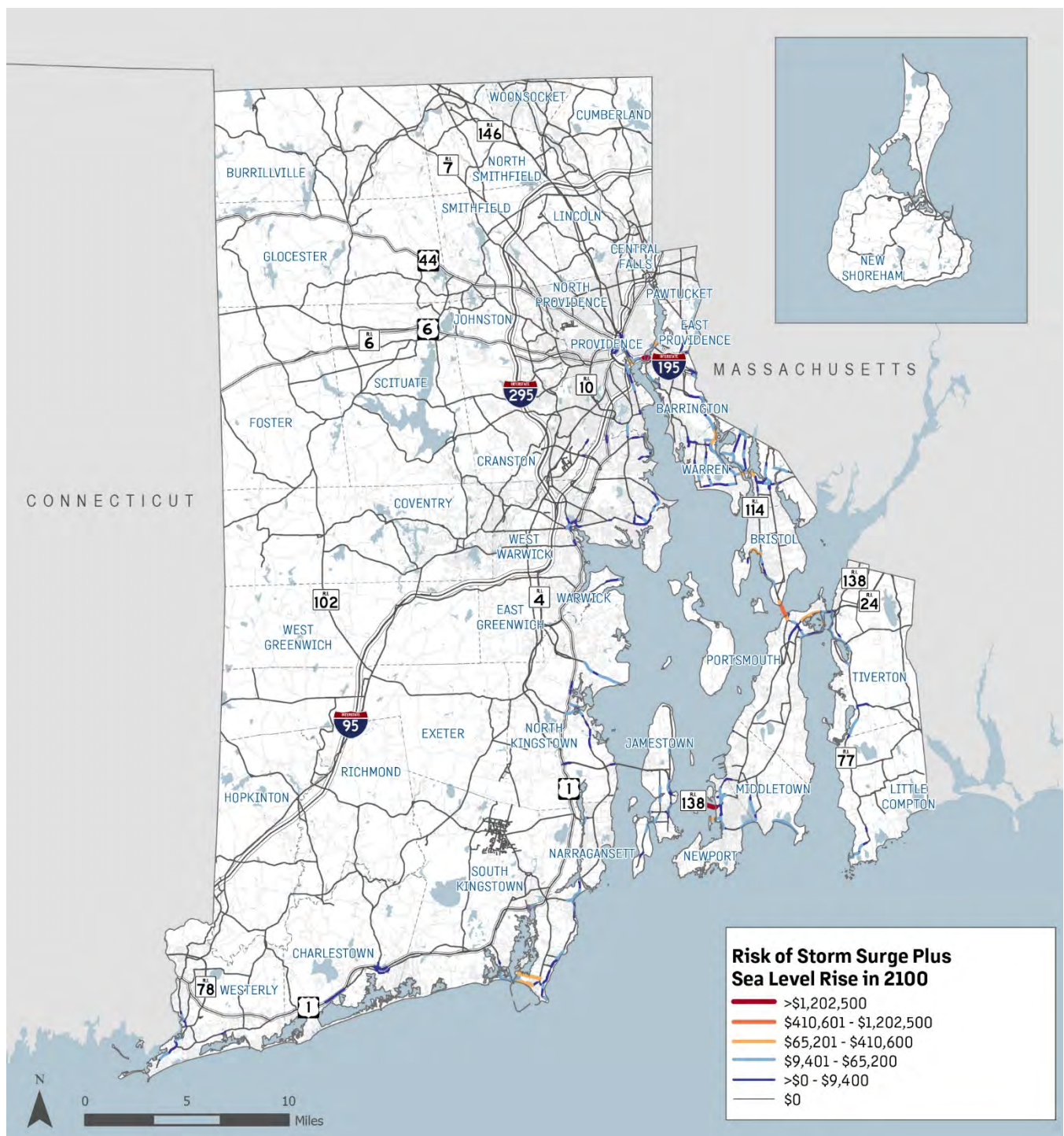




Figure E.7 Annual Risk of Flooding





Figure E.8 2035 Composite Risk





Figure E.9 2050 Composite Risk

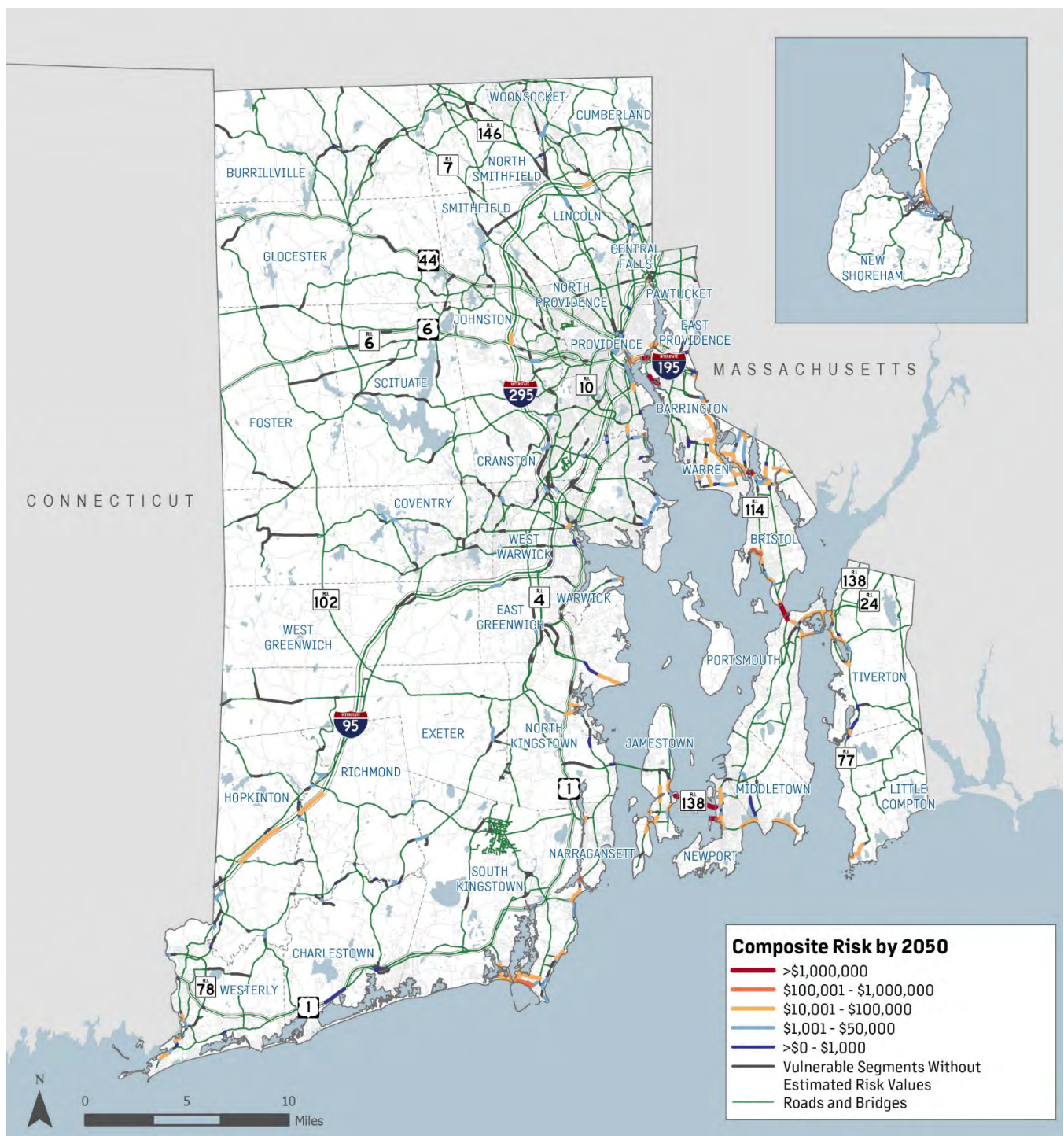
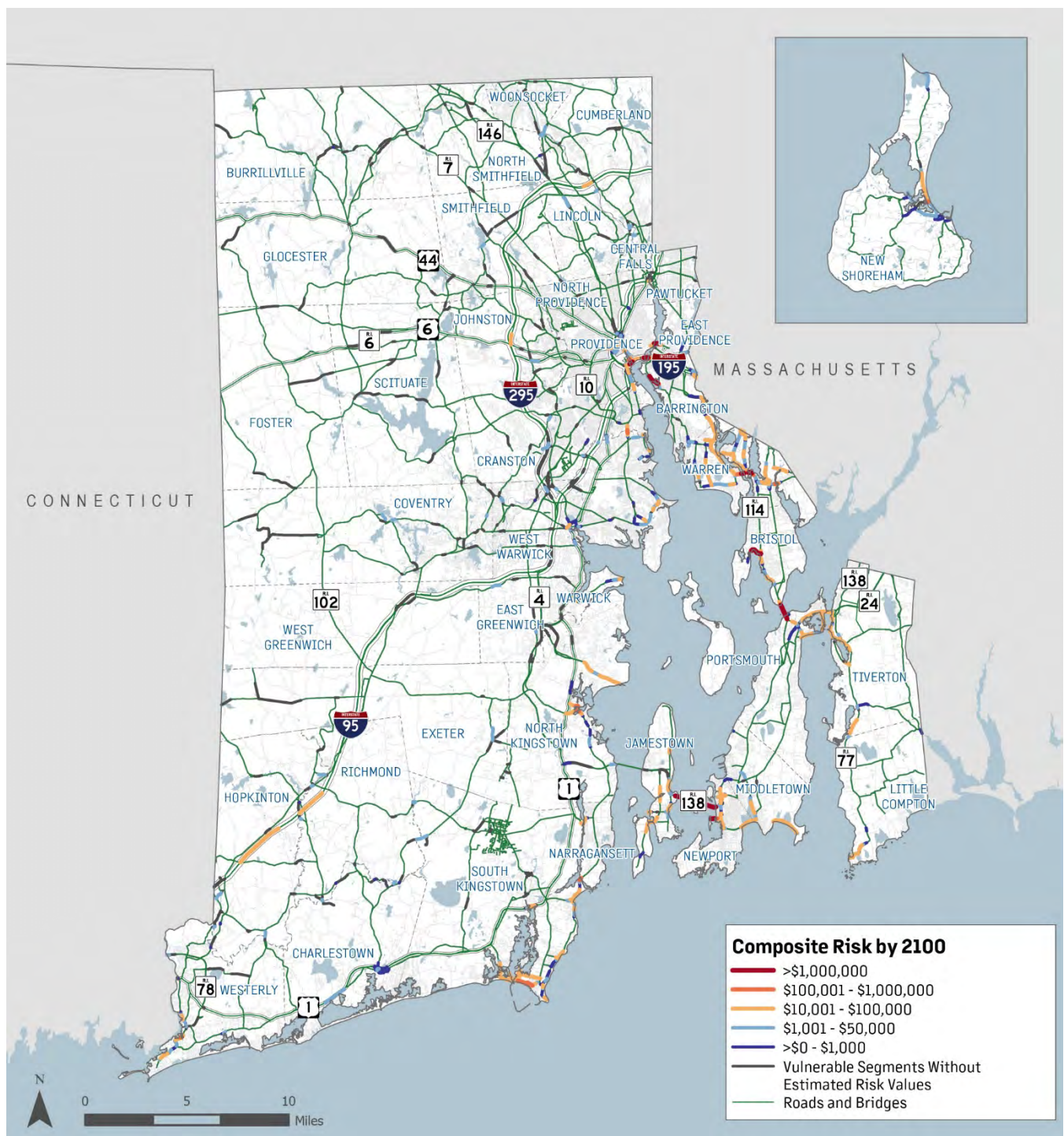




Figure E.10 2100 Composite Risk





APPENDIX I ADAPTATION STRATEGIES

I.1 Strategies for Roadway, Sidewalks, and Shared-Use Paths

The following adaptation strategies have been identified for roads:

I.1.1 Coastal Asset Protection Strategies

Coastal asset protection strategies address the needs and vulnerabilities of those road assets most adjacent to and vulnerable to the shoreline. In Rhode Island, these transportation assets can be found throughout much of the state's southern coast and islands, in addition to other low-lying areas. Strategies for coastal protection are expanded on below:

Natural Shorelines & Nature-Based Solutions

Natural shorelines are a broad category that encompasses options such as vegetation, edging, sills, beach nourishment, and a combination of vegetation with sand dunes. Where possible, nature-based solutions should be utilized, given the ability to protect coastlines against storm surge and threats from storms and hurricanes of up to Category 3 strength. FHWA's *Nature-Based Solutions for Coastal Highway Resilience*²¹ provides guidance for the use of nature-based solutions to improve the resilience of coastal highways to extreme events and sea level rise. The selection of each approach is dependent on several factors including exposure, wave action, and topography. Two of the most common applications of natural shorelines are briefly highlighted below:

Beach Nourishment & Dune Restoration

Sand dunes provide natural protection for coastal roads by providing a barrier between the roadway and the seaward ocean forces. Over time, natural processes slowly build sand dunes on coastal areas and then erode the formations through storm surges and wave actions. This process continues along an endless cycle if left without interference. On the other hand, replacement and maintenance actions may be needed if seaward ocean impacts are greater than expected.

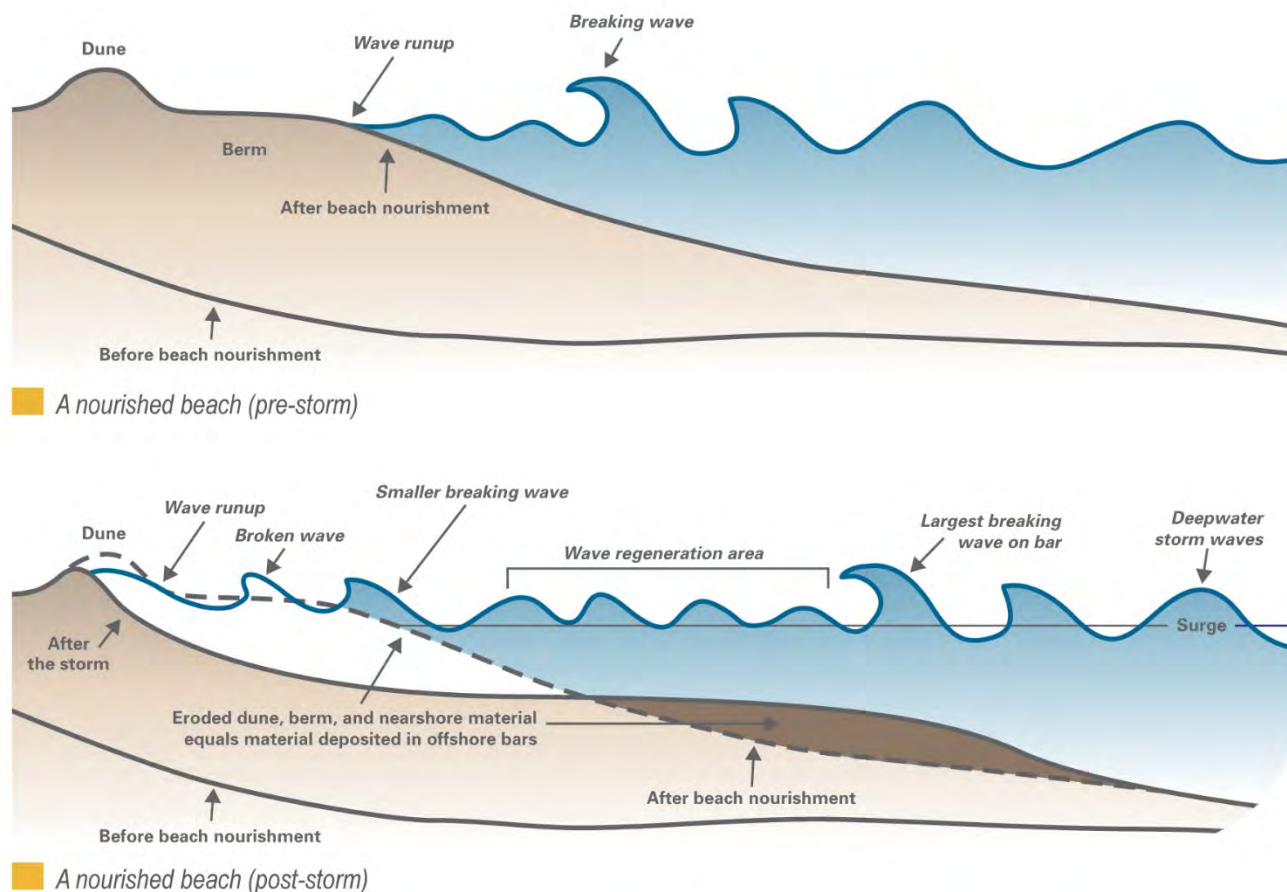
Although the design requirements for sand dunes are specific to the individual beach and road scenario, the process for restoring and creating sand dunes is standardized. Specifically, the process requires a barge to be anchored offshore where a temporary pipeline can then be extended from the barge to the shore. A large pump

²¹ Webb, B., Douglass, S., Dix, B., & Asam, S. (2018). White Paper: Nature - Based Solutions for Coastal Highway Resilience (No. FHWA - HEP - 18 - 037).



is then used to pump sand from the sea bottom onto the beach where front-end loaders are subsequently used to distribute the sand appropriately on the beach and where required into sand dunes. Costs for this approach can vary widely but are estimated at approximately over \$700,000 per 0.25 miles of coastline^{22 23}. Quantified benefits however are estimated at over three times the initial cost with a potential reduction of risk of between 30 percent and 50 percent.²⁴

Figure F.1 Beach Nourishment Process Utilizing Dunes



Source: US Army Corps of Engineers

²² California (2002). California beach restoration Study, Department of Boating and Waterways and State Coastal Conservancy, January 2002.

²³ Fournier, M., undated, 'Standards for Creating and Restoring Sand Dunes: from Massachusetts to North Carolina (ed. by Miller & Skaradek, Cape May Plant Material Center, and RPS, USDA, NRCS).

²⁴ Reguero, B. G., Beck, M. W., Bresch, D. N., Calil, J., & Meliane, I. (2018). Comparing the cost effectiveness of nature-based and coastal adaptation: A case study from the Gulf Coast of the United States. *PloS one*, 13(4), e0192132. doi:10.1371/journal.pone.0192132



Vegetation as Erosion Control

Grassy vegetation and shrubs are useful mechanisms for combatting erosion in minor to moderate wave overtopping and storm surge conditions along roadsides. Vegetation, including dune and marsh grass, as well as shrubs, and reefs appropriate to Rhode Island's ecological conditions and climate, can be combined with sand dune restoration to provide an additional level of stability to the sand dune structures. This approach also is locally dependent on conditions and soils that may not be appropriate for inland areas. Overall, the use of vegetation offers a highly effective, economically worthwhile, and environmentally beneficial approach to asset protection. Time is a key consideration for this strategy however, in that planted vegetation would likely require multiple years to mature and establish strong root structures.



Ammophila breviligulata (American Beach Grass), native to Rhode Island.

Image Source: Go Botany

Revetments and Sea Walls for Direct Asset Protection

Coastal roads that are directly exposed to wave action and surge events can be extremely susceptible to erosion on the seaward side due to increased flows during surge events. The purpose of hardening the seaward side is to provide protection against increased hydrologic action and specifically protect the roadbed from direct exposure to the elements. To accomplish this protection, the seaward side of the road embankment can be hardened using a revetment or seawall that is placed along the slope where exposure to water may occur.

The distinction between revetments and seawalls is one of functional purpose. Revetments are layers of protection on the top of a sloped surface to protect the underlying soil; while seawalls, consisting of rigid structures or rubble mounds, are designed to directly protect against large wave forces. See FHWA's *Highways in the Coastal Environment*²⁵ for more information on design considerations for revetments and seawalls.

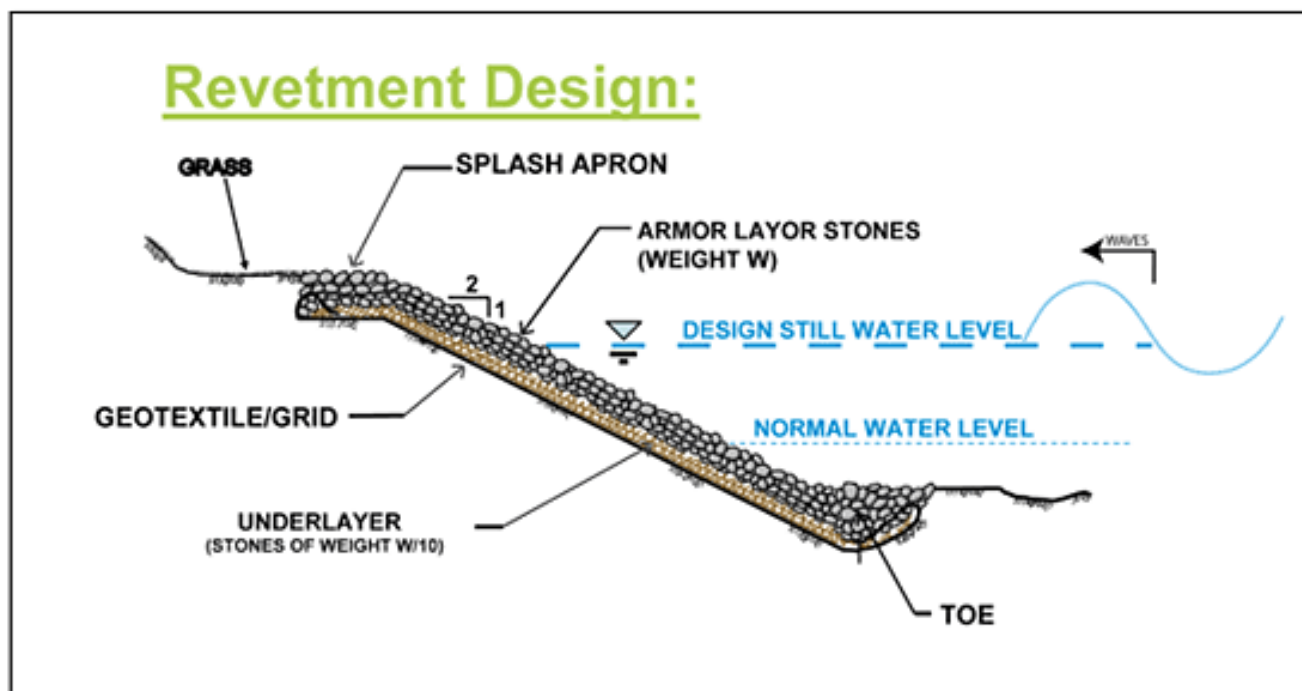
From a life-cycle perspective, revetments can provide significant benefit in that they require minimal maintenance over the design life if properly constructed and built to a level that will withstand future risks. If the revetment is constructed to a level that does not anticipate future threats, then overtopping can start to occur and cause damage to the top of the structure. Therefore, proper design analysis is required to ensure the seawall meets its required design life.

²⁵ <https://www.fhwa.dot.gov/engineering/hydraulics/pubs/hif19059.pdf>



The benefit of a sea wall system is that it provides a time-proven solution to protecting coastal assets against many different conditions including storm surge, wave action, and tidal changes. Sea walls can also provide natural areas for sea life and protection for visitors to the shore. These mechanisms have also proven to be long-lasting and require minimal maintenance in comparison to other natural solutions.

Figure F.2 Revetment Design



Source: Pilebuck

Wave Attenuation Devices

In contrast to a revetment which is a direct-asset protection strategy, wave attenuation devices (WADs) can be used to protect on-shore infrastructure from an offshore location. WADs reduce the force of waves striking the coast by dissipating energy when waves encounter them. Field measurements have shown that WADs can reduce wave height and wave energy by over 80 percent.²⁶



Installation of concrete wave attenuation devices in Tampa Bay.

Image Source: FOX 13 Tampa Bay

²⁶ <https://livingshorelinesolutions.com/living-shoreline-solutions-wave-attenuation-devices/>



There are two main commercial types of WADs. The first format is usually made with concrete and submerged to the ocean floor and can be seen in the image above. This type of WAD has minimal adverse impact on marine life due to its overall small footprint. Additionally, they act as an artificial reef to facilitate local fish populations. The second format is a floating WAD (see the image below). Floating WADs are completely portable and do not require major construction to move.

Wave attenuation devices are a newer defense against increased wave action in comparison to seawalls, as they provide an opportunity to protect significant lengths of coastline against major events such as hurricanes. The ability of the devices to reduce wave force prior to reaching shore is a significant benefit when considering strong wave forces that pose risks to transportation assets.

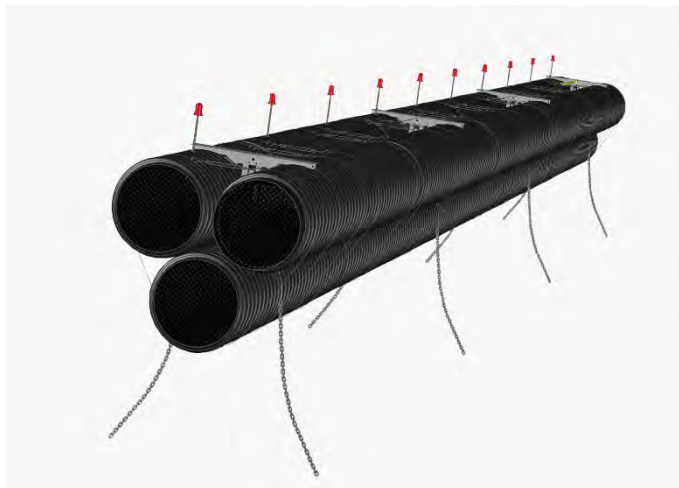


Image Source: Wahoo Docks

I.1.2 Extended Inundation-Prone Roadway Protection (Profile Raise)

In situations where extended inundation is possible due to topography, storm surge, or precipitation events, enhancing drainage may not be sufficient to avoid damages to critical roads, including those serving as emergency routes. In these instances, raising the profile of the most critical or entire segments of the road may be required to ensure adequate protection against adverse impacts. This approach will likely have significant impacts on the local area drainage functionality. However, in cases where a road is critical for emergency operations, these considerations should be weighed against the essential nature of the road in facilitating emergency operations. To analyze the benefits of elevating a roadway, the possible storm surge or other inundation scenario must be analyzed to determine the appropriate height to raise the profile. The insight gathered from Chapter 4 and Chapter 5, which discusses inundation levels and risk through 2100 provides a foundation for this decision-making process.

The cost-benefit of raising the profile focuses on the comparison of projected damages and the initial cost of raising the profile. The investment cost is focused on the initial outlay for raising the profile. Subsequent to the initial cost, the maintenance of the road returns to the typical expenditures incurred with any road on an annual basis. Additionally, once the road is raised, there is no further cost that is needed to maintain the raised profile. As a result, the one-time investment can then be offset by the protection offered to the road itself as well as the surrounding structures.



I.1.3 Asset Protection Strategies

Inundation and storm surge can cause multiple damage scenarios for road surfaces. Issues including wash boarding, alligator cracking, and transverse cracking are only a few of the potential impacts that the movement and settling of water over a road surface can create. Enhancing the surface and/or the subbase will allow a road to better withstand inundation or water movement.

Hot Mix Asphalt Upgrades

The road surface of a typical hot mix asphalt (HMA) asset is comprised of several asphalt courses. Three solutions of varying levels of investment are identified to address enhanced road surface needs, based on the severity of expected inundation and storm surge:

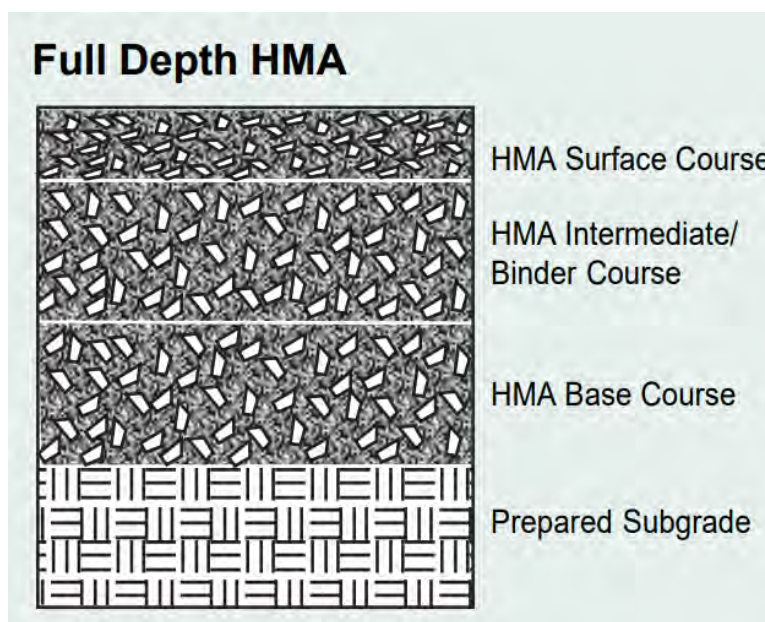
- **Enhance Road Surface:** Through the application of an additional 2 inches of course materials, or to enhance the binder course with larger aggregate to enable greater base drainage.
- **Enhanced Sub-Surface:** Through the enhancement of sub-surface thickness by between 4 and 6, inches depending on engineering requirements, in order to provide greater resistance to water infiltration cracking.
- **Complete Rebuild:** To be considered for scenarios where substantial road infrastructure improvements are necessary, or planned for other reasons in addition to resilience needs.

From a cost-benefit perspective, the overall category of enhancing a road surface has a benefit of strengthening the road and extending its design lifespan. Depending on the combination of solutions selected, the degree of enhancement to design lifespan will vary, with more substantial enhancements providing the greatest likelihood of lifespan increase. These considerations should be included in the overall planning and prioritization process of the road enhancement.

Enhanced Shoulder & Medians

Damages to pavements and roads from surge and inundation can be reduced in specific

Figure F.3 Hot Mix Asphalt Layers



Source: FHWA HMA Pavement Mix Type Selection Guide



circumstances by armoring (hardening) the sides or shoulders of roadways and/or of roadway medians. The armoring of roadway shoulders and sides will vary depending on the specific circumstances. Roads which have coastal exposure should consider the use of armoring that can withstand high velocity flows. This type of armoring includes sheet piling and gabions. The sheet piling should be located on the shoulder where supercritical flows are most likely to occur. Buried gabions can be used to resist overtopping flows that may be lower but parallel to the road during a storm event.

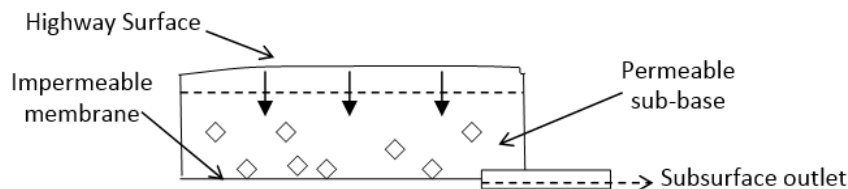
A concrete revetment system is another option to reduce erosion from overtopping. In this case, the system should be comprised of heavy blocks, vertical and horizontal interlocking cables and anchors to resist hydraulic forces from overtopping. In areas where the road is inland and will experience less intense flows, hardening of shoulders may include changing the surface of the shoulder to concrete paving to enhance protection, using riprap in vulnerable areas to divert flows away from the road surface and base, and using piling in select areas to protect key points such as intersections.

The cost perspective on medians is similar to that of increasing drainage. Inadequate drainage will increase erosion on the sides of the road as well as at the median. The 10% to 15% increase in damage can also be seen at the median. However, the enhanced median will provide additional benefits besides the protection from erosion. The advantages to drainage and stormwater will increase as reflected in the benefits provided by swales. This dual benefit creates an advantageous scenario for medians that exceeds many other options.²⁷

Permeable Pavement

Permeable pavements, also referred to as porous pavements, are loadbearing, durable road surfaces that have an underlying layered structure which temporarily stores water prior to infiltration from soil or drainage to a controlled outlet. The advantage of such a pavement system is that it can help to reduce runoff volume during periods of peak flow and minimize flooding. Permeable pavement is primarily utilized for locations that are gently sloping at most, and along roadways with low traffic volumes, axle loads, and travel speeds (less than 30 mph). The design elements associated with the construction and maintenance of porous pavements include initial grading, paving, and excavation of up to four feet of soil. Once excavated, a sight well, stone fill, and filter fabric are installed. Finally, the area is seeded and landscaped appropriately.

Figure F.4 Permeable Pavement Cross-Section



²⁷ <https://highways.dot.gov/public-roads/autumn-2022/03>



The cost-benefit of permeable pavements encompasses a broad range of elements. The most notable component of this solution is reductions in runoff into the stormwater system of up to 50 percent, depending on the location, infrastructure conditions, and soil composition^{28 29}.

²⁸ <https://sustainabletechnologies.ca/app/uploads/2015/01/PP-Tech-Brief-Final.pdf>

²⁹ https://stormwater.pca.state.mn.us/index.php?title=MS4_fact_sheet_-_Pervious_Pavements



I.2 Bridge Adaptation Strategies

The following adaptation strategies have been identified for roads:

Improve Flow Under Structure

Strategies to improve flow under the bridge and culvert structure are applicable in scenarios where the opening under the structure is too small or blocked. This can lead to increased flooding and damage/stress to decks, railings, and trusses. By improving the flow under the structure, this damage can be mitigated. This can be accomplished through the following methods:

- **Replace Multiple Spans with a Single-Span Deck:** Particularly relevant for longer bridges, this strategy reduces the number of trusses required to support the structure. Although this is an effective method of increasing flow, it also requires significant design considerations, including substructure and profile enhancements. Impacts to water velocity and scour displacement will also need to be assessed, making this a more resource-intensive adaptation strategy.
- **Elevate Bridge Deck:** This strategy is useful for bridges prone to overtopping and involves raising the bridge deck over a pre-determined flood stage. Overall, bridge deck elevation is a highly effective adaptation strategy, especially when designed to the correct elevation that protects against storm surge and flood levels. Similar to span replacement however, bridge deck elevating would require significant design considerations, including pier and abutment reconstruction. There may also be impacts from scour displacement, as well as additional resource requirements if the bridge needs to be lengthened to accommodate the raised deck.
- **Increase Bridge Length:** For culverts and bridges vulnerable to inundation, increasing the bridge length can increase flow by providing additional space for water to flow under. Increasing bridge length involves the physical expansion of the structure, including related design considerations.
- **Construct a Relief Opening:** For bridges, this involves the construction of an additional adjacent relief opening, including in the form of a culvert, at locations with the heaviest flow.

Construct Erosion & Scour Countermeasures

Erosion & scour adaptation strategies involve countermeasures applied to bridge approach slabs and abutments. Over time, increased flows can cause accelerated damage to the structural components. Related mitigation strategies seek to add protection and fortification to these components:

- **Install Riprap:** Riprap consists of loose stone, which as part of this adaptation strategy, is placed along bridge approaches, abutments, and piers to reduce erosion during high-velocity water flow. Although the



riprap will likely need to be professionally designed, this may be a relatively low-cost strategy for bridge protection.

- **Construct Bridge Wingwalls:** Construction of bridge wingwalls as extensions of abutments can significantly decrease or eliminate erosion. This involves angling wingwalls in the direction of the current. In comparison to the installation of ripraps, wingwall construction may require more resources and design considerations given the need for potentially deeper bridge foundations and permanent sheet pile installation to avoid scour.
- **Construct Spur Dikes:** Spur dikes are embankments designed to direct flood flows into a bridge opening and are constructed angularly in a straight or elliptical shape. Spur dikes can be considered larger versions of wingwalls, achieving the goal of decreasing or eliminating erosion at a larger scale.
- **Realign Piers & Abutments:** Piers and abutments can be realigned to become parallel or angled to water flow, which can reduce flow. This strategy may require significant design considerations, indicating that replacement may be more effective.
- **Increase Footing Depth:** Footings are the foundations of abutments and are constructed into the ground. As a result of storm surge, scour can damage footings and weaken the structural integrity of the bridge. Increasing the footing depth can help mitigate this damage depending on flood flow velocities and flow depth.
- **Install Flow Detectors:** Flow detectors are v-shaped structures placed at the foot of piers and abutments to reduce flow velocity to protect from scour. Constructed out of concrete, these structures are an effective method of reducing impacts from storm surge.

Reduce Debris Damage

The following strategies focus specifically on reducing debris damage and accumulation from storm surge, which can lead to increased flooding and weakened structural integrity.

- **Install Debris Deflectors:** Concrete debris deflectors, or fins, can be installed in the upstream direction piers and abutments. The deflectors have an angled shape specifically to deflect debris away from the piers and abutments. The bridge decks must be tall enough to allow for potential increases in floating debris.
- **Install Semicircular or Triangular Endnoses:** Endnoses are another adaptation strategy for reducing debris damage. Unlike debris deflectors, endnoses are made of metal sheets and tend to consist of a slightly more compact design. Similarly, the bridge decks must be tall enough to allow for potential increases in floating debris.
- **Install Batters:** Batters are steel plate barriers placed on the upstream end of concrete piers and abutments and consist of a “V” or semicircular design to deflect floating debris. These barriers are



welded or bolted to the structure and have the most compact design compared to deflectors and endnoses.

- **Minimize Below-Deck Framing:** For truss bridges and other structures relying on below-deck framing, redesign and removal can be an effective strategy to reduce debris runoff. On the other hand, this adaptation strategy may require significant design considerations, and it may be more cost-effective to undertake a complete rebuild.
- **Construct Debris Catchments:** This adaptation strategy involves the construction of catchments, such as barriers or low-height dams on smaller branch streams upstream from the bridge or culvert. This is particularly useful for scenarios in which debris impacting one or more bridges or culverts stems from defined origins, especially smaller streams and tributaries. As with each debris damage reduction adaptation strategy, it will be important to remove debris from these structures as it builds up to ensure safety and effectiveness.
- **Install Debris Sweepers:** Debris sweepers are cylinders with vertical vanes that rotate and float up and down with the water surface to sweep debris away from bridge piers and through bridge openings. These sweepers can be attached to the structure or driven into the streambed, the latter option being more most effective, and are most practical in locations where flow speeds are higher.

Relocation

Bridge relocation to a site with less exposure to damaging flows can be cost effective in areas with repeated and increased exposure to hazards of flooding and storm surge. This measure is used most often where more frequent floods regularly overtop the bridge deck, where the flood level is above the roadway surface for a short distance, and when the necessary right of way is available

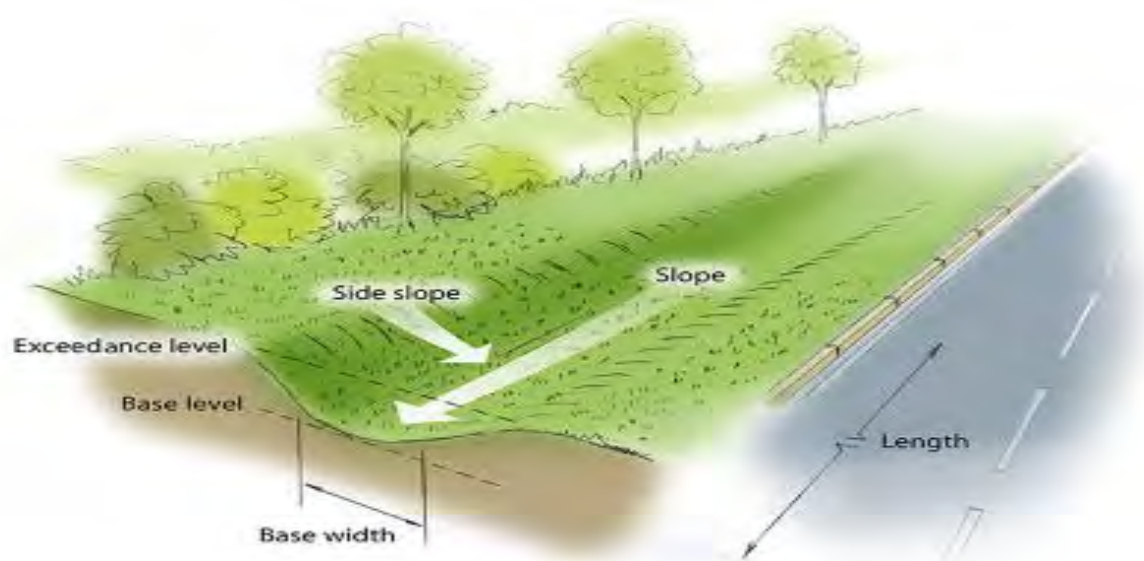
I.3 Drainage Pipes / Stormwater Treatment Unit Adaptation Strategies

Increased Swales or Ditch Capacity

Increasing the size of drainage swales or, in specific instances, drainage pipes, will allow the system to drain a greater capacity of water away from the road. In this option, the existing drainage structures, including both ditches or piping, will need to be resized to handle the increased volume of water that is projected from the inundation or surge events.



Figure G.5 Swale Design



Source: InfoDrainage

Drainage swales are a traditional method for moving water away from a road base, holding water before it enters a storm sewer system, and reducing the flow of water due to a flood event. The expansion of swales provides additional capacity in the system and thus increases the protection against flood events. Beyond initial costs, planning, and land attainment, there are few downsides to this solution, especially in areas where water enters the system on a regular basis to reduce the opportunity for standing water to serve as insect breeding areas. In areas where there is appropriate width next to a road, swales are a preferred solution to controlling flood events. Economic benefits of this type of solution result from a combination of the reduced damage caused by inundation and the increased control of the water flow entering the stormwater system. These benefits can be substantial in areas where regular flooding occurs, and inundation of roads is a regular threat. See FHWA's *Urban Drainage Design Manual*³⁰ for additional information on swale and ditch structural design.

Increased Retention or Detention Pond Capacity

For instances where greater holding capacity is required above roadside swales/piping, retention or detention ponds should be considered if the area is available to construct or expand such structures. The structures will provide a level of protection against inundation causing both surface and base damage including both erosion and surface damage. Where there is area to install such a system, ponds have proven over time to significantly reduce flooding due to overwhelmed systems. Ponds can also serve to enhance the natural environment by

³⁰ <https://www.fhwa.dot.gov/engineering/hydraulics/pubs/10009/10009.pdf>



providing homes to wildlife and providing resting areas for birds such as ducks and cranes as they traverse longer areas.

Figure G.6 Detention Pond Design



Source: Manuel Builders

The cost-benefit considerations for retention and detention ponds focus primarily on initial construction costs. These structures can be a significant investment in terms of both the cost of construction as well as the land required to support the structure. Maintenance is also required for the structures to ensure proper drainage out of the pond as well as drainage structures leading to the pond. This balancing of considerations should be offset by the significant benefit these ponds provide in terms of holding water that would otherwise inundate adjacent roads and property.

Depressed or Raised Medians

Medians can be depressed and used as an equivalent to a swale on the side of the road for drainage, or raised for use as an additional barrier to slowing the movement of the water across the roadway. The depression of the median will provide an intermediate barrier between the two sets of traffic lanes to decrease the potential impact of flooding. This level of depression will depend on a combination of drainage requirements and safety standards. The raising of the median would require enhancing the depth of the base and then placing vegetation on top to



provide a natural barrier to the flow of water across the roadway. This enhancement will allow the median to act as a separator between the lanes and reduce the amount of flow or depth of the water inundating the roads and entering the drainage swales. The use of the median as a flood control barrier or drainage component has a long-term benefit of reducing damage to road surfaces as well as to stormwater systems. However, this approach does require some annual maintenance considerations to ensure that proper growing conditions remain present.



[Greendale Grange Ave Bioswale](#), 2010; Aaron Volkening, CC BY 2.0.

Green Stormwater Infrastructure

An additional approach to addressing drainage threats is to focus on green infrastructure. Green infrastructure can take multiple forms, and is appealing from both an engineering and design perspective, given increased drainage capabilities, reduced impervious surface coverage, as well as aesthetic benefits. The strategy, centered around increased incorporation of bioswales, includes the use of planter boxes, green streets (incorporation of green areas into street and sidewalk designs), and green parking infrastructure (incorporation of permeable pavers and plants into impervious surface parking).



Bioswale adjacent to Kent Hospital

Image Source: Rhode Island Department of Environmental Management

The cost-benefit of green infrastructure varies across case histories and locations. There are additional maintenance costs to green infrastructure, but early implementation studies have demonstrated that life-cycle payback in benefits can outweigh the additional maintenance costs, based on reductions in stormwater ponding by up to 70 percent^{31 32 33}. On the other hand, these infrastructure improvements may also require increased coordination at the local scale given the need for implementation along local streets and private right of ways.

Pipe & Treatment System Resilience Strategies

In addition to the above strategies to manage stormwater, there is a general need to ensure resilience across the physical infrastructure components, including the network of treatment units, bypass structures, junctions, and pipes. Resilience strategies directly applied to these components primary consist of using durable materials and maintaining these infrastructure systems. As discussed in Chapter 4, this also includes making sure that the most critical portions of the stormwater pipes are cleaned frequently. For stormwater treatment units, this includes making sure that the most critical portions of the system are maintained to top condition

³¹ Economides, Christopher (2014). "Green Infrastructure: Sustainable Solutions in 11 Cities Across the US," Columbia University Water Center.

³² US EPA (2013). "Case Studies Analyzing the Economic Benefits of Low Impact Development and Green Infrastructure Programs," EPA 841-R-13-004.

³³ <https://www.epa.gov/green-infrastructure/performance-green-infrastructure>



APPENDIX J POTENTIAL RESILIENCE PROJECTS

Figure H.1 Potential Resilience Projects - 2035

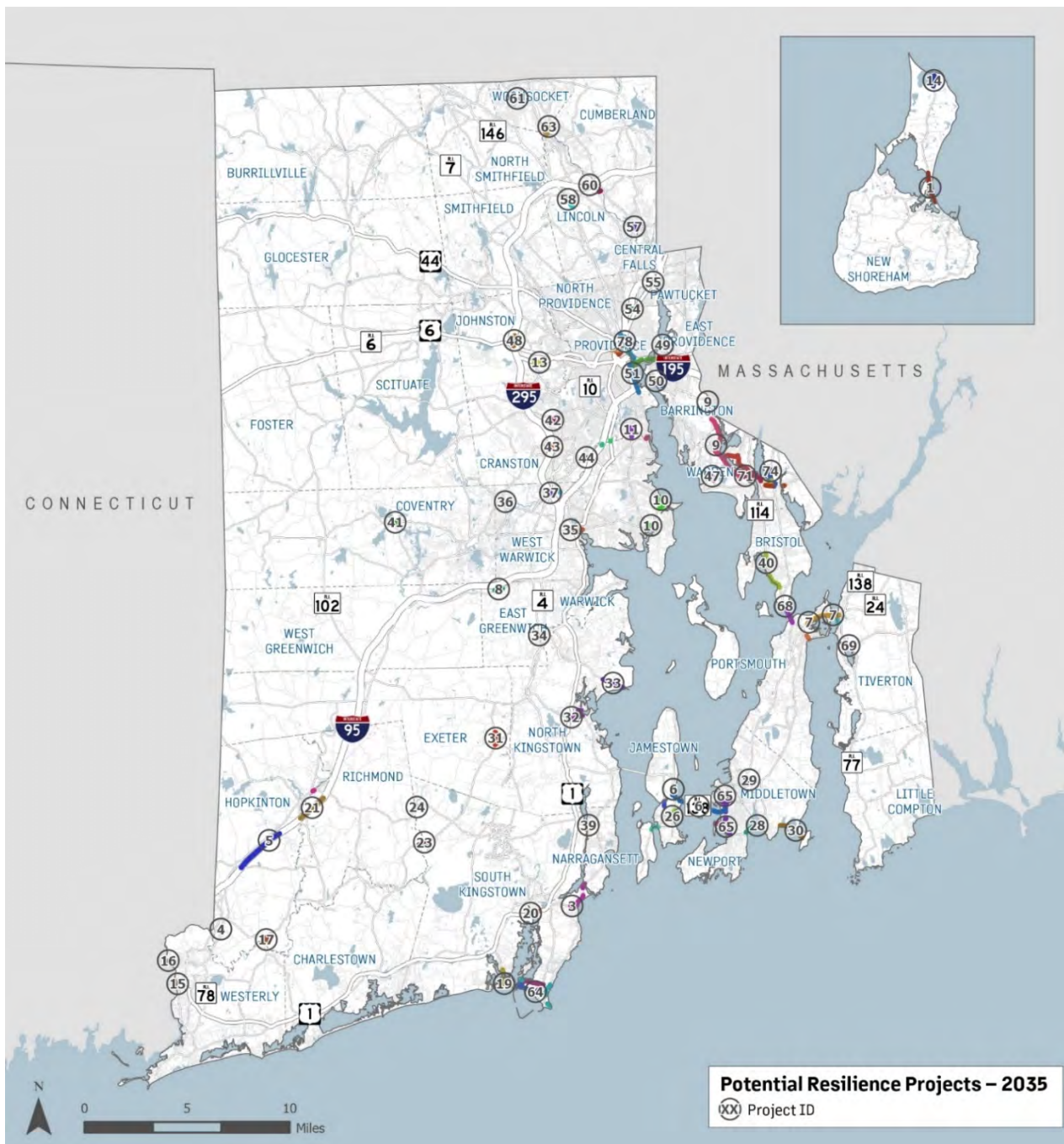




Figure H.2 Potential Resilience Projects - 2050

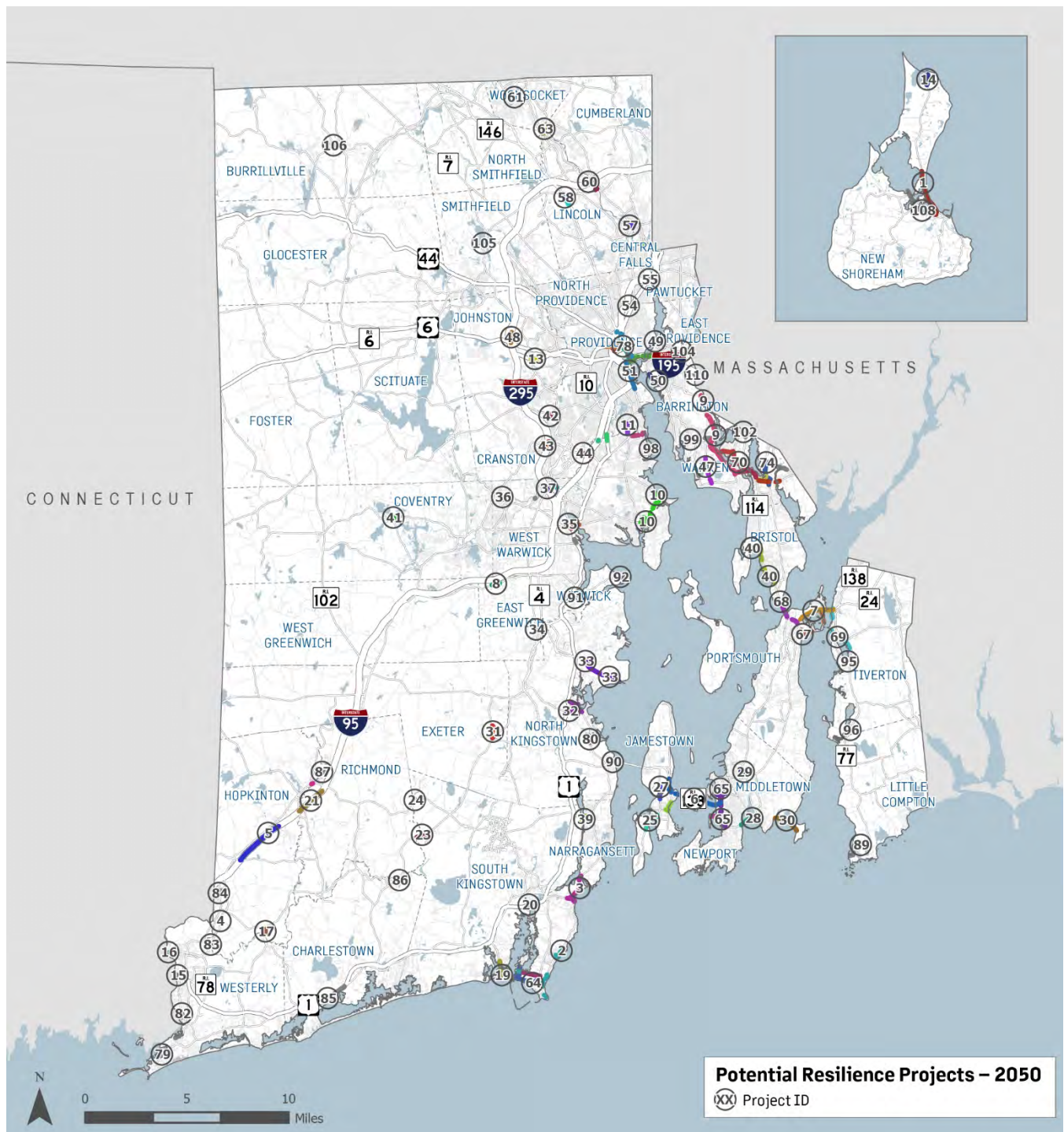




Figure H.3 Potential Resilience Projects - 2100

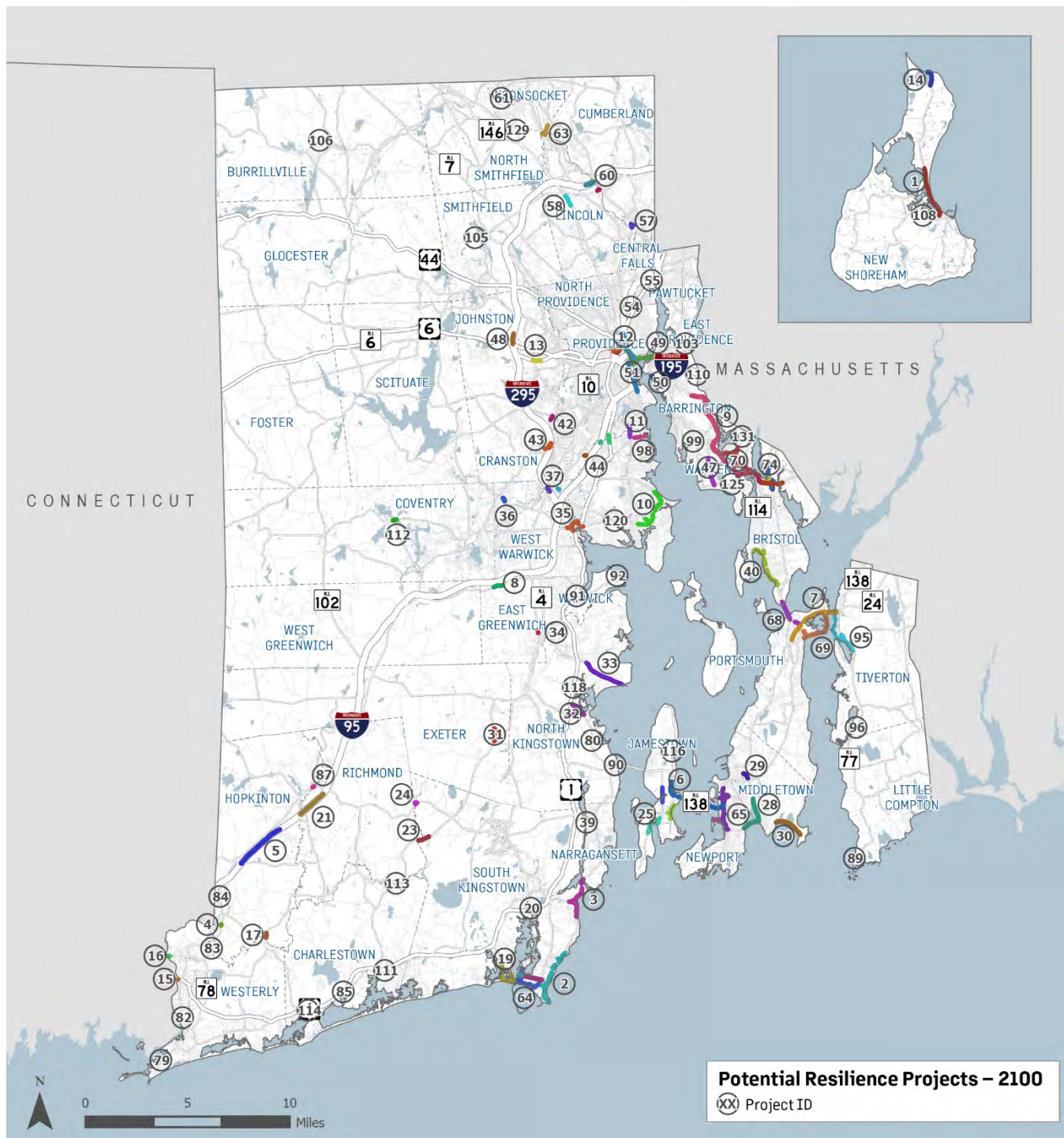




Table H.1 Potential Resilience Project List – Roads and Bridges

ID	Road Names	Criticality	Risk			Risk of No-Action (\$)				Project Length (Mile)			Rank			Hazards	Work type
			2035	2050	2100	By 2024	By 2035	By 2050	By 2100	2035	2050	2100	2035	2050	2100		
68	FERRY RD	High	High	High	High	\$1,076,282	\$12,915,385	\$326,299,584	\$2,244,357,399	1.23	1.23	1.67	1	1	1	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
52	I 195 W	High	Low	High	High	\$187,947	\$2,255,368	\$147,694,047	\$1,185,650,161	1.82	1.82	2.26	57	2	2	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
71	MAIN ST	High	Medium	High	High	\$229,436	\$2,753,226	\$58,523,934	\$547,715,685	1.61	1.61	1.87	9	3	3	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
66	GOAT ISLAND CONN	Medium	Medium	Medium	High	\$149,679	\$1,796,151	\$65,308,579	\$456,409,158	0.69	0.69	0.69	25	51	4	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
40	HOPE ST	High	Low	Medium	High	\$18,461	\$221,528	\$3,815,767	\$408,518,322	1.49	1.49	2.7	61	30	5	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
6	STATE HWY 138 W	High	Low	Medium	High	\$56,009	\$672,103	\$47,891,318	\$342,543,086	4.33	4.33	5.08	59	27	6	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
49	HENDERSON BRIDGE	Medium	Medium	High	High	\$26,569	\$318,822	\$3,348,743	\$302,576,135	0.67	0.67	1.13	28	18	7	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation



ID	Road Names	Criticality	Risk			Risk of No-Action (\$)				Project Length (Mile)			Rank			Hazards	Work type
			2035	2050	2100	By 2024	By 2035	By 2050	By 2100	2035	2050	2100	2035	2050	2100		
50	VETERANS MEMORIAL PKWY	Medium	High	Medium	High	\$12,646	\$151,756	\$14,645,904	\$105,465,301	0.82	0.82	0.99	52	52	8	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
3	BEACH ST	Medium	High	High	High	\$87,479	\$1,049,745	\$13,451,419	\$90,401,315	2.46	2.46	3.96	6	16	9	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
32	PHILLIPS ST	High	Medium	High	High	\$37,463	\$449,553	\$2,121,564	\$72,897,891	1.21	1.21	2	11	5	10	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
9	WAMPANOAG TRL	High	Low	Medium	High	\$141,876	\$1,702,509	\$7,221,659	\$40,571,619	8.05	8.05	12.23	58	28	11	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
65	AMERICAS CUP AVE	High	Medium	Medium	High	\$115,990	\$1,391,881	\$5,337,231	\$28,339,668	4.83	4.83	5.93	10	29	12	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
11	WARWICK AVE	High	Low	High	High	\$4,212	\$50,547	\$639,542	\$25,434,227	0.58	0.58	0.74	63	9	13	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
72	CHILD ST	Medium	Medium	Medium	High	\$21,110	\$253,319	\$1,203,185	\$23,498,866	0.53	0.53	1.12	29	53	14	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
64	SAND HILL COVE RD	Medium	Medium	High	High	\$92,818	\$1,113,814	\$4,550,388	\$22,762,611	2.49	2.49	3.22	26	17	15	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation



ID	Road Names	Criticality	Risk			Risk of No-Action (\$)				Project Length (Mile)			Rank			Hazards	Work type
			2035	2050	2100	By 2024	By 2035	By 2050	By 2100	2035	2050	2100	2035	2050	2100		
19	SUCCOTASH RD	Low	High	High	High	\$45,726	\$548,717	\$3,837,030	\$19,893,932	1.01	1.01	2.34	48	73	16	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
5	I 95 N	High	High	High	High	\$265,304	\$3,183,648	\$6,857,088	\$19,101,887	20.16	20.16	20.16	2	4	17	FLOODING	Inland Flood Risk Mitigation
67	BOYDS LN	Low	High	High	High	\$10,802	\$129,620	\$1,934,914	\$17,316,831	0.27	0.27	3	54	78	18	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
78	SOUTH WATER ST	High	Low	Medium	High	\$1,200	\$14,401	\$2,163,774	\$16,521,555	1.31	1.31	1.9	69	31	19	SLR, SS	Coastal Flood Risk Mitigation
28	MEMORIAL BLVD	Low	Medium	High	High	\$70,012	\$840,148	\$2,870,765	\$14,866,289	1.99	1.99	3.39	27	74	20	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
69	RIVERSIDE DR	Low	High	High	High	\$29,859	\$358,302	\$1,960,968	\$13,829,821	0.67	0.67	3.44	49	77	21	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
30	SACHUEST POINT RD	Low	High	High	High	\$62,669	\$752,028	\$2,659,044	\$13,308,017	1.53	1.53	1.97	47	75	22	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
81	WATCH HILL RD	Low		High	High			\$470,098.46	\$12,822,014.20	0	0	1.08		82	23	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
70	NEW MEADOW RD	Medium	High	High	High	\$55,310	\$663,725	\$2,346,157	\$12,760,447	2.43	2.43	3.11	7	19	24	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation



ID	Road Names	Criticality	Risk			Risk of No-Action (\$)				Project Length (Mile)			Rank			Hazards	Work type
			2035	2050	2100	By 2024	By 2035	By 2050	By 2100	2035	2050	2100	2035	2050	2100		
7	STATE HWY 24 N	High	Low	High	High	\$40,791	\$489,494	\$1,842,934	\$10,880,331	3.15	3.15	5.82	60	6	25	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
2	OCEAN RD	High	Low	Medium	Medium	\$482	\$5,784	\$1,160,800	\$10,599,111	2.04	2.04	5.1	71	32	26	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
1	CORN NECK RD	Low	High	High	High	\$25,897	\$310,765	\$1,983,454	\$10,466,990	0.98	0.98	1.71	50	76	27	SLR, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
10	WEST SHORE RD	Medium	Low	Medium	High	\$1,874	\$22,493	\$995,582	\$8,878,164	0.77	0.77	3.02	65	54	28	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
18	GALILEE ESCAPE RD	High	Low	High	High	\$608	\$7,293	\$908,197	\$7,640,797	2.78	2.78	2.78	70	8	29	SLR, SS	Coastal Flood Risk Mitigation
25	BEAVERTAIL RD	Low	High	High	High	\$20,595	\$247,142	\$1,137,726	\$6,980,973	0.49	0.49	1.5	51	79	30	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
74	KICKEMUIT RD	High	Low	Medium	High	\$15,491	\$185,896	\$1,025,542	\$6,716,948	1.19	1.19	2.43	62	33	31	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
89	SAKONNET POINT RD	Low		High	High			\$811,603.71	\$6,257,221.76	0	0	1.16		80	32	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
47	MIDDLE HWY	Medium	Medium	Medium	High	\$6,907	\$82,887	\$737,882	\$5,963,827	0.51	0.51	1.66	37	57	33	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation



ID	Road Names	Criticality	Risk			Risk of No-Action (\$)				Project Length (Mile)			Rank			Hazards	Work type
			2035	2050	2100	By 2024	By 2035	By 2050	By 2100	2035	2050	2100	2035	2050	2100		
51	ALLENS AVE	High	Low	Medium	High	\$433	\$5,195	\$646,908	\$5,444,012	1.88	1.88	1.96	72	34	34	SLR, SS	Coastal Flood Risk Mitigation
102	SOWAMS RD	Low		High	High			\$142,048.35	\$5,126,835.25	0	0	1.75		94	35	SS	Coastal Flood Risk Mitigation
26	CONANICUS AVE	Medium	Medium	Medium	High	\$20,660	\$247,917	\$845,807	\$5,079,878	0.54	0.54	1	31	55	36	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
27	NORTH RD	Medium	Medium	High	High	\$8,009	\$96,105	\$677,464	\$4,549,404	0.27	0.27	0.92	33	20	37	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
12	NORTH MAIN ST	High	Low	Medium	Medium	\$1,555	\$18,660	\$521,935	\$4,290,570	1.05	1.05	2.37	67	36	38	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
73	MARKET ST	Medium	Medium	Medium	High	\$19,696	\$236,353	\$823,213	\$4,122,386	0.48	0.48	0.64	32	56	39	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
55	I 95 N	High	Medium	Medium	High	\$7,291	\$87,488	\$646,050	\$3,842,360	0.08	0.08	0.11	16	35	40	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
79	WATCH HILL RD	Low		High	High			\$590,557.76	\$3,571,721.23	0	0	0.94		81	41	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
39	BRIDGETOWN RD	Medium	Medium	Medium	High	\$20,836	\$250,037	\$730,787	\$3,552,968	0.34	0.34	0.53	30	58	42	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation



ID	Road Names	Criticality	Risk			Risk of No-Action (\$)				Project Length (Mile)			Rank			Hazards	Work type
			2035	2050	2100	By 2024	By 2035	By 2050	By 2100	2035	2050	2100	2035	2050	2100		
92	IVES RD	Low		High	High			\$243,680.82	\$3,536,760.90	0	0	0.57		88	43	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
20	COMMODORE PERRY HWY	High	Low	Medium	High	\$276	\$3,312	\$411,458	\$3,210,044	0.59	0.59	0.59	73	37	44	SLR, SS	Coastal Flood Risk Mitigation
77	FOUNTAIN ST	High	Low	Medium	High	\$253	\$3,037	\$377,636	\$3,048,484	0.76	0.76	0.89	74	38	45	SLR, SS	Coastal Flood Risk Mitigation
21	I 95 S	High	High	High	High	\$40,818	\$489,812	\$1,054,979	\$2,938,870	2.89	2.89	2.89	3	7	46	FLOODING	Inland Flood Risk Mitigation
33	0	High	Low	Low	High	\$181	\$2,169	\$271,973	\$2,780,988	1.42	1.42	2.76	76	99	47	SLR, SS	Coastal Flood Risk Mitigation
131	NEW MEADOW RD	Low			High				\$2,655,053.09	0	0	0.77			48	SS	Coastal Flood Risk Mitigation
101	SCHOOLHOUSE RD	Low		High	High			\$403,120.95	\$2,616,820.81	0	0	0.81		84	49	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
35	CENTERVILLE RD	High	Low	Medium	Medium	\$1,282	\$15,388	\$312,906	\$2,508,896	0.49	0.49	1.49	68	39	51	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
82	BEACH ST	Low		High	High			\$282,280.09	\$2,377,092.54	0	0	0.58		87	51	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
125	NAYATT RD	Low			High				\$2,315,906.44	0	0	0.83			52	SLR, SS	Coastal Flood Risk Mitigation
100	WASHINGTON RD	Low		High	High			\$107,776.85	\$2,296,535.71	0	0	0.64		98	53	SS	Coastal Flood Risk Mitigation
110	HIGHLAND AVE	Medium		High	High			\$199,820.17	\$2,283,219.89	0	0	0.24		22	54	SLR, SS	Coastal Flood Risk Mitigation
75	POST RD	Medium	Medium	Medium	High	\$1,616	\$19,398	\$315,263	\$2,234,400	0.12	0.12	0.57	43	59	55	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation



ID	Road Names	Criticality	Risk			Risk of No-Action (\$)				Project Length (Mile)			Rank			Hazards	Work type
			2035	2050	2100	By 2024	By 2035	By 2050	By 2100	2035	2050	2100	2035	2050	2100		
76	I 95 S	High	Low	Medium	Medium	\$2,261	\$27,126	\$287,069	\$2,136,269	0.73	0.73	0.94	64	40	56	SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
53	ON RAMP TO US HWY 6 W	High	Low	Low	Medium	\$168	\$2,021	\$251,658	\$2,118,128	0.84	0.84	0.84	77	100	57	SLR, SS	Coastal Flood Risk Mitigation
85	POST RD	High		Low	High			\$5,731.07	\$1,544,611.71	0	0	3.89		103	58	SS	Coastal Flood Risk Mitigation
91	MAIN ST	High		Low	Low			\$159,681.86	\$1,387,521.59	0	0	0.43		101	59	SS	Coastal Flood Risk Mitigation
60	I 295 N	High	Medium	High	High	\$19,105	\$229,256	\$493,782	\$1,375,534	0.86	0.86	0.86	12	10	60	FLOODING	Inland Flood Risk Mitigation
107	OCEAN AVE	Low		High	High			\$403,235.96	\$1,371,685.25	0	0	0.39		83	61	SLR, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
13	US HWY 6 E	High	High	High	High	\$18,888	\$226,653	\$488,175	\$1,359,916	0.93	0.93	0.93	4	11	62	FLOODING	Inland Flood Risk Mitigation
96	MAIN RD	Medium		Medium	High			\$83,377.52	\$1,208,328.75	0	0	1.29		63	63	SS	Coastal Flood Risk Mitigation
63	STATE HWY 99 S	High	Medium	Medium	Medium	\$9,718	\$116,616	\$251,172	\$1,014,438	0.67	0.67	1	13	41	64	FLOODING	Inland Flood Risk Mitigation
95	NANAQUAKET RD	Low		High	High			\$120,361.41	\$1,012,673.67	0	0	0.24		96	65	SLR, SS	Coastal Flood Risk Mitigation
48	I 295 N	High	High	High	High	\$13,916	\$166,991	\$359,673	\$1,001,946	0.68	0.68	0.68	5	12	66	FLOODING	Inland Flood Risk Mitigation
14	CORN NECK RD	Low	High	High	High	\$9,247	\$110,963	\$238,997	\$942,030	0.34	0.34	0.48	55	89	67	SLR, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
109	0	Low		High	High			\$111,151.11	\$936,143.12	0	0	0.53		97	68	SLR, SS	Coastal Flood Risk Mitigation
88	NARRAGANSETT AVE	Low		High	High			\$201,699.91	\$931,442.31	0	0	0.11		91	69	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
98	NARRAGANSETT PKWY	Medium		Medium	High			\$110,021.34	\$925,848.42	0	0	0.27		61	70	SS	Coastal Flood Risk Mitigation



ID	Road Names	Criticality	Risk			Risk of No-Action (\$)				Project Length (Mile)			Rank			Hazards	Work type
			2035	2050	2100	By 2024	By 2035	By 2050	By 2100	2035	2050	2100	2035	2050	2100		
108	OCEAN AVE	Low		High	High			\$330,918.66	\$922,297.65	0	0	0.48		85	71	SLR, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
31	SOUTH COUNTY TRL	Medium	High	High	High	\$11,924	\$143,085	\$308,183	\$858,510	0.86	0.86	0.86	53	86	72	FLOODING	Inland Flood Risk Mitigation
23	SOUTH COUNTY TRL	Medium	High	High	High	\$9,731	\$116,767	\$251,498	\$700,601	0.72	0.72	0.72	8	21	73	FLOODING	Inland Flood Risk Mitigation
43	I 295 S	High	Medium	Medium	High	\$5,086	\$61,035	\$242,940	\$676,762	0.33	0.33	0.65	17	42	74	FLOODING	Inland Flood Risk Mitigation
116	EAST SHORE RD	Low			High				\$675,653.52	0	0	0.24			75	SS	Coastal Flood Risk Mitigation
58	OLD LOUISUISSE T PIKE	Low	High	High	High	\$9,237	\$110,843	\$238,738	\$665,057	0.7	0.7	0.7	56	90	76	FLOODING	Inland Flood Risk Mitigation
57	ANN AND HOPE WAY	Medium	Medium	Medium	High	\$7,598	\$91,181	\$208,439	\$580,651	0.34	0.34	0.35	34	60	77	FLOODING	Inland Flood Risk Mitigation
16	STATE HWY 78 S	High	Medium	High	High	\$8,039	\$96,465	\$207,770	\$579,124	0.13	0.13	0.13	14	13	78	SLR, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
45	I 95 N	High	Medium	High	High	\$7,782	\$93,390	\$201,148	\$560,340	0.07	0.07	0.07	15	14	79	FLOODING	Inland Flood Risk Mitigation
41	FLAT RIVER RD	Medium	Medium	High	High	\$7,227	\$86,719	\$186,780	\$520,316	0.23	0.23	0.23	35	23	80	FLOODING	Inland Flood Risk Mitigation
8	DIVISION RD	Medium	Medium	High	High	\$7,083	\$85,000	\$183,076	\$509,997	0.53	0.53	0.53	36	24	81	FLOODING	Inland Flood Risk Mitigation
84	WELLSTOWN RD	Low		High	High			\$177,417.21	\$494,233.68	0	0	0.51		92	82	FLOODING	Inland Flood Risk Mitigation
44	STATE HWY 37 W	High	Low	Medium	High	\$36	\$427	\$53,205	\$447,543	0.08	0.08	0.08	78	45	83	SLR, SS	Coastal Flood Risk Mitigation
126	WATERMAN AVE	Medium			Medium				\$434,563.05	0	0	0.4			84	SS	Coastal Flood Risk Mitigation
87	SKUNK HILL RD	Low		High	High			\$155,153.57	\$432,213.54	0	0	0.44		93	85	FLOODING	Inland Flood Risk Mitigation
29	VALLEY RD	Medium	Medium	High	High	\$5,881	\$70,570	\$151,998	\$423,423	0.44	0.44	0.44	38	25	86	FLOODING	Inland Flood Risk Mitigation



ID	Road Names	Criticality	Risk			Risk of No-Action (\$)				Project Length (Mile)			Rank			Hazards	Work type
			2035	2050	2100	By 2024	By 2035	By 2050	By 2100	2035	2050	2100	2035	2050	2100		
46	ELMWOOD AVE	High	Medium	Medium	High	\$1,553	\$18,637	\$41,001	\$401,564	0.02	0.02	0.47	22	48	87	SLR, SS, FLOODING	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
105	PLEASANT VIEW AVE	Low		High	High			\$136,124.96	\$379,205.25	0	0	0.24		95	88	FLOODING	Inland Flood Risk Mitigation
17	ALTON BRADFORD RD	Medium	Medium	High	High	\$4,862	\$58,345	\$125,667	\$350,071	0.31	0.31	0.31	39	26	89	FLOODING	Inland Flood Risk Mitigation
128	JOHN ST	Low			High				\$300,480.78	0	0	0.3			90	FLOODING	Inland Flood Risk Mitigation
37	I 295 S	High	Medium	High	High	\$4,069	\$48,831	\$105,173	\$292,983	0.29	0.29	0.29	18	15	91	FLOODING	Inland Flood Risk Mitigation
42	PHENIX AVE	Medium	Medium	Medium	Medium	\$3,339	\$40,073	\$100,064	\$278,750	0.22	0.22	0.26	40	62	92	FLOODING	Inland Flood Risk Mitigation
112	OLD FLAT RIVER RD	Low			High				\$246,696.63	0	0	0.37			93	FLOODING	Inland Flood Risk Mitigation
24	USQUEPAUG H RD	Medium	Medium	Medium	High	\$1,551	\$18,612	\$40,088	\$242,457	0.11	0.11	0.25	44	66	94	FLOODING	Inland Flood Risk Mitigation
129	GREENVILLE RD	Low			High				\$242,249.76	0	0	0.24			95	FLOODING	Inland Flood Risk Mitigation
104	TAUNTON AVE	High		Low	High			\$853.26	\$232,636.05	0	0	0.45		105	96	SS	Coastal Flood Risk Mitigation
36	MAIN ST	High	Medium	Medium	High	\$3,192	\$38,308	\$82,509	\$229,846	0.23	0.23	0.23	19	43	97	FLOODING	Inland Flood Risk Mitigation
80	BOSTON NECK RD	High		Low	Low			\$689.19	\$190,261.43	0	0	1.24		106	98	SS	Coastal Flood Risk Mitigation
99	WILLETT AVE	High		Low	Low			\$108.16	\$189,893.37	0	0	0.85		110	99	SS	Coastal Flood Risk Mitigation
113	SHERMAN AVE	Low			High				\$186,125.99	0	0	0.19			100	FLOODING	Inland Flood Risk Mitigation
34	SOUTH COUNTY TRL	High	Medium	Medium	High	\$2,540	\$30,479	\$65,646	\$182,872	0.08	0.08	0.08	20	44	101	FLOODING	Inland Flood Risk Mitigation
59	GEORGE WASHINGTON HWY	Medium	Medium	Medium	High	\$2,047	\$24,561	\$52,901	\$147,367	0.12	0.12	0.12	41	64	102	FLOODING	Inland Flood Risk Mitigation
93	POST RD	High		Low	Medium			\$516.57	\$139,396.90	0	0	0.34		107	103	SS	Coastal Flood Risk Mitigation
61	COURT ST	High	Low	Medium	Medium	\$1,751	\$21,010	\$45,251	\$126,057	0.08	0.08	0.08	66	46	104	FLOODING	Inland Flood Risk Mitigation



ID	Road Names	Criticality	Risk			Risk of No-Action (\$)				Project Length (Mile)			Rank			Hazards	Work type
			2035	2050	2100	By 2024	By 2035	By 2050	By 2100	2035	2050	2100	2035	2050	2100		
22	MAIN ST	Medium	Medium	Medium	High	\$1,651	\$19,817	\$42,682	\$118,901	0.11	0.11	0.11	42	65	105	FLOODING	Inland Flood Risk Mitigation
56	BROAD ST	High	Medium	Medium	High	\$1,647	\$19,770	\$42,581	\$118,619	0.07	0.07	0.07	21	47	106	FLOODING	Inland Flood Risk Mitigation
115	SWITCH RD	Low			High				\$104,691.53	0	0	0.09			107	FLOODING	Inland Flood Risk Mitigation
38	LAMBERT LIND HWY	Medium	Medium	Medium	High	\$1,409	\$16,910	\$36,422	\$101,462	0.1	0.1	0.1	45	67	108	FLOODING	Inland Flood Risk Mitigation
90	STATE HWY 138 E	High		Low	Medium			\$317.65	\$86,318.17	0	0	0.26		108	109	SS	Coastal Flood Risk Mitigation
62	HAMLET AVE	High	Medium	Medium	Medium	\$1,052	\$12,630	\$27,203	\$75,779	0.05	0.05	0.05	23	49	110	FLOODING	Inland Flood Risk Mitigation
54	SMITHFIELD AVE	High	Medium	Medium	Medium	\$996	\$11,948	\$25,734	\$71,688	0.07	0.07	0.07	24	50	111	FLOODING	Inland Flood Risk Mitigation
4	MAIN ST	Medium	Medium	Medium	Medium	\$941	\$11,295	\$24,329	\$67,773	0.06	0.06	0.06	46	68	112	FLOODING	Inland Flood Risk Mitigation
121	MAIN RD	Medium			Medium				\$58,593.65	0	0	0.12			113	SS	Coastal Flood Risk Mitigation
94	PROVIDENCE ST	Medium		Medium	Medium			\$20,943.71	\$58,343.20	0	0	0.03		69	114	FLOODING	Inland Flood Risk Mitigation
86	SOUTH COUNTY TRL	Medium		Medium	Medium			\$20,596.16	\$57,375.04	0	0	0.05		70	115	FLOODING	Inland Flood Risk Mitigation
83	MAIN ST	Medium		Medium	Medium			\$17,066.61	\$47,542.69	0	0	0.02		71	116	FLOODING	Inland Flood Risk Mitigation
103	PAWTUCKET AVE	High		Low	Low			\$147.92	\$42,156.06	0	0	0.78		109	117	SS	Coastal Flood Risk Mitigation
124	NARRAGANS ETT PKWY	Medium			Medium				\$37,008.65	0	0	0.08			118	SS	Coastal Flood Risk Mitigation
106	CHAPEL ST	Medium		Medium	Medium			\$12,725.40	\$35,449.34	0	0	0.03		72	119	FLOODING	Inland Flood Risk Mitigation
15	BROAD ST	High	Low	Low	Medium	\$196	\$2,353	\$5,068	\$14,119	0.01	0.01	0.01	75	104	120	FLOODING	Inland Flood Risk Mitigation
97	NARRAGANS ETT BLVD	High		Low	Low			\$40.39	\$11,605.15	0	0	0.19		111	121	SS	Coastal Flood Risk Mitigation
118	POST RD	High			Low				\$6,711.60	0	0	0.58			122	SS	Coastal Flood Risk Mitigation
111	POST RD	High			Low				\$5,708.47	0	0	1.36			123	SS	Coastal Flood Risk Mitigation
127	I 95 S	High			Low				\$3,258.12	0	0	0.57			124	SS	Coastal Flood Risk Mitigation
117	CODDINGTON HWY	High			Low				\$2,706.05	0	0	0.27			125	SS	Coastal Flood Risk Mitigation



ID	Road Names	Criticality	Risk			Risk of No-Action (\$)				Project Length (Mile)			Rank			Hazards	Work type
			2035	2050	2100	By 2024	By 2035	By 2050	By 2100	2035	2050	2100	2035	2050	2100		
114	POST RD	High			Low				\$2,042.59	0	0	0.4			126	SS	Coastal Flood Risk Mitigation
120	WEST SHORE RD	High			Low				\$1,577.73	0	0	0.1			127	SS	Coastal Flood Risk Mitigation
130	MAIN ST	High			Low				\$1,211.73	0	0	0.23			128	SS	Coastal Flood Risk Mitigation
119	WARWICK AVE	High			Low				\$705.53	0	0	0.06			129	SS	Coastal Flood Risk Mitigation
123	PONTIAC AVE	High			Low				\$559.69	0	0	0.07			130	SS	Coastal Flood Risk Mitigation
122	PARK AVE	High			Low				\$98.48	0	0	0.01			131	SS	Coastal Flood Risk Mitigation



Figure H.4 Potential Resilience Projects – Sidewalks

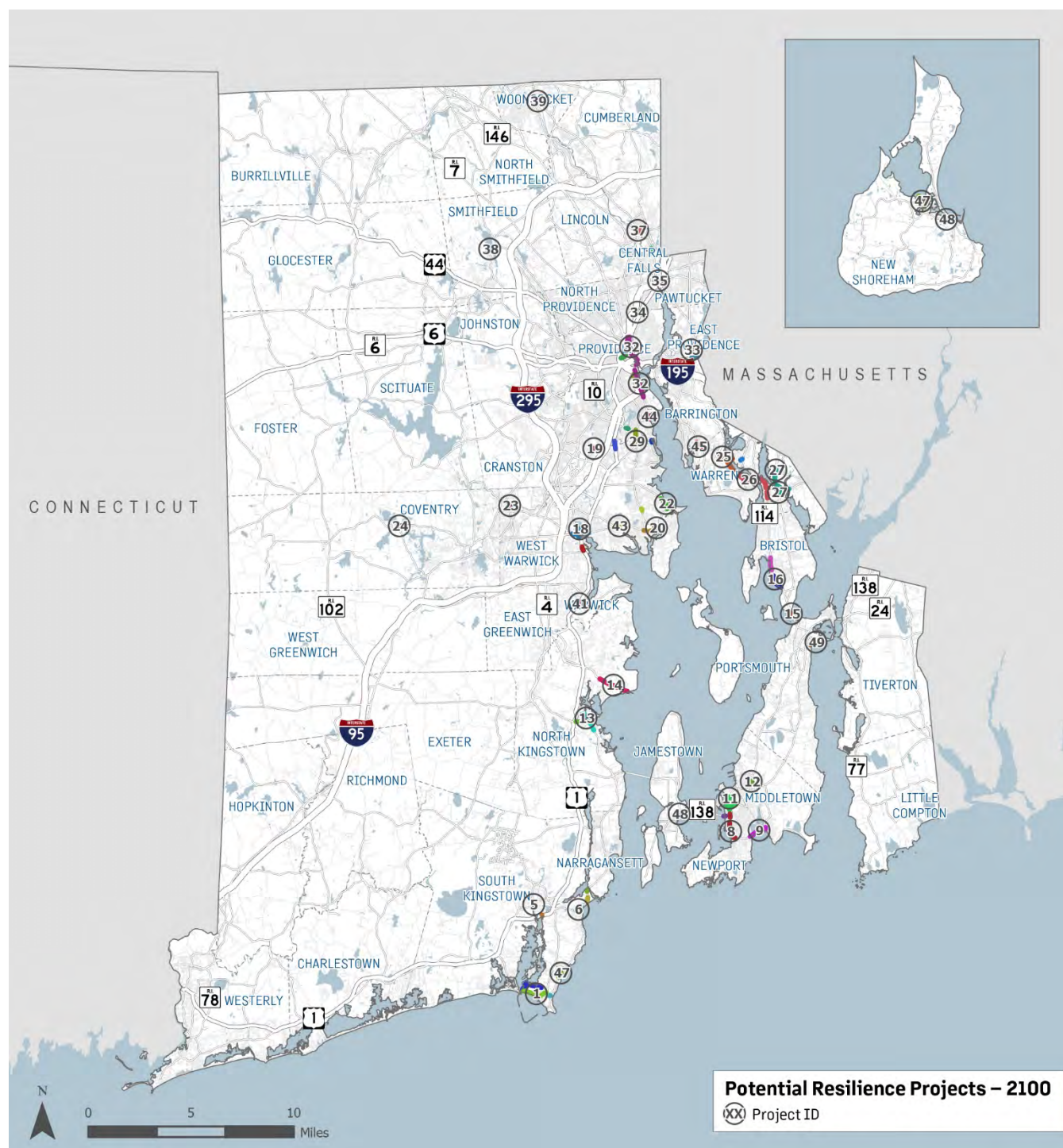




Table H.2 Potential Resilience Project List – Sidewalks

ID	Project Length (Mile)			Criticality and Risk			Cumulated Composite Risk Amount (\$)				Rank			Hazards	
	2035	2050	2100	2035	2050	2100	2024	2035	2050	2100	2035	2050	2100		Work Type
15	0.10	0.10	0.10	HighHigh	HighHigh	HighHigh	\$11,261	\$135,126.23	\$166,309.20	\$530,110.58	1	1	1	Flooding	Inland Flood Risk Mitigation
24	0.20	0.20	0.20	MediumHigh	MediumHigh	MediumHigh	\$13,803	\$165,636.02	\$203,859.71	\$649,802.84	2	2	2	Flooding	Inland Flood Risk Mitigation
28	0.10	0.10	0.10	MediumMedium	MediumMedium	MediumMedium	\$5,532	\$66,389.84	\$68,250.34	\$1,872.66	3	3	3	SS,SLR	Coastal Flood Risk Mitigation
38	0.30	0.30	0.30	LowHigh	LowHigh	LowHigh	\$30,897	\$370,760.25	\$456,320.31	\$1,454,520.99	4	4	4	Flooding	Inland Flood Risk Mitigation
26	1.90	1.90	1.90	HighLow	HighLow	HighLow	\$243,422	\$2,921,059.97	\$3,465,727.59	\$7,647,057.18	5	5	5	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
25	1.70	1.70	1.70	HighLow	HighLow	HighLow	\$27,405	\$328,863.30	\$420,414.78	\$1,312,240.16	6	6	6	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
11	1.31	1.31	1.31	HighLow	HighLow	HighLow	\$16,714	\$200,570.76	\$261,776.06	\$807,673.56	7	7	7	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
1	1.10	1.10	1.10	HighLow	HighLow	HighLow	\$16,354	\$196,247.14	\$252,738.89	\$787,293.84	8	8	8	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
8	1.43	1.43	1.43	HighLow	HighLow	HighLow	\$7,351	\$88,216.48	\$121,040.58	\$364,136.03	10	9	9	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
9	0.40	0.40	0.40	HighLow	HighLow	HighLow	\$6,602	\$79,224.23	\$104,849.14	\$322,091.43	11	10	10	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
3	0.79	0.79	0.79	HighLow	HighLow	HighLow	\$5,478	\$65,731.64	\$86,477.74	\$266,351.43	13	13	11	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation



ID	Project Length (Mile)			Criticality and Risk			Cumulated Composite Risk Amount (\$)				Rank			Hazards	
	2035	2050	2100	2035	2050	2100	2024	2035	2050	2100	2035	2050	2100		Work Type
6	0.43	0.43	0.43	HighLow	HighLow	HighLow	\$4,874	\$58,485.25	\$75,378.30	\$234,251.75	14	14	12	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
13	0.20	0.20	0.20	HighLow	HighLow	HighLow	\$3,641	\$43,694.14	\$56,819.80	\$176,028.66	15	15	13	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
27	0.90	0.90	0.90	HighLow	HighLow	HighLow	\$2,908	\$34,894.30	\$54,924.41	\$154,022.47	16	16	14	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
23	0.20	0.20	0.20	HighLow	HighLow	HighLow	\$2,352	\$28,227.24	\$34,741.21	\$110,737.61	17	17	15	Flooding	Inland Flood Risk Mitigation
17	0.41	0.41	0.41	HighLow	HighLow	HighLow	\$1,312	\$15,741.31	\$28,934.56	\$75,946.32	20	19	16	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
29	0.50	0.50	0.50	HighLow	HighLow	HighLow	\$1,484	\$17,808.81	\$25,450.30	\$75,458.08	19	21	17	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
7	0.10	0.10	0.10	HighLow	HighLow	HighLow	\$1,274	\$15,288.54	\$21,178.17	\$63,718.11	21	22	18	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
10	0.12	0.12	0.12	HighLow	HighLow	HighLow	\$26	\$306.47	\$30,490.50	\$42,805.66	28	18	19	SS, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
12	0.10	0.10	0.10	HighLow	HighLow	HighLow	\$883	\$10,597.71	\$13,043.34	\$41,575.65	22	24	20	Flooding	Inland Flood Risk Mitigation
32	3.01	3.01	3.01	HighLow	HighLow	HighLow	\$5,768	\$69,221.24	\$93,612.99	\$36,023.21	12	12	21	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
16	0.40	0.40	0.40	HighLow	HighLow	HighLow	\$551	\$6,609.90	\$10,864.76	\$29,701.62	23	25	22	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation



ID	Project Length (Mile)			Criticality and Risk			Cumulated Composite Risk Amount (\$)				Rank			Hazards	
	2035	2050	2100	2035	2050	2100	2024	2035	2050	2100	2035	2050	2100		Work Type
14	0.44	0.44	0.44	HighLow	HighLow	HighLow	\$14	\$163.34	\$16,283.65	\$24,884.75	30	23	23	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
39	0.06	0.06	0.06	HighLow	HighLow	HighLow	\$454	\$5,448.80	\$6,706.22	\$21,376.06	24	26	24	Flooding	Inland Flood Risk Mitigation
37	0.20	0.20	0.20	HighLow	HighLow	HighLow	\$367	\$4,405.17	\$5,421.75	\$17,281.82	25	28	25	Flooding	Inland Flood Risk Mitigation
18	0.42	0.42	0.42	HighLow	HighLow	HighLow	\$200	\$2,400.79	\$6,675.80	\$14,825.80	27	27	26	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
36	0.10	0.10	0.10	HighLow	HighLow	HighLow	\$271	\$3,253.64	\$4,004.48	\$12,764.29	26	29	27	Flooding	Inland Flood Risk Mitigation
31	0.58	0.58	0.58	HighLow	HighLow	HighLow	\$7,738	\$92,856.86	\$97,205.57	\$5,543.06	9	11	28	SS,SLR	Coastal Flood Risk Mitigation
22	0.30	0.30	0.30	HighLow	HighLow	HighLow	\$3	\$32.41	\$3,229.09	\$4,939.69	31	30	29	SS, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
20	0.35	0.35	0.35	HighLow	HighLow	HighLow	\$2	\$24.49	\$2,442.64	\$3,900.77	32	31	30	SS, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
30	0.10	0.10	0.10	HighLow	HighLow	HighLow	\$2	\$19.07	\$1,898.26	\$2,663.38	33	32	31	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
5	0.20	0.20	0.20	HighLow	HighLow	HighLow	\$2	\$18.67	\$1,848.81	\$1,848.81	34	33	32	Flooding	Inland Flood Risk Mitigation
34	0.10	0.10	0.10	HighLow	HighLow	HighLow	\$24	\$291.49	\$358.76	\$1,143.54	29	36	33	Flooding	Inland Flood Risk Mitigation
33	0.03	0.03	0.03	HighLow	HighLow	HighLow	\$1	\$7.26	\$722.50	\$1,016.08	35	34	34	SS, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
35	0.33	0.33	0.33	HighLow	HighLow	HighLow	\$2,068	\$24,815.77	\$25,738.86	\$795.06	18	20	35	SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation



ID	Project Length (Mile)			Criticality and Risk			Cumulated Composite Risk Amount (\$)				Rank			Hazards	
	2035	2050	2100	2035	2050	2100	2024	2035	2050	2100	2035	2050	2100		Work Type
21	0.10	0.10	0.10	HighLow	HighLow	HighLow	\$0	\$4.83	\$480.26	\$674.24	36	35	36	SS, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
4	0.01	0.01	0.01	HighLow	HighLow	HighLow	\$0	\$1.31	\$130.49	\$183.20	37	37	37	SS, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation



Figure H.5 Potential Resilience Projects – Shared-Use Paths

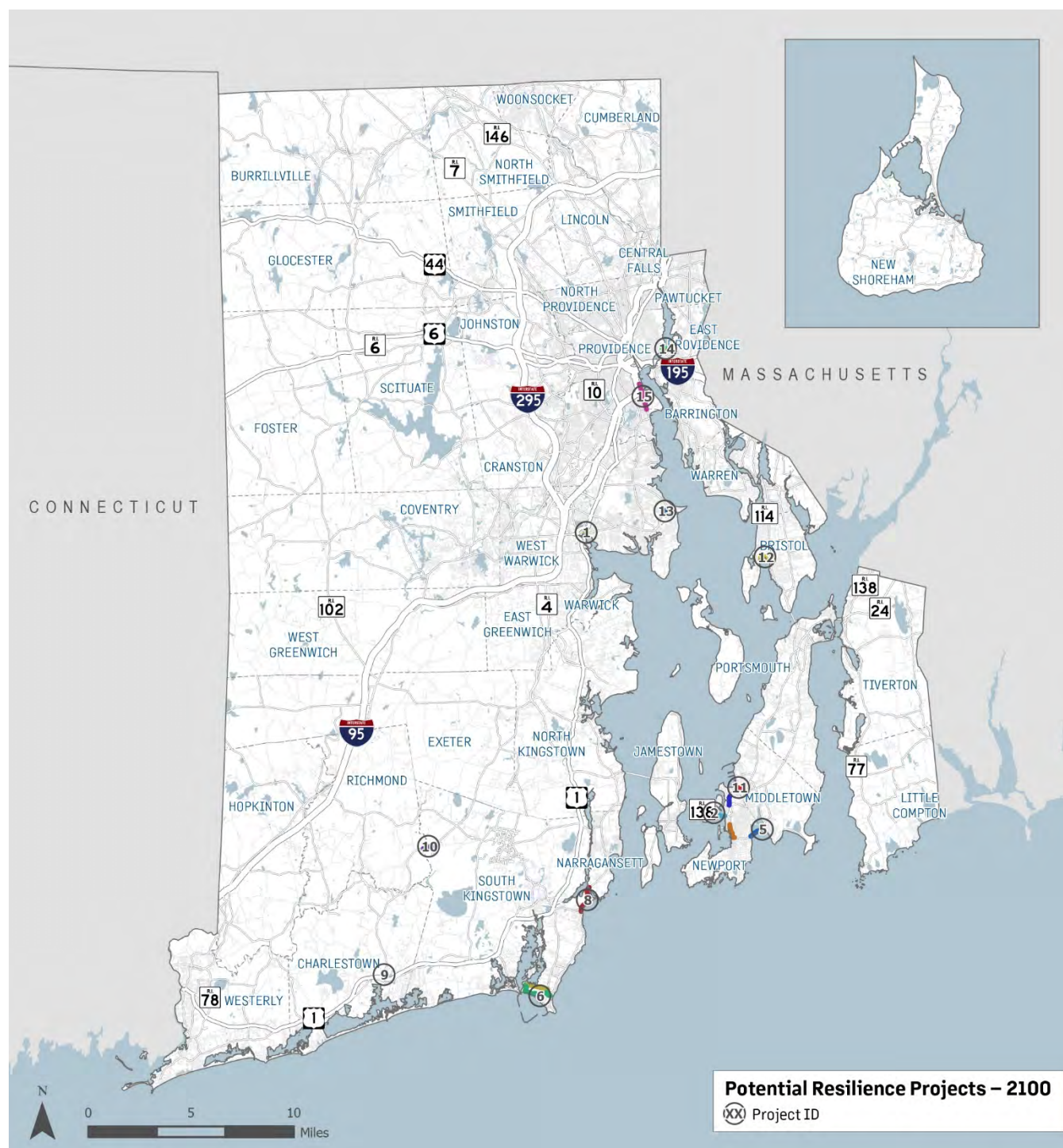




Table H.3 Potential Resilience Project List – Shared-Use Paths

ID	Project Length (Mile)			Criticality and Risk			Cumulated Composite Risk Amount (\$)				Rank			Hazards	
	2035	2050	2100	2035	2050	2100	2024	2035	2050	2100	2035	2050	2100		Work Type
6	0.08	1.54	1.80	High High	High High	High High	\$12,684.79	\$152,217.53	\$24,655.31	\$1.80	1	9	1	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
2	1.13	1.13	1.13	High High	High High	High High	\$10,781.80	\$129,381.59	\$9,952.43	\$1.13	2	9	2	Flooding	Inland Flood Risk Mitigation
13			0.29			Medium High				\$0.29			3	SS, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
14			0.34			Medium High				\$0.34			4	Flooding	Inland Flood Risk Mitigation
3	2.29	0.39	0.45	High Low	High Low	High Medium	\$1,295.50	\$15,545.95	\$4,643.60	\$0.45	4	4	5	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
10			0.54			Medium Medium				\$0.54			6	Flooding	Inland Flood Risk Mitigation
8		0.63	1.32		Low High	Low High			\$10,031.17	\$1.32		5	7	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
12			0.55			Low High				\$0.55			8	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
4	0.39	2.29	2.29	High Low	High Low	High Low	\$13,688.31	\$164,259.73	\$31,526.28	\$2.29	3	4	9	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
5	0.24	0.24	1.33	High Low	High Low	High Low	\$1.69	\$20.31	\$2,022.79	\$1.33	5	4	10	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
7	1.11	1.11	1.11	High Low	High Low	High Low	\$0.002	\$0.03	\$4,703.94	\$1.11	7	4	11	SS, SLR, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation



ID	Project Length (Mile)			Criticality and Risk			Cumulated Composite Risk Amount (\$)				Rank			Hazards	
	2035	2050	2100	2035	2050	2100	2024	2035	2050	2100	2035	2050	2100		Work Type
1	1.54	0.08	0.08	High Low	High Low	High Low	\$0.56	\$6.66	\$664.63	\$0.08	6	4	12	SS, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
15			1.29			High Low				\$1.29			13	SS	Coastal Flood Risk Mitigation
11			0.19			High Low				\$0.19			14	SS, Flooding	Coastal Flood Risk Mitigation and Inland Flood Risk Mitigation
9			0.18			High Low				\$0.18			15	SS	Coastal Flood Risk Mitigation



Figure H.6 Potential Resilience Projects – Stormwater Treatment Units

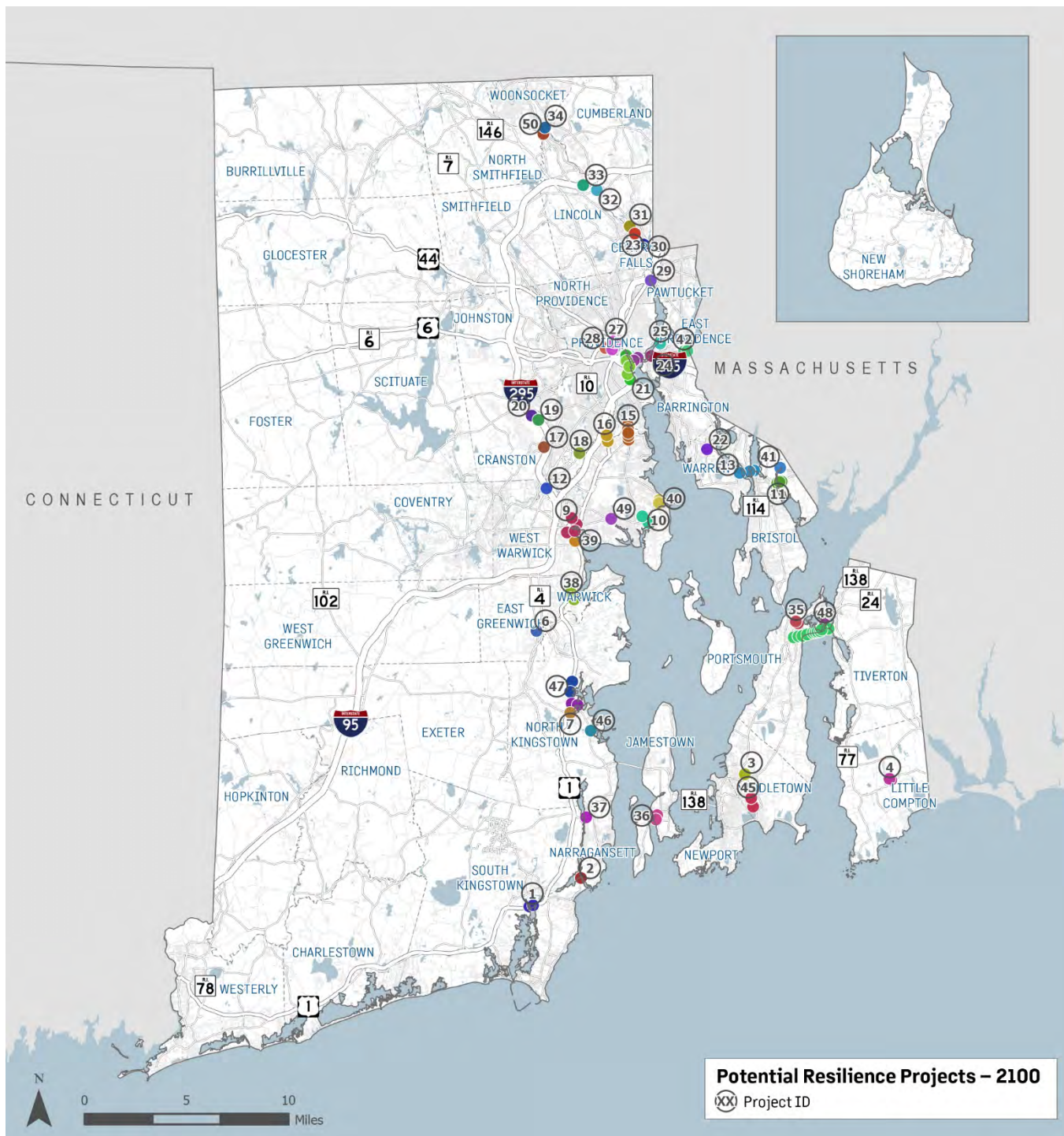




Table H.4 Potential Resilience Project List – Stormwater Treatment Units

Project ID	Asset ID	Criticality and Risk			Cumulated Composite Risk Amount (\$)				Rank			Hazards
		2035	2050	2100	2024	2035	2050	2100	2035	2050	2100	
1	1746	HighLow	HighHigh	HighHigh	\$33,855	\$406,264.32	\$2,363,237.78	\$203,642,252.76	20	1	1	SS, SLR, FLOODING
26	306	HighLow	HighHigh	HighHigh	\$300	\$3,600.39	\$502,038.82	\$20,242,308.47	24	2	2	SS, SLR
27	401	HighLow	HighMedium	HighHigh	\$1,239	\$14,871.55	\$468,607.57	\$3,808,895.67	22	13	3	SS, FLOODING
24	1930	HighHigh	HighHigh	HighHigh	\$10,959	\$131,513.17	\$409,305.91	\$3,021,699.69	2	4	4	SS, SLR, FLOODING
13	304	HighLow	HighMedium	HighHigh	\$154	\$1,851.64	\$230,577.87	\$2,586,075.75	25	14	5	SS
9	149	HighLow	HighMedium	HighHigh	\$663	\$7,951.05	\$213,079.08	\$1,730,550.97	23	15	6	SS, SLR, FLOODING
21	234	HighLow	HighMedium	HighHigh	\$63	\$755.51	\$94,080.89	\$1,583,642.74	28	17	7	SS
33	824	HighHigh	HighHigh	HighHigh	\$19,105	\$229,255.75	\$493,781.61	\$1,375,534.47	1	3	8	FLOODING
7	1031	HighLow	HighMedium	HighHigh	\$51	\$617.84	\$76,937.26	\$1,294,349.71	30	18	9	SS
10	48	HighLow	HighMedium	HighHigh	\$69	\$823.09	\$130,254.02	\$1,121,686.28	27	16	10	SS
14	578	HighLow	HighMedium	HighHigh	\$62	\$741.81	\$55,606.42	\$936,774.44	29	21	11	SS
20	1192	HighHigh	HighHigh	HighHigh	\$9,399	\$112,793.64	\$242,940.16	\$676,761.87	3	5	12	FLOODING
34	384	HighMedium	HighHigh	HighHigh	\$7,246	\$86,956.19	\$187,290.25	\$521,737.11	9	6	13	FLOODING
1	470	HighLow	HighMedium	HighHigh	\$40	\$474.10	\$59,038.18	\$423,698.36	31	20	14	SS, SLR
28	165	HighLow	HighHigh	HighHigh	\$80	\$960.91	\$119,658.55	\$335,605.38	26	7	15	SS
31	783	HighMedium	HighHigh	HighHigh	\$4,508	\$54,094.52	\$116,511.28	\$324,567.12	4	8	16	FLOODING
12	1533	HighMedium	HighHigh	HighHigh	\$4,069	\$48,830.54	\$105,173.47	\$292,983.23	5	9	17	FLOODING
6	1686	HighMedium	HighMedium	HighHigh	\$2,540	\$30,478.65	\$65,646.32	\$182,871.90	6	19	18	FLOODING
18	1842	HighLow	HighMedium	HighHigh	\$36	\$427.26	\$53,204.69	\$149,181.03	32	22	19	SS
11	612	MediumMedium	MediumMedium	MediumHigh	\$9,514	\$114,165.76	\$488,640.18	\$19,481,829.06	13	25	20	SS, SLR, FLOODING
25	1201	MediumMedium	MediumHigh	MediumHigh	\$3,721	\$44,646.88	\$904,664.36	\$14,826,664.30	11	10	21	SS, SLR, FLOODING
8	1045	LowHigh	LowHigh	MediumHigh	\$9,088	\$109,050.36	\$467,299.54	\$13,139,391.90	19	34	22	SS, SLR, FLOODING



Project ID	Asset ID	Criticality and Risk				Cumulated Composite Risk Amount (\$)			Rank			
		2035	2050	2100	2024	2035	2050	2100	2035	2050	2100	Hazards
43	716		MediumMedium	MediumHigh			\$226,056.92	\$1,901,527.89	#N/A	26	23	SS
35	606		MediumMedium	MediumHigh			\$157,433.34	\$1,324,978.73	#N/A	27	24	SS
40	44		MediumMedium	MediumHigh			\$121,874.34	\$1,025,173.00	#N/A	28	25	SS
2	1276	MediumMedium	MediumHigh	MediumHigh	\$2,761	\$33,136.27	\$137,954.43	\$766,943.81	12	12	26	SS, SLR, FLOODING
3	901	MediumMedium	MediumHigh	MediumHigh	\$5,881	\$70,570.47	\$151,997.93	\$423,422.81	10	11	27	FLOODING
44	709		MediumMedium	MediumHigh			\$30,554.58	\$257,481.77	#N/A	30	28	SS
32	1063	MediumMedium	MediumMedium	MediumHigh	\$2,047	\$24,561.14	\$52,900.92	\$147,366.84	14	29	29	FLOODING
37	1274		MediumMedium	MediumHigh			\$10,977.20	\$105,738.02	#N/A	31	30	SS
38	368		HighLow	HighMedium			\$80,084.22	\$1,411,776.91	#N/A	39	31	SS
42	1394		HighLow	HighMedium			\$1,452.64	\$348,796.56	#N/A	40	32	SS
16	1773	HighMedium	HighMedium	HighMedium	\$1,553	\$18,637.42	\$40,571.69	\$320,072.46	7	23	33	SS, SLR, FLOODING
30	1286	HighLow	HighMedium	HighMedium	\$1,647	\$19,769.87	\$42,581.25	\$118,619.20	21	22	34	FLOODING
39	66		HighLow	HighMedium			\$258.29	\$69,698.45	#N/A	41	35	SS
29	691	HighMedium	HighMedium	HighMedium	\$849	\$10,185.14	\$21,937.23	\$61,110.84	8	24	36	FLOODING
22	708	#N/A	#N/A	MediumMedium		#N/A	#N/A	\$27,843.80	#N/A	#N/A	37	SS
5	732	LowHigh	LowHigh	LowHigh	\$9,761	\$117,137.91	\$1,324,196.15	\$7,866,643.25	18	33	38	SS, SLR, FLOODING
4	718	LowHigh	LowHigh	LowHigh	\$63,088	\$757,056.85	\$2,397,495.30	\$6,694,406.73	15	32	39	SS, SLR, FLOODING
36	271		LowHigh	LowHigh			\$369,801.86	\$1,296,112.62	#N/A	35	40	SS
41	427		LowHigh	LowHigh			\$133,504.13	\$1,123,094.83	#N/A	38	41	SS, SLR
17	1183	LowHigh	LowHigh	LowHigh	\$9,399	\$112,793.64	\$242,940.16	\$676,761.87	17	36	42	FLOODING
19	1902	LowHigh	LowHigh	LowHigh	\$9,399	\$112,793.64	\$242,940.16	\$676,761.87	16	37	43	FLOODING
48	605			LowHigh				\$552,382.00	#N/A	#N/A	44	SS, SLR
45	76			LowHigh				\$453,344.78	#N/A	#N/A	45	SS
50	1088			LowHigh				\$177,956.96	#N/A	#N/A	46	FLOODING
23	1004	#N/A	#N/A	LowHigh		#N/A	#N/A	\$123,925.51	#N/A	#N/A	47	FLOODING
47	1054			HighLow				\$2,396.05	#N/A	#N/A	48	SS



Project ID	Asset ID	Criticality and Risk				Cumulated Composite Risk Amount (\$)			Rank			
		2035	2050	2100	2024	2035	2050	2100	2035	2050	2100	Hazards
49	64			HighLow				\$1,577.74	#N/A	#N/A	49	SS
46	1016			HighLow				\$322.40	#N/A	#N/A	50	SS



Figure H.7 Potential Resilience Projects - – Stormwater Pipes

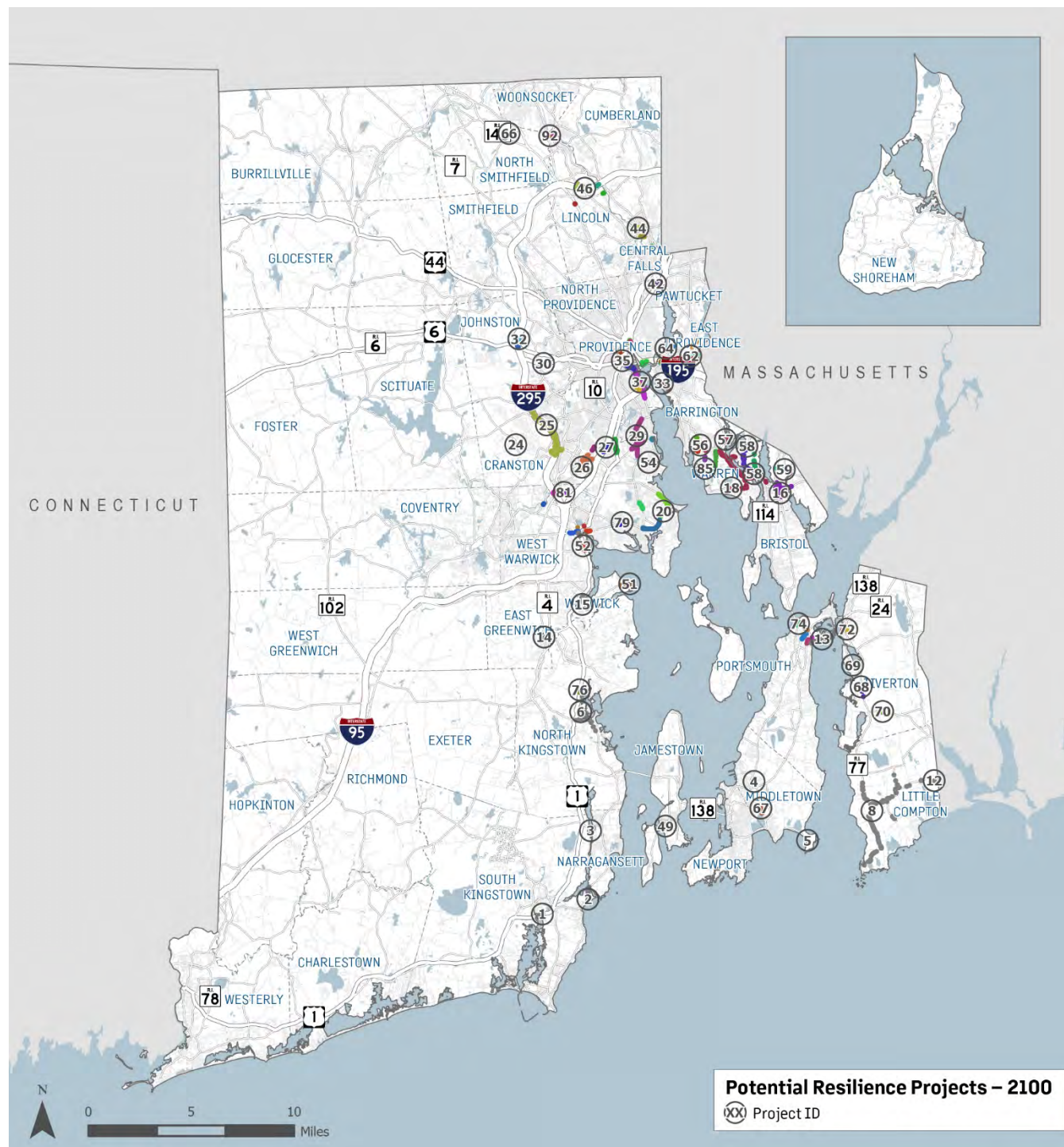




Table H.5 Potential Resilience Project List – Stormwater Pipes

Project ID	Object ID	Criticality and Risk				Cumulated Composite Risk Amount (\$)			Rank			Hazards
		2035	2050	2100	2024	2035	2050	2100	2035	2050	2100	
29	3987	HighLow	HighHigh	HighHigh	\$395,908	\$4,750,901.39	\$44,835,364.11	\$2,342,333,061.33	23	2	1	SS, SLR, FLOODING
38	4692	HighLow	HighHigh	HighHigh	\$3,985	47818.19784	\$6,276,798.54	\$277,056,417.07	36	5	2	SS, SLR
37	4686	HighLow	HighLow	HighHigh	\$15,424	185086.2258	\$24,471,021.40	\$217,316,072.09	28	37	3	SS
25	2892	HighHigh	HighHigh	HighHigh	\$2,754,045	\$33,048,537.94	\$71,181,466.32	\$198,291,227.61	1	1	4	FLOODING
6	1059	HighLow	HighHigh	HighHigh	\$98,247	\$1,178,963.13	\$5,181,945.93	\$168,082,401.39	25	7	5	SS, SLR, FLOODING
18	1874	HighLow	HighHigh	HighHigh	\$10,913	\$130,955.91	\$15,859,391.00	\$137,173,743.42	30	3	6	SS
36	4192	HighLow	HighHigh	HighHigh	\$5,456	65477.56627	\$8,153,677.55	\$69,680,280.46	34	4	7	SS
26	3237	HighLow	HighMedium	HighHigh	\$4,166	\$49,989.04	\$6,224,949.31	\$52,362,540.51	35	23	8	SS
20	2443	HighLow	HighLow	HighHigh	\$2,164	\$25,962.19	\$2,763,411.34	\$28,273,274.45	38	38	9	SS
39	4693	HighLow	HighLow	HighHigh	\$6,636	79633.22499	\$2,582,863.57	\$21,007,386.72	32	39	10	SS, FLOODING
35	4173	HighLow	HighLow	HighHigh	\$1,529	18353.11474	\$2,285,445.05	\$19,355,460.92	39	40	11	SS
89	5000	HighLow	HighLow	HighHigh	\$1,216	205594.1557	1822950.086	\$15,756,710.35	27	41	12	SS
27	3974	HighHigh	HighHigh	HighHigh	\$217,910	\$2,614,919.81	\$5,632,134.97	\$15,689,518.85	2	6	13	FLOODING
41	4697	HighLow	HighLow	HighHigh	\$538	6452.951452	\$1,308,475.36	\$11,335,633.48	44	43	14	SS
22	2634	HighLow	HighHigh	HighHigh	\$793	9513.845336	1184723.739	\$9,967,257.56	40	9	15	SS, SLR
28	3981	HighLow	HighLow	HighHigh	\$65,231	\$782,771.50	\$1,694,267.47	\$9,477,368.22	2	42	16	SS, SLR, FLOODING
19	1965	HighLow	HighMedium	HighHigh	\$741	\$8,892.44	\$1,107,488.22	\$9,355,776.68	41	24	17	SS
15	1725	HighLow	HighHigh	HighHigh	\$641	\$7,686.07	\$957,118.05	\$8,059,483.04	42	12	18	SS
21	2589	HighLow	HighLow	HighHigh	\$602	\$7,219.00	\$898,954.70	\$7,584,673.82	43	44	19	SS
1	604	HighLow	HighMedium	HighHigh	\$410	\$4,916.46	\$612,228.48	\$5,149,895.48	45	25	20	SS
30	3988	HighHigh	HighHigh	HighHigh	\$47,219	\$566,631.71	\$1,220,437.52	\$3,399,790.25	3	8	21	FLOODING



Project ID	Object ID	Criticality and Risk				Cumulated Composite Risk Amount (\$)			Rank			
		2035	2050	2100	2024	2035	2050	2100	2035	2050	2100	Hazards
44	4700	HighMedium	HighHigh	HighHigh	\$46,796	561557.1355	\$1,209,507.68	\$3,369,342.81	11	10	22	FLOODING
31	3989	HighHigh	HighHigh	HighHigh	\$41,748	\$500,972.84	\$1,079,018.43	\$3,005,837.04	4	11	23	FLOODING
40	4694	HighLow	HighLow	HighHigh	\$172	2069.508941	\$258,055.88	\$2,262,228.00	46	46	24	SS
7	1087	HighLow	HighLow	HighHigh	\$5,629	\$67,552.36	\$346,305.84	\$2,170,607.71	33	45	25	SS, SLR, FLOODING
45	4701	HighHigh	HighHigh	HighHigh	\$18,474	221685.6623	477476.8111	\$1,330,113.97	6	13	26	FLOODING
48	4704	HighHigh	HighHigh	HighHigh	\$17,133	687767.2359	442818.1816	\$1,233,564.93	7	14	27	FLOODING
17	1824	HighLow	HighHigh	HighHigh	\$89	\$1,072.10	\$133,504.13	\$1,123,190.12	47	17	28	SS, SLR
32	4148	HighHigh	HighHigh	HighHigh	\$13,916	\$166,990.95	\$359,672.81	\$1,001,945.68	5	15	29	FLOODING
23	2846	HighLow	HighHigh	HighHigh	\$12,208	146491.613	315520.3971	\$878,949.68	29	16	30	FLOODING
14	1539	HighLow	HighMedium	HighHigh	\$7,620	\$91,435.95	\$196,938.97	\$548,615.70	31	26	31	FLOODING
34	4172	HighLow	HighLow	HighHigh	\$27	\$318.26	\$39,632.17	\$334,216.87	48	48	32	SS
43	4699	HighLow	HighLow	HighHigh	\$3,295	39539.73289	\$85,162.50	\$237,238.40	37	47	33	FLOODING
8	1089	MediumHigh	MediumHigh	MediumHigh	\$5,141,678	\$61,700,133.38	\$194,225,253.02	\$902,270,079.80	8	18	34	SS, SLR, FLOODING
16	1822	MediumHigh	MediumHigh	MediumHigh	\$22,243	\$266,913.36	\$909,178.75	\$35,270,113.15	10	21	35	SS, SLR, FLOODING
2	728	MediumMedium	MediumHigh	MediumHigh	\$197,185	\$2,366,223.41	\$7,425,749.36	\$34,649,567.64	12	19	36	SS, SLR, FLOODING
3	794	MediumHigh	MediumHigh	MediumHigh	\$179,128	\$2,149,539.72	\$5,071,343.55	\$22,475,276.84	9	20	37	SS, SLR, FLOODING
4	856	MediumMedium	MediumHigh	MediumHigh	\$29,404	\$352,852.34	\$759,989.66	\$2,117,114.04	13	22	38	FLOODING
42	4698	HighLow	HighLow	HighMedium	\$290,030	3480355.856	\$28,088,778.32	\$170,166,511.45	24	36	39	SS, SLR, FLOODING
33	4149	LowHigh	LowHigh	LowHigh	\$67,587	\$811,038.06	\$101,207,338.75	\$729,436,241.16	17	27	40	SS, SLR
46	4702	LowHigh	LowHigh	LowHigh	\$401,198	4814370.651	10369413.71	\$28,886,223.91	14	28	41	FLOODING
9	1219	LowHigh	LowHigh	LowHigh	\$126,176	\$1,514,113.70	\$4,794,990.61	\$22,141,597.05	16	29	42	SS, SLR, FLOODING



Project ID	Object ID	Criticality and Risk				Cumulated Composite Risk Amount (\$)			Rank			
		2035	2050	2100	2024	2035	2050	2100	2035	2050	2100	Hazards
5	920	LowHigh	LowHigh	LowHigh	\$94,632	\$1,135,585.28	\$3,596,242.96	\$16,606,197.79	15	30	43	SS, SLR, FLOODING
10	1240	LowHigh	LowHigh	LowHigh	\$63,088	\$757,056.85	\$2,397,495.30	\$11,070,798.53	18	31	44	SS, SLR, FLOODING
11	1290	LowHigh	LowHigh	LowHigh	\$63,088	\$757,056.85	\$2,397,495.30	\$11,070,798.53	19	32	45	SS, SLR, FLOODING
12	1415	LowHigh	LowHigh	LowHigh	\$63,088	\$757,056.85	\$2,397,495.30	\$11,070,798.53	20	33	46	SS, SLR, FLOODING
13	1531	LowHigh	LowHigh	LowHigh	\$26,926	\$323,113.21	\$1,023,255.26	\$4,725,036.46	21	34	47	SS, SLR, FLOODING
24	2853	LowHigh	LowHigh	LowHigh	\$9,399	\$112,793.64	\$242,940.16	\$676,761.87	22	35	48	FLOODING



APPENDIX K RESILIENCE OPPORTUNITIES IN STIP

The tables below show the resilience opportunities in STIP

Table I.1 Resilience Opportunities in STIP

Roads and Bridges											
TIP ID	Project name	From	To	Project Status	Start Year	End Year	Project Description	Criticality Risk Rating 2035	Criticality Risk Rating 2050	Criticality Risk Rating 2100	Hazard
3102	Bridge Group 19 -- I-95 Ramps, Providence	Ramp CA Civic Center, Ramp DA Civic Center	Ramp CA Civic Center, Ramp DA Civic Center	Planning	2026	2031	This project will rehabilitate several of the ramps to and from I-95 near the Providence Viaduct.	High Low	High Low	High Medium	SS
3358	Bridge Group 02 -- I-95 Bridges, Providence South	O'Connel Street & Public Street	O'Connel Street & Public Street	Planning	2026	2031	Bridges #058401, 065101, 065201, 065401, and 065501: Minor rehab, including structural repairs to one or more deck or superstructure components. Bridges #057201 and 057301: Major rehab, including full deck replacement.	High Low	High Medium	High High	SS
3296	Bridge Group 08 -- I-295 Bridges in Warwick	West Natick Road Bridges	West Natick Road Bridges	Planning	2023	2029	Replacement of the West Natick Road NB and SB Bridges, #072501 and #072521. Preservation work will be performed on Bridges #072101, #072121, #072201, and #072221. This work will extend the useful service life of the structures, which may include join	High Medium	High High	High High	FLOODING
3012	Bridge Group 07_H -- Historic Westerly Bridges	Pawcatuck	Pawcatuck	Planning	2028	2031	Historic Minor Rehabilitation, including structural repairs to one or more deck or superstructure components. This project may require additional funding beyond 2031. This project includes a reconstruction of Sherman Avenue from RI-2 to Lewiston Avenue	High Low	High Low	High Medium	FLOODING



Roads and Bridges											
TIP ID	Project name	From	To	Project Status	Start Year	End Year	Project Description	Criticality Risk Rating 2035	Criticality Risk Rating 2050	Criticality Risk Rating 2100	Hazard
3014	Bridge Group 07A - I-95 Hopkinton	Richard Farm	Richard Farm	Planning	2026	2031	Replacement of the Hopkinton-Westerly Road Bridge (#405). Major rehabilitation of Bridge #48, New London Turnpike. Minor rehabilitation, including structural repairs to one or more deck or superstructure components, for bridges #409, #567, #568, #569,	High High	High High	High High	FLOODING
3102	Bridge Group 19 -- I-95 Ramps, Providence	Ramp BC Civic Center	Ramp BC Civic Center	Planning	2026	2031	This project will rehabilitate several of the ramps to and from I-95 near the Providence Viaduct.	High Low	High Medium	High Medium	SS
3253	Bridge Group 22C_H -- Historic Pawtuxet River East	Elmwood Ave Bridge	Elmwood Ave Bridge	Planning	2032	0	Historic Preservation - work to extend the useful service life of the Elmwood Avenue bridge, which may include joint replacement, bridge washing, zone painting, deck repairs, minimum to moderate concrete or steel superstructure repairs, moderate substruction	High Medium	High Medium	High High	SS, SLR, FLOODING
3231	Bridge Group 24C_H -- Ashton Viaduct	Ashton Viaduct	Ashton Viaduct	Planning	2025	2031	Historic Minor Rehabilitation of Bridge #275, the Ashton Viaduct, including structural repairs to one or more deck or superstructure components. Preservation of the nearby Kelly House bridge (#179).	Medium Medium	Medium Medium	Medium High	FLOODING
3007	Bridge Group 04_R- I-95/RT 10	Ramp CB	Ramp CB	Planning	2022	2028	This project consolidates work previously included under TIPIDs 3040, 3041, 3229, 9521, and 9523 into a single project. All project assets and funding under these TIPIDs are being incorporated into this TIPID, and treatments to bridges are being updated	High Medium	High High	High High	FLOODING



Roads and Bridges											
TIP ID	Project name	From	To	Project Status	Start Year	End Year	Project Description	Criticality Risk Rating 2035	Criticality Risk Rating 2050	Criticality Risk Rating 2100	Hazard
3014	Bridge Group 07A - I-95 Hopkinton	Canonchet	Canonchet	Planning	2026	2031	Replacement of the Hopkinton-Westerly Road Bridge (#405). Major rehabilitation of Bridge #48, New London Turnpike. Minor rehabilitation, including structural repairs to one or more deck or superstructure components, for bridges #409, #567, #568, #569,	High High	High High	High High	FLOODING
3012	Bridge Group 07_H -- Historic Westerly Bridges	Bradford	Bradford	Planning	2028	2031	Historic Minor Rehabilitation, including structural repairs to one or more deck or superstructure components. This project may require additional funding beyond 2031. This project includes a reconstruction of Sherman Avenue from RI-2 to Lewiston Avenue	Medium Medium	Medium High	Medium High	FLOODING
1408	RI-114, Main St. (Water St. to Wheaton St.)	114, Main St. (Water St. to Wheaton St.)	114, Main St. (Water St. to Wheaton St.)	Planning	2029	2031	This line item involves resurfacing of the roadway, limited sidewalk replacement and handicapped ramp installation. This line item may require additional funding beyond 2031.	High Low	High Medium	High Medium	SS
1419	RI-136 Market Street (RI-114 to Schoolhouse Road) and Kickemuit Rd (Everett St to Market St)	136 Market Street (RI114 to Schoolhouse Road) and Kickemuit Rd (Everett St to Market St)	136 Market Street (RI114 to Schoolhouse Road) and Kickemuit Rd (Everett St to Market St)	Planning	2028	2031	This line item involves resurfacing of the roadway, limited sidewalk replacement and handicapped ramp installation. RIDOT will coordinate with the town of Warren when designing this project to account for sea level rise concerns. This line item may requ	High Medium	High High	High High	SS, SLR, FLOODING
1474	Massasoit Ave Sidewalk (Woodward Ave. to Arvin Ave.)			Planning	2026	2029	New and upgraded sidewalks along Massasoit Avenue from Woodward Avenue to Arvin Avenue. This project will include drainage improvements if necessary, and will be coordinated with TIPID 1473.	Low High	Low High	Low High	SS, SLR, FLOODING



Roads and Bridges											
TIP ID	Project name	From	To	Project Status	Start Year	End Year	Project Description	Criticality Risk Rating 2035	Criticality Risk Rating 2050	Criticality Risk Rating 2100	Hazard
5005	East Bay Bike Path Bridge Replacement (Barrington River and Warren Bridges)			Planning	2022	2026	This project will replace two East Bay Bike Path bridges over the Barrington River. This line item will also make safety improvements to accommodate an alternative on-road path prior to the replacement of these structures. This project received a \$5,00	Medium Medium	Medium High	Medium High	SS, SLR, FLOODING
5072	Purgatory Road Sidewalk Installation (Aquidneck Ave. to Paradise Ave.)			Planning	2022	2023	Install a sidewalk along Purgatory Road between the Atlantic Beach District and Second Beach. This project received a \$960,000 USDOT earmark in 2022.	Medium Medium	Medium High	Medium High	SS, SLR, FLOODING
5106	Farewell Street Pavement Preservation and Sidewalk (America's Cup to Van Zandt)			Planning	2025	2029	This project will resurface RI-238 Farewell Street in conjunction with sidewalk improvements. These improvements entail sidewalk repairs, new granite curbing, and additional accommodations for ADA accessibility.	High Medium	High High	High High	SS, SLR, FLOODING
5120	Post Road and West Main Rd - Curbing and Sidewalks (RI-1A to Camp Ave, US-1 to Brown St)	Curbing and Sidewalks	Curbing and Sidewalks	Planning	2022	2026	This project includes installing sidewalks on both sides of Post Road in North Kingstown between Camp Ave and West Main Street, and West Main Street from Post Road to Brown Street. Drainage improvements may be included as needed. The work on West Main	Medium Medium	Medium High	Medium High	SS, SLR, FLOODING
1318	RI-122, Mendon Rd (I-295 to Lincoln T/L)	122, Mendon Rd (I295 to Lincoln T/L)	122, Mendon Rd (I295 to Lincoln T/L)	Planning	2022	2029	This line item involves resurfacing of the roadway, limited sidewalk replacement, limited sidewalk extension and handicapped ramp installation. Improvements to stormwater may be included as part of this project. Additional safety improvements may also apply.	Medium Medium	Medium Medium	Medium High	FLOODING



Roads and Bridges												
TIP ID	Project name	From	To	Project Status	Start Year	End Year	Project Description	Criticality Risk Rating 2035	Criticality Risk Rating 2050	Criticality Risk Rating 2100	Hazard	
9556	RI-122, Lonsdale Ave (Central Falls T/L to Cumberland T/L)	122/Lonsdale Ave (Central Falls T/L to Cumberland T/L)	122/Lonsdale Ave (Central Falls T/L to Cumberland T/L)	Planning	2025	2029	This project includes mill and overlay over the roadway with limited areas of reconstruction. Limited sidewalk replacement and wheelchair ramp improvements are included. This line item may also include improvements to stormwater drainage.	High Medium	High Medium	High High	FLOODING	
5375	2022 Intersection & Crosswalks: East/South	Route 1A at Dunes Club	Route 1A at Dunes Club	Scoping	2022	2024	This line item includes crosswalks, wheelchair ramps, pedestrian crossing devices, left turn lanes, and other work to improve intersections and crosswalks for the East and South regions of the state.	High Medium	High High	High High	SS	
1274	I-295 (I-95 to RI-5)	6 E from I295 to Glenbridge Ave	6 E from I295 to Glenbridge Ave	Planning	2022	2026	This line item involves resurfacing of Interstate 295 from Interstate 95 in Warwick to RI-5 Greenville Avenue and US-6 West from I-295 to Glenbridge Ave. Improvements to safety and stormwater may be included as part of this project.	High High	High High	High High	FLOODING	
3210	Bridge Group 50 -- Goat Island	Goat Island Causeway	Goat Island Causeway	Planning	2026	2031	Major rehabilitation work, superstructure, and/or total bridge replacement. This project will also resurface the approach to the Goat Island Bridge from the bridge to America's Cup Avenue.	Medium Medium	Medium Medium	Medium High	SS, SLR, FLOODING	
3016	Bridge Group 14 -- Route 99	Blackstone River and Second Ave	Blackstone River and Second Ave	Scoping	2022	2026	Minor Rehabilitation, including structural repairs to one or more deck or superstructure components.	High Medium	High Medium	High Medium	FLOODING	
3102	Bridge Group 19 -- I-95 Ramps, Providence	Ramp AC Civic Center	Ramp AC Civic Center	Planning	2026	2031	This project will rehabilitate several of the ramps to and from I-95 near the Providence Viaduct.	High Low	High Medium	High Medium	SS	



Roads and Bridges											
TIP ID	Project name	From	To	Project Status	Start Year	End Year	Project Description	Criticality Risk Rating 2035	Criticality Risk Rating 2050	Criticality Risk Rating 2100	Hazard
3300	Bridge Group 51B - RI-37 C-3	Cranston Canyon	Cranston Canyon	Scoping	2022	2027	Replacement of bridges #072801, 072821, 062101, and 062201. Bridge #083101 will be replaced with the new structure being wider to support an additional lane, alongside installation of a high-friction surface treatment (HSFT). Bridge #062001: Preservation	High Medium	High Medium	High High	FLOODING
3154	Bridge Group 76 -- I-Way Preservation	Ramp WS, Ramp WES	Ramp WS, Ramp WES	Planning	2023	2028	This project will repaint and preserve the eight bridges and ramps that form the I-Way linking I-95 and I-195 in Providence. The application of paint and minor repairs will extend the useful life of the structures in a state of good repair.	High Low	High Medium	High High	SS
3154	Bridge Group 76 -- I-Way Preservation	Ramp WS, Ramp WES	Ramp WS, Ramp WES	Planning	2023	2028	This project will repaint and preserve the eight bridges and ramps that form the I-Way linking I-95 and I-195 in Providence. The application of paint and minor repairs will extend the useful life of the structures in a state of good repair.	High Low	High Medium	High High	SS
3154	Bridge Group 76 -- I-Way Preservation	Ramp NE, Ramp SE	Ramp NE, Ramp SE	Planning	2023	2028	This project will repaint and preserve the eight bridges and ramps that form the I-Way linking I-95 and I-195 in Providence. The application of paint and minor repairs will extend the useful life of the structures in a state of good repair.	High High	High High	High High	SS
3154	Bridge Group 76 -- I-Way Preservation	Ramp NP, Ramp WP	Ramp NP, Ramp WP	Planning	2023	2028	This project will repaint and preserve the eight bridges and ramps that form the I-Way linking I-95 and I-195 in Providence. The application of paint and minor repairs will extend the useful life of the structures in a state of good repair.	High Low	High Medium	High High	SS



Roads and Bridges											
TIP ID	Project name	From	To	Project Status	Start Year	End Year	Project Description	Criticality Risk Rating 2035	Criticality Risk Rating 2050	Criticality Risk Rating 2100	Hazard
1275	I-295 (RI-5 to MA S/L)	295 (RI146 to Massachusetts S/L)	295 (RI146 to Massachusetts S/L)	Planning	2022	2027	This line item involves resurfacing of Interstate 295 from RI-5 Greenwood Avenue to the Massachusetts State Line. Improvements to safety and stormwater may be included as part of this project.	High Medium	High High	High High	FLOODING
1290	RI-114, Wampanoag Trail (East Shore Expressway to County Rd)	114, Wampanoag Trail (East Shore Expressway to Federal Rd)	114, Wampanoag Trail (East Shore Expressway to Federal Rd)	Planning	2024	2025	This line item involves resurfacing the roadway. This project was funded with August Redistribution funding in 2021 and is currently underway.	Medium Medium	Medium Medium	Medium High	SS, SLR, FLOODING
1299	Hope Street and Ferry Road Resurfacing and Sidewalks - Phase 2 (Metacom to Constitution, Columban to Ferry, Thames to Warren Town Line)	114, Hope St and Main St (Washington St to Warren T/L)	114, Hope St and Main St (Washington St to Warren T/L)	Planning	2025	2029	This project will resurface RI-114 Hope Street from Wood St. to Constitution Street and from Thames St. to Warren town line. Sidewalks along those limits will also be repaired or replaced as needed to fit ADA compliance standards. Sidewalk may be exten	High Medium	High High	High High	SS, SLR, FLOODING
1281	US-6 (RI-10 to I-95)	6 (RI10 to I95)	6 (RI10 to I95)	Planning	2026	2028	This line item involves resurfacing of the roadway.	High Low	High Low	High Medium	SS
1324	Veterans Memorial Parkway (Warren Ave to South Broadway)			Planning	2022	2025	This line item involves resurfacing of the roadway and improvements to highway lighting fixtures. Additional improvements to safety and stormwater may be included as part of this project.	Medium Medium	Medium High	Medium High	SS, SLR, FLOODING
1297	County Rd (Warren Bridge to Federal Rd.) and Rumstick Rd. (Nayatt Rd. to County Rd.)	103/114, County Rd (Warren Bridge to Federal Rd.)	103/114, County Rd (Warren Bridge to Federal Rd.)	Planning	2025	2031	This line item involves resurfacing the roadway, rehabilitation of sidewalk, and handicapped ramp installation.	High Medium	High High	High High	SS, SLR, FLOODING
9519	I-195 (I-95 to East Providence T/L)	195 (I95 to East Providence T/L)	195 (I95 to East Providence T/L)	Planning	2031	2031	This line item involves mill and overlay of the roadway.	High Low	High Medium	High High	SS



Roads and Bridges											
TIP ID	Project name	From	To	Project Status	Start Year	End Year	Project Description	Criticality Risk Rating 2035	Criticality Risk Rating 2050	Criticality Risk Rating 2100	Hazard
3007	Bridge Group 04_R- I-95/RT 10	95 (RI10 to I195)	95 (RI10 to I195)	Planning	2022	2028	This project consolidates work previously included under TIPIDs 3040, 3041, 3229, 9521, and 9523 into a single project. All project assets and funding under these TIPIDs are being incorporated into this TIPID, and treatments to bridges are being updated	High Medium	High High	High High	FLOODING
3166	Bridge Group 15F - Barrington Bridge	Barrington Bridge	Barrington Bridge	Planning	2024	2027	Minor Rehabilitation, including structural repairs to one or more deck or superstructure components.	High High	High High	High High	SS, SLR, FLOODING
3276	Bridge Group 08E - Blackstone Viaduct	Blackstone Viaduct NB, SB	Blackstone Viaduct NB, SB	Planning	2026	2031	Minor rehabilitation, including structural repairs to one or more deck or superstructure components, of two bridges carrying I-295 NB and SB over the Blackstone River in Cumberland. Minor rehab will also be performed to the Old River Road Bridge (#074	High Medium	High High	High High	FLOODING
1274	I-295 (I-95 to RI-5)	295 (I95 to RI5)	295 (I95 to RI5)	Planning	2022	2026	This line item involves resurfacing of Interstate 295 from Interstate 95 in Warwick to RI-5 Greenville Avenue and US-6 West from I-295 to Glenbridge Ave. Improvements to safety and stormwater may be included as part of this project.	High Medium	High Medium	High High	FLOODING
9505	I-95 (Hayward St. to Walcott St.)	95 (Hayward St. to Walcott St.)	95 (Hayward St. to Walcott St.)	Planning	2022	2029	This line item involves mill and overlay of the roadway. Additional improvements to traffic safety and stormwater drainage may be included as part of this project.	High Medium	High Medium	High High	FLOODING
1362	RI-1A, Boston Neck Rd (Sprague Bridge to Bridgetown Rd)	1A, Boston Neck Rd (Sprague Bridge to Bridgetown Rd)	1A, Boston Neck Rd (Sprague Bridge to Bridgetown Rd)	Planning	2022	2029	This line item involves resurfacing of the roadway, limited sidewalk replacement, limited sidewalk extension and handicapped ramp installation. Improvements to active transportation, including the installation of a bike lane on the northbound shoulder	Medium Medium	Medium High	Medium High	SS, SLR



Roads and Bridges											
TIP ID	Project name	From	To	Project Status	Start Year	End Year	Project Description	Criticality Risk Rating 2035	Criticality Risk Rating 2050	Criticality Risk Rating 2100	Hazard
1576	RI-2 - South County Trail (RI-138 to RI-102)	2 South County Trail (RI138 to RI	2 South County Trail (RI138 to RI	Planning	2022	2031	This line item involves resurfacing the roadway. Active transportation improvements, including a bicycle lane on the northbound shoulder, may be included as part of this project. Additional improvements to traffic safety and stormwater drainage may apply	Medium Medium	Medium Medium	Medium High	FLOODING
9557	RI-2, South County Trail (Charlestown T/L to RI-138)	2, South County Trail (Charlestown T/L to South Kingstown T/L)	2, South County Trail (Charlestown T/L to South Kingstown T/L)	Planning	2022	2031	This line item involves mill and overlay of the roadway. This project may include improvements to active transportation, including the possible installation of a bicycle lane along the northbound shoulder of the roadway.	Medium High	Medium High	Medium High	FLOODING
9520	I-95 (Connecticut S/L to Baker Pines Rd.)	95 (Connecticut S/L to Baker Pines Rd.)	95 (Connecticut S/L to Baker Pines Rd.)	Planning	2022	2025	This line item involves mill and overlay of the roadway. Additional improvements to traffic safety, including guardrail, may be included with this project alongside stormwater drainage improvements.	High High	High High	High High	FLOODING
9537	RI-117, Flat River Road (Hill Farm Rd to Read School House Road)	117, Flat River Road (Hill Farm Rd to Read School House Road)	117, Flat River Road (Hill Farm Rd to Read School House Road)	Planning	2031	2031	This line item involves mill and overlay of the roadway with limited wheelchair ramp improvements. This line item may require additional funding beyond 2031.	Medium Medium	Medium High	Medium High	FLOODING
3223	Wickford Village Sidewalk and Resiliency Enhancements			Planning	2028	2031	This project will address various transportation needs in the area surrounding Wickford Village in North Kingstown. It will include rehabilitation of historic bridge structures, resurfacing of the roadway, and sidewalk improvements to improve pedestrian experience.	High Medium	High Medium	High High	SS, SLR, FLOODING



Roads and Bridges												
TIP ID	Project name	From	To	Project Status	Start Year	End Year	Project Description	Criticality Risk Rating 2035	Criticality Risk Rating 2050	Criticality Risk Rating 2100	Hazard	
3300	Bridge Group 51B - RI-37 C-3	I295N (RI37 to US	I295N (RI37 to US	Scoping	2022	2027	Replacement of bridges #072801, 072821, 062101, and 062201. Bridge #083101 will be replaced with the new structure being wider to support an additional lane, alongside installation of a high-friction surface treatment (HSFT). Bridge #062001: Preservation	High Medium	High Medium	High High	FLOODING	
5283	Quonset Ferry Terminal			Planning	2022	2022	Construct a passenger ferry terminal at Rhode Island Fast Ferry's docks in the Quonset Business Park.	High Low	High High	High High	SS, SLR	
5391	Pawtucket Bridge Lighting and Bridge Strike Detection Systems	Pawtucket River Bridge	Pawtucket River Bridge	Planning	2023	2023	This line item includes improvements to lighting systems on the Pawtucket Bridge alongside early warning overheight vehicle detection systems for additional bridges throughout the state. These projects were previously included in TIPID 12113 and TIPID 7	High Medium	High Medium	High High	FLOODING	
1404	RI-77 Main Rd. (Old Main Rd. to RI-24) and Highland Rd. (Bridgeport Rd. to RI-77 Main Rd.)	77 Main Road (Middle Ave to RI24 Ramps)	77 Main Road (Middle Ave to RI24 Ramps)	Planning	2022	2026	This project will reclaim Highland Road from Bridgeport Road to RI-77 Main Road and perform a mill and overlay of RI-77 Main Road from Old Main Road to the RI-24 ramps. Resurfacing and sidewalk work along RI-77 Main Road was previously included in T1	Low High	Low High	Low High	SS, SLR, FLOODING	
7402	Chip Seal, Surface Seal, and Thin Overlay - 2022	RI77 (RI179 East Road to Quaket Creek Culvert)	RI77 (RI179 East Road to Quaket Creek Culvert)	Planning	2022	2022	This line item involves the pavement preventive maintenance activities of chip seal, surface seal and thin overlay for 2022.	Medium Medium	Medium High	Medium High	SS	
5377	2023 Intersection & Crosswalks	Allens Ave at Globe Street	Allens Ave at Globe Street	Planning	2022	2025	This line item includes crosswalks, wheelchair ramps, pedestrian crossing devices, left turn lanes, and other work to improve intersections and crosswalks statewide.	High Low	High Medium	High High	SS	



Roads and Bridges											
TIP ID	Project name	From	To	Project Status	Start Year	End Year	Project Description	Criticality Risk Rating 2035	Criticality Risk Rating 2050	Criticality Risk Rating 2100	Hazard
9992	Corridor - Jamestown	Intersection Safety Improvement at Conanicus Ave and Bay View Drive	Intersection Safety Improvement at Conanicus Ave and Bay View Drive	Planning	2022	2031	This project includes repairs to two bridges and North Road in Jamestown. Bridges #113 and 289: Replacement. In addition to bridge work, roadway resurfacing and reclaim along North Road and East Shore Road will be included.	Medium Medium	Medium High	Medium High	SS
5005	East Bay Bike Path Bridge Replacement (Barrington River and Warren Bridges)			Planning	2022	2026	This project will replace two East Bay Bike Path bridges over the Barrington River. This line item will also make safety improvements to accommodate an alternative on-road path prior to the replacement of these structures.	Medium Medium	Medium Medium	Medium High	SS, SLR, FLOODING
5377	2023 Intersection & 5 Crosswalks		5	Planning	2022	2025	This line item includes crosswalks, wheelchair ramps, pedestrian crossing devices, left turn lanes, and other work to improve intersections and crosswalks statewide.	High Low	High Medium	High High	SS
5412	2026 Statewide Safety Improvements	Squantum to Park Ave	Squantum to Park Ave	Planning	2025	2028	This line item includes installation of median guardrail, regrading to prevent median related roll-overs and median cross-over crashes related to roadway departure. Work locations will be identified through the Strategic Highway Safety Planning Process.	High Low	High High	High High	SS



APPENDIX L NOTICE OF FUNDING OPPORTUNITY FOR FISCAL YEARS 2022 AND 2023 PROMOTING RESILIENT OPERATIONS FOR TRANSFORMATIVE, EFFICIENT, AND COST-SAVING TRANSPORTATION (PROTECT) PROGRAM

U.S. DEPARTMENT OF TRANSPORTATION

Federal Highway Administration

Notice of Funding Opportunity for Fiscal Years 2022 and 2023 Promoting Resilient Operations for Transformative, Efficient, and Cost-Saving Transportation (PROTECT) Program

AGENCY: U.S. Department of Transportation (DOT), Federal Highway Administration (FHWA).

ACTION: Notice of Funding Opportunity

SUMMARY:

The purpose of this Notice of Funding Opportunity (NOFO) is to solicit applications for FHWA's Promoting Resilient Operations for Transformative, Efficient, and Cost-Saving Transportation (PROTECT) Discretionary Program Grants and will result in the distribution of up to \$848 million.

Funds made available for Federal Fiscal Years (FY) 2022 and 2023 for the PROTECT Discretionary Grant Program are to be awarded on a competitive basis to make transportation infrastructure and service more resilient to climate change and extreme weather events under 23 United States Code (U.S.C.) § 176. This notice describes the application requirements, selection and evaluation criteria, applicable program and Federal requirements, and available technical assistance during the grant solicitation period.

DATES:

Applications must be submitted electronically through [grants.gov](https://www.grants.gov) no later than 11:59 p.m., Eastern Standard Time, on Friday, August 18, 2023 (the "application deadline"). Applicants are encouraged to submit applications in advance of the application deadline; however, applications will not be evaluated, and awards will not be made, until after the application deadline.

FHWA plans to conduct outreach regarding PROTECT Discretionary Program Grants in the form of virtual meetings on May 8, 2023 and May 11, 2023, 1 PM Eastern Time. To join the Webinar, follow the instructions posted on

<https://www.fhwa.dot.gov/environment/protect/discretionary>. The Webinars will be recorded and posted on FHWA's PROTECT Discretionary Grant Program Website at <https://www.fhwa.dot.gov/environment/protect/discretionary>.

ADDRESSES:

Applications must be submitted electronically through [grants.gov](https://www.grants.gov). Refer to Assistance Listing: 20.205 – Highway Planning and Construction.

FOR FURTHER INFORMATION CONTACT:

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Agreement Officer
Office of Acquisition and Grants Management
Federal Highway Administration
U.S. Department of Transportation
1200 New Jersey Ave., SE.
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Alternate:

Robin Hobbs
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Office hours are from 7:30 a.m. to 4:00 p.m., Eastern Time, Monday through Friday, except Federal holidays.

FHWA will not review applications in advance, but FHWA staff are available for technical questions. In addition, FHWA will post answers to questions and requests for clarifications at [Grants.gov](https://www.grants.gov) under this NOFO's page. To ensure applicants receive accurate information about eligibility or the program, the applicant is encouraged to contact FHWA directly at PROTECTdiscretionary@dot.gov, rather than through intermediaries or third parties, with questions.

SUPPLEMENTARY INFORMATION:

Each section of this notice contains information and instructions relevant to the application process for PROTECT Discretionary Grant Program grants. The applicant should read this notice in its entirety in order to submit all necessary information for complete and competitive applications.

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NOTE: FHWA uses www.grants.gov for receipt of all applications. Applicants must register and use the system to submit applications electronically. Applicants are encouraged to register in advance of the submission deadline and to register to receive notifications of updates/amendments to this Notice. Approval of user registrations for the site may take multiple weeks. It is the Applicant’s responsibility to monitor for any updates to this Notice.

SECTION A – PROGRAM DESCRIPTION

1. Overview

Section 11405 of the Bipartisan Infrastructure Law (BIL), enacted as the Infrastructure Investment and Jobs Act (Pub. L. 117-58, Nov. 15, 2021), established the PROTECT Formula and Discretionary Grant programs, which are codified at 23 U.S.C. § 176. The purpose of the PROTECT Formula and Discretionary Grant programs is to plan for and strengthen surface transportation to be more resilient to natural hazards, including climate change, sea level rise, flooding, extreme weather events, and other natural disasters through formula and competitive discretionary grants. FHWA issued guidance for the formula portion of PROTECT in July 2022. This NOFO is for the discretionary portion of PROTECT. It will result in the distribution of up to \$848 million for the PROTECT Discretionary Grant Program through a competitive discretionary grant process.¹ FHWA intends to hold future solicitations for funding available in FY 2024-2026.

The vision of the PROTECT Discretionary Grant Program is to fund projects that address the climate crisis by improving the resilience of the surface transportation system, including highways, public transportation, ports, and intercity passenger rail. Projects selected under this program should be grounded in the best available scientific understanding of climate change risks, impacts, and vulnerabilities. They should support the continued operation or rapid recovery of crucial local, regional, or national surface transportation facilities. Furthermore, selected projects should utilize innovative and collaborative approaches to risk reduction, including the use of natural infrastructure, which is explicitly eligible under the program. Also called nature-based solutions, these strategies include conservation, restoration, or construction of riparian and streambed treatments, marshes, wetlands, native vegetation, stormwater bioswales, breakwaters, reefs, dunes, parks, urban forests, and shade trees. They reduce flood risks, erosion, wave damage, and heat impacts while also creating habitat, filtering pollutants, and providing recreational benefits. Projects in the PROTECT Discretionary Grant Program have the potential to demonstrate innovation and best practices that State and local governments in other parts of the country can consider replicating.

By funding projects that improve resilience to natural hazards and climate change impacts, the PROTECT Discretionary Grant Program aims to reduce damage and disruption to the transportation system, improve the safety of the traveling public, and improve equity by addressing the needs of disadvantaged communities that are often the most vulnerable to hazards. FHWA will seek to award projects to communities that demonstrate a strong need for the funding; the program includes set asides for rural communities and Indian Tribes.

¹ Funding for the PROTECT Discretionary Grants Program comes from two sources: 1) funding authorized as contract authority from the Highway Trust Fund, and 2) funding appropriated from the General Fund in annual appropriations acts. Funding authorized from the Highway Trust Fund is reduced based on the imposition of the obligation limitation contained in the annual appropriations acts. The BIL authorized \$250 million annually from the Highway Trust Fund for FY 2022 and FY 2023. An additional \$400 million was appropriated in FY 2022 and FY 2023 as shown in detail in Section B. Currently, \$300 million is authorized annually for FYs 2024, 2025, and 2026.

For the PROTECT Discretionary Grant Program, like the PROTECT Formula Program, grant funds can only be used for activities that are primarily for the purpose of resilience or inherently resilience related.

There are four categories of funding under the PROTECT Discretionary Grant Program. One category is for Planning Grants. The other three categories are for Resilience Improvement, Community Resilience and Evacuation Routes, and At-Risk Coastal Infrastructure projects, and throughout this NOFO are collectively referred to as Resilience Grants.

Planning Grants may be used to develop Resilience Improvement Plans in accordance with 23 U.S.C § 176(e)(2) for States and MPO; resilience planning, predesign, design, or the development of data tools to simulate transportation disruption scenarios, including vulnerability assessments; technical capacity building to facilitate the ability of the eligible entity to assess the vulnerabilities of its surface transportation assets and community response strategies under current conditions and a range of potential future conditions; or evacuation planning and preparation. (23 U.S.C. § 176(d)(3)).

Resilience Improvement Grants may be used to improve the ability of an existing surface transportation asset to withstand one or more elements of a weather event or natural disaster, or to increase the resilience of surface transportation infrastructure from the impacts of changing conditions, such as sea level rise, flooding, wildfires, extreme weather events, and other natural disasters. (23 U.S.C. § 176(d)(4)(A)).

Community Resilience and Evacuation Route Grants may be used for activities that strengthen and protect evacuation routes that are essential for providing and supporting evacuations caused by emergency events including activities that will improve evacuation routes, provide safe passage during an evacuation, and reduce the risk of damage to evacuation routes as a result of future emergency events. For routes that inadequately facilitate evacuations, including the transportation of emergency responders and recovery resources, activities include expanding capacity through installation of communication and intelligent transportation system equipment and infrastructure, counterflow measures, or shoulders, in addition to constructing new or redundant evacuation routes, acquiring evacuation route or traffic incident management equipment or signage, or ensuring access or service to critical destinations, including hospitals and other medical or emergency services facilities, major employers, critical manufacturing centers, ports and intermodal facilities, utilities, and Federal facilities. (23 U.S.C. § 176(d)(4)(B)).

At-Risk Coastal Infrastructure Grants may be used for activities to strengthen, stabilize, harden, elevate, relocate or otherwise enhance the resilience of highway and non-rail infrastructure, including: bridges, roads, pedestrian walkways, and bicycle lanes, and associated infrastructure, such as culverts and tide gates to protect highways that are subject to, or face increased long-term future risks of, a weather event, a natural disaster, or changing conditions, including coastal flooding, coastal erosion, wave action, storm surge, or sea level rise, in order to improve transportation and public safety and to reduce costs by avoiding larger future maintenance or rebuilding costs. (23 U.S.C. § 176(d)(4)(C)).

Together, these four funding categories will make surface transportation assets, communities, and coastal infrastructure more resilient to current and future weather events and natural disasters, such as severe storms, flooding, drought, levee and dam failures, wildfire, rockslides, mudslides or landslides, sea level rise, extreme weather, including extreme temperature, and earthquakes, as well as to allow for continued operation or rapid recovery of surface transportation systems.

Future NOFOs under the PROTECT Discretionary Grant Program may prioritize projects that are awarded planning grants funded under this NOFO for Resilience Grant awards.

Applicants may use one application to submit the same project for multiple funding categories if the project meets the eligibility criteria for multiple categories. For instance, a project to improve resilience of a coastal evacuation route could be eligible under all three Resilience Grant categories. Applicants would need to be sure to submit all the information needed for each category under which they are applying. Applicants should indicate which funding category or categories they are applying, however FHWA reserves the discretion to determine which funding category from which to make an award for a project, including a category for which the applicant did not apply.

Different funding amounts are available for the different categories, as shown below.

Category	Approximate Funding Available this Round
<u>Planning Grants</u>	Up to \$45 million
<u>Resilience Improvement Grants</u>	Up to \$638 million
<u>Community Resilience and Evacuation Route Grants</u>	Up to \$45 million
<u>At-Risk Coastal Infrastructure Grants</u>	Up to \$120 million

For more information about the eligible activities under each grant category see Section C of this NOFO.

FHWA will award FY 2022 and FY 2023 PROTECT Discretionary Grant Program grants to eligible applicants consistent with applicable statutory criteria and priorities, merit criteria, and the Administration priorities described within this NOFO.

Eligible Applicants

The PROTECT Discretionary Grant Program provides broad applicant eligibility for all levels of government to be direct recipients of funds. States, MPOs, local governments, and Indian Tribes can apply directly to FHWA. A Federal land management agency may be eligible if the agency applies jointly with a State or group of States. (See Section C for a complete list of eligible entities). FHWA encourages all eligible entities regardless of size, location, and experience administering Federal funding awards to apply for Federal assistance for Planning Grants and Resilience Grants under this NOFO.

To assist applicants, FHWA has developed application checklists which applicants are encouraged to use. These checklists can be found under the related documents tab for this NOFO on Grants.gov. The checklists are provided solely as a prompt for applicants and are not required for the grant application.

Alignment with Administration Priorities and Strategic Goals

The PROTECT Program (Formula and Discretionary) aligns with Biden-Harris Administration policies and priorities, including DOT's Strategic Plan goals of Safety, Economic Strength and Global Competitiveness, Equity, Climate and Sustainability, Transformation, and Organizational Excellence. The program will be implemented in line with Administration and Departmental policies and orders including the America the Beautiful Initiative; the Interagency [Memorandum of Understanding \(MOU\) on Promoting Equitable Access to Nature in Nature-Deprived Communities](#) signed by DOT; DOT Climate Action Plan; DOT Equity Action Plan; Executive Order (E.O.) 13985, Advancing Racial Equity and Support for Underserved Communities Through the Federal Government (86 FR 7009); E.O. 12898, Federal Actions to Address Environmental Justice in Minority Populations and Low-Income Populations (59 FR 7629); E.O. 14008, Tackling the Climate Crisis at Home and Abroad (86 FR 7619); and E.O. 14030, Climate Related Financial Risk (86 FR 27967). Also, consistent with E.O. 14030 and E.O. 13690, Establishing a Federal Flood Risk Management Standard and a Process for Further Soliciting and Considering Stakeholder Input (80 FR 6425), project applicants should be aware that DOT is in the process of developing guidance and considering updates to its floodplain requirements, including redefining the appropriate flood hazard area to account for future climate conditions.

The Department is committed to advancing safe, efficient transportation, including in the PROTECT Discretionary Grant Program. The National Roadway Safety Strategy (NRSS), issued January 27, 2022, commits DOT to respond to the current crisis in roadway fatalities by “taking substantial, comprehensive action to significantly reduce serious and fatal injuries on the Nation’s roadways” in pursuit of the goal of achieving zero roadway deaths through a Safe System Approach. The outcomes that are anticipated from the projects funded by PROTECT Discretionary Grant Program should align with the NRSS.

The Department seeks to fund projects under the PROTECT Discretionary Grant Program that reduce greenhouse gas emissions in the transportation sector, incorporate evidence-based climate resilience measures and features, reduce the lifecycle greenhouse gas emissions from the project materials, avoid adverse environmental impacts to air or water quality, wetlands, and endangered species, and address the disproportionate negative environmental impacts of transportation on disadvantaged communities, consistent with E.O. 14008, Tackling the Climate Crisis at Home and Abroad (86 FR 7619).

The Department seeks to award projects under the PROTECT Discretionary Grant Program that will create proportional impacts to all populations in a project area, remove transportation related disparities to all populations in a project area, and increase equitable access to project benefits, consistent with E.O. 13985, Advancing Racial Equity and Support for Underserved Communities Through the Federal Government (86 FR 7009).

The Department also seeks to award projects that address equity and environmental justice, particularly for disadvantaged communities that are marginalized, underserved, and overburdened by pollution and climate-related risks, and lack access to nature's benefits, consistent with E.O. 14008, Tackling the Climate Crisis at Home and Abroad (86 FR 7619).

The Department intends to use the PROTECT Discretionary Grant Program to support the creation of good-paying jobs with the free and fair choice to join a union and the incorporation of strong labor standards and training and placement programs, especially registered apprenticeships, in project planning stages, consistent with E.O. 14025, Worker Organizing and Empowerment (86 FR 22829), and E.O. 14052, Implementation of the Infrastructure Investment and Jobs Act (86 FR 64335). FHWA also intends to use the PROTECT Discretionary Grant Program to support wealth creation, consistent with the DOT's Equity Action Plan through the inclusion of local inclusive economic development and entrepreneurship such as the utilization of Disadvantaged Business Enterprises (DBE), Minority-owned Businesses, Women-owned Businesses, or 8(a) firms.

Please see Section E of this NOFO for more information on the FY 2022 and FY 2023 PROTECT Discretionary Grant Program selection criteria, the process for selecting projects, and how FHWA will evaluate applications in regard to these Administration priorities and goals. In addition, Section F.3 describes progress and performance reporting requirements for selected projects and the Administration's goals as appropriate.

2. Additional Information

The PROTECT Discretionary Grant Program is described in the Federal Assistance Listings under the assistance listing program title "Highway Planning and Construction" and assistance listing number 20.205.

FHWA is committed to considering project funding decisions holistically among the various discretionary grant programs available in BIL. FHWA also recognizes that applicants may be seeking funding from multiple discretionary grant programs and opportunities. An applicant may seek the same award amounts from multiple FHWA discretionary opportunities or seek a combination of funding from multiple DOT opportunities.

The applicant should identify any other DOT programs and opportunities they intend to apply for (or utilize if the Federal funding is already available to the applicant), and what award amounts they will be seeking, in the appropriate sections of this NOFO, including Section D.2.b.II. .

SECTION B – FEDERAL AWARD INFORMATION

1. Award Amount Available

For FY 2022 and FY 2023, up to \$848 million² is available for PROTECT Discretionary Grant Program awards under this notice. Of this amount, up to \$45 million is available for Planning Grants; up to \$638 million is available for Resilience Improvement Grants; up to \$45 million is available for Community Resilience and Evacuation Route Grants; and up to \$120 million is available for At-Risk Coastal Infrastructure Grants. Subject to the availability of funds, an eligible entity may request, and the Secretary may distribute funds for a grant under this subsection on a multiyear basis, as the Secretary determines to be necessary. Any award under this NOFO will be subject to the availability of funding.

2. Award Size

For FY 2022 and FY 2023 PROTECT Discretionary Grant Program Planning Grants, FHWA requests applications for a minimum award size of \$100,000 and no maximum award size.

For FY 2022 and FY 2023 PROTECT Discretionary Grant Program Resilience Improvement, Community Resilience and Evacuation Routes, and At-Risk Coast Infrastructure grants FHWA requests applications for a minimum award size of \$500,000 and no maximum award size. FHWA anticipates awarding between 30 to 40 Resilience Improvement, Community Resilience and Evacuation Routes, and At-Risk Coast Infrastructure grants.

If an applicant seeks an award amount that is less than the minimum, the application should include a justification for the requested amount. FHWA reserves the discretion to consider award sizes below the minimum award size listed above. Consideration will include the total amount of eligible PROTECT Discretionary Grant Program grant funds requested and consideration of the evaluation and selection criteria described in Section E.

² PROTECT Discretionary Grant Program funds from the Highway Account of the Highway Trust Fund are contract authority, which is reduced based on the imposition of the Federal-aid obligation limitation contained in the annual appropriations acts. Accordingly, the amount of \$250,000,000 authorized for each of FYs 2022 and 2023 is reduced to \$228,250,000 and \$219,500,000, respectively. Additional funds were appropriated from the General Fund in the FY 2022 and FY 2023 annual appropriations acts to the PROTECT Discretionary Grant Program, which are not subject to the obligation limitation (Consolidated Appropriations Act, 2023 (Public Law 117-103) and Consolidated Appropriations Act, 2023 (Public Law 117-328)).

3. Statutory Funding Provisions

a. Rural Set-Aside

Of the amounts made available for each fiscal year, the Secretary shall use not less than 25 percent for grants to projects located in areas that are outside an urban area with a population over 200,000. (23 U.S.C. § 176(d)(5)(H)(ii)).

b. Tribal Set-Aside

Of the amounts made available for each fiscal year, the Secretary shall use not less than 2 percent for grants to Indian Tribes (as defined in section 23 U.S.C. § 207(m)(1)) for eligible PROTECT Discretionary Grant Program Planning Grants, Resilience Improvement Grants, Community Resilience and Evacuate Routes Grants, and At-Risk Coastal Grants. (23 U.S.C. § 176(d)(5)(H)(iii)).

c. Unutilized Amounts

For any fiscal year, if the Secretary determines that the amount of the Rural Set-Aside or Tribal Set-Aside will not be fully utilized for PROTECT Discretionary Grant Program grants, the Secretary may reallocate the unutilized funds to provide grants to other PROTECT Discretionary Grant Program eligible entities. (23 U.S.C. § 176(d)(5)(H)(iv)).

d. Limitation on Intercity Rail Passenger Transportation

Limit on certain activities: not more than 25 percent will be used for a facility or service for intercity rail passenger transportation (as defined in 49 U.S.C. § 24102). (23 U.S.C. § 176(d)(5)(G)(ii)).

4. Availability of Funds

Selected projects will be funded with either FY 2022 or FY 2023 funds. Awards funded with FY 2022 funds must be obligated, via a signed grant agreement between FHWA and the recipient, by September 30, 2025. Awards funded with FY 2023 funds must be obligated via a signed grant agreement by September 30, 2026. Funding expenditure deadlines for FY 2022 and 2023 funds will be documented in the grant agreement depending on the fund source. To determine whether a project selected to receive a PROTECT Discretionary Grant Program award will be funded with FY 2022 or FY 2023 funds, FHWA will consider the estimated project start date provided in the application. Funds will be assigned after announcement of award during the grant agreement process.

Obligation for a PROTECT Discretionary Grant Program grant occurs when a selected applicant enters a project agreement with FHWA, and FHWA authorizes the project to proceed. For construction, this is generally after the applicant has satisfied applicable administrative requirements, including transportation planning and environmental review requirements and compliance with 23 Code of Federal Regulations (CFR) 635 subpart C “Physical Construction

Authorization” for State department of transportation (State DOT) applicants or 2 CFR 200.318 – 327 for non-State DOT applicants.

5. Type of Award

If the successful applicant is a State DOT or a State DOT will serve as a pass-through entity to a non-State DOT entity, PROTECT Discretionary Grant Program funds will be awarded upon the execution of a project agreement, which is a type of grant agreement for administration of funds to a State DOT in the Fiscal Management Information System (FMIS).

If the successful applicant is a non-State DOT, funds will be awarded upon the execution of a grant agreement with FHWA. A determination will be made if the successful applicant will administer the project or if the successful applicant included a request for an applicable State DOT, or FHWA Federal Lands Highway Division for Indian Tribe projects, to administer the project.

6. Period of Performance

If a grant recipient will be a State DOT or if a State DOT will serve as a pass-through to a non-State DOT entity awarded a grant, the DOT Payment System will be “Current Bill” in FMIS and the start of the period of performance will begin on the date PROTECT Discretionary Grant Program funds are obligated in FMIS and end on the project end date in FMIS.

If a grant recipient will be a non-State DOT entity, the DOT Payment System will be “DELPHI eInvoicing” and the start of the period of performance will begin on the date the grant agreement is executed by FHWA, which will obligate the PROTECT Discretionary Grant Program funds, and end according to the period of performance listed in the grant agreement.

SECTION C – ELIGIBILITY INFORMATION

To be selected for a PROTECT Discretionary Grant Program grant, an applicant must meet the eligibility requirements under this section including Eligible Entity; Eligible Project; Statutory Cost Sharing or Match requirements; and Statutory Applicant Requirements. Applications that do not meet these threshold eligibility requirements will not be evaluated under the criteria in Section E.1.

1. Eligible Applicants

- a. Planning Grants, Resilience Improvement Grants, and Community Resilience and Evacuation Route Grants have the same statutory rules for all applicants eligible to apply. At-Risk Coastal Grants have different statutory rules for applicant eligibility.

Eligible Applicants
Planning Grants, Resilience Improvement Grants, and
Community Resilience and Evacuation Route Grants
(23 U.S.C. § 176(d)(2))

1. A State or political subdivision of a State. (State includes the District of Columbia and Puerto Rico)
2. An MPO.
3. A unit of local government.
4. A special purpose district or public authority with a transportation function, including a port authority.
5. An Indian Tribe (as defined in 23 U.S.C. § 207(m)(1)).
6. A Federal land management agency that applies jointly with a State or group of States.
7. A multi-State or multijurisdictional group of entities described in (1) through (6).

Eligible Applicants
At-Risk Coastal Infrastructure Grants
23 U.S.C. § 176(d)(4)(C)(i)(I-VIII)

1. A State (including the U.S. Territories Virgin Islands, Guam, American Samoa, and the Commonwealth of the Northern Mariana Islands) in, or bordering on, the Atlantic, Pacific, or Arctic Ocean, the Gulf of Mexico, Long Island Sound, or one or more of the Great Lakes.
2. A political subdivision of a State described in (1.) above.
3. An MPO in a State described in (1.) above.
4. A unit of local government in a State described in (1.) above.
5. A special purpose district or public authority with a transportation function, including a port authority, in a State described in (1.) above.
6. An Indian Tribe in a State described in (1.) above.
7. A Federal land management agency that applies jointly with a State or group of States described in (1.) above.
8. A multi-State or multijurisdictional group of entities described in (1.) through (7.) above.

b. Joint Applications for Any PROTECT Discretionary Grant Program Funding Category

Multiple States or a multijurisdictional group that submits a joint application should identify a lead applicant as the primary point of contact. Joint applications should include a description of the roles and responsibilities of each applicant and should be signed or include a letter of support by each applicant. The applicant that will be responsible for financial administration of the project and the recipient of a PROTECT Discretionary Grant Program award must be an eligible applicant.

2. Cost Sharing or Matching

For PROTECT Discretionary Grant Program Planning Grants, the Federal share of an eligible planning activity shall be 100 percent. (23 U.S.C. § 176(d)(5)(E)(i)).

For PROTECT Discretionary Grant Program Resilience Grants, the Federal share of an eligible project shall not exceed 80 percent of the total cost of the project. (23 U.S.C. § 176(d)(5)(E)(ii)(I)). If an Indian Tribe (as defined in 23 U.S.C. § 207(m)(1)) receives a Resilience Improvement Grant and Community Resilience and Evacuation Route Grant or At-Risk Coastal Infrastructure Grants, on the determination of the Secretary of DOT, the Federal share of an eligible project may be up to 100 percent. (23 U.S.C. § 176(d)(5)(E)(ii)(II)).

Per 23 U.S.C. § 176(d)(5)(E)(iii), an eligible entity may use Federal funds other than a grant under the PROTECT Discretionary Grant Program to meet the non-Federal cost share requirement. However, the non-Federal share requirements for those other Federal funds must still be met. In addition to applicable Federal share requirements, eligibility requirements must also be met for all funds involved on the PROTECT Discretionary Grant Program project. Matching funds are subject to the same Federal requirements as PROTECT Discretionary Grant Program funds described in Section F.2. Unless otherwise authorized by statute, the cost-share may not be counted as non-Federal share for both a PROTECT Discretionary Grant Program project and another Federal program.

For the PROTECT Discretionary Grant Program, like the PROTECT Formula Program, grant funds can only be used for activities that are primarily for the purpose of resilience or inherently resilience related. As such, only the incremental cost of making assets more resilient are eligible for PROTECT funding. (For more information, see “Promoting Resilient Operations for Transformative, Efficient, and Cost Saving Transportation (PROTECT) Formula Program Implementation Guidance” July 29, 2022, p.14 available at (https://www.fhwa.dot.gov/environment/sustainability/resilience/policy_and_guidance/protect_formula.pdf)). The PROTECT Discretionary Grant Program project cost cannot include any previously incurred costs.

Applicants should use the following equation when determining the cost share for their Resilience grant project:

(PROTECT Grant + Other Funds for the Project) = Total PROTECT Discretionary Grant Program Project Cost

PROTECT Grant must be $\leq 0.80 \times$ Total PROTECT Discretionary Grant Program Project Cost

However, the non-Federal share shall be reduced as provided under 23 U.S.C. § 176(e)(1)(B)(i) and (ii) to a maximum reduction of 10 percent³ under 23 U.S.C. § 176(e)(1)(B)(iii)(I).

The non-Federal share of the cost of a project carried out using PROTECT Formula Program funds may be reduced by seven percentage points if the State has developed a Resilience Improvement Plan and prioritized the project on that Resilience Improvement Plan (23 U.S.C. 176(e)(1)(B)(i)); and three percentage points if a State Resilience Improvement Plan developed in accordance with Section 176(e) is incorporated (directly or by reference) into the metropolitan transportation plan under 23 U.S.C. 134 or the statewide long-range transportation plan under 23 U.S.C. 135, as applicable. (23 U.S.C.176(e)(1)(B)(ii)).

For example, a State may have a resilience project eligible under PROTECT that is prioritized in its Resilience Improvement Plan developed under Section 176(e). Provided the State meets all procedural requirements for the Resilience Improvement Plan, the scenario may result in a split of 87 percent PROTECT Formula Program funds and 13 percent non-Federal funds. In a variation of the same scenario, if the State also incorporates the Resilience Improvement Plan into the long-range statewide transportation plan, it may result in a split of 90 percent of PROTECT Formula Program funds and a non-Federal share of 10 percent.

Reduction of Non-Federal Share for Resilience Grants			
Project is Prioritized in a Resilience Improvement Plan (23 U.S.C. § 176(e)(1)(B)(i))		Resilience Improvement Plan is Incorporated in Other Planning (23 U.S.C. § 176(e)(1)(B)(ii))	
7% Reduction		3% Reduction	
Eligible Entity (Applicant)	Requirements	Eligible Entity (Applicant)	Requirements
State or MPO	1. Developed a Resilience Improvement Plan in accordance with 23 U.S.C. § 176(e)(2); and	State or MPO	The Resilience Improvement Plan developed in accordance with 23 U.S.C. § 176(e)(2) has been incorporated into the

³ A reduction of the non-Federal share under 23 U.S.C. § 176(e)(1)(B) shall not reduce the non-Federal share of the costs of a PROTECT Discretionary Grant Program Resilience Grant funded project to an amount that is less than zero. (23 U.S.C. § 176(e)(1)(E)(B)(iii)(II)).

	2. The project is prioritized on the Resilience Improvement Plan developed in (1.) above.		metropolitan transportation plan under 23 U.S.C. § 134 or the long-range statewide transportation plan under 23 U.S.C. § 135, as applicable.
Non-State or Non-MPO	Non-State or Non-MPO is located in a State or area served by a MPO that has: a. Developed a Resilience Improvement Plan in accordance with 23 U.S.C. § 176(e)(2); and b. The project is prioritized on the Resilience Improvement Plan developed in (1.) above.	Non-State or Non-MPO	Non-State or Non-MPO is located in a State or area served by a MPO that has incorporated the Resilience Improvement Plan developed in accordance with 23 U.S.C. § 176(e)(2) into the metropolitan transportation plan under 23 U.S.C. § 134 or the long-range statewide transportation plan under 23 U.S.C. § 135, as applicable.

For each project that receives a grant under this notice, FHWA expects the project to be completed using at least the amount of matching funds that was specified in the application.

See Section D.2.b.II for information about documenting cost sharing in the application.

3. Other

a. Eligible Projects

Eligible activities are different for each of the four types of grants under the PROTECT Discretionary Grant Program. Example projects can be found in the [PROTECT Formula Program Implementation Guidance](#).

Eligible Projects	
PROTECT Discretionary Grant Program Funding Category	Project Activities
Planning Grant 23 U.S.C. § 176(d)(3)	1. In the case of a State or MPO, developing a Resilience Improvement Plan under 23 U.S.C. § 176(e)(2); 2. Resilience planning, predesign, design, or the development of data tools to simulate transportation disruption scenarios, including vulnerability assessments; 3. Technical capacity building by the eligible entity to facilitate the ability of the eligible entity to assess the vulnerabilities of the surface transportation assets and community response strategies of the eligible entity under current conditions and a range of potential future conditions; or 4. Evacuation planning and preparation.
Resilience Improvement Grant 23 U.S.C. § 176(d)(4)(A)(ii)	One or more construction activities to improve the ability of an existing surface transportation asset to withstand one or more elements of a weather event or natural disaster, or to increase the resilience of surface transportation infrastructure from the impacts of changing conditions, such as sea level rise, flooding, wildfires, extreme weather events, and other natural disasters including: 1. Resurfacing, restoration, rehabilitation, reconstruction, replacement, improvement, or realignment of an existing surface transportation facility eligible for assistance under this title; 2. The incorporation of natural infrastructure; 3. The upgrade of an existing surface transportation facility to meet or exceed a design standard adopted by FHWA; 4. The installation of mitigation measures that prevent the intrusion of floodwaters into surface transportation systems; 5. Strengthening systems that remove rainwater from surface transportation facilities;

	6. Upgrades to and installation of structural stormwater controls; 7. A resilience project that addresses identified vulnerabilities described in the Resilience Improvement Plan of the eligible entity, if applicable; 8. Relocating roadways in a base floodplain to higher ground above projected flood elevation levels, or away from slide prone areas; 9. Stabilizing slide areas or slopes; 10. Installing riprap; 11. Lengthening or raising bridges to increase waterway openings, including to respond to extreme weather; 12. Increasing the size or number of drainage structures; 13. Installing seismic retrofits on bridges; 14. Adding scour protection at bridges; 15. Adding scour, stream stability, coastal, and other hydraulic countermeasures, including spur dikes; 16. Vegetation management practices in transportation rights-of-way to improve roadway safety, prevent against invasive species, facilitate wildfire control, and provide erosion control; and 17. Any other protective features, including natural infrastructure, as determined by the Secretary.
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Community Resilience and Evacuation Route Grant 23 U.S.C. § 176(d)(4)(B)(ii)	One or more projects that strengthen and protect evacuation routes that are essential for providing and supporting evacuations caused by emergency events, including a project that- 1. Is a Resilience Improvement Project under 23 U.S.C. § 176(d)(4)(A)(ii), if that eligible activity will improve an evacuation route; 2. Ensures the ability of the evacuation route to provide safe passage during an evacuation and reduces the risk of damage to evacuation routes as a result of future emergency events, including restoring or replacing existing evacuation routes that are in poor condition or not designed to meet the anticipated demand during an emergency event, and including steps to protect routes from mud, rock, or other debris slides; 3. If the eligible entity notifies⁴ the Secretary that existing evacuation routes are not sufficient to adequately facilitate evacuations, including the transportation of emergency responders and recovery resources, expands the capacity of evacuation routes to swiftly and safely accommodate evacuations, including installation of- (a) Communications and intelligent transportation system equipment and infrastructure; (b) Counterflow measures; or (c) Shoulders;
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⁴ See [PROTECT Formula Program Implementation Guidance](https://www.fhwa.dot.gov/environment/sustainability/resilience/policy_and_guidance/protect_formula.pdf) (https://www.fhwa.dot.gov/environment/sustainability/resilience/policy_and_guidance/protect_formula.pdf).

	4. Is for the construction of new or redundant evacuation routes, if the eligible entity notifies the Secretary that existing evacuation routes are not sufficient to adequately facilitate evacuations, including the transportation of emergency responders and recovery resources; 5. Is for the acquisition of evacuation route or traffic incident management equipment or signage; or 6. Will ensure access or service to critical destinations, including hospitals and other medical or emergency service facilities, major employers, critical manufacturing centers, ports and intermodal facilities, utilities, and Federal facilities.
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b. Statutory Eligibility Criteria for At-Risk Coastal Infrastructure Grants

- 1. Eligible Project.** The Secretary shall provide At-Risk Coastal Infrastructure grants for projects (I) that address the risks from a current or future weather event or natural disaster, including coastal flooding, coastal erosion, wave action, storm surge, or sea level change; and (II) that reduce long-term infrastructure costs by avoiding larger future maintenance or rebuilding costs. (23 U.S.C. § 176(d)(4)(C)(iv)).

FHWA will determine whether an application for an At-Risk Coastal Infrastructure grant meets the project eligibility projects by (1) addressing how the project will use data or evidence (qualitative or quantitative) that the proposed project will effectively improve resilience to current and future weather events and natural disasters and changing conditions, and (2) validating the documentation provided by the applicant demonstrating that the proposed project to be carried out by the eligible entity will reduce long-term infrastructure costs by avoiding larger future maintenance or rebuilding costs.

Documentation for (II) above should detail projected costs associated with build and no-build case scenarios. Applicants are encouraged to establish a clear baseline from which the incremental cost reduction of a proposed project can be derived. Maintenance and rebuilding costs associated with existing and future hazard risk and those that would accrue apart from hazard risks, altogether, should be provided. The build case should account for the non-hazard maintenance costs associated with the project as well as the expected reduction in maintenance or rebuilding costs. For projects involving the construction of new infrastructure, total non-hazard maintenance costs would be zero in the base case, so net non-hazard maintenance costs would typically be positive, reflecting the ongoing expenditures needed to maintain the new asset over its lifecycle. For projects intended to replace, reconstruct, or rehabilitate existing infrastructure, however, the net change in non-hazard maintenance costs under the proposed project will often be negative, as newer infrastructure requires less frequent and less costly maintenance to keep it in service than would an aging, deteriorating asset.

- ii. Eligible Activities.** Eligible activities for eligible At-Risk Coastal Infrastructure projects are: strengthening, stabilizing, hardening, elevating, relocating, or otherwise enhancing the resilience of highway and non-rail infrastructure, including bridges, roads, pedestrian walkways, and bicycle lanes, and associated infrastructure, such as culverts and tide gates to protect highways, that are subject to, or face increased long-term future risks of, a weather event, a natural disaster, or changing conditions, including coastal flooding, coastal erosion, wave action, storm surge, or sea level rise, in order to improve transportation and public safety and to reduce costs by avoiding larger future maintenance or rebuilding costs. (23 U.S.C. § 176(d)(4)(C)(iii)).

c. System Resilience for Resilience Grants

A PROTECT Discretionary Grant may include the use of natural infrastructure or the construction or modification of storm surge, flood protection, or aquatic ecosystem restoration elements functionally connected to a transportation improvement, such as:

1. Increasing marsh health and total area adjacent to a highway right-of-way to promote additional flood storage;
2. Upgrades to and installing of culverts designed to withstand 100-year flood events;
3. Upgrades to and installation of tide gates to protect highways; and
4. Upgrades to and installation of flood gates to protect tunnel entrances.

d. Eligible Facilities

A PROTECT Discretionary Grant Program Planning Grant or Resilience Grant may be awarded only for the following facilities:

1. a highway⁵ project eligible for assistance under this title;
2. a public transportation facility or service eligible for assistance under chapter 53 of title 49;
3. a facility or service for intercity rail passenger transportation (as defined in section 24102 of title 49); or
4. a port facility, including a facility that:
 - a. connects a port to other modes of transportation;
 - b. improves the efficiency of evacuations and disaster relief; or
 - c. aids transportation. (23 U.S.C. § 176(d)(5)(C)).

e. Statutory Requirements for Resilience Projects in Floodplains

If a PROTECT Discretionary Grant Program project is proposed to be carried out, in whole or in part, within a base floodplain, under 23 U.S.C. § 176(d)(5)(B)(ii) the application must include the following:

1. Identification of the floodplain in which the project is to be located and disclose that information to the Secretary in the application; and
2. Indication, whether, if selected, the eligible entity will implement one or more components of the risk mitigation plan under section 322 of the [Robert T. Stafford Disaster Relief and Emergency Assistance Act](#) (42 U.S.C. § 5165) with respect to the area.

Transportation infrastructure should not be built in floodplain areas if it can be avoided. If it cannot be avoided then the transportation infrastructure should be built to minimize impacts to floodplain function (to reduce risk to others) as well as built to protect transportation safety.

⁵ The term “highway” includes a road, street, and parkway; a right-of-way, bridge, railroad-highway crossing, tunnel, drainage structure including public roads on dams, sign, guardrail, and protective structure, in connection with a highway; and a portion of any interstate or international bridge or tunnel and the approaches thereto, the cost of which is assumed by a State transportation department, including such facilities as may be required by the U.S. Customs and Immigration Services in connection with the operation of an international bridge or tunnel. 23 U.S.C. § 101(a)(11)

Nature-based solutions may be one option for reducing impacts to the floodplain while protecting infrastructure.

f. Eligible Project Costs

For the PROTECT Discretionary Grant Program, like the PROTECT Formula Program, grant funds can only be used for activities that are primarily for the purpose of resilience or inherently resilience related. The focus should be on the incremental cost of making transportation assets more resilient. However, to the extent that the primary purpose of a project is to address resilience, it could be eligible to be fully funded (up to the applicable Federal share). Final determinations regarding whether an improvement is eligible for PROTECT Discretionary Grant Program funding are made by FHWA. Other project components that are not inherently resilience-focused should generally be funded from other sources, including other Title 23 programs.

Eligible project costs are different for planning grants vs. resilience grants.

Eligible Project Costs	
Planning Grants (23 U.S.C. § 176(d)(5)(F)(ii))	Resilience Grants (23 U.S.C. § 176(d)(5)(F)(i))
Development phase activities, including planning, feasibility analysis, revenue forecasting, environmental review, preliminary engineering and design work, other preconstruction activities, and other activities consistent with carrying out an eligible PROTECT Discretionary Grant Program Planning project.	1. Development phase activities, including planning, feasibility analysis, revenue forecasting, environmental review, preliminary engineering and design work, and other preconstruction activities; and 2. Construction, reconstruction, rehabilitation, and acquisition of real property (including land related to the project and improvements to land), environmental mitigation, construction contingencies, acquisition of equipment directly related to improving system performance, and operational improvements.

g. Limitation on Award

Limitations	
Planning Grants	Resilience Grants (23 U.S.C. § 176(d)(5)(G)(i) and (ii))
No limitations.	1. New Construction. No more than 40 percent of the amount of a PROTECT Discretionary Grant Program Resilience Grant may fund construction of new capacity; and 2. Planning Activities. No more than 10 percent of the amount of a PROTECT Discretionary Grant Program

	Resilience Grant may fund eligible development phase activities, including planning, feasibility analysis, revenue forecasting, environmental review, preliminary engineering and design work, and other preconstruction activities described under 23 U.S.C. § 176(d)(5)(F)(i).
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h. Definition of Urban and Rural Communities

For purpose of this notice, the PROTECT Discretionary Grant Program defines a rural area as any area outside of an Urbanized Area with a population over 200,000. (23 U.S.C. § 176(d)(5)(H)(ii)). An Urbanized Area is defined under 23 U.S.C. § 101(36) as an “[a]rea with a population of 50,000 or more designated by the Bureau of the Census.” For the PROTECT Discretionary Grant Program rural set-aside, Congress established a population threshold of 200,000 for eligibility out of urbanized areas with a population of 50,000. U.S. Census Bureau population data is available at <https://data.census.gov>.

SECTION D – APPLICATION AND SUBMISSION INFORMATION

1. Address to Request Application Package

All application materials may be found on Grants.gov at <http://www.grants.gov>.

Once at Grants.gov, select the Search Grants tab. Then enter one of the following:

- Opportunity Number: 693JJ323NF00013
- Opportunity Name: PROTECT Discretionary Grant Program
- Assistance Listing Number: 20.205 – Highway Planning and Construction

When at one of these pages, select the Opportunity, which will open to a page with several tabs. The first tab is a synopsis of the opportunity. Select the Application Package tab to download the forms needed to submit an FY 2022 and FY 2023 PROTECT Discretionary Grant Program grant applications.

For a TDD, please call (202) 366-3993. If potential applicants are unable to download the application package from the internet, they may send a written request for a paper copy to the below address. Requests should be sent to:

Robert Miller
Agreement Officer
Office of Acquisition and Grants Management
Federal Highway Administration
U.S. Department of Transportation
1200 New Jersey Ave., SE.
Washington, DC 20590
Email: PROTECTdiscretionary@dot.gov (preferred)
Phone: (202) 366-9167

Alternate:

Robin Hobbs
Agreement Officer/Team Leader
Office of Acquisition and Grants Management
Federal Highway Administration
U.S. Department of Transportation
1200 New Jersey Ave., SE.
Washington, DC 20590
Email: PROTECTdiscretionary@dot.gov (preferred)
Phone: (202) 366-4004

Office hours are from 7:30 a.m. to 4:00 p.m., Eastern Time,
Monday through Friday, except Federal holidays.

2. Content and Form of Application Submission

All applications must be submitted electronically through grants.gov.

The application package must consist of the following in this order:

a. Application Forms:

1. Standard Form 424

Note: Applicants may leave fields 5a, 5b, 6, 7, and 13 blank on the form.

Note: A number of separate PDF flat files must be attached in Item 15 to provide required project information that is not included in the body of the SF-424.

Instructions for adding Item 15 Attachments:

Click on “Add Attachments” in Item 15 to open the first pop-up window. Click “Add Attachment” and a second pop-up File Explorer/Directory window will appear, from which you can choose files to attach. Attachments can be added one at a time or all at once by holding down the CTRL key and selecting multiple files. Select “Open” to add the selected files as attachments.

Click “Done” to finalize the attachments.

Click “View Attachments” to see a list of files that have been added as attachments.

Required attachment(s): the Project Narrative must be added as Attachments to Item 15 of the SF-424:

2. Required Forms for Non-Construction Projects

Applicants proposing non-construction-related projects must submit the following forms: Standard Form 424A (Budget Information for Non-Construction Programs); and Standard Form 424B (Assurances for Non-Construction Programs).

Note: The Standard Form 424A should be supported by a budget narrative explaining each element of cost.

3. Required Forms for Construction Projects

Applicants proposing construction-related projects must submit the following forms:

Standard Form 424C (Budget Information Construction Programs); and Standard Form 424D (Assurances for Construction Programs).

Note: The Standard Form 424C should be supported by a budget narrative explaining each element of cost.

4. Mandatory Forms for All Applications

Grants.gov Lobbying Form; and
Disclosure of Lobbying Activities form (SF-LLL).

Note: All mandatory forms are available for download at
<https://www.grants.gov/web/grants/forms/sf-424-mandatory-family.html>.

b. Project Narrative

FHWA recommends that the project narrative follow the outline below. Following the outline will also assist evaluators in locating relevant information.

I. Basic Project Information – Description, Location, and Parties	See D.2.b.I
II. Grant Funds, Sources and Uses of all Project Funding	See D.2.b.II
III. Merit Criteria	See D.2.b.III
IV. Benefit-Cost Analysis	See D.2.b.IV
V. FHWA Priority Considerations	See D.2.b.V

The project narrative should provide information, including a table of contents, maps, and graphics as appropriate, necessary for FHWA to determine that the project satisfies project requirements described in Section C for the grant program and to assess the selection criteria specified in Section E.1.

To the extent practicable, applicants should provide supporting data and documentation in a form that is directly verifiable by FHWA. FHWA may, but is not required to, request additional information, including additional data, to clarify supporting data and documentation submitted in an application, but FHWA encourages applicants to submit the most relevant and complete information they can provide. FHWA also encourages applicants, to the extent practicable, to provide data in a form that is publicly available or verifiable. To ensure a fair and unbiased evaluation of applications submitted under this notice, FHWA will not request additional information to perfect incomplete applications.

FHWA recommends that the project narrative be prepared with standard formatting preferences (i.e., a single-spaced document, using a standard 12-point font such as Times New Roman, with 1-inch margins). The project narrative should not exceed 25 pages in length. Appendices may include documents supporting assertions or conclusions made in the 25-page project narrative and do not count towards the 25-page limit. If possible, Website links to supporting documentation should be provided rather than copies of these supporting materials. If supporting documents are submitted, applicants should clearly identify within the project narrative the

relevant portion of the project narrative that each supporting document supports. FHWA recommends using appropriately descriptive final names (e.g., “Project Narrative,” “Maps,” “Memoranda of Understanding and Letters of Support,” etc.) for all attachments. FHWA recommends applications include the following sections:

I. Basic Project Information – Project Description, Location, and Parties

The applicant should provide a concise description of the project including a discussion of the project’s history and a description of any previously incurred costs. The applicant may use this section to place the project into a broader context of other infrastructure investments being pursued by the project sponsor.

The applicant should describe in the application the project location, including a detailed geographical description of the proposed project, a map of the project’s location and connections to existing transportation infrastructure, and geospatial data describing the project location (shapefiles will be accepted under 200MB file size). The application should also identify:

- (a) If the project is located in a [Transportation Disadvantaged Census Tract](https://usdot.maps.arcgis.com/apps/dashboards/d6f90dfcc8b44525b04c7ce748a3674a) as identified in the arcgis online tool (<https://usdot.maps.arcgis.com/apps/dashboards/d6f90dfcc8b44525b04c7ce748a3674a>);
- (b) If there is an applicable Resilience Improvement Plan under 23 U.S.C. § 176(e)(2), and if so, whether the proposed project is included and whether the project is prioritized in the applicable plan.. (23 U.S.C. § 176(d)(4)(A)(iii)(I) and 23 U.S.C. § 176(e)(1)(B));
- (c) the floodplain in which the project is to be located and whether, if selected, the eligible entity will implement one or more components of the risk mitigation plan under section 322 of the Robert T. Stafford Disaster Relief and Emergency Assistance Act (42 U.S.C. 5165) with respect to the area (23 U.S.C. § 176(d)(5)(B)(ii)) (as applicable); and
- (d) in a detailed description, the projected changes in development patterns, demographics, and extreme weather events, identify the evidence and analysis upon which the project changes are based. Include verifiable information about the source of the information. Development pattern changes could come from a State DOT, MPO, Regional Planning Agency or local planning agency (applicable for a Community Resilience and Evacuation Route grant).

Information under (b) may be used for FHWA’s internal data tracking.

To the extent possible, FHWA encourages, but is not requiring, applicants to identify the degree to which the project may contribute to the functioning and growth of the economy, including the extent to which the project addresses congestion or freight connectivity, bridges service gaps in rural areas, or promotes greater land-use productivity, including main street revitalization or locally-driven density decisions that support equitable commercial and mixed-income residential development.

This section of the application should provide details about the lead applicant. The details should include the lead applicant's demonstrated experience with receipt and expenditure of Federal-aid highway program funds under title 23, U.S.C. If an application is rated highly under other criteria, but the applicant has no or limited experience with the receipt and expenditure of Federal-aid highway program funds, FHWA may contact the applicant prior to final selection of FY 2022 and FY 2023 awards to discuss technical resources that may be available to assist the applicant in obtaining the capacity and expertise to ensure compliance with all applicable Federal requirements and timely delivery of the project. The applicant should describe in detail all the other public and private parties who are involved in delivering the project, including a specific description of the role of each entity in delivering the project.

II. Grant Funds, Sources and Uses of all Project Funding

Project budgets should show how different funding sources will share in each activity and present those data in dollars and percentages. The budget should identify other Federal funds the applicant is applying for or has been awarded, if any, that the applicant intends to use. Funding sources should be grouped into three categories: non-Federal, PROTECT Discretionary Grant Program, and other Federal with specific amounts from each funding source.

The applicant should describe the PROTECT Discretionary Grant Program eligible project budget in detail and the plans for covering the full cost of the project from all sources. The applicant should provide the following:

- All Federal funds to be used for future eligible costs of the project, including the requested PROTECT Discretionary Grant Program amount, other Federal grants that have been awarded to the project or for which the project intends to apply in the future, and any Federal formula funds that have already been programmed for the project or are planned to be programmed for the project.
- For each category of Federal funds to be used for future eligible project costs, the amount, nature, and source of any required non-Federal match for those funds.

Grant funds can only be used for activities that are primarily for the purpose of resilience or inherently resilience related.

Detailed Budget: If the project contains components, the budget should separate the costs of each project component. If the project will be completed in phases, the budget should separate the costs of each phase. The budget should be detailed enough to demonstrate that the project satisfies the PROTECT Discretionary Grant Program statutory cost-sharing requirements described in Section C.2 and those associated with each category of Federal funding. The application should include information showing that the applicant has budgeted sufficient contingency amounts to cover unanticipated cost increases. In addition to the information enumerated above, this section should provide complete information on how all project funds may be used. For example, if a source of funds is available only after a condition is satisfied, the application should identify that condition and describe the applicant's control over whether it is satisfied. Similarly, if a source of funds is available for expenditure only during a fixed period,

the application should describe that restriction. Complete information about project funds will ensure that FHWA’s expectations for award execution align with any funding restrictions unrelated to FHWA, even if an award differs from the applicant’s request.

III. Merit Criteria

The applicant should describe how the proposal meets the Merit Criteria listed in Section E.1.

IV. Economic Analysis

This section outlines the requirements and describes the recommended approach for the completion and submission of a benefit-cost analysis (BCA) and other required costing information for the PROTECT Discretionary Grant Program.

Economic Analysis Requirements		
PROTECT Discretionary Grant Program Funding Category	Requirements	Purpose
Planning Grant 23 U.S.C. § 176(d)(3)	Benefit Cost Analysis NOT required.	N/A
Resilience Improvement Grant 23 U.S.C. § 176(d)(4)(A)(iii)	Benefit Cost Analysis required, unless the project is in a qualifying Resilience Improvement Plan (23 U.S.C. § 176(e)(2)).	Project Prioritization
Community Resilience and Evacuation Route Grant 23 U.S.C. § 176(d)(4)(B)(iii)	Benefit Cost Analysis required.	Project Prioritization
At-Risk Coastal Infrastructure Grant 23 U.S.C. § 176(d)(4)(C)(iv)	Benefit Cost Analysis NOT required however, must demonstrate that the project would reduce long-term infrastructure costs by avoiding larger future maintenance or rebuilding costs.	Project Eligibility

The purpose of the BCA is to enable FHWA to evaluate the project’s cost effectiveness by comparing its expected benefits to its expected costs, relative to a no-build scenario for the purpose of project prioritization. Projects requiring a BCA (see Economic Analysis Requirements table), should provide a narrative description of their BCA in their project narrative, as well as all relevant files used for their BCA, including any spreadsheet files

(unlocked) and technical memos describing the analysis. The spreadsheets and technical memos should present the calculations in sufficient detail and transparency to allow the analysis to be reproduced by FHWA evaluators.

The BCA should carefully document the assumptions and methodology used to produce the analysis, including a description of the baseline, the sources of data used to project the outcomes of the project, and the values of key input parameters. The analysis should provide present value estimates of a project's benefits and costs relative to a no-build baseline. To calculate present values, applicants should apply a real discount rate of 7 percent per year to the project's streams of benefits and costs, which should be stated in constant-dollar terms. The costs and benefits that are compared in the BCA must cover the same project scope.

Any benefits claimed for the project, both quantified and unquantified, should be clearly tied to the expected outcomes of the project including climate resilience benefits such as damages avoided. Projected benefits may accrue to both users of the facility and those who are affected by its use (such as through changes in emissions of greenhouse gases and other pollutants, access to services and green spaces, or availability of affordable housing or more affordable transportation choices). Usage forecasts applied in estimating future benefits should account for any additional demand induced by the improvements to the facility. While benefits should be quantified wherever possible, applicants are encouraged to describe other categories of benefits in the BCA that are more difficult to quantify and/or value in economic terms (e.g., benefits associated with nature-based solutions or for documenting benefit of evacuation during emergency events).

The BCA should include the full costs of developing, constructing, operating, and maintaining the proposed project compared to those costs for the existing facility, and the expected timing or schedule for costs in each of these categories. The BCA should include the costs that may accrue to those affected by the proposed project's construction or use. The BCA may also include the present discounted value of any remaining service life of the asset at the end of the analysis period.

Detailed guidance from DOT on estimating benefits and costs, together with recommended economic values for converting them to dollar terms and discounting to their present values, is available on the DOT Website (see <https://www.transportation.gov/mission/office-secretary/office-policy/transportation-policy/benefit-cost-analysis-guidance>).

Applications for At Risk Coastal Infrastructure grants must demonstrate that the project will reduce long-term infrastructure costs by avoiding larger future maintenance or rebuilding costs. See Section C.3.b. (Statutory Eligibility Criteria for At-Risk Coastal Infrastructure Grants) of this notice for information on required documentation.

Economic Analysis Flexibility for Indian Tribes

FHWA is providing additional flexibility to Indian Tribes regarding the information required in a BCA. At their discretion, Indian Tribe applicants may elect to provide raw data with their application to support the need for a project, without additional analysis. The applicant should

provide clear identification of the problem to be solved by the project (asset risk); estimation of the costs of the proposed project; current and projected future asset use (e.g., vehicle miles traveled, ridership, freight movement); and projection of the expected impacts of the project. An important benefit of projects that improve the resilience of existing transportation assets is travel time savings, for example from avoiding or reducing the amount of time an asset is closed. For such projects, applicants are encouraged to provide information that will enable FHWA to calculate travel time savings from a project. Such information may include the average amount of additional travel time from using the next best highway route. FHWA will use this data to develop estimates (given the data provided) of benefits and costs.

V. FHWA Priority Considerations

The applicant should describe how the proposal meets FHWA FY 2022 FY 2023 Priority Considerations listed in Section E.1.

3. UNIQUE ENTITY IDENTIFIER (UEI) AND SYSTEM FOR AWARD (SAM)

Effective April 4, 2022, the Data Universal Numbering System (DUNS) number is no longer required for entities doing business with the Federal Government and has been replaced by the SAM UEI. As of that date, applicants are required to use a UEI issued during the SAM.gov registration process. Active registrants in SAM.gov have had their SAM UEI automatically assigned, and it is currently viewable within SAM.gov. There is no additional action necessary for registered entities to take to obtain their SAM UEI.

Each applicant is required to:

- a. Be registered in SAM before submitting their application;
- b. Provide a valid UEI in their application; and
- c. Continue to maintain an active SAM registration with current information at all times during which the applicant has an active Federal award or an application or plan under consideration by a Federal awarding agency.

Please see <https://www.grants.gov/web/grants/applicants/applicant-faqs.html#UEI> for more information on the transition from DUNS to SAM UEI, including what UEI to enter into the UEI field on grants.gov and on application package forms.

FHWA may not make a Federal award to an applicant until the applicant has complied with all applicable UEI and SAM requirements. If an applicant has not fully complied with the requirements by the time the Federal awarding agency is ready to make a Federal award, the Federal awarding agency may determine that the applicant is not qualified to receive a Federal award and use that determination as a basis for making a Federal award to another applicant.

NOTE TO APPLICANTS: The SAM.gov requires the registrant to provide a UEI number to complete the registration. These processes can take several weeks to complete so they should be started well before the application deadline.

4. SUBMISSION DATES AND TIMES

Applications must be submitted electronically through grants.gov no later than 11:59 p.m., Eastern Standard Time on August 18, 2023 (the “application deadline”).

5. INTERGOVERNMENTAL REVIEW

An application under this NOFO is not subject to the Intergovernmental review under E.O. 12372.

6. FUNDING RESTRICTIONS

- a. Pre-PROTECT Discretionary Grant Program award costs will not be reimbursed.
- b. Grant funds may not be used to support or oppose union organizing, whether directly or as an offset for other funds.
- c. For more information on Eligible Project Costs see Section C.3.e.

7. OTHER SUBMISSION REQUIREMENTS

a. Scalable Project Options

Applicants are encouraged to identify scaled funding options in case insufficient funding is available to fund a project at the full requested amount. If an applicant advises that a project is scalable, the applicant must provide an appropriate minimum funding amount that will fund an eligible project that achieves the objectives of the program and meets all relevant program requirements. The applicant must provide a clear explanation of how the project budget would be affected by a reduced award. FHWA may award a lesser amount whether or not a scalable option is provided.

b. Submission Location

Applications must be submitted to *Grants.gov*.

c. Consideration of Applications

Only applicants who comply with all submission deadlines described in this notice and electronically submit valid applications through *Grants.gov* will be eligible for award. Applicants are strongly encouraged to make submissions in advance of the deadline.

d. Late Applications

Applicants experiencing technical issues with Grants.gov that are beyond the applicant's control must contact PROTECTdiscretionary@dot.gov prior to the application deadline with the username of the registrant and details of the technical issue experienced. The applicant must provide:

- (1) Details of the technical issue experienced;
- (2) Screen capture(s) of the technical issues experienced along with corresponding Grants.gov "Grant tracking number;"
- (3) The "Legal Business Name" for the applicant that was provided in the SF-424;
- (4) The Point of Contact name submitted in the SF-424;
- (5) The UEI associated with the application; and
- (6) The Grants.gov Help Desk Tracking Number.

To ensure a fair competition of limited discretionary funds, the following conditions are not valid reasons to permit late submissions: (1) failure to complete the registration process before the deadline; (2) failure to follow Grants.gov instructions on how to register and apply as posted on its Website; (3) failure to follow all instructions in this NOFO; and (4) technical issues experienced with the applicant's computer or information technology environment.

After FHWA reviews all information submitted and contacts the Grants.gov Help Desk to validate reported technical issues, FHWA staff will contact late applicants to approve or deny a request to submit a late application through Grants.gov. FHWA will not accept appeals of FHWA's decision to approve or deny a request for a late application. If the reported technical issues cannot be validated, late applications will be rejected as untimely.

e. Compliance with Section 508 of the Rehabilitation Act of 1973

FHWA encourages applicants to submit documents that are compliant with Section 508 of the Rehabilitation Act of 1973. Section 508 guidelines are available at <https://www.access-board.gov/ict/>.

f. Sharing of Application Information

The Department may share application information within the Department or with other Federal Agencies if the Department determines that sharing is relevant to the respective program's objectives.

SECTION E – APPLICATION REVIEW INFORMATION

1. Criteria

FHWA will evaluate applications using the criteria below. The criteria for Planning grants are different from Resilience grants.

For Planning Grants, the merit criteria are:

1. Program Alignment
2. Schedule and Budget
3. Public Engagement, Partnerships and Collaboration.
4. Innovation

For Resilience Grants, the merit criteria are:

1. Vulnerability and Risk
2. Criticality to Community
3. Design Elements
4. Public Engagement, Partnerships and Collaboration
5. Equity and Justice⁴⁰
6. Climate Change and Sustainability
7. Schedule and Budget
8. Innovation

The Review Team will consider whether the application narrative is responsive to the merit criteria using the rating scale below:

Rating Scale	High	Medium	Low	Non-Responsive
Description	The application is substantively and comprehensively responsive to the criterion. It makes a strong case, including sufficient detail and evidence, that it will advance all of the areas described in the criterion descriptions.	The application is moderately responsive to the criterion. It makes a moderate case, including some detail and evidence, that it will advance all or some of the areas described in the criterion descriptions.	The application is minimally responsive to the criterion. It makes a weak case, including insufficient detail or evidence, about advancing the areas described in the criterion descriptions.	The narrative indicates the proposal is counter to the criterion or does not contain sufficient information. It does not advance or may negatively impact criterion goals.

a. Planning Grants

Criterion #1: Program Alignment

FHWA will evaluate the alignment of the proposed Planning Grant activity with the purpose of PROTECT Discretionary Grant Program as described in 23 U.S.C. § 176(d)(3). Because there are four statutorily distinct categories of eligible Planning grant activities, FHWA will evaluate the program alignment of each planning grant application against the specific planning grant activity the applicant identified in the application.

For all Planning grant applications, FHWA will evaluate whether the application includes a description of how completion of the Planning Grant will contribute to the applicant's near- and long-term system resilience plans. For example, for a proposed grant for technical capacity building, FHWA will evaluate how the applicant will maintain the technical capacity it needs to assess the vulnerabilities of surface transportation assets and community response strategies, including the development of high-quality technical capacity building programs with supportive services to help train, place, and retain people with the technical capacity needed for the long-term. The planning activity should support multimodal transportation networks that enable all people to reach their desired destination safely, affordably, and with a comparable level of efficiency and ease. The need for the PROTECT Discretionary Grant Program Planning project should be of critical importance in supporting the continued operation or rapid recovery of crucial local, regional, or national surface transportation assets and facilities.

Resilience Improvement Plan: The application provides detailed information about how the planning project will:

- identify short- and long-range planning activities and investments with respect to the multimodal resilience of surface transportation;
- include input from Tribes or Federal Land managers as well as local governments; and
- include a systemic approach to transportation system resilience and critical needs including a risk-based assessment of vulnerabilities of transportation assets and systems to current and future weather events and natural disasters.

Resilience Planning, Predesign, Design, or Data Tools: The application:

- provides a detailed description that identifies the type and source of relevant climate data sets, information resources, and decision-support tools that will be utilized to inform the multimodal planning process;
- describes in detail how models and assessments will utilize data on past and future conditions and address critical needs;
- describes how the assessment will consider protective services offered by existing natural systems (for example flood protection offered by wetlands) and/or how the assessment will consider options for using nature-based solutions to provide protective services;
- provides a detailed plan of how the applicant will coordinate the project with other existing plans such as State and local hazard mitigation plans;
- if the planning project is to develop a tool: 1) identifies and describes in detail the disciplines that will utilize the tool and how the tool will be developed to apply easily

- to these disciplines (end users); and 2) demonstrates in detail how the data or tool format is most useful including a discussion about GIS applications or incorporation into currently used models; and
- if the planning project is to develop predesign or design elements: 1) describes methods for developing resilience solutions; 2) describes how nature-based solutions will be considered.

Technical Capacity Building: The application:

- provides a detailed description of the training programs the applicant intends to implement;
- includes a detailed assessment of current staff knowledge on climate change vulnerability and resilience and capability to utilize that knowledge with actions to strengthen current practices; and
- provides a detailed description of the applicant's near- and long-term plan to ensure it maintains the technical capacity that will be developed with the use of PROTECT Planning funds including the development of high-quality technical training programs with support services, to help train, place, and retain people.

Evacuation Planning and Preparation: The application:

- documents potential future disasters or conditions that demonstrate the need for multimodal evacuation routes as part of community resilience;
- includes verifiable information about community growth (past and future), critical, needs, and anticipated future traffic volumes and accommodation of transit and active transportation modes;
- provides a detailed discussion of considerations that will be to current and future conditions in the facility design;
- explains in detail how the project will leverage planning information from State and local level government such as hazard mitigation and emergency management planning; and
- includes a detailed post-project description of how the Planning Grant will contribute to the applicant's near- and long-term system resilience plans.

Criterion #2: Schedule and Budget

FHWA will rate the responsiveness to the Schedule and Budget criterion as follows:

- The Application presents a detailed schedule and budget and amount of funds for each major project activity, milestone, and deliverable.
- The application includes a detailed budget that shows the total project cost, including the amount requested from the PROTECT Discretionary grant program and all other funding sources and amounts. (Other funding sources, as appropriate, include Other Federal funds; State funds; Tribal funds; local funds; and other funds such as private funds. If the funding was received, please provide the date of award and how the funds have been or are expected to be used on the project.)
- The project schedule presented in the application includes the anticipated start and end dates for each activity, as well as a detailed post-Planning grant description of

how the Planning Grant will contribute to short, intermediate, and long-term system resilience.

- The schedule and budget are feasible, reasonable, and commensurate with the work described in the application.

Criterion #3: Public Engagement, Partnerships and Collaboration

FHWA will rate the responsiveness to the Public Engagement, Partnerships, and Collaboration criterion, as follows:

- The application provides specific strategies and actions to be incorporated in the planning process to ensure meaningful public engagement, partnering, and collaboration.
- The planning process will engage a wide range of applicable partners and stakeholders and incorporates opportunities to for public collaboration. Example partners include community-based organizations, public agencies (e.g., local, regional, State, and Federal), and relevant sectors (e.g., emergency management, environmental, planning, floodplain management, health, housing and development, private.
- The application outlines an engagement process that will involve a range of relevant experts in diverse areas and demonstrate inclusion of disadvantaged populations or communities that have or will likely experience a disproportionate impact from a natural disaster or catastrophic event.
- The application demonstrates how input provided will be considered during the planning grant implementation process.

Criterion #4 Innovation

FHWA will rate the responsiveness to the Innovation criterion, as follows:

- The planning project employs innovative or novel partnerships, technologies, and techniques that will be used to analyze risks and risk reduction strategies; or
- The planning project analyzes the potential use of nature-based solutions and/or builds technical capacity for the use of nature-based solutions and access to nature; or
- The planning project utilizes partnerships that demonstrate potential best practices in cost-sharing and use of technical expertise for the best outcome of the project (for example, involvement of partners with expertise in nature-based solutions such as natural resource management agencies); or
- The planning project demonstrates and documents innovative techniques and best practices that other parts of the country can consider replicating.

Overall Planning Grant Application Rating

FHWA's First-level review team will assign an overall application merit rating using the following methodology:

Overall Merit Rating	Individual Criteria Ratings
Highly Recommended	• ‘High’ ratings on at least three [3] of four [4] criteria; and • ‘High’ or ‘Medium’ rating on Equity and Justice⁴⁰ (Indian Tribes are assumed ‘High’ for Equity and Justice⁴⁰); and • Zero ‘Non-Responsive’ ratings
Recommended	• At least three [3] ‘Medium’ or ‘High’ ratings; and • Zero ‘Non-Responsive’ ratings.
Not Recommended	• One or more ‘Non-Responsive’ ratings

FHWA FY 2022 and FY 2023 Planning Grant Priority Considerations

After completing the merit review, among projects of similar merit, FHWA will prioritize projects that demonstrate:

- Exceptional benefits under merit criteria #3 Public Engagement, Partnerships and Collaboration; or
- Strong need for funding; but for the award of a PROTECT Discretionary Grant Program Planning Grant, the applicant would not be able to begin or complete the activity. For State DOT applicants, this could be demonstrated by showing that the need exceeds the amount provided under the PROTECT Formula Program.

b. Resilience Improvements, Community Resilience and Evacuation Routes, and At-Risk Coastal Infrastructure Grants

FHWA will evaluate Resilience Grant applications using the following Merit Criteria for FY 2022 and FY 2023.

Criterion #1: Vulnerability and Risk

FHWA will evaluate the degree to which the application:

1. Describes in detail with verifiable underlying data and science-based information the relevant current or future weather events, natural disasters, or changing conditions that the project area is exposed to that the project will address (e.g., severe storms, flooding, drought, levee and dam failures, wildfire, rockslides, mudslides, sea level rise, extreme weather, including extreme temperatures, and earthquakes);
2. Assesses, in detail with verifiable data and information, each element, as applicable, of vulnerability - (2a) exposure, (2b) sensitivity, and (2c) adaptive capacity (see table below); and

3. Demonstrates that there is high risk to the transportation asset or system (risk is a function of the likelihood of negative impact and the consequence should the impact occur).

For Community Resilience and Evacuation Route grant applications, the application should address current and future vulnerabilities to an evacuation route because of future occurrence or recurrence of emergency events that are likely to occur in the geographic area in which the evacuation route is located; and projected changes in development patterns, demographics, and extreme weather events based on the best available evidence and analysis.

To assist applicants, the table below defines exposure, sensitivity, and adaptive capacity and provides instructions on the information applicants should address in the application.

Vulnerability Elements	
Element	Instructions
Exposure means the degree to which an asset or system is located in an area experiencing direct effects of climate change, extreme weather events, natural disasters, and changing conditions.	Describe how the project: (1) will address current or future weather events, natural disasters, or changing conditions that the project area is exposed to (e.g., severe storms, flooding, drought, levee and dam failures, wildfire, rockslides, mudslides, sea level rise, extreme weather, extreme temperature, earthquakes); and (2) will reduce the impacts, risks, and consequences of exposure.
Sensitivity means how an asset or system responds to, or is affected by, exposure to a climate-related or other natural hazard stressor including but not limited to flooding, landslides, lava and wildfires.	Describe how the project will seek to reduce the sensitivity of the asset or system to the stressors identified. This may include a description of how the project would improve the ability of the asset or system to reduce or avoid potential impacts from stressors. For example, sensitivity data and information could include thresholds (empirical or observed) to potential stressors. The sensitivity data could include the asset age or condition and vary with the applicable climate stressor thresholds.
Adaptive Capacity means the ability of a transportation asset or system to adjust, repair, or flexibly respond to damage caused by climate variability, extreme weather, or other natural disaster.	Describe how the project will improve the ability of the asset or system to withstand, adjust to, or respond to impacts. For example, an application could describe how raising a bridge and associated pedestrian, bicycle, and green infrastructure would allow for continued operation of the system during heavy precipitation events. Raising the bridge could also reduce travel time because of detour route use, which could be lengthy and is often a measure of asset adaptive capacity. Fast Recovery also utilizes efficient emergency response, inspection, and evacuation. A project utilizing nature-based solutions, such as restoration of natural dunes to protect from

Vulnerability Elements	
Element	Instructions
	wave damage, could improve the ability of the asset to recover more quickly and reduce loss of functionality and community impact.

Criterion #2: Criticality to Community

FHWA will evaluate criticality as follows:

1. The application demonstrates the critical need for the project, based on the current or future events the project will address. Critical need is defined as the importance of the project in supporting the continued operation or rapid recovery of crucial local, regional, or national surface transportation assets and facilities served by those assets in the community. Critical need could be demonstrated through indicators such as average daily traffic, evacuation needs, or freight volume.
2. The application identifies critical facilities and destinations that the project provides continued access to. Example facilities and destinations include homes, businesses hospitals and other medical or emergency service facilities, major employers, critical manufacturing centers, ports, intermodal facilities or nexus points, utilities, and Federal facilities.
3. The project is timely in addressing risks to the community. Consequences are severe if resilience improvements are not made in the near term.
4. The project is critical to supporting community functions (such as access for disadvantaged populations, economic activities, emergency operations during evacuations).

Criterion #3: Design Elements

FHWA will evaluate proposed project design elements to specifically address relevant current or future weather events, natural disasters, or changing conditions over the anticipated service life. Applications should identify the anticipated service life of the facility or asset and identify design element(s), as applicable to the affected mode(s) of transportation, to address current and future vulnerabilities over that service life. Applications should document the source of relevant evidence-based climate data, downscaled projections, and scenarios used or proposed for use as basis for climate resilience design measures or features. Applicants are strongly encouraged to use best-available climate data sets, information resources, and decision-support tools (including FHWA and other Federal resources) to assess the climate-related vulnerability and risk and to develop or deploy resilience solutions to address those risks over the anticipated infrastructure service life.

1. The application describes in detail how the anticipated service life of the facility or asset was determined.

2. The design elements will reduce current and future vulnerabilities and risks. The application identifies the design element that will address each current or future vulnerability identified under Criterion #1 up to the anticipated service life of the facility or asset. If the project is in pre-design or design, the application identifies and provides a detailed discussion about what methodology(ies) will be used to identify the measures or features to address the current and future vulnerabilities. If the project is post-design, the application provides a detailed description about how a determination was made that the design element identified above will reduce current and future vulnerabilities.
3. The project considers and uses, as appropriate, nature-based solutions. Examples of nature-based solutions include wetland buffers, marsh breakwaters, sea grass plantings, dune restoration, landscaping such as greenscaping, rain gardens and bioswales, riverbank protection such as vegetated erosion control or native material revetments and adding stream sinuosity.
4. The project will improve safety for all users of the transportation asset or system by reducing current and future vulnerabilities. The applicant has assessed safety needs and how PROTECT Discretionary Grant funds would contribute to improving safety through the anticipated service life of the asset or system.
5. The application includes a detailed maintenance plan for the project describing how the funded design elements will be maintained for the anticipated life of the facility, in order to ensure they continue to reduce the vulnerabilities identified earlier. The plan identifies which entity will be responsible for maintaining the project, as well as estimated maintenance costs over the life of the project, and sources to fund those costs.

Visit <https://www.fhwa.dot.gov/environment/protect/discretionary> for a list of resources and tools to help assess vulnerabilities, identify critical assets, and assess the resilience of projects to climate change, extreme weather, and other natural hazards. The questions in the table below are provided to help applicants develop their response to the design elements criterion.

FHWA's evaluation of design elements as described herein as part of the review and selection of projects under this notice is not and should not be interpreted as FHWA approval or concurrence of final project design. The purpose of this criterion is to evaluate whether and to what degree an applicant is evaluating design elements to reduce future vulnerabilities. This criterion is intended to evaluate elements that would be considered during the design phase of the project. Approval of the final and specific design elements that will be used in the construction of the project will follow title 23 or 2 CFR part 200 requirements depending on the type of applicant.

Example Questions for Applicants to Consider in Developing their Response to the Design Elements Criterion
Can the facility be sited or relocated outside of an area with a current or projected future hazard or climate impact (e.g., 100-year flood, increasing flood return intervals)? Where relevant, does the project reduce impacts to floodplain function?
If applicable, is the project sized to accommodate increases in sea level? If yes, what sea level rise projection will be used and how will this be incorporated into project design?
Does the project take into account increased future stormwater flows, ex., the need for additional drainage infrastructure or upsized culverts, elevated bridges and approaches?
Will certain design elements increase ability of the asset to continue to serve its primary function, and/or support emergency evacuation and post disaster relief, during and after weather events and natural disasters? How?
Does the project add redundancy to the transportation system? And does this redundancy enhance emergency evacuations?
Will the project use a nature-based solution? Or explain why a nature-based solution would not be a solution. (e.g., wetland buffers, marsh breakwaters, reefs, sea grass plantings, dune restoration, adding stream sinuosity, etc.)
Does the project employ the use of innovative or novel technologies, such as improved roadway subgrades, heat-resistant pavement mixes, natural infrastructure or permeable pavement?

Criterion #4: Public Engagement, Partnerships and Collaboration

The application provides a detailed public engagement, partnerships and collaboration plan that either addresses each plan component (1) through (5) below or provides a detailed explanation why certain plan components are not applicable for the project.

1. How public engagement will be conducted demonstrating engagement of diverse input such as community-based organizations during project planning and consideration of how input provided will be considered during the project decision-making process;
2. Partnerships and collaboration with community stakeholders;
3. Partnerships and collaboration with other agencies (e.g., State, local, regional, Federal);
4. Partnerships and collaboration across relevant sectors (e.g., emergency management, environmental, planning, floodplain management, health, housing and development, private sector); and
5. For (1) through (4) above, as applicable, the roles of these entities in the planning, design, construction, and operation of the project.

Criterion #5: Equity and Justice⁴⁰

Applicants should demonstrate how meaningful public involvement⁶, inclusive of disadvantaged communities, will occur throughout a project's life cycle. Projects should demonstrate, to the extent possible, that outcomes should target at least 40 percent of benefits towards disadvantaged communities, including communities underserved by affordable transportation or overburdened. Applicants should address how project benefits will apply to disadvantaged communities to (1) reduce the impacts of emergency events, (2) improve access to critical community services, (3) connect Americans to good-paying jobs, (4) reduce current or potential burdens, and (5) improve access to resources and quality of life.

The application:

1. includes an equity assessment that provides a detailed discussion about whether and, if so, how the project will create proportional impacts and remove transportation related disparities to all populations in a project area;
2. demonstrates in detail how meaningful public engagement with disadvantaged communities will occur throughout a project's life cycle;
3. provides a detailed discussion about each project benefit 1-5 discussed under this criterion and how it is a benefit of the project or not;
4. demonstrates in detail that the outcome of the project targets no less than 40 percent of benefits 1-5 towards disadvantaged communities.

Applications from Indian Tribes are not required to provide information on equity and Justice⁴⁰. FHWA will consider project applications from Indian Tribes to inherently benefit a Justice⁴⁰ community and will rate them high for this criterion.

Applications should address equity and environmental justice, particularly for communities that have experienced disproportionate impacts from emergency events or from past resilience improvement activities. The [Climate and Economic Justice Screening Tool](#) (CEJST), a new tool by the White House Council on Environmental Quality (CEQ), aims to help Federal agencies identify disadvantaged communities as part of the Justice⁴⁰ Initiative. Applicants should use CEJST as the primary tool to identify disadvantaged communities (Justice 40 communities). For the purpose of assessing the components of project benefits and burdens, applicants should also use [DOT's transportation disadvantage tool](#).

Criterion #6: Climate Change and Sustainability

The application uses data-driven and evidence-based methods to demonstrate that the project will:

1. Significantly reduce greenhouse gas emissions in the transportation sector;
 - a. Utilizing fiscally responsible land use;

⁶ <https://www.transportation.gov/priorities/equity/promising-practices-meaningful-public-involvement-transportation-decision-making>

- b. Increasing the use of energy-efficient modes of transportation like transit, rail, and active transportation; transitioning to clean vehicles and fuels, including electrification; or
 - c. Incorporating carbon-reducing uses of the rights-of-way or other carbon reduction strategies.
- 2. Incorporate evidence-based climate resilience measures or features such as constructing or upgrading infrastructure using the Federal Flood Risk Management Standard and monitoring performance of climate resilience measures.
- 3. Reduce the lifecycle greenhouse gas emissions from the project material; such as by:
 - a. reducing the emissions from the manufacturing, installation, maintenance, and/or disposal of those materials; or
 - b. including incentives for low carbon materials in project procurement.
- 4. Address the disproportionate negative environmental impacts of transportation on disadvantaged communities such as considering the benefits and burdens a project may create, and what communities would be most affected; and
- 5. Avoid adverse environmental impacts to air or water quality, wetlands, and endangered species, such as through reduction in Clean Air Act criteria pollutants and greenhouse gases, improved stormwater management, or improved habitat connectivity.

Criterion #7: Schedule and Budget

FHWA will rate the responsiveness to the Schedule and Budget criterion as follows:

- 1. The application presents a detailed schedule and budget and amount of funds for each major project activity, milestone, and deliverable.
- 2. For the PROTECT Discretionary Grant Program, like the PROTECT Formula Program, grant funds can only be used for activities that are primarily for the purpose of resilience or inherently resilience related. The budget should detail the incremental cost of making transportation assets more resilient. Other project components that are not inherently resilience-focused should generally be funded from other sources, including other Title 23 programs. To the extent that the primary purpose of a project is to address resilience, it would be eligible to be fully funded (up to the applicable Federal share)
- 3. The application includes a detailed budget that shows the total project cost, including the amount requested from the PROTECT Discretionary Grant Program and all other funding sources and amounts. (Other funding sources, as appropriate, include other Federal funds; State funds; Tribal funds; local funds; and other funds such as private funds. If the funding was received, please provide the date of award and how the funds have been or are expected to be used on the project.)
- 4. The project schedule presented in the application includes the anticipated start and end dates for each major activity or milestone.
- 5. The schedule and budget are feasible, reasonable, and commensurate with the work described in the application.

Criterion # 8: Innovation

FHWA will rate the responsiveness to the Innovation criterion as follows:

1. The project employs innovative or novel partnerships, technologies, or techniques that will be used to mitigate risk of structural failure and/or increase reliability of the surface transportation asset throughout its anticipated service life.
2. The project uses nature-based solutions to improve resilience. Examples of nature-based solutions to manage stormwater include permeable pavement, stormwater harvest and reuse, and landscaping (greenspaces, rain gardens, bioswales, etc) to absorb and slow the flow of stormwater. These stormwater nature-based solutions reduce downstream flooding and filter pollutants. Nature-based solutions such as river training, bank protection, and riparian treatments, can be used in riverine environments near highways and rail to stabilize slopes, reduce erosion, and reduce flooding of transportation infrastructure. Additional examples of nature-based solutions include wetland buffers, marsh breakwaters, stormwater bioswales, sea grass plantings, dune restoration, and adding stream sinuosity.
3. The project utilizes partnerships that demonstrate potential best practices in cost-sharing and use of technical expertise for the best outcome of the project (for example, involvement of partners with expertise in nature-based solutions such as natural resource management agencies).
4. The project demonstrates and documents innovative techniques and best practices that other parts of the country can consider replicating.

Overall Merit Criteria Rating

FHWA's first-level review team will assign an overall merit criteria rating on the following basis:

Overall Merit Rating	Individual Criteria Ratings
Highly Recommended	• 'High' ratings on at least six [6] of eight [8] criteria; and • 'High' or 'Medium' rating on Equity and Justice⁴⁰ (Indian Tribes are assumed 'High' for Equity and Justice⁴⁰); and • Zero 'Non-Responsive' ratings
Recommended	• At least five [5] 'Medium' or 'High' ratings, and • Zero 'Non-Responsive' ratings.

Not Recommended	Fewer than five [5] ‘Medium’ or ‘High’ ratings or • One or more ‘Non-Responsive’ ratings
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The Economic Analysis Review and Statutory Prioritization

Economic Analysis and Statutory Prioritization for Resilience Improvement, and Community Resilience and Evacuation Routes

FHWA will conduct an economic analysis for certain types of projects that receive an overall rating of Highly Recommended or Recommended, including for Resilience Improvement projects which are not on an applicable Resilience Improvement Plan under 23 U.S.C. § 176(e)(2), and Community Resilience and Evacuation Route projects that receive an overall rating of Highly Recommended or Recommended, FHWA will consider the costs and benefits of these projects seeking PROTECT Discretionary Grant Program funding in determining whether a project is cost effective. In addition, FHWA will consider: (1) the need to address the vulnerabilities of surface transportation assets of the eligible entity with a high risk or, and impacts associated with, failure because of the impacts of weather events, natural disasters, or changing conditions for Resilience Improvement projects; and (2) take into account current and future vulnerabilities to an evacuation route; and projected changes in development patterns, demographics, and extreme weather events based on the best available evidence and analysis for Community Resilience and Evacuation Route projects.

To the extent possible, FHWA will rely on the applicant’s submission of well-supported BCA analysis results described in the project narrative. FHWA acknowledges that some aspects of calculating resilience benefits are difficult to quantify (e.g., benefits associated with nature-based solutions or for documenting benefit of evacuation during emergency events). Applicants should nonetheless discuss these types of benefits qualitatively.

For Resilience Improvement projects which are not on the applicable Resilience Improvement Plan and Community Resilience and Evacuation Routes projects, FHWA will consider a project’s benefits as compared to its costs to determine whether a project is cost effective and assign an economic analysis rating. To the extent possible, FHWA will rely on quantitative, evidence-based, and data-supported analysis in this assessment. FHWA will also consider qualitative descriptions of benefits that are more difficult to quantify and/or value in monetary terms.

Based on FHWA’s assessment, FHWA will assign an economic analysis rating of high, medium-high, medium, medium-low, or low according to the following table:

Rating	Description
High	The project’s benefits will exceed its costs. with a benefit-cost ratio of at least 1.5

Rating	Description
Medium-High	The project's benefits will exceed its costs
Medium	The project's benefits are likely to exceed its costs
Medium-Low	The project's costs are likely to exceed its benefits
Low	The project's costs will exceed its benefits

Resilience Improvement projects that are not on the applicable Resilience Improvement Plan and receive an economic analysis rating of Medium, Medium-High, or High will receive a determination that the benefits of the eligible activity proposed to be carried out by an eligible entity exceed the cost of the activity under 23 U.S.C. § 176(d)(4)(A)(iii)(I)(aa).

Community Resilience and Evacuation Routes projects that receive an economic analysis rating of Medium, Medium-High, or High will receive a determination that the eligible activity is cost-effective after taking into account:

- current and future vulnerabilities to an evacuation route because of future occurrence or recurrence of emergency events that are likely to occur in the geographic area in which the evacuation route is located and
- projected changes in development patterns, demographics, and extreme weather events based on the best available evidence and analysis under 23 U.S.C. § 176(d)(4)(B)(iii).

Resilience Improvement projects will be given priority if they receive an economic analysis rating of Medium, Medium-High, or High and FHWA determines that there is a need to address the vulnerabilities because of the high risk of, or impacts associated with failure because of, the impacts of weather events, natural disasters, or changing conditions. However, projects that receive a rating of Medium-Low or Low may be selected for an award if the project demonstrates clear potential benefits to resiliency, connectivity, community engagement, quality of life for disadvantaged communities, particularly in geographically remote or less populated areas.

Resilience Improvement projects that are on the applicable Resilience Improvement Plan will be prioritized. Applicants should consult with their State DOT or MPO regarding applicable Resilience Improvement Plans.

FHWA FY 2022 and FY 2023 Priority Considerations for Resilience Improvements, Community Resilience and Evacuation Routes, and At-Risk Coastal Infrastructure Grants

After completing the merit review, among projects of similar merit, FHWA will prioritize projects that meet one or more of the following:

1. Exceptional benefits under merit criteria #5 Equity and Justice⁴⁰
2. Workforce Development, Job Quality, and Wealth Creation

Application reflects one or more of the following:

- Include project labor agreements;⁷
- Describe how the project will expand strong labor standards, including not only compliance with prevailing wage requirements but also non-construction labor provisions that are relevant to the project such as transit, railroad, and Buy America protections;
- Commit to registered apprenticeship positions and use apprentices on the funded project, sometimes called an apprenticeship utilization requirement (e.g., requiring that a percentage of all labor hours will be performed by registered apprentices);
- Include high-quality workforce development programs with supportive services to help train, place, and retain people in good-paying jobs or registered apprenticeship;
- Demonstrate clear utilization of local and economic hiring preferences that ensure workers on the project come from economically disadvantaged communities⁸;
- Track and publish aggregate workforce data, including information on demonstrating that employment opportunities are available to historically underserved workers in their communities;
- Identify training programs that are diverse, including pre-apprenticeship and apprenticeship readiness programs, and explain how individuals from these programs are considered for and hired for apprenticeship slots and other jobs on the project;

⁷ These agreements may include pre-hire collective bargaining agreements between unions and contractors that govern terms and conditions of employment for all workers on a construction project.

⁸ These preferences may include goals for contractors based on geographic areas, geographic areas defined by economic distress, or individual characteristics of workers that make them economically disadvantaged. Project sponsors ask contractors to report data on project work hours, disaggregated across the different groups covered in local and economic hiring preferences. Alternatively, the project applicant may have documented agreements or other policy or ordinance in place to hire from economically disadvantaged groups.

- Include Local inclusive economic development and entrepreneurship such as the utilization of DBE, Minority-owned Businesses, Women-owned Businesses, or 8(a) firms;
- Expand access to goods and job opportunities through new or improved freight access; or
- Describe a State/regional/local comprehensive plan to promote equal opportunity, including removing barriers to hire and preventing harassment on work sites, and ensure that plan demonstrates action to create an inclusive environment with a commitment to equal opportunity, including:
- Affirmative efforts to remove barriers to Equal Employment Opportunity (EEO) above and beyond complying with Federal law and proactive partnerships with the U.S. Department of Labor's (DOL) Office of Federal Contract Compliance Programs (OFCCP) to promote compliance with the requirements of E.O. 11246, EEO;
- No discrimination in the use of criminal background screens and taking affirmative steps to recruit and include those with prior convictions, in accordance with the Fair Chance Act and equal opportunity requirements;
- Efforts to prevent harassment based on race, color, religion, sex, sexual orientation, gender identity, ability, and national origin;
- The inclusion of training on anti-harassment and third-party reporting procedures, and robust anti-retaliation measures, covering employees and contractors.

3. Construction Readiness

Priority may be given to a project that is post-design with incorporation of evidence-based climate change measures and features to address future vulnerabilities. Projects should be ready to proceed to construction within 10 months of selection to receive a PROTECT Discretionary Grant Program grant. FY 2022 funds may be prioritized for construction ready projects because of the need to be obligated before the end of 2025.

4. Funding Needs

FHWA will assess whether, but for the award of a PROTECT Discretionary Grant Program Resilience Grant, the applicant would not be able to begin or complete the activity. For State DOT applicants, this could be demonstrated by showing that the need exceeds the amount provided under the PROTECT Formula Program.

2. Review and Selection Process

FHWA will conduct an application intake and eligibility review by providing an initial eligibility screening based on the statutory eligibility criteria in Section C. Only eligible projects will be evaluated against the Merit Criteria in Section E.

The PROTECT Discretionary Grant Program application evaluation process consists of (1) a technical review, economic analysis review with statutory prioritization, and (2) an FHWA and the Office of the Secretary (OST) Senior Review.

The Technical Review

The evaluation of eligible applications will be conducted by a Technical Review Team who may consist of individuals from: (1) FHWA Headquarters (HQ) and division offices; (2) Federal Transit Administration; (3) Maritime Administration; and (4) OST. The Technical Review Team will evaluate each eligible application against the Merit Criteria in section E.1. and assign an overall application rating.

Planning Grant Applications. After assigning an overall rating to each Planning grant application, the Technical Review Team will evaluate and prepare a narrative on each application's responsiveness to FHWA Priority Considerations.

Resilience Grants. After assigning an overall rating to each Resilience Improvement and Community Resilience and Evacuation Routes grant application, the Technical Review Team will evaluate and prepare a narrative on each application's responsiveness to FHWA Priority Considerations.

The Economic Analysis Review

The economic analysis evaluation of Highly Recommended and Recommended Resilience Improvement and Community Resilience and Evacuation Route applications will be conducted by the Economic Analysis Review Team. The Economic Analysis Review Team may consist of individuals from: (1) FHWA Office of Transportation Policy Studies; (2) FHWA Resource Center; (3) other FHWA HQs and division offices; (4) and OST. The Economic Analysis Team will assess the benefit-cost analysis of Community Resilience and Evacuation Route projects, and Resilience Improvement projects not on the applicable Resilience Improvement Plan.

Projects that receive a rating of Medium-Low or Low may be selected for an award if the project demonstrates clear potential benefits to resilience, connectivity, community engagement, quality of life for economically disadvantaged communities, particularly in geographically remote or rural areas.

Applications Recommended for Consideration by the FHWA Senior Review Team (SRT)

The Technical Review Team will send to the FHWA SRT the overall project rating for each eligible Planning Grant and the evaluation of responsiveness to FHWA Priority Considerations. For the Resilience Improvement, and Community Resilience and Evacuation Route grant applications, the Technical Review Team will send to the FHWA SRT the Highly Recommended and Recommended projects that meet the Statutory Priority Criteria, the results of the economic analysis review, and the evaluation of responsiveness to FHWA Priority Considerations.

For the At-Risk Coastal Infrastructure applications, the Technical Review Team will send to the FHWA SRT the Highly Recommended and Recommended projects and the evaluation of responsiveness to FHWA Priority Considerations.

FHWA Senior Review Team Process and Project Selection

The FHWA SRT consists of senior FHWA officials who have been requested to serve by the FHWA Administrator. The SRT, with the FHWA Administrator's participation in the review process, advises the FHWA Administrator which projects the FHWA Administrator should select for funding.

Planning Grants: FHWA SRT will determine which Highly Recommended Planning projects to advance to the Administrator. FHWA SRT may also recommend for selection a Planning project that is rated Recommended over a Highly Recommended project. The SRT may recommend a Planning grant project that is rated Recommended over a Highly Recommended project, if the Recommended project is responsive to the Priority Considerations.

Resilience Grants: The FHWA SRT will determine which Highly Recommended Resilience Grant projects to advance to the Administrator. FHWA SRT may also recommend for selection a Resilience Grant project that is rated Recommended over a Highly Recommended project. The SRT may recommend a Resilience grant project that is rated Recommended over a Highly Recommended project if the Recommended project is particularly responsive to the Priority Considerations.

FHWA will use not less than 2 percent for grants to Indian Tribes. The FHWA SRT may recommend for funding all Highly Recommended applications under the set-aside and then recommend for funding additional Recommended applications up to the minimum amount. The FHWA reserves the flexibility to increase the minimum Tribal set-aside to fund all Highly Recommended Tribal projects and additional Recommended Tribal projects selected to receive funding. When selecting among applications, the FHWA SRT may consider a project's merit criteria ratings as well as responsiveness to the Priority Considerations. Any remaining Recommended applications will be considered for funding with the remaining non-Tribal applicants under review by the SRT.

FHWA Administrator will determine which Highly Recommended Planning Grant and Resilience Grant projects, and any Recommended Planning Grant and Resilience Grant projects, after considering the FHWA SRT recommendations including the responsiveness to FHWA Priority Considerations. The FHWA Administrator may consider options for reduced awards and ability to award certain eligible applications under any of the four grant types. To the extent possible, all applications will be considered for any of the four grant programs regardless of which grant is applied for.

The FHWA Administrator makes final project selections that best address the PROTECT Discretionary Grant Program goals for Planning, Resilience Improvements, Community Resilience and Evacuation Routes, and At-Risk Coastal Infrastructure Grants, Secretarial Statutory Selection Priorities, and FHWA Priority Considerations.

3. Additional Information

- a. FHWA may, during the selection process, enter into discussions with an applicant that may include mutually agreeing upon a lesser amount of a potential award than originally requested in the application, if necessary, because of the quantity, size and scope of the applications received in response to this notice and the results of the application review process. Discussions may include scalable project options as described under Section D.7.a. of this notice.
- b. Prior to award, each selected applicant will be subject to a risk assessment as required by 2 CFR 200.206. FHWA must review and consider any information about the applicant that is in the designated integrity and performance system accessible through SAM (currently the Federal Awardee Performance and Integrity Information System (FAPIIS)). An applicant may review information in FAPIIS and comment on any information about itself that a Federal awarding agency previously entered. FHWA will consider comments by the applicant, in addition to the other information in FAPIIS, in making a judgment about the applicant's integrity, business ethics, and record of performance under Federal awards when completing the review of risk posed by applicants.

SECTION F – FEDERAL AWARD ADMINISTRATION INFORMATION

1. FEDERAL AWARD NOTICES

Following the evaluation outlined in Section E, the Secretary will announce awarded projects by posting a list of selected projects at <https://www.fhwa.dot.gov/environment/protect/discretionary>. Notice of selection is not authorization to begin performance or to incur costs for the proposed project. Following that announcement, FHWA will contact the point of contact listed in the SF-424 to initiate negotiation of the project agreement for authorization. Recipients of PROTECT Discretionary Grant Program grant awards will not receive lump-sum cash disbursements at the time of award announcement or obligation of funds. Instead, FHWA will reimburse PROTECT Discretionary Grant Program funds to recipients only after a project agreement has been executed, allowable expenses are incurred, and valid requests for reimbursement are submitted. Unless authorized by FHWA in writing after announcement of FY 2022 and FY 2023 PROTECT Discretionary Grant Program awards, any costs that a recipient incurs before FHWA executes a project agreement for that recipient's project are ineligible for reimbursement and are ineligible match for cost share requirements.

2. ADMINISTRATIVE AND NATIONAL POLICY REQUIREMENTS

All awards will be administered pursuant to the Uniform Administrative Requirements, Cost Principles and Audit Requirements for Federal Awards found in 2 CFR Part 200, as adopted by FHWA at 2 CFR Part 1201. Applicable Federal laws, rules, and regulations set forth in title 23, U.S.C., and title 23 of the CFR, shall apply to awards provided under this program. In connection with any program or activity conducted with or benefiting from funds awarded under this notice, recipients of funds must comply with all applicable requirements of Federal law, including, without limitation, the Constitution of the United States; the conditions of performance, non-discrimination requirements, and other assurances made applicable to the

award of funds in accordance with regulations of DOT and FHWA; and applicable Federal financial assistance and contracting principles promulgated by the Office of Management and Budget. In complying with these requirements, recipients, in particular, must ensure that no concession agreements are denied, or other contracting decisions made on the basis of speech or other activities protected by the First Amendment. If FHWA determines that a recipient has failed to comply with applicable Federal requirements, FHWA may terminate the award of funds and disallow previously incurred costs, requiring the recipient to reimburse any expended award funds.

In particular, E.O. 14005 directs the Executive Branch Departments and agencies to maximize the use of goods, products, and materials produced in, and services offered in, the United States through the terms and conditions of Federal financial assistance awards. If selected for an award, grant recipients must be prepared to demonstrate how they will maximize the use of domestic goods, products, and materials in constructing their project. Any grant projects involving vehicle acquisition must involve only vehicles that comply with applicable Federal Motor Vehicle Safety Standards (FMVSS) and Federal Motor Carriers Safety Regulations (FMCSR), or vehicles that are exempt from FMVSS or FMCSR in a manner that allows for the legal acquisition and deployment of the vehicle or vehicles.

A. Program Requirements

i. Critical Infrastructure Security and Resilience

It is the policy of the United States to strengthen the security and resilience of its critical infrastructure against both physical and cyber threats. Each applicant selected for Federal funding under this notice must demonstrate, prior to the signing of the grant agreement, effort to consider and address physical and cyber security risks relevant to the transportation mode and type and scale of the project. Projects that have not appropriately considered and addressed physical and cyber security and resilience in their planning, design, and project oversight, as determined by FHWA and the U.S. Department of Homeland Security, will be required to do so before receiving funds for construction, consistent with Presidential Policy Directive 21 - Critical Infrastructure Security and Resilience and the National Security Presidential Improving Cybersecurity for Critical Infrastructure Control Systems.”

ii. Domestic Preference Requirements

As expressed in E.O. 14005, Ensuring the Future Is Made in All of America by All of America’s Workers (86 FR 7475), the executive branch should, consistent with law, maximize the use of goods, products, and materials produced in, and services offered in, the United States. Funds made available under this notice are subject to the domestic preference requirements at Pub. L. No 117-58, div. G §§ 70901–70927. FHWA expects all applicants to comply with those requirements.

The applicability of Federal requirements to a project may be affected by the scope of the National Environmental Policy Act (NEPA) reviews for that project. For example, under 23 U.S.C. § 313(g), Buy America requirements apply to all contracts that are eligible for assistance

under title 23, U.S.C., and are carried out within the scope of the NEPA finding, determination, or decision regardless of the funding source of such contracts if at least one contract is funded with Title 23 funds. As another example, Americans with Disabilities Act (ADA) regulations apply to all projects funded under this Notice.

iii. Civil Rights and Title VI

As a condition of a grant award, grant recipients should demonstrate that the recipient has a plan for compliance with civil rights obligations and nondiscrimination laws, including Title VI of the Civil Rights Act of 1964 and implementing regulations (49 CFR § 21), the ADA, and Section 504 of the Rehabilitation Act, all other civil rights requirements, and accompanying regulations. This should include a current Title VI Plan, completed Community Participation Plan, and a plan to address any legacy infrastructure or facilities that are not compliant with ADA standards. FHWA Office of Civil Rights may work with awarded grant recipients to ensure full compliance with Federal civil rights requirements.

iv. Federal Contract Compliance

As a condition of grant award and consistent with E.O. 11246, EEO (30 FR 12319, and as amended), all federally-assisted contractors are required to make good faith efforts to meet the goals of 6.9 percent of construction project hours being performed by women, in addition to goals that vary based on geography for construction work hours and for work being performed by people of color. Under Section 503 of the Rehabilitation Act and its implementing regulations, affirmative action obligations for certain contractors include an aspirational employment goal of 7 percent workers with disabilities.

The DOL's OFCCP is charged with enforcing E.O. 11246, Section 503 of the Rehabilitation Act of 1973, and the Vietnam Era Veterans' Readjustment Assistance Act of 1974. OFCCP has a Mega Construction Project Program through which it engages with project sponsors as early as the design phase to help promote compliance with non-discrimination and affirmative action obligations. The OFCCP will identify projects that receive an award under this notice and are required to participate in OFCCP's Mega Construction Project Program from a wide range of federally-assisted projects over which OFCCP has jurisdiction and that have a project cost above \$35 million. Additional information on how OFCCP makes their selections for participation in the Mega Construction Project Program is outlined under "Scheduling" on DOL's Website: <https://www.dol.gov/agencies/ofccp/faqs/construction-compliance>.

v. Climate Change and Environmental Justice

Each applicant selected for PROTECT grant funding must demonstrate effort to consider climate change and environmental justice impacts. Projects that have not sufficiently considered climate change and environmental justice in their planning, as determined by FHWA, will be required to do so before receiving funds, consistent with E.O. 14008, Tackling the Climate Crisis at Home and Abroad (86 FR 7619).

vi. Equity and Barriers to Opportunity.

Each applicant selected for PROTECT grant funding must demonstrate effort to improve equity and reduce barriers to opportunity. Projects that have not sufficiently considered equity and barriers to opportunity in their planning, as determined by FHWA, will be required to do so before receiving funds, consistent with E.O. 13985, Advancing Racial Equity and Support for Underserved Communities Through the Federal Government (86 FR 7009).

vii. Labor and Workforce

Each applicant selected for PROTECT grant funding must demonstrate, to the full extent possible consistent with the law, an effort to create good-paying jobs with the free and fair choice to join a union and incorporation of high labor standards. To the extent that applicants have not sufficiently considered job quality and labor rights in their planning, as determined by DOL, the applicants will be required to do so before receiving funds, consistent with E.O. 14025, Worker Organizing and Empowerment (86 FR 22829), and E.O. 14052, Implementation of the Infrastructure Investment and Jobs Act (86 FR 64335).

3. REPORTING

a. Progress Reporting on Grant Activities

Each applicant selected for PROTECT Discretionary Grant Program grant funding must submit semi-annual progress reports and Federal Financial Reports (SF-425) to monitor project progress and ensure accountability and financial transparency in the PROTECT Discretionary Grant Program.

b. Performance and Program Evaluation

As a condition of grant award, grant recipients may be required to participate in an evaluation undertaken by FHWA or another agency or partner. The evaluation may take different forms such as an implementation assessment across grant recipients, an impact and/or outcomes analysis of all or selected sites within or across grant recipients, or a benefit/cost analysis or assessment of return on investment. FHWA may require applicants to collect data elements to aid the evaluation and/or use information available through other reporting. As a part of the evaluation, as a condition of award, grant recipients must agree to: (1) make records available to the evaluation contractor or FHWA staff; (2) provide access to program records, and any other relevant documents to calculate costs and benefits; (3) in the case of an impact analysis, facilitate the access to relevant information as requested; (4) access to right-of-way to contractor or FHWA staff for long-term data and observation collection; and (5) follow evaluation procedures as specified by the evaluation contractor or FHWA staff.

FHWA is required (See 23 U.S.C. 176(f)(1)(B)) to establish impact and effectiveness metrics and evaluation procedures for the PROTECT Discretionary Grant Program. FHWA will select a representative sample of projects to evaluate their effectiveness and impact. Projects selected as part of this representative sample may have additional reporting requirements. FHWA will

publish the planned metrics in the Federal Register for public comment. These eventual metrics will not affect award decisions and should not impact applicant's proposals.

Recipients and subrecipients are also encouraged to incorporate program evaluation including associated data collection activities from the outset of their program design and implementation to meaningfully document and measure their progress towards meeting an agency priority goal(s). Title I of the Foundations for Evidence-Based Policymaking Act of 2018 (Evidence Act), Pub. L. No. 115-435 (2019) urges Federal awarding agencies and Federal assistance recipients and subrecipients to use program evaluation as a critical tool to learn, to improve equitable delivery, and to elevate program service and delivery across the program lifecycle. Evaluation means "an assessment using systematic data collection and analysis of one or more programs, policies, and organizations intended to assess their effectiveness and efficiency." 5 U.S.C. § 311. Credible program evaluation activities are implemented with relevance and utility, rigor, independence and objectivity, transparency, and ethics (OMB Circular A-11, Part 6 Section 290).

For grant recipients receiving an award, evaluation costs are allowable costs (either as direct or indirect), unless prohibited by statute or regulation, and such costs may include the personnel and equipment needed for data infrastructure and expertise in data analysis, performance, and evaluation. (2 CFR Part 200).

c. PROTECT Discretionary Grant Program Project Specific Reporting Requirements

Each applicant selected for PROTECT Discretionary Grant Program grant funding must collect and report to FHWA information on the project's performance based on performance indicators FHWA identifies related to program goals (e.g., increase resiliency, improve evacuation routes, strengthen at-risk coastal infrastructure project) and other information as requested by FHWA. Performance indicators should include measurable goals or targets that FHWA will use internally to determine whether the project meets program goals, and grant funds achieve the intended long-term outcomes of the PROTECT Discretionary Grant Program. FHWA will work with the grant recipient to establish a recommended two to four performance measures that enable FHWA to measure and evaluate the outcomes of the individual grant, relative to the program goals (identified in Section E.1.). Performance reporting continues for several years after project construction is completed. Monitoring and measurement costs can be included in project budgets.

d. Reporting of Matters Related to Recipient Integrity and Performance

If the total value of a selected applicant's currently active grants, cooperative agreements, and procurement contracts from all Federal awarding agencies exceeds \$10,000,000 for any period of time during the period of performance of this Federal award, then the applicant during that period of time must maintain the currency of information reported to the SAM that is made available in the designated integrity and performance system (currently FAPIIS) about civil, criminal, or administrative proceedings described in paragraph 2 of this award term and condition. This is a statutory requirement under section 872 of Public Law 110-417, as amended (41 U.S.C. § 2313). As required by section 3010 of Public Law 111-212, all information posted

in the designated integrity and performance system on or after April 15, 2011, except past performance reviews required for Federal procurement contracts, will be publicly available.

e. Other

FHWA reserves the right to request additional information, if deemed needed, to better understand the status of the project. The successful applicant will provide additional financial reporting beyond the semi-annual reporting, if such statements are necessary to address FHWA's Stewardship and Oversight responsibility of the funds. The successful applicant also agrees to allow periodic project inspections and FHWA will provide notice for such inspections.

SECTION G – FEDERAL AWARDING AGENCY CONTACTS

For questions concerning this NOFO and the PROTECT Discretionary Grant Program, please contact:

Robert Miller
Agreement Officer
Office of Acquisition and Grants Management
Federal Highway Administration
U.S. Department of Transportation
1200 New Jersey Ave., SE.
Washington, DC 20590
Email: PROTECTdiscretionary@dot.gov (preferred)
Phone: (202) 366-9167

Alternate:

Robin Hobbs
Agreement Officer/Team Leader
Office of Acquisition and Grants Management
Federal Highway Administration
U.S. Department of Transportation
1200 New Jersey Ave., SE.
Washington, DC 20590
Email: PROTECTdiscretionary@dot.gov (preferred)
Phone: (202) 366-4004
A TDD is available for individuals who are deaf or hard of hearing at (202) 366-3993.

Office hours are from 7:30 a.m. to 4:00 p.m., Eastern Time, Monday through Friday, except Federal holidays.

SECTION H – OTHER INFORMATION

1. Protection of Confidential Business Information.

All information submitted as part of or in support of any application shall use publicly available data or data that can be made public and methodologies that are accepted by industry practice and standards, to the extent possible. If the applicant submits information that the applicant considers to be a trade secret or confidential commercial or financial information, the applicant must provide that information in a separate document, which the applicant may cross-reference from the application narrative or other portions of the application. For the separate document containing confidential information, the applicant must do the following: (1) State on the cover of that document that it "Contains Confidential Business Information (CBI)"; (2) mark each page that contains confidential information with "CBI"; (3) highlight or otherwise denote the confidential content on each page; and (4) at the end of the document, explain how disclosure of the confidential information would cause substantial competitive harm. FHWA will protect confidential information complying with these requirements to the extent required under applicable law. If FHWA receives a Freedom of Information Act (FOIA) request for the information that the applicant has marked in accordance with this section, FHWA will follow the procedures described in its FOIA regulations at 49 CFR 7.29. Only information that is in the separate document, marked in accordance with this section, and ultimately determined to be confidential under § 7.29 will be exempt from disclosure under FOIA.

2. Publication/Sharing of Application Information.

Following the completion of the selection process and announcement of awards, FHWA may publish a list of all applications received along with the names of the applicant organizations and funding amounts requested. Except for the information properly marked as described in Section H.1., FHWA may make application narratives publicly available or share application information within FHWA or with other Federal agencies if FHWA determines that sharing is relevant to the respective program's objectives.

3. FHWA Feedback on Applications

Debriefs by FHWA are available to applicants not selected for award to receive information about the evaluation of their application.



APPENDIX M FHWA APPROVAL



U.S. Department
of Transportation
**Federal Highway
Administration**

Rhode Island Division

380 Westminister Street
Suite 601
Providence, RI 02903
(401) 528-4541
RIDIVFHWA@dot.gov

September 16, 2024

Mr. Peter Alviti, Jr., P.E.
Director
Rhode Island Department of Transportation
Two Capitol Hill
Providence, RI 02903-1124

Attention: Pamela Cotter, Administrator of Planning
Christos S. Xenophontos, Assistant Director for Administrative Services

Subject: Rhode Island Department of Transportation Resilience Improvement Plan

Dear Director Alviti:

The Federal Highway Administration (FHWA) hereby confirms that the Rhode Island Department of Transportation (RIDOT) Resilience Improvement Plan dated June 24, 2024, meets all the minimum requirements provided in 23 U.S.C. 176(e)(2).

We have noted that Rhode Island intends to incorporate the 2024 Resilience Improvement Plan (RIP) into the 2050 long-range statewide transportation plan and metropolitan transportation plan (Rhode Island's Long Range Transportation Plan or LRTP).

Below are two requests for future updates to the plan and to aid our office in future reviews.

- Future updates: Division requests a 30-day notification to review any updates to the plan, including time to review the integration of the RIP with the LRTP. The Division will provide a written response to acknowledge that the RIP has been incorporated into the LRTP to allow opportunity to reduce the non-Federal cost participation on projects identified in the Resilience Improvement Plan.
- Project approval in FHWA's Fiscal Management Information System (FMIS): Please include in the FMIS "notes" field a note identifying which projects are asking for a reduced federal share.¹ The note should include a reference to the date FHWA determined the plan to be complete and the date the plan was incorporated into the LRTP.²

¹ A 7% reduction in the non-Federal share can be applied if the state has developed a Resilience Improvement Plan in accordance with 23 U.S.C. 176(e) and prioritized the project in the plan.

² A 3% reduction in the non-Federal share can be applied if the state incorporates the Resilience Improvement Plan into the long-range statewide transportation plan under 23 U.S.C. 135, or an MPO incorporates the plan into its metropolitan transportation plan under 23 U.S.C. 134.

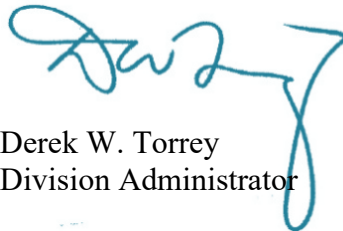
Mr. Peter Alviti, Jr., P.E.
September 16, 2024
Page 2

We would like to commend you and your staff for the broad participation in development of the RIP. The plan provides excellent information, methodology, and rationale for future climate projection data. It also includes well-documented evaluations of asset criticality, vulnerability, and potential climate threats and the risk-based assessment is a strong example of Promoting Resilient Operations for Transformative, Efficient, and Cost-Saving Transportation (PROTECT) Program requirements and best practices.

The RIDOT's RIP establishes a systemic approach to addressing climate and resilience needs and sets guidelines for how to better manage the uncertainty of climate risks to the transportation network. We look forward to seeing the RIP integrated with short and long-term planning efforts and the development of investment strategies to fund priority projects listed in the plan.

If you have any questions or would like to discuss this project in more detail, please contact me or Michael D'Alessandro, Planning and Environmental Program Manager, at (401) 330-5229 or michael.dalessandro@dot.gov.

Sincerely,



Derek W. Torrey
Division Administrator



RI
dot
Driven to get you there